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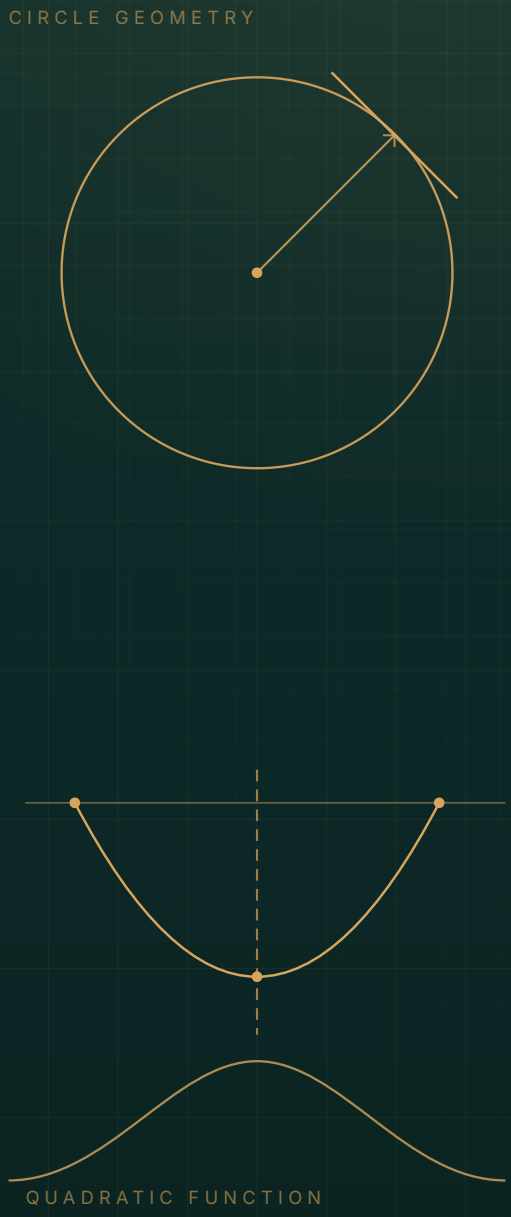
Mathematics

4

STUDENT COURSEBOOK

MYP 4 · YEAR 4 · THIRTEEN CHAPTERS AND A CAPSTONE IN FOUR PARTS

A concept-driven inquiry course in number, algebra, functions, shape and space, statistics and probability. You model, justify and apply, then reason, communicate and evaluate.



NUMBER FOUNDATIONS: EXPONENTS, SURDS AND ACCURACY ·
ALGEBRAIC EXPRESSIONS AND FACTORISATION ·
LINEAR EQUATIONS, INEQUALITIES AND SYSTEMS ·
FUNCTIONS, COORDINATES AND LINEAR MODELS ·
QUADRATIC EXPRESSIONS AND EQUATIONS ·
QUADRATIC FUNCTIONS AND GRAPHICAL REASONING ·
PATTERNS, SEQUENCES AND GENERALISATION ·
GROWTH AND FINANCIAL MATHEMATICS ·
CONGRUENCE, SIMILARITY AND TRANSFORMATIONS ·
PYTHAGORAS, TRIGONOMETRY AND MEASUREMENT ·
SURFACE AREA AND VOLUME ·
STATISTICS: DISTRIBUTIONS AND DATA ·
PROBABILITY: COMBINED EVENTS ·
CAPSTONE: DESIGNING A SUSTAINABLE SPACE

HOW THIS BOOK WORKS

Every chapter opens with a task. You investigate, notice a pattern, state a mathematical conclusion and justify it before you meet the formal explanation.

MYP ASSESSMENT CRITERIA

CRITERION A	Knowing and understanding
CRITERION B	Investigating patterns
CRITERION C	Communicating
CRITERION D	Applying mathematics in real-life contexts

SADIKOU MOUKODDIM

MYP 4 · STUDENT COURSEBOOK

AUTHOR SADIKOU MOUKODDIM

About this edition

This is the second edition of the MYP 4 mathematics coursebook. It was rebuilt from the pre-rewrite MYP 4 audit, which recorded the chapter scope, the key and related concepts, the global contexts and the inquiry questions of the earlier edition, and identified where the teaching had to be strengthened.

The edition keeps that chapter scope and the visual identity of the collection, and rebuilds the teaching sequence around inquiry: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, extensive visual practice, and application, assessment and reflection.

The earlier edition's course map and its chapter openers disagreed on the key concept and the global context in eight chapters. The opener labels are used here, because they are the ones the teaching actually develops, and each reconciliation is recorded in the alignment review. The one confirmed defect in the earlier text — a probability game described as always having a winner — is corrected in Chapter 13.

All classroom data sets are simulated and are labelled as such. Real data would require provenance, so no claim in this book rests on unverified statistics.

No publisher, ISBN, print specification or commercial print-readiness claim is made in this file. The book is produced as an A4 review copy: fonts are embedded, pagination is continuous and every page has been checked for overflow.

MYP 4 · STUDENT COURSEBOOK

How to use this book

Each chapter follows the same sequence, so you always know what is coming next. The sequence is deliberate: you investigate before you are told the rule, because a rule you have discovered is a rule you can defend — and defending it is what Year 4 asks of you.

The chapter sequence

1. **Engagement.** A short situation that raises the question the chapter will answer.
2. **Inquiry questions.** Factual, conceptual and debatable questions that frame the chapter.
3. **Investigation.** A task you complete before the formal teaching: record evidence in a table and look for a pattern.
4. **Observation and discovery.** What you noticed, stated in your own words and compared with a partner.
5. **Mathematical conclusion.** The rule that the evidence supports, written as a general statement you can justify.
6. **Formal development.** The explanation, definitions and notation, with the reasoning made explicit.
7. **Worked examples.** Problem, visual representation, mathematical reasoning, step-by-step solution, final answer and interpretation.
8. **Practice.** Separated into Criterion A, B, C and D practice, with visual representations throughout.
9. **Application, assessment and reflection.** Real-context tasks, criterion-based assessment practice, and a reflection that returns to the statement of inquiry.

MYP 4 · STUDENT COURSEBOOK

How to use this book

The four MYP mathematics criteria

A · Knowing and understanding

Apply knowledge to familiar and unfamiliar problems, and check results.

B · Investigating patterns

Investigate relationships, describe patterns, formulate a conjecture, test it and justify the generalisation — at Year 4 level a justification is a general argument, not a list of cases.

C · Communicating

Use correct mathematical language, notation and complete representations, and organise reasoning into a coherent structure that a reader can follow.

D · Applying mathematics in real-life contexts

Apply mathematics to authentic situations, identify relevant information, and comment on the accuracy and sense of the result.

Working with this book

- Practice is separated by criterion so that you can see which skill a task is training. Criterion B practice is additional genuine pattern investigation, not a repeat of the teaching.
- Visuals are part of the mathematics. Where a task supplies a diagram, grid, table, number line or model, use it; where it asks you to sketch, your drawing is part of the answer.
- Answers and student-facing reasoning are collected at the back of the book, with a formula bank, a glossary and references.
- Every chapter ends with project connections: a Service as Action opportunity, a Community Project connection and a Personal Project connection for MYP 5.
- The book closes with a capstone chapter: a single sustained modelling task that uses mathematics from every part of the year.

MYP 4 · STUDENT COURSEBOOK

MYP mathematics and this coursebook

The IB Middle Years Programme asks students to inquire into concepts through global contexts, to develop approaches to learning (ATL) skills, and to act on what they learn through service as action. In mathematics this means more than practising procedures: students investigate patterns, justify generalisations, communicate with precise language and apply mathematics to situations that matter to them.

MYP mathematics is organised into four branches — number, algebra, geometry and trigonometry, and statistics and probability — and the programme recommends at least 50 hours of teaching per subject each year. This coursebook covers the Year 4 content of those branches across thirteen chapters and a capstone, grouped into five parts.

Year 4 is the bridge level. The register for this year records the same four branches as every other year, and the chapters here follow the branch recorded for each of them: number and abstract reasoning, thinking with models, spatial reasoning, and reasoning with data, with an integrated capstone at the end.

Assessment in MYP mathematics uses four equally weighted criteria, each with a maximum of 8: A knowing and understanding, B investigating patterns, C communicating, and D applying mathematics in real-life contexts. This book trains each criterion explicitly, and says so on the page where the training happens.

MYP 4 · STUDENT COURSEBOOK

MYP mathematics and this coursebook

Projects, service and action

Service as Action

Each chapter includes a service opportunity with a community need, a mathematical connection, a proposed action, an expected impact and a student reflection. Completing an exercise is not service: service requires action that benefits others and reflection on that action.

Community Project

MYP students completing the programme in Year 3 or Year 4 complete a community project. Each chapter offers a student-led project idea with a research question, the mathematics involved, an action or outcome, beneficiaries and evidence. Follow your school's requirements and supervision.

Personal Project (MYP 5)

Students completing MYP in Year 5 complete a personal project. Each chapter offers an adaptable MYP 5-oriented extension with a personal interest, a learning goal, a product, the mathematics involved, a process and evidence. It is deliberately different from the community project and from service as action.

What this book does not claim

- Programme requirements quoted here follow the official IB MYP curriculum, mathematics and projects pages and the MYP general regulations. Everything else — the chapter sequence, the investigations, the practice sets and the project ideas — is author-designed material built to serve those requirements, and is not an IB publication.
- No ISBN, publisher, imprint or print specification is claimed. The file is an A4 review copy with embedded fonts and continuous pagination, produced for proofing and classroom review rather than for commercial print production.

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Application and assessment · Review · Projects and service

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Application and assessment · Review · Projects and service

Every chapter follows the same sequence: engagement, inquiry questions, investigation, observation, mathematical conclusion, formal development, worked examples, Criterion A–D practice, application, assessment and reflection, and project connections.

CHAPTER 1 · PART I · NUMBER AND ABSTRACT REASONING

01

Number Foundations: Exponents, Surds and Accuracy

The same number can be written as a fraction, a decimal, a power or a surd. Each form is exact or approximate, and each makes a different operation easy.

ENGAGE

A engineer specifies a bolt as 12.7 mm across. A machinist measures it as 12.73 mm. A calculator gives 12.732 051… mm. Which of these is the bolt, and which is the measurement?

Framing the inquiry

Key concept: Form
Related concepts: Simplification · Equivalence · Approximation · Quantity
Global context: Scientific and technical innovation
Global context exploration: Precision, accuracy and measurement

Statement of inquiry: The form in which a quantity is written — exact, approximate, or as a power — determines both the precision that can be claimed and the operations that can be performed, and simplification is the art of choosing the form that answers the question.

Questions to guide us

- **Factual:** What do the index laws say, and how is a number written in standard form?
- **Conceptual:** Why does the form of a quantity decide which operations are easy, and what does an error interval tell you that a rounded value cannot?
- **Debatable:** When a measurement is reported to the nearest millimetre, is the rounded value a lie, a convenience, or the most honest statement available?

ATL · CRITICAL THINKING. In order for a student to decide how exact an answer must be, students must compare two forms of the same quantity, identify what each form hides, and justify the choice of form for a stated purpose.

CHAPTER 1 · INVESTIGATION 1.1

Number systems and proof

INQUIRY QUESTION

How can a claim about all numbers be shown to be true rather than tested?

YOUR TASK

Take six numbers of your own: two rational, two irrational, and two you are unsure about. Add and multiply each pair, and decide for each result whether it is rational or irrational. Then try to state a rule that predicts the kind of the answer before you calculate it.

pair	operation	result	rational?
1/2 + 1/3	add	5/6	yes
1/2 + √2	add	irrational	no
√2 × √8	multiply	4	yes
√2 × √3	multiply	√6	no
√2 × 0	multiply	0	yes
√2 + (−√2)	add	0	yes

CHAPTER 1 · INVESTIGATION 1.1

Number systems and proof

What do you notice?

- Adding two rationals always gave a rational.
- Adding a rational and an irrational always gave an irrational.
- Multiplying two irrationals sometimes gave a rational and sometimes an irrational, so no single rule exists.
- Zero is the exception that breaks every multiplication rule, because anything times zero is zero.

> **Observation.** The kind of a result is fixed by the operation and the kinds of the inputs, except for multiplication, where the surds can combine into a perfect square.

MATHEMATICAL CONCLUSION

The rationals are closed under addition, subtraction, multiplication and division by a non-zero rational; the irrationals are closed under none of them, and this is why exact form matters when a calculation must stay exact.

Your own words

CHAPTER 1 · 1.1

Number systems and proof

The number system is built in stages: the counting numbers, then the integers, then the rationals, then the irrationals. Each stage exists because the previous one could not answer a question that arose naturally.

A rational number is one that can be written as a fraction of two integers. Its decimal either terminates or recurs. An irrational number cannot be written as a fraction, and its decimal neither terminates nor recurs. Equivalence between two forms of a number means that they name the same point, not that they look alike.

Proof at this level means a general argument. To show that the sum of two rationals is rational, write them as a/b and c/d and add: (ad + bc)/bd, which is again a fraction of integers. One argument covers every pair, so no list of examples is needed.

A counterexample is enough to destroy a universal claim. Finding ten examples that work is not enough to establish one, and this asymmetry is the whole logic of generalisation.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.1 · WORKED EXAMPLE 1

Number systems and proof

WORKED EXAMPLE 1.1.1

Problem. Show that the sum of two rational numbers is rational, and find an irrational number whose sum with 1 is rational.

number	form	kind
3/4	fraction	rational
0.25	terminating decimal	rational
√2	surd	irrational
π	non-repeating decimal	irrational

Form decides kind, and kind decides which operations stay inside the syste

Mathematical reasoning. Write both numbers as fractions and add; the result is again a fraction, which is the definition of rational.

1. Let the numbers be a/b and c/d with integers a, b, c, d and $b, d \neq 0$.
2. Their sum is $a/b + c/d = (ad + bc)/bd$.
3. $ad + bc$ and bd are integers and $bd \neq 0$, so the sum is rational.
4. For the second part, $\sqrt{2} - 1$ is irrational and $1 + (\sqrt{2} - 1) = \sqrt{2}$, so no such example exists with a rational result unless the irrational part is removed.

Final answer. the sum of two rationals is always rational; an irrational plus a rational is always irrational

Interpretation. One algebraic argument settles infinitely many cases, which is why proof is worth more than testing.

CHAPTER 1 · 1.1 · WORKED EXAMPLE 2

Number systems and proof

WORKED EXAMPLE 1.1.2

Problem. A student claims that the product of two irrational numbers is always irrational. Investigate.

first	second	product	kind of product
√2	√8	4	rational
√2	√3	√6	irrational
π	1/π	1	rational

Two irrationals can multiply to a rational or to an irrational.

Mathematical reasoning. A universal claim needs one counterexample to fall, and here there are several.

1. $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$, which is rational.
2. So the claim is false as stated.
3. But $\sqrt{2} \times \sqrt{3} = \sqrt{6}$, which is irrational, so no simple replacement claim works either.
4. The honest statement is that the product of two irrationals may be rational or irrational, and each case must be checked.

Final answer. the claim is false: $\sqrt{2} \times \sqrt{8} = 4$

Interpretation. A counterexample ends a universal claim, and finding one is a complete proof of its falsity.

CHAPTER 1 · INVESTIGATION 1.2

Integer and rational exponents

INQUIRY QUESTION

Why do the index laws hold for every exponent, not only for the ones we can count?

YOUR TASK

Build a table of powers of 2 from 2^{-3} to 2^4 and a matching table of powers of 3. Use only your tables to work out $2^{-3} \times 2^5$, $3^4 \div 3^6$ and $(2^3)^{2/3}$. Then explain why the answers must be what they are.

expression	using the laws	value
$2^{-3} \times 2^5$	$2^{-3+5} = 2^2$	4
$3^4 \div 3^6$	$3^{4-6} = 3^{-2}$	1/9
$(2^3)^{2/3}$	$2^{3 \times 2/3} = 2^2$	4
$2^{-3} \div 2^{-3}$	2^0	1
$(3^{-2})^{-1}$	3^2	9

CHAPTER 1 · INVESTIGATION 1.2

Integer and rational exponents

What do you notice?

- Every answer came from adding, subtracting or multiplying exponents, never from calculating the powers.
- A negative exponent always produced a reciprocal.
- A zero exponent always produced 1, which is what the quotient law requires.
- The fractional exponent undid the integer exponent exactly, because $3 \times 2/3 = 2$.

Observation. The index laws are the only rules needed, and they work for negative, zero and fractional exponents without any new idea.

MATHEMATICAL CONCLUSION

The index laws hold for all rational exponents because they are the unique extension that keeps $a^m \times a^n = a^{m+n}$ true, and a zero or negative exponent is a consequence of that requirement rather than a separate rule.

Your own words

CHAPTER 1 · 1.2

Integer and rational exponents

For positive integers the index laws are definitions of repeated multiplication: $a^m \times a^n$ means m copies of a times n copies of a, which is m + n copies.

The zero index is fixed by the quotient law. If $a^m \div a^n = a^{m-n}$ must hold when m = n, then $a^0 = a^m \div a^m = 1$, and this forces $a \neq 0$ because division by zero is undefined.

The negative index follows the same way. a^{-n} must satisfy $a^n \times a^{-n} = a^0 = 1$, so $a^{-n} = 1/a^n$. The laws are not memorised; they are the only extension that keeps them consistent.

A rational index is a root written as a power. a^(1/n) is the number whose nth power is a, and a^(m/n) = $(\sqrt[n]{a})^m$. This unifies roots and powers into one notation.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.2 · WORKED EXAMPLE 1

Integer and rational exponents

WORKED EXAMPLE 1.2.1

Problem. Evaluate 8^(2/3) × 8^(1/3) ÷ 8^(−1).

step	working	value
8^(2/3)	cube root then square	4
× 8^(1/3)	4 × 2	8
÷ 8^(−1)	8 ÷ 1/8	64

one base throughout, so the exponents add and subtract

Mathematical reasoning. All three factors have base 8, so the exponents combine by the product and quotient laws.

1. $8^{(2/3)} \times 8^{(1/3)} = 8^{(2/3 + 1/3)} = 8^{1}$.
2. $8^{1} \div 8^{(-1)} = 8^{(1 - (-1))} = 8^{2}$.
3. $8^{2} = 64$.
4. Check: $8^{(2/3)} = 4$, $8^{(1/3)} = 2$, and $4 \times 2 \div (1/8) = 8 \times 8 = 64$.

Final answer. 64

Interpretation. The laws turn a calculation with roots and reciprocals into one line of arithmetic on exponents.

CHAPTER 1 · 1.2 · WORKED EXAMPLE 2

Integer and rational exponents

WORKED EXAMPLE 1.2.2

Problem. Solve 4^(x+1) = 8^(x−1).

side	as a power of 2
4^(x+1)	$(2^2)^{(x+1)} = 2^{(2x+2)}$
8^(x−1)	$(2^3)^{(x-1)} = 2^{(3x-3)}$

Both sides are powers of the same base.

Mathematical reasoning. Rewrite both sides with a common base, then equate the exponents because a one-to-one power function has one input per output.

1. $4 = 2^2$ and $8 = 2^3$.
2. So 2^(2x+2) = 2^(3x−3).
3. Equal powers of the same base have equal exponents: $2x + 2 = 3x - 3$.
4. Solving gives $x = 5$.
5. Check: $4^6 = 4096$ and $8^4 = 4096$.

Final answer. x = 5

Interpretation. A common base converts an exponential equation into a linear one, which is the standard first move.

CHAPTER 1 · INVESTIGATION 1.3

Standard form and estimation

INQUIRY QUESTION

Why is a × 10^{n} the most useful way to write a very large or very small quantity?

YOUR TASK

Find four quantities from science or everyday life: one very large, one very small, one a ratio, and one you can only estimate. Write each in standard form, then estimate a calculation that combines two of them and state the precision of your answer.

quantity	standard form	kind
distance to the Sun	1.5×10^{11} m	very large
width of a virus	1×10^{-7} m	very small
cells in the body	3.7×10^{13}	very large
body cells ÷ virus width	3.7×10^{20}	ratio

CHAPTER 1 · INVESTIGATION 1.3

Standard form and estimation

What do you notice?

- Every large quantity had a positive exponent and every small one a negative exponent.
- The ratio had an exponent equal to the difference of the two exponents.
- Rounding to one significant figure before dividing kept the arithmetic mental.
- The precision of the estimate was set by the least precise input.

Observation. Standard form separates the digits of a quantity from its size, so operations on very large and very small quantities become operations on small numbers and on exponents.

MATHEMATICAL CONCLUSION

A quantity is written in standard form when its size and its digits must be handled separately, and the precision claimed is decided by the least precise quantity in the calculation.

Your own words

CHAPTER 1 · 1.3

Standard form and estimation

Standard form writes a number as $a \times 10^n$ with $1 \le a < 10$ and n an integer. It separates the size of a quantity from its digits, so 3.2×10^8 and 3.2×10^{-8} can be compared at a glance.

Multiplying in standard form multiplies the coefficients and adds the exponents: $(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}$, then adjust so the coefficient lies in range.

Estimation is a deliberate choice of precision. Rounding every quantity to one significant figure before calculating gives an answer whose error can be bounded, which is what an engineer needs before a detailed calculation.

Significant figures count measured meaning from the first non-zero digit. A leading zero is never significant, and a trailing zero is significant only if the quantity is known to that precision.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 1 · 1.3 · WORKED EXAMPLE 1

Standard form and estimation

WORKED EXAMPLE 1.3.1

Problem. A red blood cell is about 0.000 007 5 m across and a grain of sand about 0.000 35 m. Write both in standard form and estimate how many cells fit across one grain.

object	ordinary form	standard form
red blood cell	0.0000075	7.5×10^{-6}
grain of sand	0.00035	3.5×10^{-4}
ratio	0.00035 ÷ 0.0000075	4.7×10^{1}

Standard form makes the sizes and their ratio readable.

Mathematical reasoning. Write each quantity as a coefficient times a power of ten, then divide the coefficients and subtract the exponents.

1. $0.000\,007\,5 = 7.5 \times 10^{-6}$ m.
2. $0.000\,35 = 3.5 \times 10^{-4}$ m.
3. The ratio is $(3.5 \div 7.5) \times 10^{\wedge}(-4 - (-6)) = 0.47 \times 10^{2}$.
4. Adjusting to standard form: 4.7×10^{1}, so about 47 cells fit across one grain.

Final answer. 7.5×10^{-6} m and 3.5×10^{-4} m; about 47 cells fit across

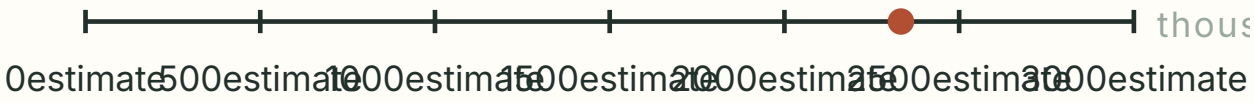
Interpretation. Standard form turns a comparison of six zeros into a comparison of two exponents.

CHAPTER 1 · 1.3 · WORKED EXAMPLE 2

Standard form and estimation

WORKED EXAMPLE 1.3.2

Problem. Estimate $(4.9 \times 10^{6}) \div (2.1 \times 10^{-3})$ and state the error interval of your estimate.



Mathematical reasoning. Round each coefficient to one significant figure, then divide the coefficients and subtract the exponents.

1. $4.9 \times 10^{6} \approx 5 \times 10^{6}$ and $2.1 \times 10^{-3} \approx 2 \times 10^{-3}$.
2. The estimate is $(5 \div 2) \times 10\text{^}(6 - (-3)) = 2.5 \times 10^{9}$.
3. The true value is about 2.333×10^{9}, so the estimate is about 7% high.
4. Stating one significant figure signals that the answer is an estimate, not a measurement.

Final answer. about 2.5×10^{9}, roughly 7% above the true value

Interpretation. An estimate is only useful if its precision is stated, because the point of estimating is to know how good the answer is.

CHAPTER 1 · INVESTIGATION 1.4

Surds and exact values

INQUIRY QUESTION

When is √12 a better answer than 3.464, and what does exact form let you do that a decimal cannot?

YOUR TASK

Draw a spiral of right-angled triangles, each with one leg of length 1 and the other leg the previous hypotenuse. Record each hypotenuse exactly and to four decimal places, then find the point at which the decimal form stops being useful.

step	exact	to 4 d.p.	kind
1	√1	1.0000	rational
2	√2	1.4142	irrational
3	√3	1.7321	irrational
4	√4 = 2	2.0000	rational
5	√5	2.2361	irrational
6	√6	2.4495	irrational

CHAPTER 1 · INVESTIGATION 1.4

Surds and exact values

What do you notice?

- Only the perfect squares gave rational hypotenuses.
- The decimal form of √2 repeats no pattern, so it can never be written exactly as a decimal.
- The exact form stays short while the decimal form grows without end.
- Adding the next triangle used the exact form, not the rounded one.

Observation. Exact form stays finite and computable where decimal form does not, and the two forms are the same number.

MATHEMATICAL CONCLUSION

An irrational quantity is best left in exact form while a calculation is in progress, and rounded once at the end, because the exact form carries no error and the decimal form carries an unbounded one.

Your own words

CHAPTER 1 · 1.4

Surds and exact values

A surd is an exact irrational root left in root form, such as $\sqrt{2}$ or $3\sqrt{5}$. It is exact because it names the number rather than approximating it.

Surds simplify by extracting square factors: $\sqrt{(ab)} = \sqrt{a} \times \sqrt{b}$, so $\sqrt{12} = \sqrt{(4 \times 3)} = 2\sqrt{3}$. Simplification makes two surds comparable, because $2\sqrt{3}$ and $\sqrt{12}$ are the same number in different forms.

Rationalising a denominator multiplies by the conjugate so that the denominator becomes rational: $1/(\sqrt{3}) = \sqrt{3}/3$, and $1/(2 + \sqrt{3}) = 2 - \sqrt{3}$, because $(2 + \sqrt{3})(2 - \sqrt{3}) = 1$.

Exact form is not pedantry. A calculation carried out in exact form and rounded once at the end has less error than the same calculation rounded at every step, and the difference grows with the number of steps.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

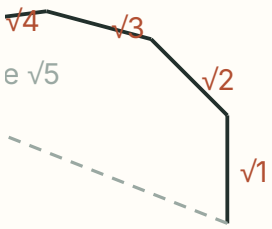
CHAPTER 1 · 1.4 · WORKED EXAMPLE 1

Surds and exact values

WORKED EXAMPLE 1.4.1

Problem. Simplify √72 + √50 − √18 and hence estimate its value.

√1, √2, √3, √4, √5 as hypotenuses



Mathematical reasoning. Extract the square factor from each surd, then collect like terms.

1. √72 = √(36 × 2) = 6√2.
2. √50 = √(25 × 2) = 5√2.
3. √18 = √(9 × 2) = 3√2.
4. So the expression is 6√2 + 5√2 − 3√2 = 8√2.
5. Since √2 ≈ 1.414, the value is about 11.31.

Final answer. 8√2, which is about 11.31

Interpretation. Three unrelated-looking surds collapse into one term once they are written in the same form, which is what simplification is for.

CHAPTER 1 · 1.4 · WORKED EXAMPLE 2

Surds and exact values

WORKED EXAMPLE 1.4.2

Problem. Rationalise the denominator of $6/(\sqrt{5}-1)$.

step	expression
start	$6/(\sqrt{5}-1)$
multiply by the conjugate	$6(\sqrt{5}+1)/((\sqrt{5}-1)(\sqrt{5}+1))$
denominator	$5-1=4$
simplified	$(3/2)(\sqrt{5}+1)$

The conjugate removes the surd from the denominator.

Mathematical reasoning. Multiplying by $(\sqrt{5}+1)/(\sqrt{5}+1)$ is multiplying by 1, so the value is unchanged while the denominator becomes rational.

1. $(\sqrt{5}-1)(\sqrt{5}+1)=5-1=4$, by the difference of squares.
2. So $6/(\sqrt{5}-1)=6(\sqrt{5}+1)/4$.
3. Simplifying the fraction gives $(3/2)(\sqrt{5}+1)$.
4. Numerically that is about 4.854, and $6/(\sqrt{5}-1)\approx 6/1.236\approx 4.854$.

Final answer. $(3/2)(\sqrt{5}+1)$

Interpretation. Rationalising does not change the number; it changes the form so that the value can be read or compared more easily.

CHAPTER 1 · INVESTIGATION 1.5

Accuracy, bounds and error

INQUIRY QUESTION

What does a rounded value actually promise, and how do the promises combine?

YOUR TASK

Measure the same object three times with three different instruments: a ruler to the nearest millimetre, a tape to the nearest centimetre, and a digital caliper to 0.01 mm. Record each reading, find its error interval, and decide which instrument is good enough for a stated purpose.

instrument	reading	error interval	relative error
ruler	12.4 cm	±0.05 cm	0.4%
tape	12 cm	±0.5 cm	4%
caliper	12.43 cm	±0.005 cm	0.04%

CHAPTER 1 · INVESTIGATION 1.5

Accuracy, bounds and error

What do you notice?

- All three instruments measured the same object and gave three different-looking answers.
- The error interval, not the number of digits shown, is what distinguishes them.
- The caliper's relative error is ten times smaller than the ruler's.
- The best instrument depends on the purpose, not on the instrument.

Observation. An instrument's precision is a property of its smallest division, and the error interval is the honest statement of what a reading claims.

MATHEMATICAL CONCLUSION

A measurement is reported with its error interval, and the fitness of a measurement is judged against the purpose it must serve rather than against the number of digits it happens to show.

Your own words

CHAPTER 1 · 1.5

Accuracy, bounds and error

A value given to the nearest unit lies within half a unit of the true value. That interval is the error interval, and it is what the rounded value actually claims. Approximation is therefore a claim with a width, not a single number.

Bounds combine by the operation. The upper bound of a sum uses the upper bounds of both terms; the upper bound of a difference uses the upper bound of the first and the lower bound of the second.

Truncation is not rounding. Truncating 3.97 to one decimal place gives 3.9, while rounding gives 4.0, and the two have different error intervals.

Relative error compares the error with the quantity itself. A 1 mm error matters in a 10 mm part and hardly at all in a 10 m bridge, so absolute error alone never says whether a measurement is good enough.

> FORMAL DEVELOPMENT
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> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

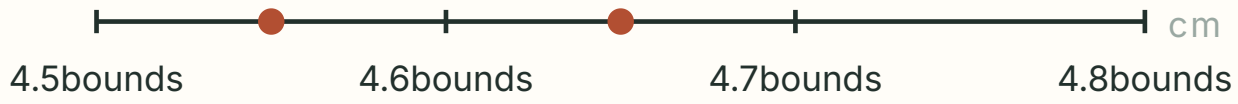
In your own words

CHAPTER 1 · 1.5 · WORKED EXAMPLE 1

Accuracy, bounds and error

WORKED EXAMPLE 1.5.1

Problem. A length is recorded as 4.6 cm to one decimal place. State its error interval, and find the error interval of 3 × this length.



Mathematical reasoning. The recorded value is within half of the last unit, and multiplication by a positive constant scales the interval.

1. The last unit is 0.1 cm, so half of it is 0.05 cm.
2. The length lies in [4.55, 4.65) cm.
3. Tripling gives [13.65, 13.95) cm.
4. Rounded to one decimal place the answer is between 13.6 and 13.9 cm, which is a wide interval for a single recorded measurement.

Final answer. the length is in [4.55, 4.65) cm and three times it is in [13.65, 13.95) cm

Interpretation. Multiplication magnifies absolute error, so a measurement that is fine on its own can be poor after it is scaled up.

CHAPTER 1 · 1.5 · WORKED EXAMPLE 2

Accuracy, bounds and error

WORKED EXAMPLE 1.5.2

Problem. A rectangular card measures 12.4 cm by 8.6 cm, each to one decimal place. Find the error interval of its area.

quantity	lower bound	upper bound
length	12.35	12.45
width	8.55	8.65
area	105.5925	107.6925

The extreme areas come from the extreme dimensions.

Mathematical reasoning. The largest area uses both upper bounds and the smallest uses both lower bounds, because both dimensions are positive.

1. Length is in [12.35, 12.45) cm and width in [8.55, 8.65) cm.
2. Minimum area = $12.35 \times 8.55 = 105.5925$ cm^2.
3. Maximum area = $12.45 \times 8.65 = 107.6925$ cm^2.
4. So the area is in [105.5925, 107.6925) cm^2, about 106 cm^2 to three significant figures.

Final answer. the area lies between 105.5925 cm² and 107.6925 cm²

Interpretation. The area is known to about 1%, even though each side is known to about 0.4%, because the relative errors add.

CHAPTER 1 · INVESTIGATION 1.6

Investigation: how exact is exact?

INQUIRY QUESTION

When is an exact form necessary, and when does it cost more than it is worth?

YOUR TASK

Choose one calculation that matters to you — a recipe scaled up, a journey cost, a savings target, or a measurement for a project. Carry it out three ways: exactly throughout, rounded at every step, and rounded once at the end. Compare the three answers and decide which you would report.

method	answer	error	cost
exact throughout	14060.80	0	most work
rounded each step	14060.80	0 here	least work
rounded once at end	14060.80	0 here	medium

CHAPTER 1 · INVESTIGATION 1.6

Investigation: how exact is exact?

What do you notice?

- With three years of compounding all three methods agreed.
- With thirty years they would not, and the exact form would win.
- The cost of exactness is arithmetic effort, not accuracy.
- The cost of rounding early is accuracy that cannot be recovered.

> **Observation.** Exactness is free in accuracy and expensive in effort; rounding is the reverse, and the trade-off is decided by the number of steps and the precision required.

MATHEMATICAL CONCLUSION

An exact form is used while a calculation is in progress and a rounded form is used to report its result, because the exact form carries no error and the rounded form communicates the precision actually achieved.

Your own words

CHAPTER 1 · 1.6

Investigation: how exact is exact?

An exact form is necessary when a later step would amplify the error of a rounded one, or when the answer must be checked symbolically rather than numerically.

An exact form costs more when the calculation is long, the numbers are awkward, and the required precision is low. In that case rounding early is efficient and the error is bounded.

The decision is a modelling decision: it depends on the required precision, the number of steps, and whether the result will be compared with a measurement.

A calculation carried out exactly and rounded once has less error than the same calculation rounded at every step, and the difference is the accumulated rounding error.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

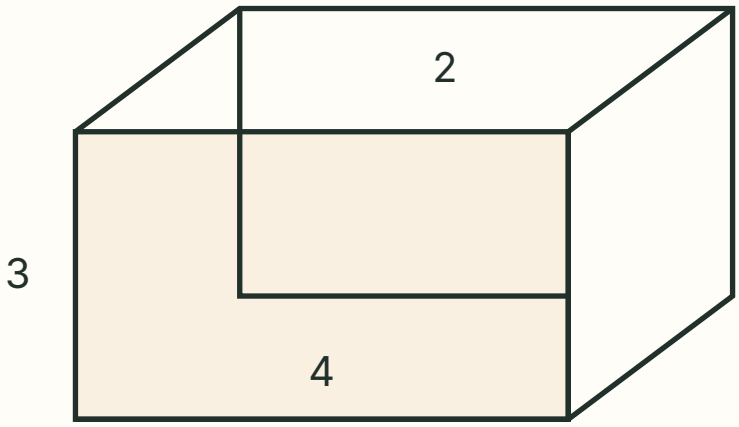
In your own words

CHAPTER 1 · 1.6 · WORKED EXAMPLE 1

Investigation: how exact is exact?

WORKED EXAMPLE 1.6.1

Problem. A cube has side 3.7 cm, measured to one decimal place. Find the error interval of its volume, first exactly and then with each side rounded to 3.7 from the start.



side 3.7 cm

Mathematical reasoning. The exact calculation keeps the bounds symbolic; the rounded calculation loses them immediately.

1. Side is in [3.65, 3.75) cm.
2. Exact: volume is in $[3.65^3, 3.75^3) = [48.627\,125, 52.734\,375)$ cm^3.
3. Rounded: volume = $3.7^3 = 50.653$ cm^3, which sits inside the interval but says nothing about it.
4. The interval is about 8% of the value, which is large for a measurement to one decimal place.

Final answer. the volume lies between 48.627 125 cm³ and 52.734 375 cm³

Interpretation. A single rounded value hides an interval that a power has magnified, because the relative error is multiplied by the exponent.

CHAPTER 1 · 1.6 · WORKED EXAMPLE 2

Investigation: how exact is exact?

WORKED EXAMPLE 1.6.2

Problem. A bank pays 4% interest on a deposit of 12 500 dollars for 3 years, compounded annually. Compare the answer calculated exactly with the answer rounded at each step.

year	exact balance	rounded to cents
0	12500.00	12500.00
1	13000.00	13000.00
2	13520.00	13520.00
3	14060.80	14060.80

Here the two agree, because the rounding is small and the number of steps

Mathematical reasoning. Compounding multiplies by the same factor each year, so the exact form is initial × multiplier^{n}.

1. The multiplier is 1.04, so the balance is 12 500 × 1.04^{3}.
2. Exactly that is 12 500 × 1.124 864 = 14 060.80 dollars.
3. Rounded to cents at each step gives the same figure here.
4. With a larger number of years the two methods would separate, and the exact form would be the honest one.

Final answer. 14 060.80 dollars, and the two methods agree at three years

Interpretation. Rounding at each step is harmless while the accumulated error is below the precision required, and the modeller's job is to know when that stops being true.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

1.A1 FLUENCY

Write $0.\overline{45}$ as a fraction.

1.A2 FLUENCY

Classify $\sqrt{16}$, $\sqrt{20}$ and $\sqrt[3]{8}$ as rational or irrational.

1.A3 REASONING

Show that $\sqrt{2} + \sqrt{3}$ is irrational.

1.A4 FLUENCY

Evaluate 27^(2/3).

1.A5 FLUENCY

Solve 2^(x+2) = 32.

1.A6 FLUENCY

Simplify $(3x^2y)^3 \div (9x^4y)$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

1.A7 FLUENCY

Write 0.000 000 42 in standard form.

1.A8 FLUENCY

Simplify $\sqrt{48}$.

1.A9 FLUENCY

Rationalise $5/\sqrt{7}$.

1.A10 FLUENCY

A mass is 3.40 kg to two decimal places. State its error interval.

1.A11 REASONING

Explain why 1/3 cannot be written exactly as a decimal.

1.A12 REASONING

Find the error interval of the area of a rectangle with sides 5.2 m and 3.1 m, each to one decimal place.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

1.B1 REASONING

The first four terms of a sequence of index values are 2^0, 2^1, 2^2, 2^3. Predict the next three and justify your rule.

1.B2 REASONING

Investigate whether $\sqrt{(n^2 + n)}$ is ever an integer for n a positive integer. State and justify a conclusion.

1.B3 REASONING

Build a table of $\sqrt{n}$ for n = 1 to 20 and describe the pattern of the gaps between consecutive values.

1.B4 REASONING

Conjecture a rule for the last digit of 2^n for positive integers n, test it, and justify it.

1.B5 REASONING

Investigate the values of n^(1/n) for n = 1 to 6 and predict where the maximum occurs.

1.B6 REASONING

The perimeters of the spiral triangles in section 1.4 form a sequence. Find its rule and test it on a term you did not draw.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

1.B7 REASONING

Investigate whether $n^3 - n$ is always divisible by 6 for positive integers n, and justify your conclusion.

1.B8 CHALLENGE

Conjecture a rule for the number of digits of 2^n and test it for n = 1 to 10.

1.B9 REASONING

Find the pattern in the sums 1/2, 1/2 + 1/4, 1/2 + 1/4 + 1/8 and justify the general rule.

1.B10 REASONING

Investigate the error intervals of √n when n is rounded to the nearest integer, for n from 1 to 10.

1.B11 REASONING

A pattern of tiles gives 1, 5, 13, 25 dots for figures 1 to 4. Find the rule and test it on figure 6.

CRITERION C Communicating

Criterion C practice · Communicating

1.C1 REASONING
Explain in your own words what an index law is and why it holds for negative exponents.

1.C2 FLUENCY
Write a complete solution to: simplify $\sqrt{200} - \sqrt{72}$, explaining each step.

1.C3 REASONING
Choose between 3.464 and $2\sqrt{3}$ as the answer to a calculation and justify your choice.

1.C4 FLUENCY
Set out a full solution to $(2 \times 10^{5}) \times (3 \times 10^{-8})$, labelling each step.

1.C5 REASONING
Explain the difference between rounding 3.97 to one decimal place and truncating it.

1.C6 FLUENCY
Describe, with a diagram, the error interval of a length recorded as 5.0 cm.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

1.C7 REASONING
Explain why the upper bound of a difference uses the lower bound of the second quantity.

1.C8 FLUENCY
Write the reasoning that shows $\sqrt{2} \times \sqrt{2} = 2$, using the index laws.

1.C9 REASONING
Set out the argument that the rationals are closed under addition, in a form another student could follow.

1.C10 FLUENCY
Explain what standard form is and give one large and one small example.

1.C11 REASONING
Describe the relationship between significant figures and precision.

1.C12 REASONING
Explain, with an example, why a counterexample is enough to disprove a universal claim.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

1.D1 APPLICATION

A pipe is specified as 2.50 m to the nearest centimetre. Find the error interval of its length and of three pipes joined end to end.

1.D2 APPLICATION

A sample of soil has mass 0.046 kg measured to the nearest gram. Write this in standard form and find its error interval.

1.D3 APPLICATION

A tank measures 1.2 m by 0.8 m by 0.5 m, each to one decimal place. Find the error interval of its capacity in litres and comment on whether it holds 480 litres.

1.D4 APPLICATION

A phone signal takes 0.000 000 13 s to travel 40 m. Estimate the speed and state the precision of your estimate.

1.D5 APPLICATION

A recipe for 4 people uses 0.35 kg of flour measured to the nearest 10 g. Scale it for 15 people and state the error interval.

1.D6 APPLICATION

A car's fuel consumption is 5.6 litres per 100 km, measured to one decimal place. Estimate the fuel for a 380 km trip and state the error interval.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

1.D7 REASONING

A builder needs a beam at least 3.00 m long. A supplier offers 2.98 m measured to the nearest centimetre. Is it acceptable? Justify.

1.D8 APPLICATION

A photograph is 12.4 cm by 8.6 cm, each to one decimal place. It is enlarged by a factor of 2.5. Find the error interval of the enlarged area and comment on the accuracy.

1.D9 REASONING

A bank rounds interest to the nearest cent each year. Explain whether the balance after 20 years is exact, and what a customer should ask for.

1.D10 REASONING

A nurse measures a dose as 2.5 ml to one decimal place. The safe range is 2.4 to 2.6 ml. Is the measurement safe? Justify.

1.D11 APPLICATION

A farmer's field is 120 m by 85 m, each measured to the nearest metre. Find the error interval of the area in hectares and comment on the effect on a yield estimate.

1.D12 REASONING

Explain, with a calculation of your own, when reporting a measurement to too many digits is misleading.

CHAPTER 1 · APPLICATION AND ASSESSMENT

Applying chapter 1

These tasks use index laws, standard form, surds and error intervals. State the form of every quantity you report, and say whether it is exact or approximate.

A1 · The bridge cable

A suspension bridge cable is specified as 48.6 m to one decimal place. The design load is calculated from the cross-sectional area, which is 2.4×10^{-3} m² measured to two significant figures. Find the error interval of the volume of cable needed, state it to a sensible precision, and explain whether reporting the answer to six significant figures would be honest.

Criteria A and D · carry the bounds through the calculation and comment on the precision you report.

A2 · The medicine dose

A syrup is supplied at a concentration of 4.5 mg per ml, measured to two significant figures. A patient needs a dose of 18 mg. A nurse measures the dose with a syringe marked to 0.1 ml. Find the error interval of the delivered dose, state whether it lies within 10% of 18 mg, and explain which measurement limits the accuracy.

Criteria A and C · set out the bounds of each measurement before combining them.

CHAPTER 1 · REVIEW

What you should now be able to do

Key ideas

- The rationals are closed under addition and multiplication; the irrationals are not.
- Index laws hold for negative, zero and rational exponents.
- Standard form separates the digits of a quantity from its size.
- A surd is exact; a decimal approximation is not.
- Rationalising multiplies by the conjugate and leaves the value unchanged.
- A rounded value lies within half of its last unit.
- Bounds combine according to the operation used.
- Relative error, not absolute error, decides whether a measurement is good enough.

Vocabulary

rational · irrational · surd · index · standard form · significant figures · error interval · bound · truncation · relative error · conjugate · rationalise

CHAPTER 1 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

1.R1 Simplify $\sqrt{98}$.

1.R2 Evaluate $16^{(-3/4)}$.

1.R3 Write 0.000 035 in standard form.

1.R4 A length is 7.3 cm to one decimal place. State its bounds.

1.R5 Rationalise $4/(3-\sqrt{2})$.

1.R6 Explain why $a^0 = 1$ for $a \neq 0$.

CHAPTER 1 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The form in which a quantity is written — exact, approximate, or as a power — determines both the precision that can be claimed and the operations that can be performed, and simplification is the art of choosing the form that answers the question.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of simplification, equivalence, approximation and quantity appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of precision, accuracy and measurement, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a rounded value for a decision about safety or cost, and what would you still like to ask?

CHAPTER 1 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Precision is a design decision. Choose the form that makes the quantity usable, and say what the choice costs.

Service as Action

COMMUNITY NEED	Public measurements — water volumes, medicine doses, construction sizes — are reported without their precision.
MATHEMATICAL CONNECTION	Error intervals, bounds and relative error.
PROPOSED ACTION	Collect three reported measurements, find the error interval of each, and publish what they actually claim.
EXPECTED IMPACT	A reader can tell how precise a reported number is.
STUDENT REFLECTION	Record the measurements, the intervals and what you found.

Community Project

PROJECT IDEA	A precision audit of the school's own measurements.
NEED	Timetables, budgets and lab results are quoted to different precisions without explanation.
RESEARCH QUESTION	How precise are the numbers the school publishes, and does the precision match the use?
MATHEMATICS	Bounds, error intervals, relative error and standard form.
ACTION OR OUTCOME	A one-page audit with the interval for each figure.
BENEFICIARIES	Everyone who relies on a school figure.
EVIDENCE	The audit sheet with intervals and a verdict for each figure.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cooking, sport, music or gaming.
LEARNING GOAL	To measure something you care about and state honestly how well you measured it.
PRODUCT OR OUTCOME	A measurement report with error intervals.
MATHEMATICS	Bounds, relative error and choice of form.
PROCESS	Measure three times, find the intervals, and report the precision.
EVIDENCE	The report and the intervals you found.

CHAPTER 2 · PART I · NUMBER AND ABSTRACT REASONING

02

Algebraic Expressions and Factorisation

The same expression can be expanded, factorised, or written as a single fraction. Choosing the right form is the whole skill, and every choice is reversible.

ENGAGE

A packaging designer needs a box whose surface area is $S = 2x^2 + 10x + 8$. Written this way the shape is invisible. What does the expression look like once it is factorised, and what does that tell you about the box?

Framing the inquiry

Key concept: Form
Related concepts: Equivalence · Simplification · Pattern · Representation
Global context: Scientific and technical innovation
Global context exploration: Design, optimisation and efficient form

Statement of inquiry: The form an algebraic expression is written in decides what it reveals — a factorised form shows roots, an expanded form shows coefficients, and a completed square shows a turning point — and equivalence between forms means each one can be rewritten without changing its value.

Questions to guide us

- **Factual:** What are the four methods of factorising a quadratic, and how is an algebraic fraction simplified?
- **Conceptual:** Why does factorising reveal information about a quadratic that expansion hides, and what does it mean for two expressions to be equivalent?
- **Debatable:** Is a factorised form always the most useful form of an expression, or does usefulness depend entirely on the question being asked?

ATL · TRANSFER. Students apply factorisation, the laws of indices and algebraic fractions to expressions that first appear in expanded form, and transfer the structure of a pattern they have seen numerically to an expression they have not met before.

CHAPTER 2 · INVESTIGATION 2.1

Expanding and simplifying

INQUIRY QUESTION

When do two different-looking expressions represent the same quantity?

YOUR TASK

Take the expression $(x + a)(x + b)$ and expand it for five pairs of values of a and b. Record the coefficients of x and the constant term, then predict the expansion of $(x + a)(x + b)$ for $a = 12$ and $b = -5$ without expanding.

a	b	x^2 coefficient	x coefficient	constant
2	3	1	5	6
−2	3	1	1	−6
4	−1	1	3	−4
−3	−3	1	−6	9
5	0	1	5	0

CHAPTER 2 · INVESTIGATION 2.1

Expanding and simplifying

What do you notice?

- The x^2 coefficient was always 1, because both leading terms were x.
- The x coefficient was always a + b.
- The constant term was always ab.
- The signs of a and b decided the signs of both results, exactly as multiplication of signs predicts.

Observation. The expanded form carries a + b and ab as its middle and constant terms, so the factorised form can be recovered from them.

MATHEMATICAL CONCLUSION

Expanding $(x + a)(x + b)$ gives $x^2 + (a + b)x + ab$, and factorising a monic quadratic is therefore the search for two numbers whose sum and product are the middle and constant terms.

Your own words

CHAPTER 2 · 2.1

Expanding and simplifying

Expanding multiplies out brackets using the distributive law: $a(b + c) = ab + ac$. The product of two binomials has four terms before any are collected.

Simplifying collects like terms and combines indices. Only terms with the same variables to the same powers can be combined, so $3x^2$ and $5x^2$ combine while $3x^2$ and $5x$ do not.

Equivalence is tested by substituting values or by rewriting. Two expressions are equivalent if they give the same value for every input, which is a stronger claim than agreeing at one value.

Simplification is the search for the shortest equivalent form, and it is always reversible, so no information is lost when an expression is simplified.

The product $(a + b)(c + d) = ac + ad + bc + bd$ is the engine of every expansion, and $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$ are the two special cases worth knowing.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.1 · WORKED EXAMPLE 1

Expanding and simplifying

WORKED EXAMPLE 2.1.1

Problem. Expand and simplify $(3x - 2)(2x + 5) - x(4x - 1)$.

method	working
product	$6x^2 + 15x - 4x - 10$
second product	$4x^2 - x$
subtract	$6x^2 + 11x - 10 - 4x^2 + x$
collect	$2x^2 + 12x - 10$

Four terms from the first product, two from the second.

Mathematical reasoning. Expand each product, subtract every term of the second, then collect like terms.

1. $(3x - 2)(2x + 5) = 6x^2 + 15x - 4x - 10 = 6x^2 + 11x - 10$.
2. $x(4x - 1) = 4x^2 - x$.
3. Subtracting gives $6x^2 + 11x - 10 - 4x^2 + x$.
4. Collecting like terms: $2x^2 + 12x - 10$.

Final answer. $2x^2 + 12x - 10$

Interpretation. The expanded form shows the coefficients directly, which is what makes it easy to read off the size of each term.

CHAPTER 2 · 2.1 · WORKED EXAMPLE 2

Expanding and simplifying

WORKED EXAMPLE 2.1.2

Problem. Show that $(x + 3)^2 - 9$ and $x^2 + 6x$ are equivalent, and say what each form reveals.

form	what it reveals
$(x + 3)^2 - 9$	a square shifted down by 9
$x^2 + 6x$	a quadratic with no constant term
$x(x + 6)$	roots at 0 and -6

Three forms of the same expression, three readings.

Mathematical reasoning. Expand the first expression and compare term by term with the second.

1. $(x + 3)^2 = x^2 + 6x + 9$.
2. So $(x + 3)^2 - 9 = x^2 + 6x$.
3. This is exactly the second expression, so they are equivalent.
4. The first shows a minimum at $x = -3$; the factorised form $x(x + 6)$ shows the roots.

Final answer. the two expressions are equivalent

Interpretation. Equivalence is proved by rewriting, and each form exposes a different feature of the same quantity.

CHAPTER 2 · INVESTIGATION 2.2

Common factors and grouping

INQUIRY QUESTION

How does a common factor reveal the structure of an expression?

YOUR TASK

Take ten expressions of your own: four with a common factor, four needing grouping in pairs, and two with neither. Factorise each where possible, record the method, and check every answer by expanding.

expression	method	factorised
$8x^2 + 12x$	common factor	$4x(2x + 3)$
$x^3 + x^2$	common factor	$x^2(x + 1)$
$ax + ay + bx + by$	grouping	$(x + y)(a + b)$
$2am + 2an + bm + bn$	grouping	$(m + n)(2a + b)$
$x^2 + x + 1$	neither	does not factorise

CHAPTER 2 · INVESTIGATION 2.2

Common factors and grouping

What do you notice?

- Every expression with a common factor gave a product immediately.
- Every four-term expression could be grouped once the pairs were chosen.
- The last expression had no factorisation, which is a legitimate outcome.
- Expanding each answer returned the original expression exactly.

> **Observation.** Factorising is the distributive law applied backwards, and the two routes into it are a common factor for the whole expression or a common factor for each of two groups.

MATHEMATICAL CONCLUSION

An expression is factorised by taking out the highest common factor, and when there is none for the whole expression the terms are grouped so that each group has one.

Your own words

CHAPTER 2 · 2.2

Common factors and grouping

Every term of an expression may share a factor, and taking out the highest common factor rewrites the expression as a product. This is the first thing to try, because it makes every later step smaller.

The highest common factor includes the largest number that divides every coefficient and the lowest power of every variable that appears in every term. Taking out a smaller factor leaves work still to do.

When there is no common factor for the whole expression, the terms can often be grouped in pairs so that each pair has one. Four-term expressions are the usual case, and the same bracket then appears twice.

Grouping is expanding in reverse: the distributive law is applied backwards. Checking by expanding confirms the factorisation, and the check is quick enough that it should always be done.

> FORMAL DEVELOPMENT
>
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In your own words

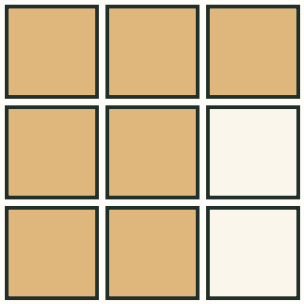
CHAPTER 2 · 2.2 · WORKED EXAMPLE 1

Common factors and grouping

WORKED EXAMPLE 2.2.1

Problem. Factorise fully $12x^3y - 18x^2y^2 + 6xy^3$.

each term shares the same shaded factor



6xy is common to all three terms

Mathematical reasoning. Find the highest common factor of the coefficients and of each variable, then divide every term by it.

1. The coefficients are 12, 18 and 6, whose highest common factor is 6.
2. The lowest power of x present in every term is x, and of y is y.
3. So the highest common factor is 6xy.
4. Dividing each term by 6xy gives $2x^2 - 3xy + y^2$.
5. Hence $12x^3y - 18x^2y^2 + 6xy^3 = 6xy(2x^2 - 3xy + y^2)$.

Final answer. $6xy(2x^2 - 3xy + y^2)$

Interpretation. Taking out the highest common factor first leaves a shorter expression, and the shorter expression is the one that can be examined for further structure.

CHAPTER 2 · 2.2 · WORKED EXAMPLE 2

Common factors and grouping

WORKED EXAMPLE 2.2.2

Problem. Factorise ax + ay + bx + by by grouping.

step	expression
group	(ax + ay) + (bx + by)
factor each pair	a(x + y) + b(x + y)
common bracket	(x + y)(a + b)

the same bracket appears in both groups

Mathematical reasoning. Group the terms in pairs, take out the factor of each pair, then take out the bracket the pairs now share.

1. Group as (ax + ay) + (bx + by).
2. The first pair has factor a and the second has factor b.
3. So the expression is a(x + y) + b(x + y).
4. Now (x + y) is common, so the expression is (x + y)(a + b).
5. Check by expanding: ax + ay + bx + by, which matches.

Final answer. (x + y)(a + b)

Interpretation. Grouping works whenever the terms can be paired so that the same bracket appears twice, and the expansion check confirms it.

CHAPTER 2 · INVESTIGATION 2.3

Quadratic factorisation

INQUIRY QUESTION

How is a quadratic turned into a product, and what does the product reveal?

YOUR TASK

Build a table of quadratics $x^2 + bx + c$ for b from −6 to 6 and c from −6 to 6, and mark which factorise over the integers. Then find the condition on b and c for factorising to be possible, and test it on a case you did not tabulate.

b	c	factorises?	factors
5	6	yes	$(x + 2)(x + 3)$
1	−6	yes	$(x − 2)(x + 3)$
2	5	no	—
−6	9	yes	$(x − 3)^2$
4	4	no	—

CHAPTER 2 · INVESTIGATION 2.3

Quadratic factorisation

What do you notice?

- Factorising worked whenever two integers had the right sum and product.
- The discriminant $b^2 - 4c$ was a perfect square in every successful case.
- $b = 2$, $c = 5$ failed because $b^2 - 4c = -16$ is negative.
- The same two numbers appear in the factors and in the discriminant.

Observation. Whether a monic quadratic factorises over the integers is decided by whether $b^2 - 4c$ is a perfect square, and the two numbers needed are half of b plus or minus half of that square root.

MATHEMATICAL CONCLUSION

Factorising over the integers is possible exactly when $b^2 - 4c$ is a perfect square, which is the first appearance of the discriminant as a test of structure rather than of calculation.

Your own words

CHAPTER 2 · 2.3

Quadratic factorisation

A monic quadratic $x^2 + bx + c$ factorises when two numbers can be found whose sum is b and whose product is c. The order of the pair does not matter, because multiplication is commutative.

When the leading coefficient is not 1, either split the middle term into two terms whose coefficients multiply to ac and add to b, or take out a common factor first if there is one.

A product is zero only when one of its factors is zero, so a factorised quadratic equation becomes two linear equations. This is the reason factorising is worth the effort.

Not every quadratic factorises over the integers, and recognising that quickly saves time. The test is whether $b^2 - 4ac$ is a perfect square, which is examined in chapter 5.

> FORMAL DEVELOPMENT
>
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In your own words

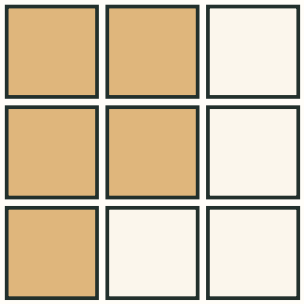
CHAPTER 2 · 2.3 · WORKED EXAMPLE 1

Quadratic factorisation

WORKED EXAMPLE 2.3.1

Problem. Factorise $2x^2 + 7x - 15$ fully and hence solve $2x^2 + 7x - 15 = 0$.

splitting the middle term into 10x and −3x



Mathematical reasoning. Split the middle term into two terms whose coefficients multiply to $2 \times (-15) = -30$ and add to 7.

1. Two numbers with product −30 and sum 7 are 10 and −3.
2. So $2x^2 + 7x - 15 = 2x^2 + 10x - 3x - 15$.
3. Grouping: $2x(x + 5) - 3(x + 5) = (2x - 3)(x + 5)$.
4. Setting each factor to zero gives $x = 3/2$ or $x = -5$.

Final answer. $(2x - 3)(x + 5)$, so $x = 1.5$ or $x = -5$

Interpretation. Factorisation converts a quadratic equation into two linear ones, which is the whole point of the technique.

CHAPTER 2 · 2.3 · WORKED EXAMPLE 2

Quadratic factorisation

WORKED EXAMPLE 2.3.2

Problem. Factorise $3x^2 + 12x$ and explain what the factorised form shows that the expanded form does not.

form	expression	what it shows
expanded	$3x^2 + 12x$	the two terms
factorised	$3x(x + 4)$	the roots 0 and −4

the factorised form answers a question the expanded form cannot

Mathematical reasoning. Take out the highest common factor, then read the roots from the product.

1. The highest common factor of $3x^2$ and $12x$ is $3x$.
2. So $3x^2 + 12x = 3x(x + 4)$.
3. Setting $3x(x + 4) = 0$ gives $x = 0$ or $x = -4$.
4. The expanded form gives no hint of either root.
5. The factorised form answers the question in one step.

Final answer. $3x(x + 4)$, with roots 0 and −4

Interpretation. The factorised form makes the roots visible, which is the information a factorisation is usually wanted for.

CHAPTER 2 · INVESTIGATION 2.4

Special products and identities

INQUIRY QUESTION

Why do the special products of a binomial deserve their own names?

YOUR TASK

Build a table of $(a + b)^2$ and $(a - b)^2$ for $a = 1$ to 5 and $b = 1$ to 5. Record the coefficients of the cross term and of the constant, then predict $(5x - 7)^2$ and $16x^2 - 49$ without expanding, and check one case by multiplication.

a	b	$(a + b)^2$	cross term	$a^2 - b^2$
1	1	$a^2 + 2ab + b^2$	2	0
1	2	$a^2 + 4ab + 4b^2$	4	−3
2	3	$4a^2 + 12ab + 9b^2$	12	−5
5	7	$25a^2 + 70ab + 49b^2$	70	−24
4	7	$16a^2 - 56ab + 49b^2$	−56	−33

CHAPTER 2 · INVESTIGATION 2.4

Special products and identities

What do you notice?

- The cross term was always 2ab, whatever a and b were.
- The constant term was always b^2.
- The difference $a^2 - b^2$ always factorised as $(a + b)(a - b)$.
- The prediction for $a = 5$, $b = 7$ was confirmed by direct multiplication.

Observation. The expansion of a squared binomial is completely determined by the two parts, and the difference of two squares is completely determined by them too.

MATHEMATICAL CONCLUSION

$(a + b)^2 = a^2 + 2ab + b^2$, $(a - b)^2 = a^2 - 2ab + b^2$ and $a^2 - b^2 = (a + b)(a - b)$ are general identities, so any specific case is a substitution rather than a new multiplication.

Your own words

CHAPTER 2 · 2.4

Special products and identities

Expanding a product of two binomials gives four terms, and in three important cases they collapse. Recognising the cases turns a multiplication into a substitution.

The perfect squares $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$ each have three terms, and the middle one is twice the product of the two parts.

The difference of two squares $a^2 - b^2 = (a + b)(a - b)$ has only two terms, and the absence of a middle term is the signal that identifies it.

These identities are equivalences, so they read in both directions. Reading them forwards expands and reading them backwards factorises, and the commonest error is to forget the cross term.

A pattern in a table of squares is one representation of these identities, and the representation as a diagram shows why the cross term must be there.

> FORMAL DEVELOPMENT
>
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In your own words

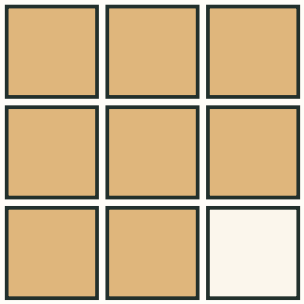
CHAPTER 2 · 2.4 · WORKED EXAMPLE 1

Special products and identities

WORKED EXAMPLE 2.4.1

Problem. Expand $(3x + 2)^2$ and explain why it is not $9x^2 + 4$.

a 3 by 3 square of side 3x + 2



$9x^2$, 12x and 4

Mathematical reasoning. Use the perfect square identity and identify the missing cross term.

1. $(3x + 2)^2 = (3x)^2 + 2(3x)(2) + 2^2 = 9x^2 + 12x + 4$.
2. The term 12x is the cross term, and it is what $9x^2 + 4$ leaves out.
3. The error is treating the square of a sum as the sum of the squares.
4. A check at x = 1 gives 25 against 13, so the claim fails.
5. One substitution is enough to refute an algebraic claim.

Final answer. $9x^2 + 12x + 4$

Interpretation. The cross term is the signature of a perfect square, and its absence is the commonest error in expanding.

CHAPTER 2 · 2.4 · WORKED EXAMPLE 2

Special products and identities

WORKED EXAMPLE 2.4.2

Problem. Factorise $9x^2 - 25$ and explain why the answer is not $(3x - 5)^2$.

form	structure
$9x^2 - 25$	difference of two squares
$(3x)^2 - 5^2$	$a^2 - b^2$ with $a = 3x$, $b = 5$
$(3x + 5)(3x - 5)$	factorised

A difference of two squares, not a perfect square.

Mathematical reasoning. Recognise $a^2 - b^2$ and apply the identity.

1. $9x^2 = (3x)^2$ and $25 = 5^2$.
2. So $9x^2 - 25 = (3x + 5)(3x - 5)$.
3. Squaring $(3x - 5)$ would give $9x^2 - 30x + 25$, which has a middle term.
4. A difference has no middle term, so the two are not the same.
5. The absence of a middle term is the signal for the identity.

Final answer. $(3x + 5)(3x - 5)$

Interpretation. Confusing a difference of squares with a perfect square is the commonest error with these identities, and the middle term is what distinguishes them.

CHAPTER 2 · INVESTIGATION 2.5

Algebraic fractions

INQUIRY QUESTION

Why must an algebraic fraction be simplified before it can be used?

YOUR TASK

Take six fractions of the form $(x^2 + bx + c)/(x^2 + dx + e)$ of your own choice. Factorise each, simplify, and record the excluded values. Then find one pair of fractions that simplify to the same expression but have different excluded values.

fraction	simplifies to	excluded
$(x^2 - 4)/(x^2 - x - 6)$	$(x + 2)/(x - 3)$	$x = 3, x = -2$
$(x^2 - 4)/(x^2 + x - 6)$	$(x - 2)/(x + 3)$	$x = -3, x = 2$
$(x^2 - 9)/(x^2 - 9)$	1	$x = 3, x = -3$
$(x^2 + 5x + 6)/(x^2 - 4)$	$(x + 3)/(x - 2)$	$x = 2, x = -2$

CHAPTER 2 · INVESTIGATION 2.5

Algebraic fractions

What do you notice?

- Cancelling changed the look of the fraction but never its domain.
- Two fractions can simplify to the same expression and still be different functions.
- A fraction that simplifies to 1 still has excluded values.
- The excluded values came from the denominator, not from the numerator.

Observation. A simplified fraction and its original are equal everywhere except at the values excluded by the original denominator, so simplification is an equivalence with holes.

MATHEMATICAL CONCLUSION

Simplifying an algebraic fraction is legitimate only when the excluded values are carried alongside the simplified form, because the denominator's zeros survive cancellation.

Your own words

CHAPTER 2 · 2.5

Algebraic fractions

An algebraic fraction is simplified by factorising numerator and denominator and cancelling common factors. Cancellation is only legal for factors, never for terms.

Multiplying fractions multiplies numerators and denominators; dividing multiplies by the reciprocal. Adding and subtracting needs a common denominator.

A fraction is undefined wherever its denominator is zero, and these excluded values must be stated. They are not removable by cancelling; they are properties of the original expression.

Simplifying first keeps expressions small and errors rare, and it is the algebraic version of choosing the efficient form.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 2 · 2.5 · WORKED EXAMPLE 1

Algebraic fractions

WORKED EXAMPLE 2.5.1

Problem. Simplify $(x^2 - 9)/(x^2 + x - 12)$ and state the excluded values.

part	factorised
numerator	$(x + 3)(x - 3)$
denominator	$(x + 4)(x - 3)$
cancelled	$(x + 3)/(x + 4)$
excluded	$x = -4, x = 3$

The common factor cancels, but the excluded value stays.

Mathematical reasoning. Factorise both parts, cancel the common factor, and state the values that made the original denominator zero.

1. $x^2 - 9 = (x + 3)(x - 3)$, a difference of squares.
2. $x^2 + x - 12 = (x + 4)(x - 3)$, because 4 and -3 have sum 1 and product -12.
3. Cancelling $(x - 3)$ gives $(x + 3)/(x + 4)$.
4. The original denominator is zero at $x = -4$ and $x = 3$, so both are excluded even though only -4 remains.

Final answer. $(x + 3)/(x + 4)$, with $x \neq -4$ and $x \neq 3$

Interpretation. Cancelling removes a factor from the expression but not from the domain, and stating the excluded values is part of the answer.

CHAPTER 2 · 2.5 · WORKED EXAMPLE 2

Algebraic fractions

WORKED EXAMPLE 2.5.2

Problem. Simplify $2/(x + 1) + 3/(x - 2)$ as a single fraction.

step	expression
common denominator	$(x + 1)(x - 2)$
rewrite	$2(x - 2)/((x + 1)(x - 2))$
rewrite	$3(x + 1)/((x + 1)(x - 2))$
add	$(2x - 4 + 3x + 3)/((x + 1)(x - 2))$
simplify	$(5x - 1)/((x + 1)(x - 2))$

One denominator, then one numerator.

Mathematical reasoning. Rewrite each fraction with the common denominator, add the numerators, and factorise the result.

1. The common denominator is $(x + 1)(x - 2)$.
2. $2/(x + 1) = 2(x - 2)/((x + 1)(x - 2))$ and $3/(x - 2) = 3(x + 1)/((x + 1)(x - 2))$.
3. Adding the numerators: $2x - 4 + 3x + 3 = 5x - 1$.
4. So the sum is $(5x - 1)/((x + 1)(x - 2))$.

Final answer. $(5x - 1)/((x + 1)(x - 2))$

Interpretation. A single fraction is easier to evaluate and to compare than two separate ones, which is why addition is always carried out this way.

CHAPTER 2 · INVESTIGATION 2.6

Rearranging formulae

INQUIRY QUESTION

Why does rearranging a formula preserve equality at every step?

YOUR TASK

Take five formulae from science, sport or home economics of your choice. For each, rearrange to make a different quantity the subject, and record the conditions introduced at each step.

formula	original subject	new subject	condition
$v = u + at$	v	t	$a \neq 0$
$A = \pi r^2$	A	r	$r \geq 0$
$P = 2(l + w)$	P	w	none
$F = ma$	F	m	$a \neq 0$
$s = d/t$	s	t	$t \neq 0$

CHAPTER 2 · INVESTIGATION 2.6

Rearranging formulae

What do you notice?

- Every rearrangement that divided by a variable introduced a condition.
- The formulae with two variables stayed symmetrical whichever was the subject.
- Making r the subject of $A = \pi r^2$ gave two roots, and only one was physically meaningful.
- Recording the conditions was as important as the rearrangement.

> **Observation.** Rearrangement is a chain of reversible operations, and each irreversible step adds a condition that must travel with the result.

MATHEMATICAL CONCLUSION

A rearranged formula is equivalent to the original only on the domain where every step is reversible, so the conditions are stated as part of the formula.

Your own words

CHAPTER 2 · 2.6

Rearranging formulae

Rearranging applies the same operation to both sides, so equality is preserved. Adding, subtracting, multiplying and dividing by a non-zero quantity are all reversible.

The subject of a formula is the quantity being found, and rearranging changes which quantity is the subject. A formula with two subjects is a relationship, not a recipe.

Squaring both sides can introduce solutions that do not satisfy the original, so it must be checked. Multiplying by an expression that could be zero can lose solutions, which is the opposite risk.

Efficient rearrangement chooses the order of operations that avoids fractions inside fractions, because each extra layer is a place where an error can hide.

The pattern of a rearrangement is the same for every formula of the same shape, so a method learned on one transfers to the next.

> FORMAL DEVELOPMENT
>
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In your own words

CHAPTER 2 · 2.6 · WORKED EXAMPLE 1

Rearranging formulae

WORKED EXAMPLE 2.6.1

Problem. Make r the subject of $A = \pi r^2 + 2\pi rh$.

step	working
start	$A = \pi r^2 + 2\pi rh$
group the r terms	$A = \pi r(r + 2h)$
divide	$A/(\pi r) = r + 2h$
isolate	$r = A/(\pi r) - 2h$

The radius appears twice, so it must be grouped first.

Mathematical reasoning. Factorise the right-hand side by r, then isolate r by reversible operations.

1. $A = \pi r^2 + 2\pi rh = \pi r(r + 2h)$.
2. Dividing by πr gives $A/(\pi r) = r + 2h$.
3. Subtracting $2h$ gives $r = A/(\pi r) - 2h$.
4. This still contains r on both sides, so the formula must instead be treated as a quadratic in r: $\pi r^2 + 2\pi hr - A = 0$.
5. Solving gives $r = -h + \sqrt{(h^2 + A/\pi)}$, taking the positive root.

Final answer. $r = -h + \sqrt{(h^2 + A/\pi)}$

Interpretation. Recognising that a rearrangement is a quadratic in disguise prevents a circular manipulation that never isolates the subject.

CHAPTER 2 · 2.6 · WORKED EXAMPLE 2

Rearranging formulae

WORKED EXAMPLE 2.6.2

Problem. Make x the subject of $(x + a)/(x + b) = c$, stating any restriction.

step	working
multiply by $x + b$	$x + a = c(x + b)$
expand	$x + a = cx + cb$
collect x	$x - cx = cb - a$
factorise	$x(1 - c) = cb - a$

Clearing the fraction makes the algebra linear.

Mathematical reasoning. Multiply by the denominator, expand, collect the terms in x, then factorise x out.

1. Multiply both sides by $(x + b)$: $x + a = c(x + b)$.
2. Expanding: $x + a = cx + cb$.
3. Collecting x terms: $x - cx = cb - a$.
4. Factorising: $x(1 - c) = cb - a$, so $x = (cb - a)/(1 - c)$.
5. This needs $c \neq 1$, and the original needs $x \neq -b$.

Final answer. $x = (cb - a)/(1 - c)$, with $c \neq 1$ and $x \neq -b$

Interpretation. Every division and every multiplication by a variable introduces a condition, and the conditions are part of the answer.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

2.A1 FLUENCY

Expand $(x + 5)(x - 3)$.

2.A2 FLUENCY

Factorise $x^2 - 7x + 12$.

2.A3 FLUENCY

Simplify $(x^2 - 16)/(x^2 - 8x + 16)$.

2.A4 FLUENCY

Make x the subject of $y = 3x - 7$.

2.A5 FLUENCY

Evaluate $(2x^3)^2$ when $x = -1$.

2.A6 FLUENCY

Expand and simplify $(x + 4)^2 - (x - 4)^2$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

2.A7 FLUENCY

Factorise $4x^2 - 49$.

2.A8 FLUENCY

Simplify $3/(x) + 2/(x + 1)$ as a single fraction.

2.A9 FLUENCY

Write $(x^{-2})^{-3}$ with a positive index.

2.A10 FLUENCY

Make a the subject of $v = u + at$.

2.A11 FLUENCY

Expand $(2x - 3y)^2$.

2.A12 FLUENCY

Simplify $(6x^3y^2)/(9x^5y)$.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

2.B1 REASONING

The perimeters of a sequence of rectangles are 6, 10, 14, 18 cm. Find the rule for the nth perimeter and test it on the fifth rectangle.

2.B2 REASONING

Investigate the areas of the same rectangles and find the rule.

2.B3 REASONING

Conjecture a rule for the number of diagonals of an n-sided polygon and test it for n = 4, 5, 6.

2.B4 REASONING

Investigate the pattern in 1^2, 2^2, 3^2 and the differences between consecutive squares.

2.B5 REASONING

Build a table of $x^2 + bx + c$ for b = 4 and c = −5, −4, −3, …, 5 and mark which factorise.

2.B6 REASONING

Conjecture a rule for the sum of the first n odd numbers and justify it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

2.B7 REASONING

Investigate whether $x^2 + x + 41$ is prime for every positive integer x, and state your conclusion.

2.B8 CHALLENGE

The sequence of expressions x, x(x + 1), x(x + 1)(x + 2) is built by multiplication. Predict the next and investigate its factors.

2.B9 REASONING

Find the pattern in the sums $1/(1\times2)$, $1/(1\times2) + 1/(2\times3)$, and justify the general rule.

2.B10 REASONING

Investigate the areas of squares drawn on the sides of a right-angled triangle with legs a and b.

2.B11 REASONING

Conjecture a rule for the last digit of n^5 for positive integers n and justify it.

CRITERION C Communicating

Criterion C practice · Communicating

2.C1 REASONING

Explain in your own words what it means for two expressions to be equivalent.

2.C2 FLUENCY

Write a complete solution to: factorise $3x^2 - 12$.

2.C3 REASONING

Choose between $(x + 2)(x + 3)$ and $x^2 + 5x + 6$ as the answer to a question about roots and justify.

2.C4 FLUENCY

Set out a full solution to: simplify $(x^2 - 25)/(x^2 - 10x + 25)$.

2.C5 REASONING

Explain why $(x + y)^2$ is not $x^2 + y^2$.

2.C6 REASONING

Describe, with an example, the difference between cancelling a factor and cancelling a term.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

2.C7 REASONING
Explain why the excluded values of a fraction must be stated.

2.C8 FLUENCY
Write the reasoning that shows $x^{-n} = 1/x^{n}$.

2.C9 REASONING
Set out the argument that the discriminant $b^2 - 4c$ decides whether a monic quadratic factorises.

2.C10 FLUENCY
Explain what the subject of a formula is and give an example.

2.C11 REASONING
Describe the relationship between factorising and solving an equation.

2.C12 REASONING
Explain, with an example, why squaring both sides of an equation can introduce false solutions.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

2.D1 APPLICATION

A rectangular garden has length $(2x + 6)$ m and width $(x - 1)$ m. Find its area in factorised form and the value of x that gives zero area.

2.D2 APPLICATION

A company's profit is $P = (x - 20)(200 - x)$ dollars for x items. Find the factorised form, the break-even values, and the number of items giving the greatest profit.

2.D3 APPLICATION

A tray's surface area is $S = 2x^2 + 14x + 24$ cm^2. Find the value of x that gives the smallest area and the area itself.

2.D4 APPLICATION

A phone plan costs $C = 20 + 0.05n$ dollars for n minutes. Rewrite this to make n the subject and find the minutes affordable for 35 dollars.

2.D5 REASONING

A photograph is enlarged by a factor of k and its area becomes $A = k^2(2x + 10)(x + 4)$. Find the factorised form and the value of k giving zero area.

2.D6 REASONING

A swimming pool's volume is $V = x(2x + 6)(x - 2)$ m^3. Find the value of x that gives the greatest volume for $0 < x < 2$ and justify.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

2.D7 REASONING

A bridge cable's tension is $T = (3x + 12)(x - 5)$ newtons. Explain whether a tension of zero is possible, and justify.

2.D8 APPLICATION

A shop's revenue is $R = x(40 - x)$ dollars. Find the price giving the greatest revenue and explain why the factorised form makes this easy.

2.D9 CHALLENGE

A field's yield is $Y = (x + 3)(x - 1)(10 - x)$ tonnes. Investigate the values of x in $1 < x < 10$ that maximise the yield and justify.

2.D10 APPLICATION

A car's value is $V = 12000(1 - r)^{n}$ dollars after n years. Rewrite to make r the subject and find the rate that halves the value in 3 years.

2.D11 APPLICATION

A cone's volume is $V = (1/3)\pi r^2 h$. Rewrite to make r the subject and explain which form is best for a given volume and height.

2.D12 REASONING

Explain, with a formula of your own, when a factorised form is misleading.

CHAPTER 2 · APPLICATION AND ASSESSMENT

Applying chapter 2

Each task asks you to choose a form and justify the choice. Show the rewriting, not just the answer.

A1 · The packaging problem

A box with a square base of side x cm and height 8 cm has surface area $S = 2x^2 + 32x + 128$ cm^2. Factorise S fully, find the value of x for which the surface area is zero, and explain why that value is not a useful answer for a designer. Then find the value of x that gives a surface area of 200 cm^2.

Criteria A and C · show the factorisation and the equation you solve, and comment on the physical meaning of each root.

A2 · The formula and the conditions

A formula for the speed v of a wave is $v = \sqrt{(T/\mu)}$, where T is tension and μ is mass per unit length. Rewrite the formula to make μ the subject, state the conditions that apply, and explain what happens to the speed when the mass per unit length is quadrupled.

Criteria A and D · square both sides, isolate μ, and check the direction of the change.

CHAPTER 2 · REVIEW

What you should now be able to do

Key ideas

- Expanding uses the distributive law; factorising reverses it.
- $(x + a)(x + b) = x^2 + (a + b)x + ab$.
- $a^2 - b^2 = (a + b)(a - b)$ is the difference of two squares.
- A product is zero only when a factor is zero.
- Algebraic fractions simplify by cancelling factors, never terms.
- Excluded values come from the original denominator and survive cancellation.
- Index laws apply to any base and any rational exponent.
- $(x + y)^2 = x^2 + 2xy + y^2$, not $x^2 + y^2$.
- Rearranging preserves equality only while every step is reversible.

Vocabulary

expand · factorise · equivalence · like terms · common factor · difference of squares · algebraic fraction · excluded value · subject · cross term · zero-product property · discriminant

CHAPTER 2 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

2.R1 Expand $(2x - 5)(x + 3)$.

2.R2 Factorise $3x^2 + 10x - 8$.

2.R3 Simplify $(x^2 - 1)/(x^2 + 2x + 1)$.

2.R4 Make t the subject of $s = ut + \frac{1}{2}at^2$.

2.R5 Evaluate $(27x^6)^{(1/3)}$ when $x = 2$.

2.R6 Explain why $x = 0$ is excluded from $1/(x^2 - x)$.

CHAPTER 2 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The form an algebraic expression is written in decides what it reveals — a factorised form shows roots, an expanded form shows coefficients, and a completed square shows a turning point — and equivalence between forms means each one can be rewritten without changing its value.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of equivalence, simplification, pattern and representation appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of design, optimisation and efficient form, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of choosing a form for convenience rather than for the question, and what would you still like to ask?

CHAPTER 2 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Efficient form is a design goal. Find an expression in the world around you and show which form answers the question being asked.

Service as Action

COMMUNITY NEED	Packaging and material use are decided without showing the algebra behind them.
MATHEMATICAL CONNECTION	Expanding, factorising and choosing an efficient form.
PROPOSED ACTION	Model the surface area of one real package in factored form and publish what the form reveals.
EXPECTED IMPACT	A reader sees why the shape of a box follows from its algebra.
STUDENT REFLECTION	Record the model, the factorisation and what it showed.

Community Project

PROJECT IDEA	An algebra audit of the school's own cost formulae.
NEED	Costs, timetables and catering figures are quoted as numbers with no visible structure.
RESEARCH QUESTION	Which forms of the school's own formulae make decisions easiest, and why?
MATHEMATICS	Expanding, factorising, rearranging and indices.
ACTION OR OUTCOME	A sheet showing three school formulae in two forms each, with a recommendation.
BENEFICIARIES	Anyone who has to use the school's figures.
EVIDENCE	The audit sheet with the forms and the reasoning.

Personal Project connection · MYP 5

PERSONAL INTEREST	Cooking, sport, gaming or music.
LEARNING GOAL	To model something you enjoy with an algebraic expression and choose the best form of it.
PRODUCT OR OUTCOME	A model with two forms and a justified choice.
MATHEMATICS	Expanding, factorising and rearranging.
PROCESS	Build the model, rewrite it in two forms, and test each against a real question.
EVIDENCE	The model and the comparison of the two forms.

CHAPTER 3 · PART II · THINKING WITH MODELS

03

Linear Equations, Inequalities and Systems

Two linear descriptions of one quantity agree at exactly one point. Several constraints acting together describe a region, and the answer is often a set of values rather than a single one.

ENGAGE

A school has 240 dollars to spend on 30 books. Some cost 6 dollars and some cost 10 dollars. How many of each can it buy, and how many answers are possible?

Framing the inquiry

Key concept: Relationships
Related concepts: Equivalence · Change · System
Global context: Fairness and development
Global context exploration: Collective decisions under constraints

Statement of inquiry: A linear equation records an equivalence between two descriptions of the same quantity, and solving it is the search for the value at which the two descriptions agree; when several constraints act at once, the feasible values are a system rather than a single number, and the solution is the region where all the constraints hold.

Questions to guide us

- **Factual:** What are the three algebraic methods for solving a pair of simultaneous linear equations, and how is a linear inequality solved?
- **Conceptual:** Why does solving a linear equation preserve equivalence, and what does the solution of a system of inequalities represent?
- **Debatable:** When a group must share a limited resource under several constraints, is there a single fair answer, or is fairness itself the thing being negotiated?

ATL · COMMUNICATION. Students move between the algebraic, tabular and graphical representations of the same linear relationship, and explain to a reader which representation answers the question asked.

CHAPTER 3 · INVESTIGATION 3.1

Linear equations and formulae

INQUIRY QUESTION

Why does every step in solving an equation preserve equivalence?

YOUR TASK

Take five linear equations of increasing difficulty: one with the unknown on one side, one with it on both, one with brackets, one with fractions, and one of your own invention. Solve each, record the sequence of steps, and check each solution in the original equation.

equation	steps used	solution	check
$5x + 3 = 18$	subtract, divide	3	$18 = 18$
$7x - 4 = 3x + 12$	collect, divide	4	$24 = 24$
$2(x + 5) = 3(x - 1)$	expand, collect	13	$36 = 36$
$x/2 + 1 = 4$	clear, subtract	6	$4 = 4$
$(x - 2)/5 = (x + 1)/3$	clear, collect	-5.5	check passes

CHAPTER 3 · INVESTIGATION 3.1

Linear equations and formulae

What do you notice?

- Every solution came from reversible steps only.
- The check always gave a true statement, which is the definition of a correct solution.
- The equations with brackets needed expansion before collecting.
- The equations with fractions needed one multiplication to clear them.

Observation. Solving a linear equation is a chain of reversible transformations, and the check confirms that the chain was reversible at every link.

MATHEMATICAL CONCLUSION

Two equations linked by reversible steps have exactly the same solution set, which is what makes algebraic solution legitimate rather than merely convenient.

Your own words

CHAPTER 3 · 3.1

Linear equations and formulae

An equation states that two expressions have the same value. Solving means finding every value of the unknown for which that statement is true, and every legal step keeps the statement true.

The legal steps are adding, subtracting, multiplying and dividing both sides by the same non-zero quantity. Each is reversible, so the solution set is unchanged at every step.

Collecting the unknown on one side and the constants on the other turns the equation into x = k. The order of the steps does not matter, but doing the same thing to both sides always does.

An equation with the unknown on both sides is solved by collecting terms, and an equation with brackets is solved by expanding first. Multiplying by a common denominator clears fractions in one step.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

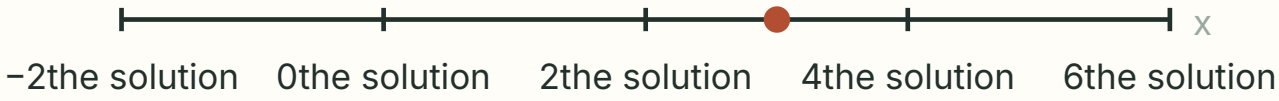
In your own words

CHAPTER 3 · 3.1 · WORKED EXAMPLE 1

Linear equations and formulae

WORKED EXAMPLE 3.1.1

Problem. Solve $3(2x - 5) - 4(x + 1) = 5$.



Mathematical reasoning. Expand, collect the unknown on one side and the constants on the other, then divide by the coefficient.

1. $3(2x - 5) = 6x - 15$ and $4(x + 1) = 4x + 4$.
2. So $6x - 15 - 4x - 4 = 5$, which is $2x - 19 = 5$.
3. Adding 19 to both sides gives $2x = 24$.
4. Dividing by 2 gives $x = 12$.
5. Check: $3(24 - 5) - 4(13) = 57 - 52 = 5$.

Final answer. $x = 12$

Interpretation. Every step was reversible, so the solution of the final equation is the solution of the original, which is why checking works.

CHAPTER 3 · 3.1 · WORKED EXAMPLE 2

Linear equations and formulae

WORKED EXAMPLE 3.1.2

Problem. Solve $(x + 2)/3 - (x - 1)/4 = 1$.

step	working
common denominator	12
multiply through	$4(x + 2) - 3(x - 1) = 12$
expand	$4x + 8 - 3x + 3 = 12$
collect	$x + 11 = 12$
solve	$x = 1$

Clearing the fractions first keeps the arithmetic small.

Mathematical reasoning. Multiply every term by the lowest common denominator, then solve the resulting linear equation.

1. The lowest common denominator of 3 and 4 is 12.
2. Multiplying every term by 12 gives $4(x + 2) - 3(x - 1) = 12$.
3. Expanding: $4x + 8 - 3x + 3 = 12$.
4. Collecting: $x + 11 = 12$, so $x = 1$.
5. Check: $3/3 - 0/4 = 1$.

Final answer. $x = 1$

Interpretation. Multiplying by a non-zero common denominator does not change the solution set, so the cleared equation is equivalent to the original.

CHAPTER 3 · INVESTIGATION 3.2

Inequalities and number lines

INQUIRY QUESTION

How does a solution change when equality is replaced by an inequality?

YOUR TASK

Take five inequalities: one with a positive coefficient, one with a negative coefficient, one compound, one with brackets, and one with fractions. Solve each, draw the solution on a number line, and test one value inside and one outside your solution set.

inequality	solution	inside	outside
$2x + 1 > 7$	$x > 3$	$x = 4$ works	$x = 2$ fails
$-3x \le 9$	$x \ge -3$	$x = 0$ works	$x = -4$ fails
$1 \le 2x - 3 < 7$	$2 \le x < 5$	$x = 3$ works	$x = 6$ fails
$4(x - 1) < 2x + 6$	$x < 5$	$x = 0$ works	$x = 6$ fails
$x/3 - 2 \ge 1$	$x \ge 9$	$x = 9$ works	$x = 8$ fails

CHAPTER 3 · INVESTIGATION 3.2

Inequalities and number lines

What do you notice?

- The negative coefficient reversed the sign in exactly one case.
- The compound inequality gave a closed left end and an open right end.
- Testing a value inside and a value outside caught every sign error.
- The solution was a set in every case, never a single number.

Observation. An inequality is solved by the same reversible steps as an equation, except that division by a negative reverses the order, and the answer is an interval rather than a value.

MATHEMATICAL CONCLUSION

The solution of a linear inequality is the set of values for which the statement is true, and it is verified by testing one value from inside the set and one from outside.

Your own words

CHAPTER 3 · 3.2

Inequalities and number lines

A linear inequality states that one expression is greater or less than another. Its solution is a set of values, usually an interval, not a single number.

Solving an inequality uses the same steps as solving an equation, with one exception: multiplying or dividing by a negative number reverses the inequality sign.

The solution is shown on a number line, with an open circle for a strict inequality and a closed circle for one that includes the endpoint, and it is written in interval notation.

Compound inequalities such as $3 < 2x + 1 \leq 9$ are solved by applying the same operation to all three parts, which keeps the two comparisons linked.

> FORMAL DEVELOPMENT
>
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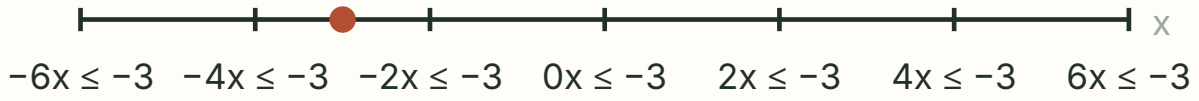
In your own words

CHAPTER 3 · 3.2 · WORKED EXAMPLE 1

Inequalities and number lines

WORKED EXAMPLE 3.2.1

Problem. Solve $5 - 2x \geq 11$ and show the solution on a number line.



Mathematical reasoning. Collect the unknown, then divide by a negative coefficient, which reverses the sign.

1. $5 - 2x \geq 11$.
2. Subtracting 5 from both sides gives $-2x \geq 6$.
3. Dividing by -2 reverses the sign: $x \leq -3$.
4. On a number line this is every value to the left of -3, including -3.
5. Check at $x = -4$: $5 + 8 = 13 \geq 11$, and at $x = 0$: $5 \geq 11$ is false.

Final answer. $x \leq -3$

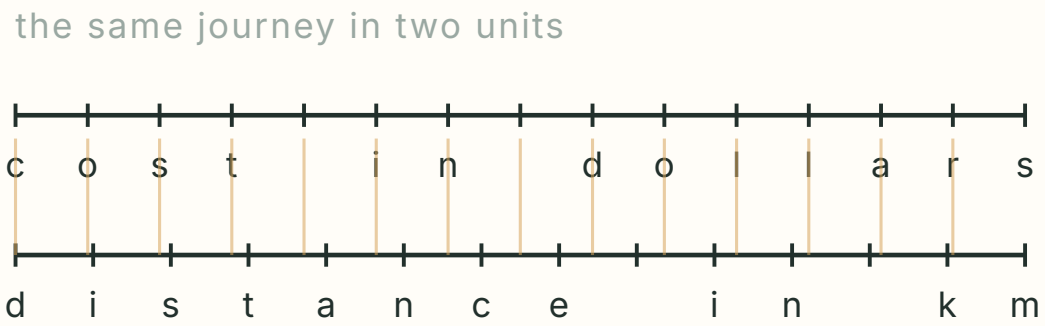
Interpretation. The reversal of the sign is the only difference between solving an inequality and solving an equation, and forgetting it is the commonest error.

CHAPTER 3 · 3.2 · WORKED EXAMPLE 2

Inequalities and number lines

WORKED EXAMPLE 3.2.2

Problem. A taxi charges 3 dollars plus 1.80 dollars per kilometre. A budget of 25 dollars is available. Find the greatest whole number of kilometres possible.



Mathematical reasoning. Model the cost as a linear function of distance, write the budget as an inequality, and solve.

1. The cost is $C = 3 + 1.8k$, where k is the distance in kilometres.
2. The budget gives $3 + 1.8k \leq 25$.
3. Subtracting 3 gives $1.8k \leq 22$.
4. Dividing by 1.8 gives $k \leq 12.22\ldots$
5. So the greatest whole number of kilometres is 12.

Final answer. 12 km

Interpretation. The continuous solution was 12.22 km, and the question asked for a whole number, so the answer was rounded down rather than to the nearest.

CHAPTER 3 · INVESTIGATION 3.3

Systems of equations: graphs and substitution

INQUIRY QUESTION

Why do two linear equations in two unknowns have one solution, and how is it found?

YOUR TASK

Take five pairs of linear equations: one with a subject already isolated, one needing rearranging first, one with parallel lines, one with the same line twice, and one of your own. Solve each by substitution and by drawing, and record what the pair of lines looks like.

pair	substitution result	lines
$y = 3,\ x + y = 8$	$x = 5,\ y = 3$	cross once
$y = 2x,\ 3x + y = 10$	$x = 2,\ y = 4$	cross once
$x + y = 4,\ x + y = 7$	no solution	parallel
$2x + 2y = 6,\ x + y = 3$	infinitely many	the same line
$3x - y = 2,\ 6x - 2y = 5$	no solution	parallel

CHAPTER 3 · INVESTIGATION 3.3

Systems of equations: graphs and substitution

What do you notice?

- Every consistent pair with different gradients gave exactly one solution.
- The parallel pairs gave no solution, because no point lies on both lines.
- The pair with proportional equations gave infinitely many solutions.
- Substitution was easier when one equation already had a subject.

> **Observation.** Two linear equations in two unknowns have one solution, no solution, or infinitely many, according to whether their lines cross, are parallel, or coincide.

MATHEMATICAL CONCLUSION

The number of solutions of a linear system is a property of the pair of lines, so the algebraic answer and the graphical answer must always agree.

Your own words

CHAPTER 3 · 3.3

Systems of equations: graphs and substitution

A pair of simultaneous linear equations is a system: two conditions that must hold at the same time. Each equation alone has infinitely many solutions, and the system has the values that satisfy both.

Graphically, each equation is a line and the solution is the point where the lines cross. The graphical method gives an estimate whose precision is limited by the grid, and it is the method that shows why a solution exists.

Substitution replaces one unknown with an expression taken from the other equation. It is the natural choice when one equation already has a subject, and it turns a system into a single equation in one unknown.

The algebraic and graphical methods must always agree, because they answer the same question. A disagreement means an error in one of them, and the check is worth doing every time.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

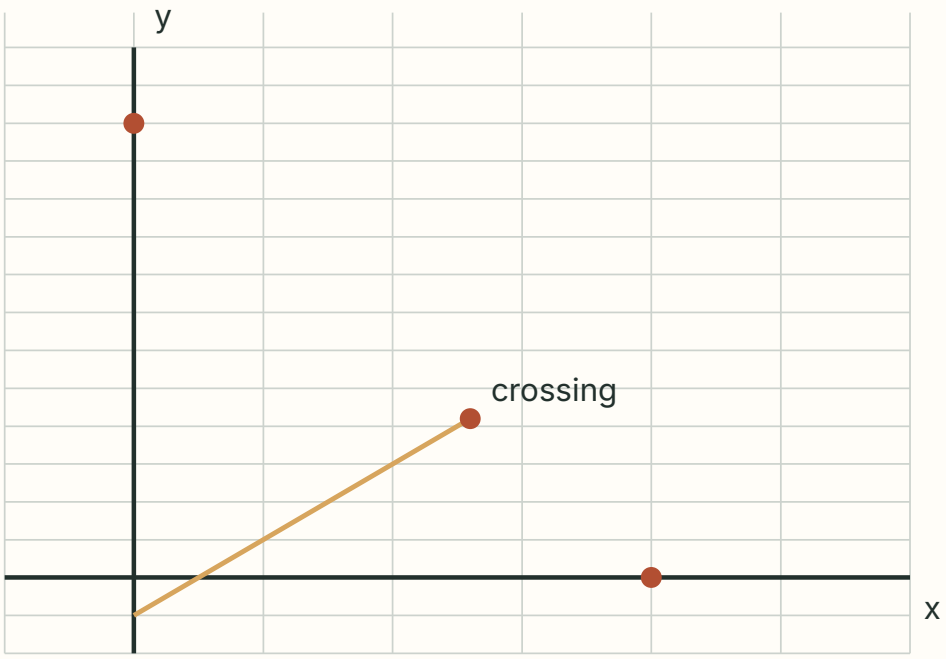
CHAPTER 3 · 3.3 · WORKED EXAMPLE 1

Systems of equations: graphs and substitution

WORKED EXAMPLE 3.3.1

Problem. Solve $y = 2x - 1$ and $3x + y = 12$ by substitution.

the lines cross at (2.6, 4.2)



Mathematical reasoning. Substitute the expression for y from the first equation into the second.

1. The first equation gives $y = 2x - 1$.
2. Substituting into the second: $3x + (2x - 1) = 12$.
3. So $5x = 13$ and $x = 2.6$.
4. Then $y = 2(2.6) - 1 = 4.2$.
5. Check in the second equation: $3(2.6) + 4.2 = 12$.

Final answer. $x = 2.6$, $y = 4.2$

Interpretation. Substitution reduces two equations to one, and the check confirms that the pair still holds.

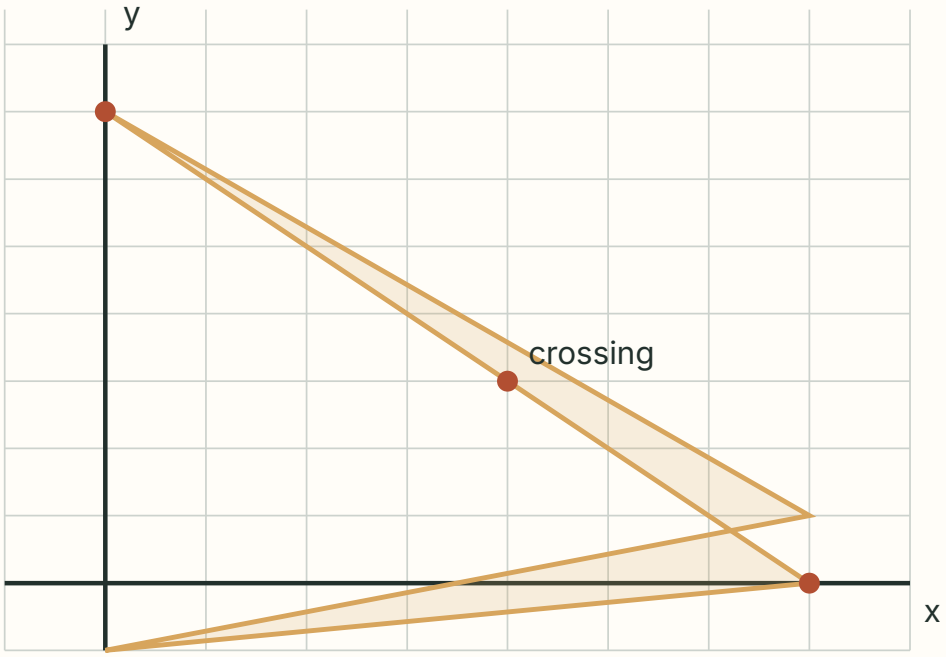
CHAPTER 3 · 3.3 · WORKED EXAMPLE 2

Systems of equations: graphs and substitution

WORKED EXAMPLE 3.3.2

Problem. Solve $x + y = 7$ and $x - y = 1$ graphically, and state the precision of your answer.

the two lines cross at (4, 3)



Mathematical reasoning. Draw both lines on the same axes and read the coordinates of the crossing point.

1. $x + y = 7$ passes through (0, 7) and (7, 0).
2. $x - y = 1$ passes through (0, −1) and (7, 6), so it has gradient 1.
3. The lines cross at (4, 3), which the grid shows directly.
4. Reading from a grid gives the answer to the precision of the grid, here about 0.1.
5. Solving algebraically gives exactly $x = 4$ and $y = 3$, confirming the reading.

Final answer. $x = 4$, $y = 3$, to the precision of the grid

Interpretation. The graphical method shows why a solution exists and gives an estimate; the algebraic method gives the exact value, and the two must agree.

CHAPTER 3 · INVESTIGATION 3.4

Elimination and special cases

INQUIRY QUESTION

Why do two linear equations in two unknowns have one solution, and when do they have none or many?

YOUR TASK

Take five pairs of linear equations: one with a clean elimination, one needing substitution, one with parallel lines, one with the same line twice, and one of your own. Solve each, and record what the pair of lines looks like as a graph.

pair	result	lines
$x + y = 5,\ x - y = 1$	$x = 3,\ y = 2$	cross once
$2x + y = 7,\ y = 3$	$x = 2,\ y = 3$	cross once
$x + y = 4,\ x + y = 7$	no solution	parallel
$2x + 2y = 6,\ x + y = 3$	infinitely many	the same line
$3x - y = 2,\ 6x - 2y = 5$	no solution	parallel

CHAPTER 3 · INVESTIGATION 3.4

Elimination and special cases

What do you notice?

- Every consistent pair with different gradients gave exactly one solution.
- The parallel pairs gave no solution, because no point lies on both lines.
- The pair with proportional equations gave infinitely many solutions.
- Substitution was easier when one equation already had a subject.

Observation. Two linear equations in two unknowns have one solution, no solution, or infinitely many, according to whether their lines cross, are parallel, or coincide.

MATHEMATICAL CONCLUSION

The number of solutions of a linear system is a property of the pair of lines, so the algebraic answer and the graphical answer must always agree.

Your own words

CHAPTER 3 · 3.4

Elimination and special cases

A pair of simultaneous linear equations is a system: two conditions that must hold at the same time. Each equation alone has infinitely many solutions, and the system has the values that satisfy both.

Elimination adds or subtracts the equations to remove one unknown, and works when a coefficient can be matched. Substitution replaces one unknown with an expression from the other equation.

Two equations with the same left-hand side but different right-hand sides describe parallel lines and have no solution. Two equations that are multiples of each other describe the same line and have infinitely many solutions.

Solving a system is equivalent to finding the point of intersection of two lines, which is why the graphical method and the algebraic method must always agree.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

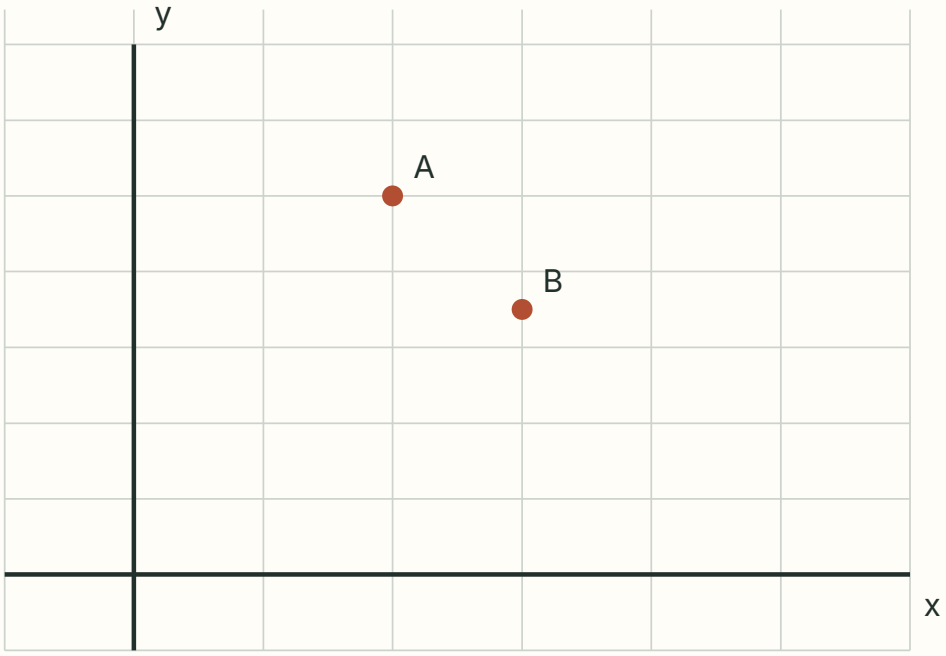
CHAPTER 3 · 3.4 · WORKED EXAMPLE 1

Elimination and special cases

WORKED EXAMPLE 3.4.1

Problem. Solve $3x + 2y = 16$ and $5x - 2y = 8$ simultaneously.

the two lines meet at (3, 3.5)



Mathematical reasoning. The coefficients of y are opposites, so adding the equations removes y in one step.

1. Adding: $(3x + 2y) + (5x - 2y) = 16 + 8$, so $8x = 24$.
2. Hence $x = 3$.
3. Substituting into the first equation: $9 + 2y = 16$, so $2y = 7$.
4. Hence $y = 3.5$.
5. Check in the second: $15 - 7 = 8$.

Final answer. $x = 3$, $y = 3.5$

Interpretation. Choosing to add rather than subtract is the whole skill of elimination, and it is decided by looking at the coefficients before starting.

CHAPTER 3 · 3.4 · WORKED EXAMPLE 2

Elimination and special cases

WORKED EXAMPLE 3.4.2

Problem. A cinema sells 200 tickets for 1560 dollars. Adult tickets cost 10 dollars and child tickets 6 dollars. Find how many of each were sold.

quantity	equation	value
adults	a	to find
children	c	to find
total tickets	a + c = 200	200
total money	10a + 6c = 1560	1560

Two conditions, two equations.

Mathematical reasoning. Write one equation for the number of tickets and one for the money, then eliminate.

1. Let a be the number of adults and c the number of children.
2. a + c = 200 and 10a + 6c = 1560.
3. Multiplying the first by 6 gives 6a + 6c = 1200.
4. Subtracting from the second gives 4a = 360, so a = 90.
5. Then c = 200 − 90 = 110.

Final answer. 90 adult tickets and 110 child tickets

Interpretation. The two conditions are independent, so the system has one solution, and the check is that both totals are matched.

CHAPTER 3 · INVESTIGATION 3.5

Modelling with systems

INQUIRY QUESTION

How does a real constraint become an equation, and what does the solution mean?

YOUR TASK

Find three real situations that can be modelled by a linear equation or a system: one cost comparison, one mixture or recipe, and one allocation. Model each, solve it, and test whether the solution is physically possible.

situation	model	solution	feasible?
two phone plans	$15 + 0.5n = 30 + 0.2n$	$n = 50$	yes
nut mix	$x + y = 500$, $8x + 5y = 3200$	$x = 233$, $y = 267$	yes
shift roster	$a + b = 40$, $6a + 4b = 200$	$a = 20$, $b = 20$	yes
over-budget trip	$x + y = 10$, $30x + 12y = 400$	$x = 11.1$	no

CHAPTER 3 · INVESTIGATION 3.5

Modelling with systems

What do you notice?

- Every feasible solution had all variables non-negative.
- The over-budget trip gave a negative value for the cheaper option, so the model said the budget was impossible.
- The mixture models always conserved both total and value.
- The model's failure was informative, not a mistake.

> **Observation.** A linear model turns a constraint into an equation, and the algebra returns a solution whose physical sense must still be checked.

MATHEMATICAL CONCLUSION

A solution is acceptable only when it satisfies every constraint of the situation, including the ones the equations do not express.

Your own words

CHAPTER 3 · 3.5

Modelling with systems

A linear model has a fixed part and a part that changes with the quantity. Writing it as $y = mx + c$ separates the two, and the intercept is the value at zero.

Break-even is where cost equals revenue, so the model is an equation. Profit is where revenue exceeds cost, so the model is an inequality.

Mixture problems are conservation problems: the total amount and the total value are both unchanged, which gives two equations.

Allocation problems divide a limited resource under several conditions, and the conditions are the equations. The solution is feasible only if every variable is non-negative, which the algebra does not know.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

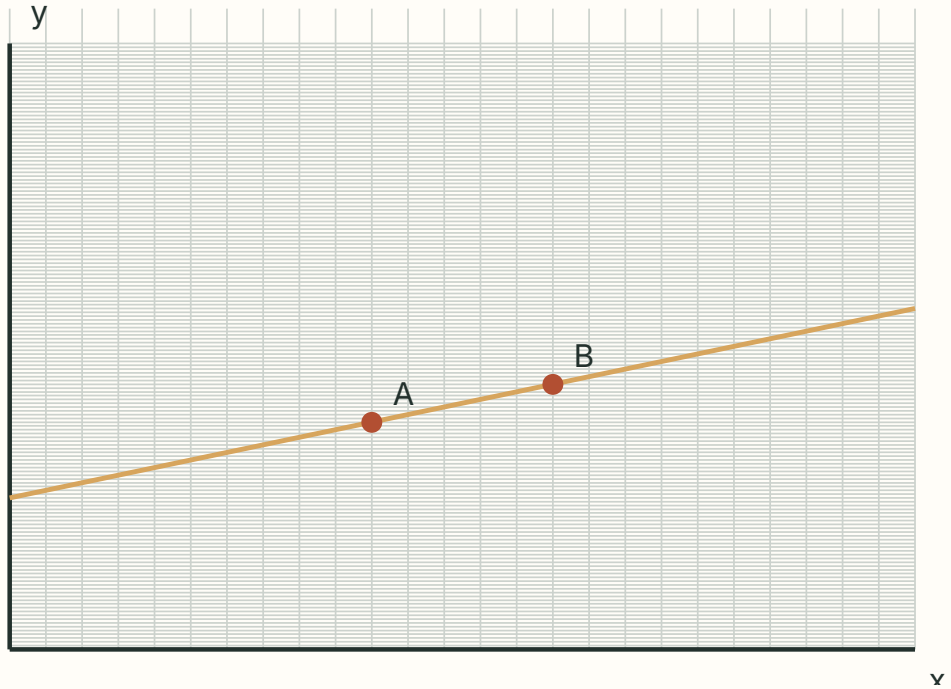
CHAPTER 3 · 3.5 · WORKED EXAMPLE 1

Modelling with systems

WORKED EXAMPLE 3.5.1

Problem. A print shop charges a setup fee of 40 dollars plus 2 dollars per poster. A rival charges 6 dollars per poster with no fee. Find the number of posters at which the two shops charge the same, and say which is cheaper for 30 posters.

the two cost lines cross at 10 posters



Mathematical reasoning. Write each cost as a linear model and solve the equation that sets them equal.

1. The first shop charges $C_1 = 40 + 2n$ and the second $C_2 = 6n$.
2. Setting them equal: $40 + 2n = 6n$.
3. So $40 = 4n$ and $n = 10$.
4. For 30 posters the first shop charges 100 dollars and the second 180 dollars.
5. So the first shop is cheaper for 30 posters.

Final answer. the shops charge the same at 10 posters; the first is cheaper for 30

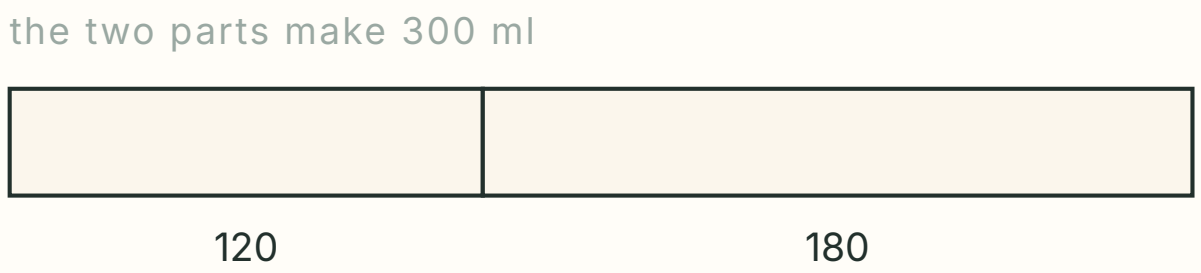
Interpretation. The break-even point is where two models agree, and the cheaper option is decided by which side of that point the order lies on.

CHAPTER 3 · 3.5 · WORKED EXAMPLE 2

Modelling with systems

WORKED EXAMPLE 3.5.2

Problem. A 40% saline solution is mixed with a 10% solution to make 300 ml of a 25% solution. Find the volume of each.



Mathematical reasoning. The total volume and the total amount of salt are both conserved, which gives two equations.

1. Let x be the volume of the 40% solution and y the volume of the 10% solution.
2. $x + y = 300$.
3. $0.40x + 0.10y = 0.25 \times 300 = 75$.
4. Multiplying the first by 0.10 and subtracting gives $0.30x = 45$.
5. So $x = 150$ and $y = 150$.

Final answer. 150 ml of each

Interpretation. The two conserved quantities give two equations, and the solution is feasible only because both volumes came out positive.

CHAPTER 3 · INVESTIGATION 3.6

Investigation: break-even and beyond

INQUIRY QUESTION

When two linear models are compared, where do they agree, and what does the comparison decide?

YOUR TASK

Choose a limited resource that matters to you — money, time, space or food. Model the sharing with one equation and at least two inequalities, draw the feasible region, and then choose a point using a stated criterion. Defend the criterion in writing.

criterion	point chosen	why
equal shares	the centre of the region	everyone gets the same
greatest need	a vertex	the largest share goes to the greatest need
least disruption	the point nearest the current share	change is costly
maximum total benefit	a vertex	a linear objective is optimised at a vertex

CHAPTER 3 · INVESTIGATION 3.6

Investigation: break-even and beyond

What do you notice?

- Every criterion chose a different point of the same region.
- The region itself was the same, so the mathematics was neutral.
- The choice of criterion was a judgement, not a calculation.
- Some criteria gave points on the boundary and some gave interior points.

> **Observation.** A model of a shared resource produces a feasible region, and the point chosen within it is decided by a criterion that comes from outside the model.

MATHEMATICAL CONCLUSION

Mathematics can show which allocations are possible and what each criterion would choose, but the choice of criterion is a decision about values that the model itself cannot make.

Your own words

CHAPTER 3 · 3.6

Investigation: break-even and beyond

Two linear models of the same quantity agree where their graphs cross, and the crossing point is found by solving the equation obtained by setting them equal. That point is the break-even point of the comparison.

On one side of the crossing point one model is larger and on the other side the other is, so the better choice depends on where the quantity being compared lies.

A comparison is only complete when the period of use is stated, because a model that is cheaper for a short period can be dearer for a long one.

A model makes the consequences of a choice visible. It does not make the choice, and a modeller who hides the assumption on which the choice rests is not being honest.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

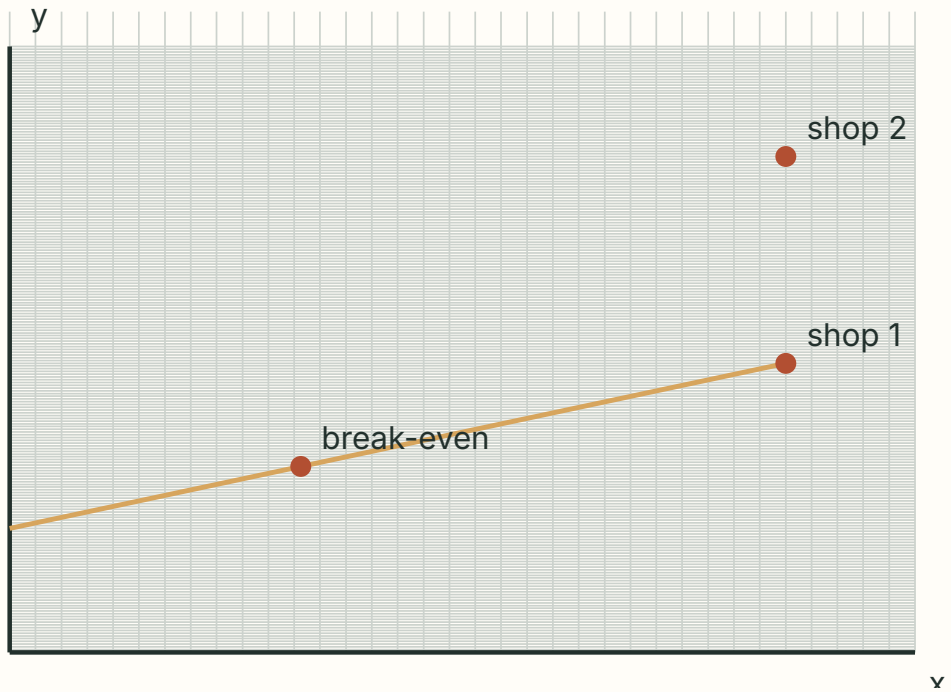
CHAPTER 3 · 3.6 · WORKED EXAMPLE 1

Investigation: break-even and beyond

WORKED EXAMPLE 3.6.1

Problem. A print shop charges 45 dollars for a call-out plus 2 dollars per poster. A rival charges 6 dollars per poster with no fee. Find the number of posters at which the two shops charge the same, and say which is cheaper for 30 posters.

the two cost lines cross at 11.25 posters



Mathematical reasoning. Write each cost as a linear model and solve the equation that sets them equal.

1. The first shop charges $C_1 = 45 + 2n$ and the second $C_2 = 6n$.
2. Setting them equal: $45 + 2n = 6n$.
3. So $45 = 4n$ and $n = 11.25$.
4. For fewer than 11.25 posters the second shop is cheaper; for more, the first is.
5. For 30 posters the first shop charges 105 dollars and the second 180 dollars, so the first is cheaper.

Final answer. the shops charge the same at 11.25 posters; the first is cheaper for 30

Interpretation. The break-even point is where two models agree, and the cheaper option is decided by which side of that point the order lies on.

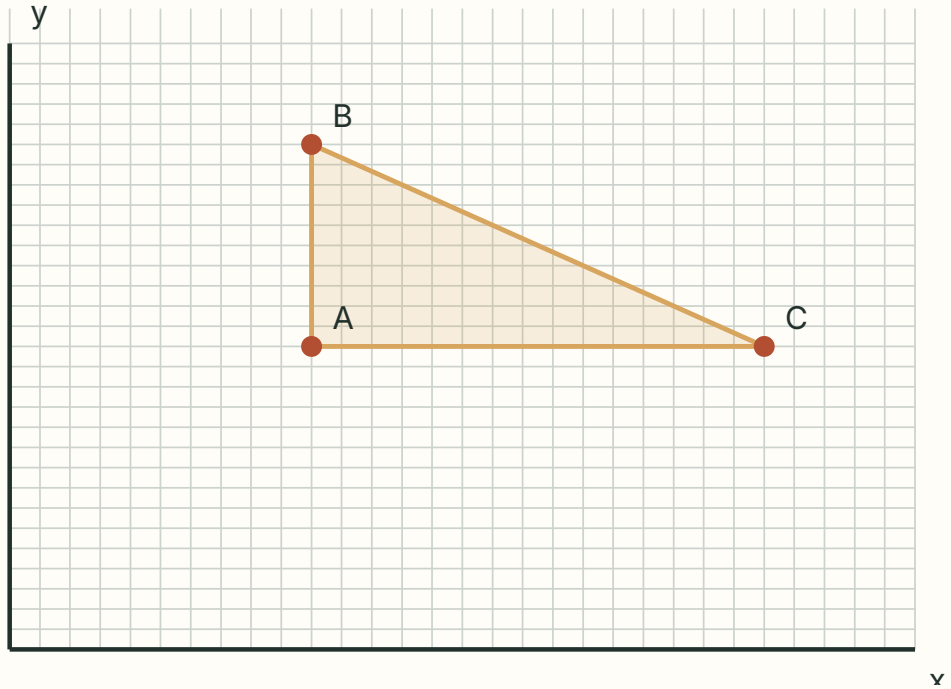
CHAPTER 3 · 3.6 · WORKED EXAMPLE 2

Investigation: break-even and beyond

WORKED EXAMPLE 3.6.2

Problem. A school has 60 hours of tutoring to share between three year groups. Year 9 needs at least 10 hours, Year 10 at least 15, and Year 11 at most 25. Model the situation and find the feasible region.

three constraints meet in a region



Mathematical reasoning. Write the total as an equation and each condition as an inequality, then find the region.

1. Let x, y and z be the hours for Years 9, 10 and 11.
2. $x + y + z = 60$, with $x \geq 10$, $y \geq 15$ and $z \leq 25$.
3. Substituting $z = 60 - x - y$ gives $x \geq 10$, $y \geq 15$ and $x + y \geq 35$.
4. The feasible region is a triangle in the (x, y) plane with vertices (10, 25), (25, 15) and (10, 15).
5. Equal shares would be (20, 20, 20), which is feasible.

Final answer. the feasible region is a triangle, and equal shares are feasible

Interpretation. The model shows which shares are possible; it does not say which is fair, because that depends on a criterion the model does not contain.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

3.A1 FLUENCY

Solve $4x - 7 = 21$.

3.A2 FLUENCY

Solve $5 - 3x = 17$.

3.A3 FLUENCY

Solve $2(x + 3) = 5(x - 1)$.

3.A4 FLUENCY

Solve $x/4 + 2 = 5$.

3.A5 FLUENCY

Solve the pair $x + y = 9$ and $x - y = 3$.

3.A6 FLUENCY

Solve the pair $2x + y = 11$ and $x - y = 1$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

3.A7 FLUENCY

Solve $3x + 4 < 19$ and show the solution on a number line.

3.A8 FLUENCY

Solve $-2x \geq 10$.

3.A9 FLUENCY

Solve the pair $y = 2x$ and $y = x + 3$.

3.A10 FLUENCY

Solve $7 - 2x \leq 1$.

3.A11 FLUENCY

Solve $(x - 1)/3 = (x + 2)/5$.

3.A12 FLUENCY

Solve $4(x - 2) = 2(x + 5)$.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

3.B1 REASONING

The number of diagonals of an n-sided polygon is $n(n-3)/2$. Find n for a polygon with 20 diagonals and justify the rejection of the second root.

3.B2 REASONING

Investigate the pattern in the sums $1 + 3 + 5 + \ldots$ for 1, 2, 3 and 4 terms, and justify the general rule.

3.B3 REASONING

Conjecture a rule for the number of handshakes among n people and test it for $n = 3, 4, 5$.

3.B4 REASONING

Investigate the sequence of perimeters of rectangles with sides n and $n + 3$ and find its rule.

3.B5 REASONING

Build a table of solutions to $x + 2y = 12$ for whole numbers and describe the pattern.

3.B6 REASONING

Conjecture a rule for the number of matches needed to make a row of n squares and justify it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

3.B7 REASONING

Investigate the pattern in the number of intersection points of n lines in general position.

3.B8 REASONING

A pattern of dots gives 4, 9, 16, 25 for figures 1 to 4. Find the rule and test it on figure 7.

3.B9 REASONING

Investigate the areas of the rectangles with perimeter 24 and find the shape of greatest area.

3.B10 REASONING

Conjecture a rule for the number of diagonals from one vertex of an n-sided polygon.

3.B11 REASONING

Investigate the pattern in $2^n - 1$ for n = 1 to 6 and describe what you find.

CRITERION C Communicating

Criterion C practice · Communicating

3.C1 REASONING
Explain in your own words what it means to solve an equation.

3.C2 FLUENCY
Write a complete solution to $3(2x - 1) = 4(x + 5)$, labelling each step.

3.C3 REASONING
Choose between solving graphically and solving algebraically for a pair of equations and justify.

3.C4 FLUENCY
Set out a full solution to the pair $3x + 2y = 12$ and $x - y = 1$.

3.C5 REASONING
Explain why multiplying an inequality by a negative number reverses the sign.

3.C6 REASONING
Describe, with an example, the difference between no solution and infinitely many solutions.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

3.C7 REASONING

Explain why the solution of a system of inequalities is a region.

3.C8 REASONING

Write the reasoning that shows two equations with the same left-hand side have no solution.

3.C9 REASONING

Set out the argument that the optimum of a linear function over a polygon occurs at a vertex.

3.C10 REASONING

Explain what the intercept of a linear model represents in a real situation.

3.C11 REASONING

Describe the relationship between the algebraic and graphical solutions of a pair of linear equations.

3.C12 REASONING

Explain, with an example, why a feasible solution must be checked against the situation.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

3.D1 APPLICATION

A phone plan costs 15 dollars plus 0.05 dollars per minute. Another costs 40 dollars with unlimited minutes. Find the number of minutes at which the two plans cost the same, and say which is cheaper for 400 minutes.

3.D2 APPLICATION

A charity has 500 dollars to spend on 40 food parcels. A small parcel costs 10 dollars and a large one 15. Find how many of each it can buy.

3.D3 APPLICATION

A farmer mixes feed at 12 dollars per kg with feed at 8 dollars per kg to make 200 kg at 10 dollars per kg. Find the mass of each.

3.D4 APPLICATION

A taxi charges 4 dollars plus 2.50 dollars per km. A bus costs 1.20 dollars per km with no fixed charge. Find the distance at which the taxi is cheaper and state the answer for a 6 km trip.

3.D5 REASONING

A print shop has 60 hours of machine time and needs at least 20 jobs of type A and at most 30 of type B. Model the situation and find the feasible range for the number of type A jobs.

3.D6 APPLICATION

A school canteen sells 120 meals for 900 dollars. A student meal costs 6 dollars and an adult meal 9 dollars. Find how many of each were sold.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

3.D7 APPLICATION

A builder has 40 m of fencing and wants a rectangular pen of area at least 96 m^2. Find the feasible range of widths and the dimensions of greatest area.

3.D8 APPLICATION

A coach company charges 200 dollars plus 12 dollars per passenger. A group of 30 has a budget of 700 dollars. Is the trip affordable, and what is the largest group possible?

3.D9 REASONING

A chemist mixes a 30% acid solution with a 60% solution to make 500 ml at 45%. Investigate whether this is possible and justify.

3.D10 APPLICATION

A farm has 100 hectares and at least 40 hectares of maize. Wheat needs 3 workers per hectare and maize 5, with 420 workers available. Model the situation and find the feasible region.

3.D11 REASONING

A bakery sells loaves at 3 dollars and rolls at 0.50 dollars. It sold 200 items for 400 dollars. Investigate whether the figures are consistent.

3.D12 REASONING

Explain, with a model of your own, when a linear model is a poor description of a situation.

CHAPTER 3 · APPLICATION AND ASSESSMENT

Applying chapter 3

Each task involves a constraint and a decision. State the constraint as an equation or inequality before solving it, and check that your answer satisfies every condition.

A1 · The shared budget

A community centre has 600 dollars and 40 m^2 of floor space to set up a study area and a play area. Each square metre of study area costs 12 dollars and each square metre of play area costs 15 dollars. The play area must be at least 10 m^2 and the study area at least 15 m^2. Model the situation, find the feasible region, and recommend one allocation with a stated reason.

Criteria A and D · state the constraints, solve the system, and justify the allocation you recommend.

A2 · The two suppliers

Two suppliers quote for printed programmes. Supplier A charges 150 dollars plus 0.80 dollars per copy; supplier B charges 60 dollars plus 1.20 dollars per copy. Find the number of copies at which the quotes are equal, say which supplier is cheaper for 300 copies, and find the largest number of copies affordable within a 900 dollar budget.

Criteria A and C · model both quotes, solve the equation, and solve the inequality.

CHAPTER 3 · REVIEW

What you should now be able to do

Key ideas

- Solving an equation uses reversible steps only.
- Dividing an inequality by a negative number reverses the sign.
- The solution of an inequality is an interval, not a value.
- Two linear equations in two unknowns have one, none, or infinitely many solutions.
- The solution of a pair of equations is the intersection of two lines.
- A system of inequalities describes a feasible region.
- The optimum of a linear model over a region occurs at a vertex.
- A feasible solution must still be physically possible.

Vocabulary

equation · inequality · simultaneous · elimination · substitution · feasible region · vertex · break-even · linear model · interval · half-plane · constraint

CHAPTER 3 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

3.R1 Solve $5x - 3 = 2x + 12$.

3.R2 Solve $3 - 4x > 11$.

3.R3 Solve the pair $2x + 3y = 16$ and $x - y = 2$.

3.R4 Find the feasible region for $x \geq 1$, $y \geq 2$ and $x + y \leq 8$.

3.R5 Solve $(2x - 1)/4 = (x + 3)/3$.

3.R6 Explain why $x + y = 2$ and $2x + 2y = 5$ have no solution.

CHAPTER 3 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A linear equation records an equivalence between two descriptions of the same quantity, and solving it is the search for the value at which the two descriptions agree; when several constraints act at once, the feasible values are a system rather than a single number, and the solution is the region where all the constraints hold.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of equivalence, change and system appear in your work?
- How is the global context of fairness and development, and its exploration of collective decisions under constraints, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a linear model for a decision about sharing, and what would you still like to ask?

CHAPTER 3 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Fairness under constraints is a mathematical question with a human answer. Model one real allocation and say which solution you would defend.

Service as Action

COMMUNITY NEED	Shared budgets and shared spaces are allocated without showing the constraints behind them.
MATHEMATICAL CONNECTION	Linear equations, inequalities and feasible regions.
PROPOSED ACTION	Model one real allocation and publish the feasible choices with the reasoning.
EXPECTED IMPACT	A group can see which choices are possible before it argues about which is best.
STUDENT REFLECTION	Record the model, the region and what you chose.

Community Project

PROJECT IDEA	A fairness audit of one school allocation.
NEED	Budgets, rooms and time are divided without stated criteria.
RESEARCH QUESTION	Which allocations are feasible, and what does each criterion choose?
MATHEMATICS	Linear models, systems and feasible regions.
ACTION OR OUTCOME	A one-page report with the model and two recommendations.
BENEFICIARIES	Everyone affected by the allocation.
EVIDENCE	The report with the region and the criteria compared.

Personal Project connection · MYP 5

PERSONAL INTEREST	Money, time or a team you belong to.
LEARNING GOAL	To model a limited resource and state a criterion for sharing it.
PRODUCT OR OUTCOME	A model with a feasible region and a defended choice.
MATHEMATICS	Linear equations and inequalities.
PROCESS	State the constraints, solve, and choose a point with a stated criterion.
EVIDENCE	The model, the region and the written defence.

CHAPTER 4 · PART II · THINKING WITH MODELS

04

Functions, Coordinates and Linear Models

A coordinate turns a location into a pair of numbers. A function turns a rule into a table, a graph and an equation, and all three describe the same relationship.

ENGAGE

A delivery rider travels 12 km in 40 minutes. Where is she after 15 minutes, how fast is she going, and which representation of the journey answers each question most easily?

Framing the inquiry

Key concept: Form
Related concepts: Representation · Change · Pattern
Global context: Orientation in space and time
Global context exploration: Maps, routes and rates

Statement of inquiry: A function is a rule that gives exactly one output for each permitted input, and the form of that rule decides what can be read from it — a graph shows change, a table shows values, and an equation shows structure — so the same relationship can be represented in several ways, each of which answers a different question.

Questions to guide us

- **Factual:** What is a function, and what are the gradient and intercept of a linear function?
- **Conceptual:** Why do the table, graph and equation of a function describe the same relationship, and what does the gradient tell you about change?
- **Debatable:** Is a graph a picture of reality or a model of it, and what is lost when a journey is drawn as a straight line?

ATL · THINKING. Students identify the key features of a linear function from its graph, its table and its equation, and explain which representation is most efficient for the question asked.

CHAPTER 4 · INVESTIGATION 4.1

Functions and function notation

INQUIRY QUESTION

What makes a rule a function, and what does function notation add?

YOUR TASK

Collect five rules from everyday life: a bus fare, a phone charge, a recipe, a sports score and one of your own. For each, state the domain, the range, and whether the rule is a function. Then write each as a formula where you can.

rule	domain	range	function?
bus fare 2.50 per ride	rides	money	yes
bus fare capped at 5.00	rides	money	yes
square of a number	all numbers	non-negative numbers	yes
square root of a number	non-negative numbers	non-negative numbers	yes
a person's height	people	heights	no, changes with time

CHAPTER 4 · INVESTIGATION 4.1

Functions and function notation

What do you notice?

- Every rule that gave one output per input was a function.
- The capped bus fare was still a function, though not a linear one.
- Height was not a function of the person, because it changes with time.
- Adding time to the domain made height a function again.

> **Observation.** Whether a rule is a function depends on the domain chosen, and the same quantity can be a function of one variable and not of another.

MATHEMATICAL CONCLUSION

A function is a rule with exactly one output for each permitted input, and the domain is part of the definition rather than an afterthought.

Your own words

CHAPTER 4 · 4.1

Functions and function notation

A function is a rule that gives exactly one output for each permitted input. The permitted inputs form the domain and the possible outputs form the range.

Function notation f(x) names the rule f and the input x together, so f(3) means the output when the input is 3. The letter is a label, not a multiplier.

A function can be given as a formula, a table, a graph or a description in words, and all four are the same function if they agree on every input of the domain.

The vertical line test decides whether a graph is a function: a vertical line that meets the graph twice would give one input two outputs, which is not allowed.

A function is a representation of a relation between two quantities, and a formula, a graph and a table are different representations of the same function. Each representation makes some questions easy and others hard, so a modeller chooses deliberately.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

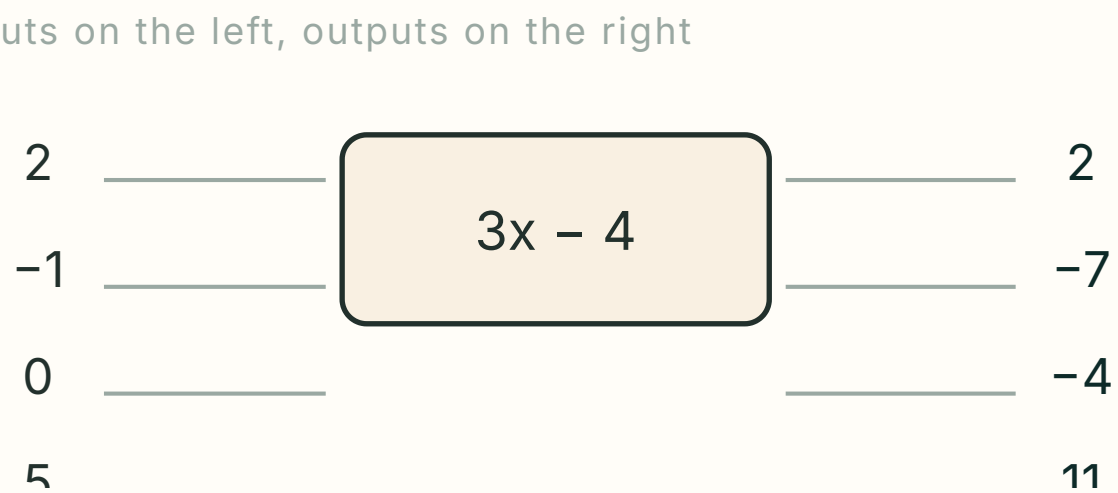
In your own words

CHAPTER 4 · 4.1 · WORKED EXAMPLE 1

Functions and function notation

WORKED EXAMPLE 4.1.1

Problem. A function is given by f(x) = 3x − 4. Find f(2), f(−1) and f(0), and find x when f(x) = 11.



Mathematical reasoning. Substitute each input into the rule, and solve the equation for the inverse question.

1. f(2) = 3(2) − 4 = 2.
2. f(−1) = 3(−1) − 4 = −7.
3. f(0) = 3(0) − 4 = −4.
4. Setting 3x − 4 = 11 gives 3x = 15, so x = 5.
5. The outputs rise by 3 for every rise of 1 in the input.

Final answer. f(2) = 2, f(−1) = −7, f(0) = −4, and f(x) = 11 at x = 5

Interpretation. Finding an output is substitution; finding the input is solving an equation, and the two questions are different.

CHAPTER 4 · 4.1 · WORKED EXAMPLE 2

Functions and function notation

WORKED EXAMPLE 4.1.2

Problem. Decide whether the relation g = {(1, 2), (2, 4), (3, 2), (4, 8)} is a function, and state its domain and range.

input	output
1	2
2	4
3	2
4	8

Each input appears once, so the relation is a function.

Mathematical reasoning. Check that no input is paired with two different outputs.

1. The inputs are 1, 2, 3 and 4, each appearing exactly once.
2. So every permitted input has exactly one output, and g is a function.
3. The domain is {1, 2, 3, 4}.
4. The range is {2, 4, 8}, because the output 2 occurs twice.
5. The range is smaller than the domain, which is allowed.

Final answer. g is a function with domain {1, 2, 3, 4} and range {2, 4, 8}

Interpretation. Two inputs may share an output, but one input may not have two, and that asymmetry is the definition.

CHAPTER 4 · INVESTIGATION 4.2

Distance and midpoint

INQUIRY QUESTION

How is the distance between two points found from their coordinates?

YOUR TASK

Take five pairs of points of your own, including one pair sharing a coordinate, one pair on a diagonal and one pair in different quadrants. For each, find the distance and the midpoint, and check the midpoint by measuring on squared paper.

pair	distance	midpoint
(0, 0), (6, 8)	10	(3, 4)
(1, 1), (1, 9)	8	(1, 5)
(−2, 3), (4, −1)	$\sqrt{52} \approx 7.21$	(1, 1)
(−5, −5), (5, 5)	$\sqrt{200} \approx 14.14$	(0, 0)
(2, −3), (8, 5)	$\sqrt{116} \approx 10.77$	(5, 1)

CHAPTER 4 · INVESTIGATION 4.2

Distance and midpoint

What do you notice?

- The pair sharing a coordinate gave a distance that was a simple difference.
- Every other pair needed Pythagoras on the two differences.
- The midpoint coordinates were always the averages.
- The midpoint always lay on the segment, as the check confirmed.

> **Observation.** The distance between two points comes from Pythagoras applied to the coordinate differences, and the midpoint comes from averaging them.

MATHEMATICAL CONCLUSION

Coordinates make distance and midpoint calculable, and the two quantities are found from the same pair of differences.

Your own words

CHAPTER 4 · 4.2

Distance and midpoint

The distance between two points is found from the differences of their coordinates. When the points share a coordinate the distance is simply the difference of the other one.

In general the horizontal and vertical differences are the legs of a right-angled triangle whose hypotenuse is the distance, so Pythagoras' theorem gives the distance formula.

The midpoint of two points is found by averaging each coordinate, and it is the point equidistant from both. It is also the centre of symmetry of the segment.

A coordinate turns a location into a pair of numbers, and that is what makes distance and midpoint calculable rather than measurable, which is the whole advantage of coordinates.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

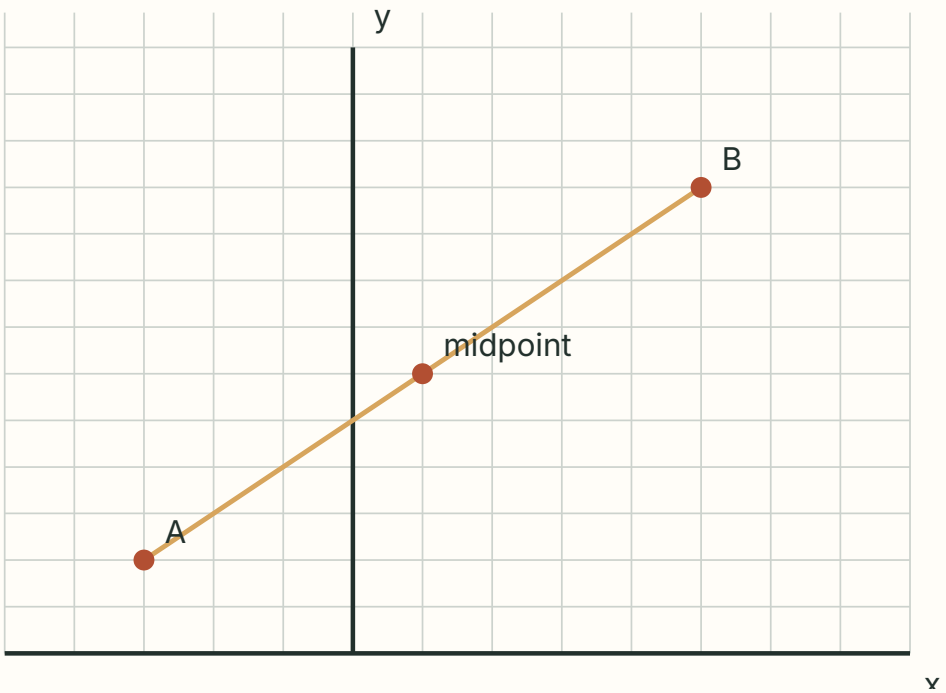
CHAPTER 4 · 4.2 · WORKED EXAMPLE 1

Distance and midpoint

WORKED EXAMPLE 4.2.1

Problem. Find the distance and the midpoint of A(−3, 2) and B(5, 10).

the segment from A to B



Mathematical reasoning. Take the differences for the legs, apply Pythagoras, and average the coordinates for the midpoint.

1. The horizontal difference is 5 − (−3) = 8 and the vertical difference is 10 − 2 = 8.
2. So the distance is $\sqrt{(8^2 + 8^2)} = \sqrt{128} = 8\sqrt{2} \approx 11.31$.
3. The midpoint is ((−3 + 5)/2, (2 + 10)/2) = (1, 6).
4. The midpoint lies on the segment, halfway along it.
5. Check: the distance from A to the midpoint is also $8\sqrt{2}/2$.

Final answer. distance $8\sqrt{2} \approx 11.31$, midpoint (1, 6)

Interpretation. One calculation gives the length of a segment that could not be measured directly, which is why coordinates are used.

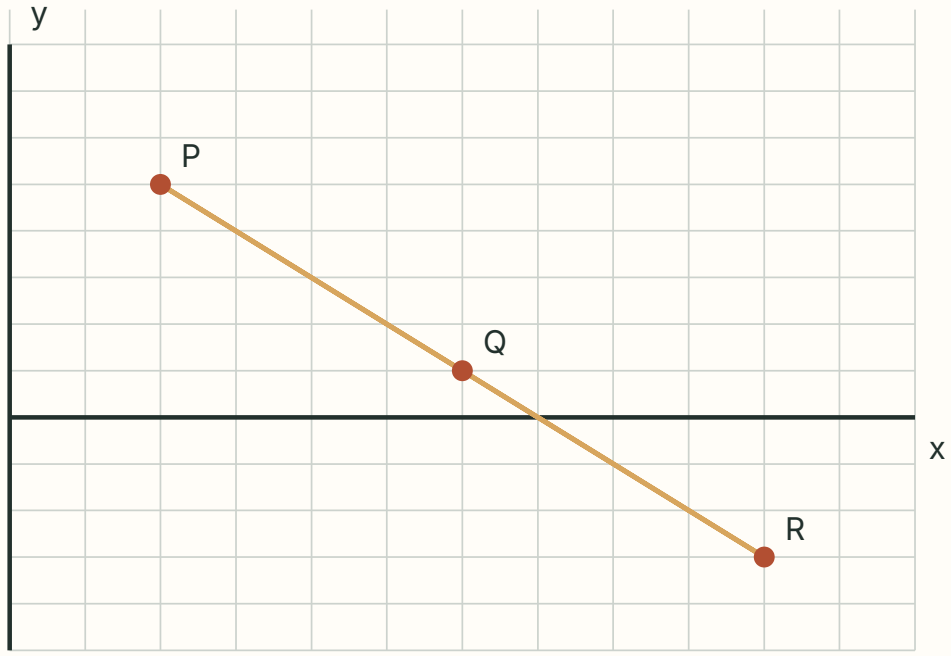
CHAPTER 4 · 4.2 · WORKED EXAMPLE 2

Distance and midpoint

WORKED EXAMPLE 4.2.2

Problem. The points P(2, 5), Q(6, 1) and R form a straight line with Q the midpoint of PR. Find R.

Q is the midpoint of PR



Mathematical reasoning. Use the midpoint formula in reverse: the midpoint is the average of the endpoints.

1. If Q is the midpoint of P and R, then Q = (P + R)/2.
2. So R = 2Q − P, coordinate by coordinate.
3. The x-coordinate is 2(6) − 2 = 10.
4. The y-coordinate is 2(1) − 5 = −3.
5. So R is (10, −3), and the three points are collinear.

Final answer. R is (10, −3)

Interpretation. The midpoint formula can be read backwards, and reading it backwards finds an endpoint from a midpoint.

CHAPTER 4 · INVESTIGATION 4.3

Gradient and equations of lines

INQUIRY QUESTION

What does the gradient of a line measure, and how is it read from an equation?

YOUR TASK

Take six linear equations of your own: three in the form $y = mx + c$ and three in another form. For each, find the gradient and the intercepts, then draw the six lines on one pair of axes and group them by gradient.

equation	gradient	y-intercept	x-intercept
$y = 2x + 1$	2	1	-0.5
$y = -2x + 1$	-2	1	0.5
$y = 0.5x - 3$	0.5	-3	6
$x + y = 5$	-1	5	5
$2x - 3y = 6$	$2/3$	-2	3
$y = 4$	0	4	none

CHAPTER 4 · INVESTIGATION 4.3

Gradient and equations of lines

What do you notice?

- Lines with the same gradient were parallel, whichever form they were written in.
- The intercepts decided where the line crossed the axes, not its direction.
- A horizontal line had a gradient of 0 and no x-intercept unless it was the axis itself.
- Rearranging every equation into the same form made the comparison immediate.

Observation. The gradient and the intercepts of a line can be read from any form of its equation once it has been rearranged, and together they fix the line completely.

MATHEMATICAL CONCLUSION

A straight line is determined by its gradient and one point, and $y = mx + c$ is the form that displays both without further work.

Your own words

CHAPTER 4 · 4.3

Gradient and equations of lines

The gradient of a line is the change in y for a unit change in x. It is the rate at which the output changes as the input changes, and it is the same everywhere on a straight line.

The equation y = mx + c gives the gradient as m and the intercept as c directly. Parallel lines have equal gradients and perpendicular lines have gradients whose product is −1.

A gradient of zero is a horizontal line, and a vertical line has no gradient because the change in x is zero. A line with an undefined gradient is not the graph of a function of x.

The gradient is the slope of a real rate: speed, cost per item, or change per year. Reading it from an equation is the fastest way to interpret a model.

A straight line is the picture of a constant pattern of change: equal steps in the input give equal steps in the output. When the steps stop being equal, the graph bends and the model is no longer linear.

> FORMAL DEVELOPMENT
>
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In your own words

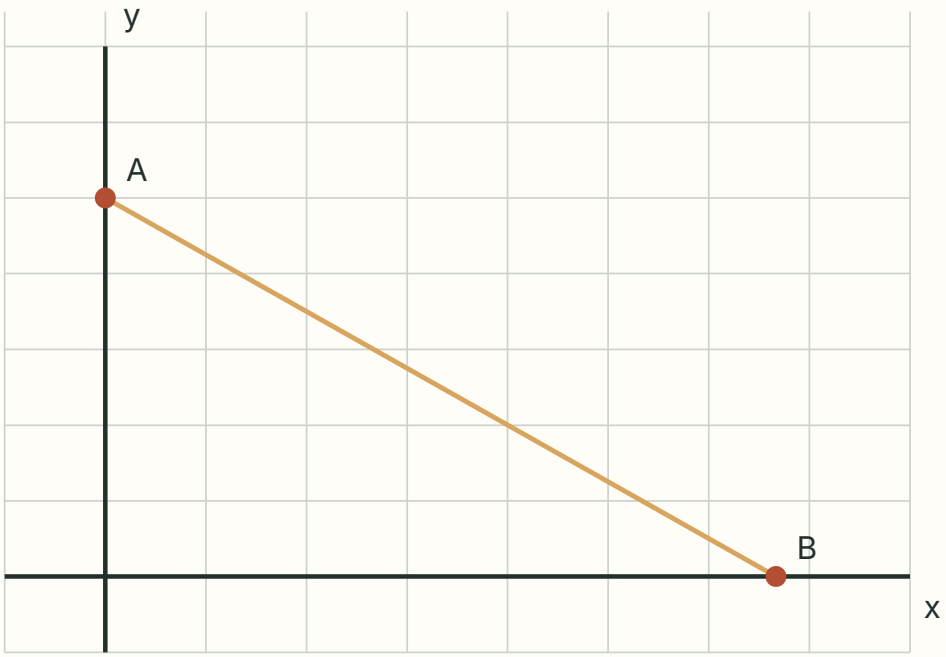
CHAPTER 4 · 4.3 · WORKED EXAMPLE 1

Gradient and equations of lines

WORKED EXAMPLE 4.3.1

Problem. A line has equation 3x + 4y = 20. Find its gradient, its intercepts, and the gradient of a line perpendicular to it.

the line 3x + 4y = 20



Mathematical reasoning. Rearrange into y = mx + c, then read the gradient and use the perpendicular rule.

1. 3x + 4y = 20 gives 4y = −3x + 20, so y = (−3/4)x + 5.
2. The gradient is −3/4 and the intercept is 5.
3. The x-intercept is at y = 0, so 3x = 20 and x = 20/3.
4. A perpendicular line has gradient 4/3, because (−3/4)(4/3) = −1.
5. Both intercepts are positive, so the line falls through the first quadrant.

Final answer. gradient −3/4, intercepts (0, 5) and (20/3, 0), perpendicular gradient 4/3

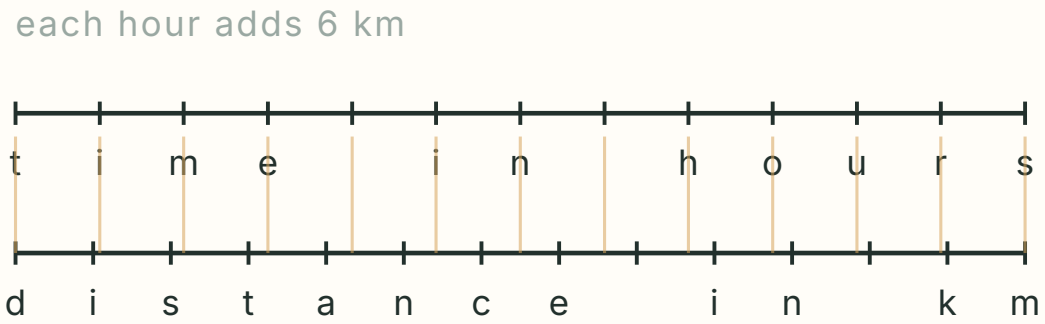
Interpretation. Rearranging into gradient-intercept form makes the two key features visible without any drawing.

CHAPTER 4 · 4.3 · WORKED EXAMPLE 2

Gradient and equations of lines

WORKED EXAMPLE 4.3.2

Problem. A cyclist's distance from home is d = 6t + 2, with t in hours. Find the distance at the start, the speed, and the time to reach 50 km.



Mathematical reasoning. Read the intercept and gradient from the equation, then solve for the required time.

1. At t = 0 the distance is d = 2 km, so she starts 2 km from home.
2. The gradient is 6, so her speed is 6 km/h.
3. Setting 6t + 2 = 50 gives 6t = 48, so t = 8 hours.
4. The straight line assumes a constant speed with no stops.
5. The model is therefore a simplification of any real journey.

Final answer. 2 km at the start, 6 km/h, 8 hours to reach 50 km

Interpretation. The two numbers in the equation are the two facts about the journey: where it began and how fast it moved.

CHAPTER 4 · INVESTIGATION 4.4

Parallel and perpendicular lines

INQUIRY QUESTION

How do the gradients of two lines decide whether they are parallel or perpendicular?

YOUR TASK

Take six lines of your own: three parallel pairs and three perpendicular pairs. For each, rearrange into gradient-intercept form, compute the product of the gradients, and record whether the pair is parallel, perpendicular or neither.

pair	gradients	product	relation
$y = 3x + 1$, $y = 3x - 4$	3, 3	9	parallel
$y = -2x$, $y = -2x + 7$	−2, −2	4	parallel
$y = x$, $y = -x + 3$	1, −1	−1	perpendicular
$y = 2x$, $y = -\frac{1}{2}x$	2, −0.5	−1	perpendicular
$y = 4x + 1$, $y = -\frac{1}{4}x + 2$	4, −0.25	−1	perpendicular

CHAPTER 4 · INVESTIGATION 4.4

Parallel and perpendicular lines

What do you notice?

- Every parallel pair had equal gradients.
- Every perpendicular pair had a product of exactly −1.
- The perpendicular gradient was always the negative reciprocal.
- A vertical and a horizontal line are perpendicular, and the product rule does not apply to them.

Observation. Parallel lines have equal gradients and perpendicular lines have gradients whose product is −1, and both facts are read from the gradient-intercept form.

MATHEMATICAL CONCLUSION

The relation between two lines is decided by their gradients: equal gradients give parallel lines and a product of −1 gives perpendicular lines.

Your own words

CHAPTER 4 · 4.4

Parallel and perpendicular lines

Two lines are parallel when they have the same gradient, and they are perpendicular when the product of their gradients is −1. Both statements follow from the meaning of gradient as change in y per unit change in x.

A vertical line has no gradient, so it is parallel to every other vertical line and perpendicular to every horizontal one. These are the two cases the product rule does not cover.

Rearranging an equation into y = mx + c makes the gradient visible, so the first step in any question about parallel or perpendicular lines is to rearrange.

The gradient of a line perpendicular to one of gradient m is −1/m, which is the negative reciprocal. This is the rule used to find the equation of a perpendicular through a given point.

> FORMAL DEVELOPMENT
>
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In your own words

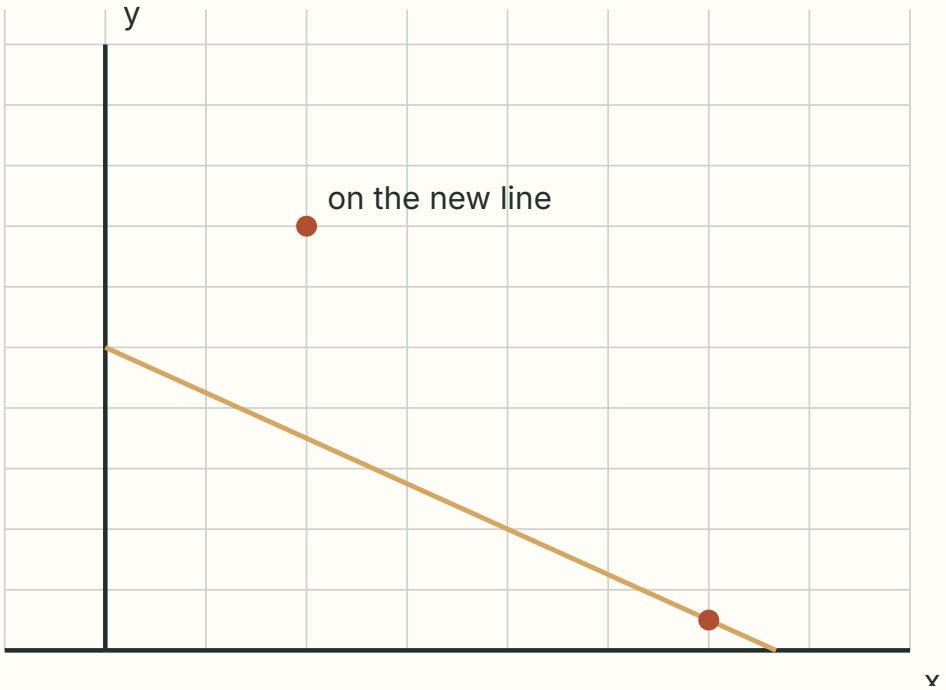
CHAPTER 4 · 4.4 · WORKED EXAMPLE 1

Parallel and perpendicular lines

WORKED EXAMPLE 4.4.1

Problem. Find the equation of the line through (2, 7) that is parallel to $3x + 4y = 20$.

the parallel line through (2, 7)



Mathematical reasoning. Find the gradient of the given line, then use it with the point to find the new line.

1. $3x + 4y = 20$ gives $y = (-3/4)x + 5$, so the gradient is $-3/4$.
2. A parallel line has the same gradient, so the new line is $y = (-3/4)x + c$.
3. Substituting (2, 7) gives $7 = (-3/4)(2) + c$, so $c = 8.5$.
4. Hence the line is $y = (-3/4)x + 8.5$, or $3x + 4y = 34$.
5. The two lines never meet, which is what parallel means.

Final answer. $y = -\frac{3}{4}x + 8.5$, or $3x + 4y = 34$

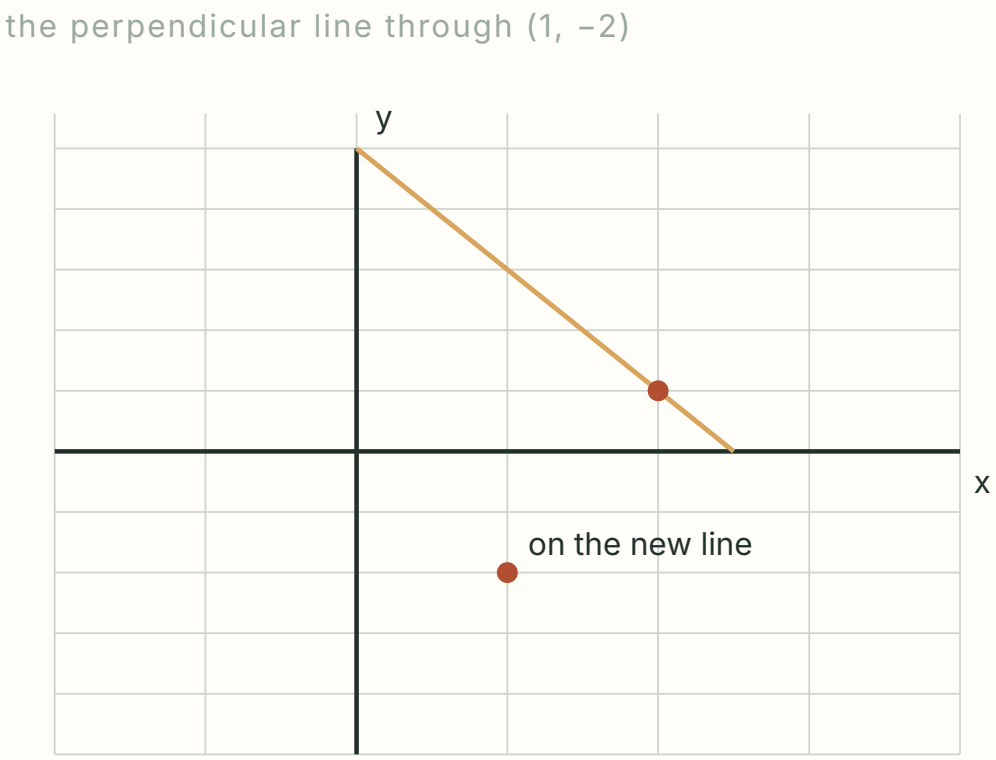
Interpretation. Parallel lines share a gradient, so one gradient and one point fix the new line completely.

CHAPTER 4 · 4.4 · WORKED EXAMPLE 2

Parallel and perpendicular lines

WORKED EXAMPLE 4.4.2

Problem. Find the equation of the line through (1, −2) that is perpendicular to y = 2x + 5.



Mathematical reasoning. Take the negative reciprocal of the gradient, then use the point.

1. The gradient of y = 2x + 5 is 2.
2. A perpendicular line has gradient −1/2.
3. So the new line is y = (−1/2)x + c.
4. Substituting (1, −2) gives −2 = (−1/2)(1) + c, so c = −1.5.
5. Hence the line is y = −½x − 1.5, and the product of the gradients is 2 × (−1/2) = −1.

Final answer. y = −½x − 1.5

Interpretation. The negative reciprocal is the only gradient that makes the product −1, and the check is the product itself.

CHAPTER 4 · INVESTIGATION 4.5

Linear models and variation

INQUIRY QUESTION

How is a real rate turned into a linear model, and what does the model assume?

YOUR TASK

Find three real rates: a speed, a cost per unit and a change per year. Model each as a linear function, use it to interpolate one value and extrapolate another, and comment on the reliability of each prediction.

rate	model	interpolation	extrapolation
water filling a tank	$V = 4t + 20$	60 litres at $t = 10$	220 litres at $t = 50$
savings	$S = 50m + 200$	500 at $m = 6$	5200 at $m = 100$
temperature	$T = -0.5t + 20$	15 at $t = 10$	-30 at $t = 100$

CHAPTER 4 · INVESTIGATION 4.5

Linear models and variation

What do you notice?

- The interpolation values were close to what really happens.
- The extrapolation values grew without limit, which reality does not allow.
- The temperature model predicted an impossible negative value.
- Every model assumed a constant rate, which is the source of the failure.

Observation. A linear model is reliable inside the range of the data and increasingly unreliable outside it, because the assumption of a constant rate is rarely true for long.

MATHEMATICAL CONCLUSION

A linear model is built from a starting value and a rate, and its predictions are trustworthy only where the rate itself can be trusted.

Your own words

CHAPTER 4 · 4.5

Linear models and variation

A linear model has the form y = mx + c, where c is the starting value and m is the rate of change. Every linear model assumes that the rate is constant.

Building a model means identifying the two quantities, choosing units, and finding the rate from two data points. The rate is the gradient of the segment joining them.

Interpolation predicts inside the range of the data and extrapolation predicts outside it. Extrapolation assumes the rate continues, which is often false.

A model is judged by its usefulness, not its truth. A linear model of a cooling cup is wrong but useful for the first few minutes.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 4 · 4.5 · WORKED EXAMPLE 1

Linear models and variation

WORKED EXAMPLE 4.5.1

Problem. A plumber charges 45 dollars for a call-out and 60 dollars per hour. Write the cost as a linear model, find the cost of a 3-hour job, and find the longest job affordable for 300 dollars.

hours	cost
0	45
1	105
2	165
3	225
4	285
5	345

The cost rises by 60 for every extra hour.

Mathematical reasoning. Identify the fixed charge as the intercept and the hourly rate as the gradient.

1. The model is $C = 60h + 45$, with h in hours.
2. For a 3-hour job, $C = 60(3) + 45 = 225$ dollars.
3. Setting $60h + 45 = 300$ gives $60h = 255$, so $h = 4.25$ hours.
4. So the longest whole-number job is 4 hours, at 285 dollars.
5. The model assumes the same rate for every hour, including overtime.

Final answer. $C = 60h + 45$; 225 dollars for 3 hours; 4 hours within 300 dollars

Interpretation. The fixed charge is the intercept and the hourly rate is the gradient, and both come straight from the wording of the situation.

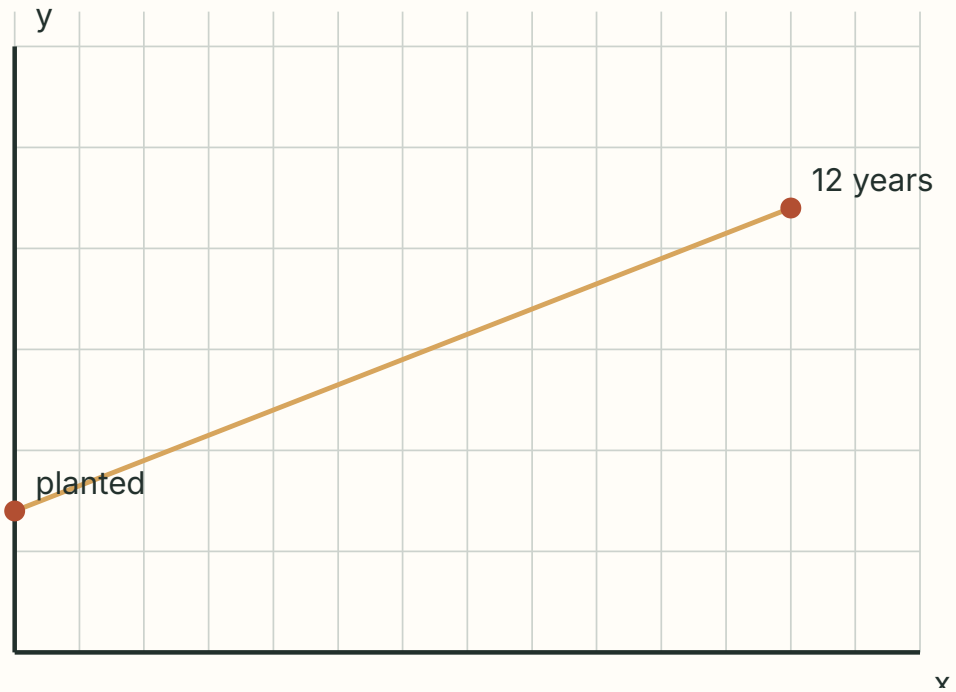
CHAPTER 4 · 4.5 · WORKED EXAMPLE 2

Linear models and variation

WORKED EXAMPLE 4.5.2

Problem. A tree is 1.4 m tall when planted and grows 0.25 m per year. Model its height, predict the height after 12 years, and comment on the model's limits.

the model is a straight line



Mathematical reasoning. Write the model in gradient-intercept form and substitute.

1. The model is h = 0.25t + 1.4, with t in years.
2. After 12 years, h = 0.25(12) + 1.4 = 4.4 m.
3. The model assumes the same growth every year.
4. In reality growth slows as the tree matures, so the model overestimates later heights.
5. It is useful for the first decade and unreliable beyond it.

Final answer. h = 0.25t + 1.4; 4.4 m after 12 years; unreliable in the long term

Interpretation. A linear model is a local description, and extrapolating it far beyond the data is where it fails.

CHAPTER 4 · INVESTIGATION 4.6

Investigation: modelling a fair fare

INQUIRY QUESTION

How is a fare modelled, and what does the model leave out?

YOUR TASK

Find two real fare structures: two bus routes, two phone plans, two delivery services or two taxis. Model each as a linear function, draw both graphs on the same axes, find the break-even point, and write a recommendation that states the journey length it assumes and what the model leaves out.

fare	flag-fall	rate	break-even	cheaper for short trips
company 1	3.00	2.20	4.29 km	yes
company 2	6.00	1.50	4.29 km	no
bus route A	0.50	0.80	—	yes
bus route B	2.00	0.40	3.75 stops	no

CHAPTER 4 · INVESTIGATION 4.6

Investigation: modelling a fair fare

What do you notice?

- Every fare had a flag-fall, so no graph passed through the origin.
- The break-even point was where the two lines crossed.
- The fare with the larger flag-fall was cheaper for long journeys only.
- Waiting time and surcharges were left out of every model.

> **Observation.** A fare is modelled by an intercept and a gradient, and the fairer fare depends on the journey length, so the recommendation must state the journey it assumes.

MATHEMATICAL CONCLUSION

A fare is a linear model whose intercept is the fixed charge and whose gradient is the rate, and comparing two fares gives a break-even point beyond which the ordering reverses.

Your own words

CHAPTER 4 · 4.6

Investigation: modelling a fair fare

A fare usually has a fixed part and a part that depends on distance or time, so it is a linear model whose intercept is the flag-fall and whose gradient is the rate per unit.

The graph of a fare against distance is a straight line that does not pass through the origin, so the fare is not proportional to the distance. The intercept is the charge for travelling no distance at all.

A fare model assumes the rate is constant, and real fares have waiting time, traffic, surcharges and shared rides, none of which a single straight line can show.

Comparing two fares means comparing their models over a stated journey, and the fairer fare depends on the distance travelled, which is why a single answer is impossible.

> FORMAL DEVELOPMENT
>
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In your own words

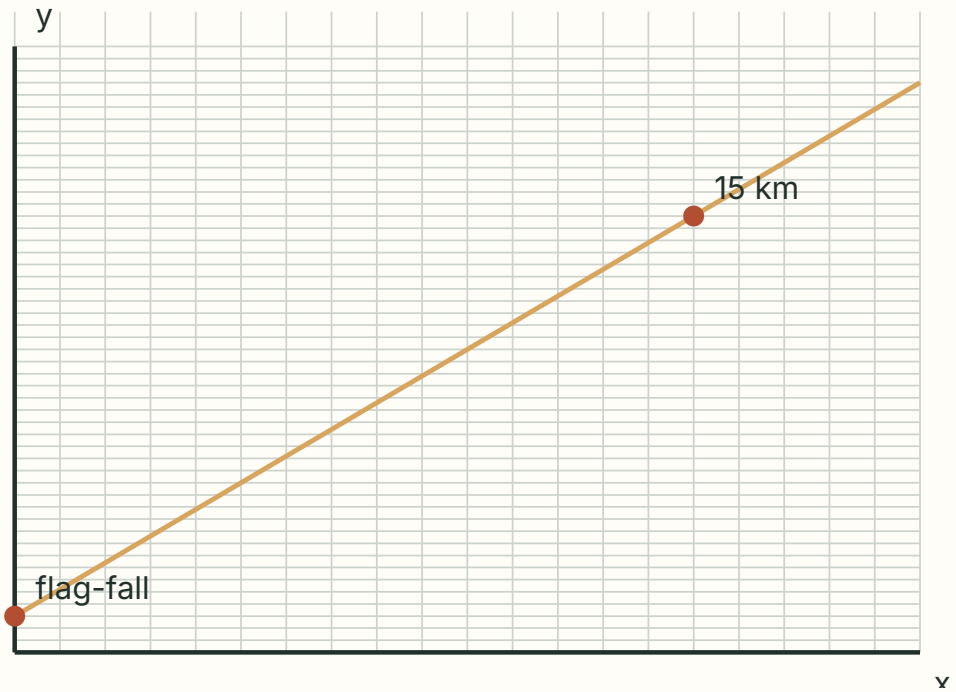
CHAPTER 4 · 4.6 · WORKED EXAMPLE 1

Investigation: modelling a fair fare

WORKED EXAMPLE 4.6.1

Problem. A taxi charges a flag-fall of 3 dollars plus 2.20 dollars per kilometre. Model the fare, draw the graph, and find the fare for a 15 km journey.

the fare rises linearly from the flag-fall



Mathematical reasoning. Read the intercept and the gradient from the wording, then substitute.

1. The model is C = 2.2d + 3, with d in kilometres.
2. The intercept of 3 dollars is the charge for travelling no distance.
3. The gradient of 2.20 is the charge per kilometre.
4. For 15 km the fare is 2.2 × 15 + 3 = 36 dollars.
5. The graph starts at (0, 3), not at the origin, so the fare is not proportional to the distance.

Final answer. C = 2.2d + 3, and the fare for 15 km is 36 dollars

Interpretation. The two numbers in the model are the two parts of the fare, and the non-zero intercept is what distinguishes a fare from a proportional charge.

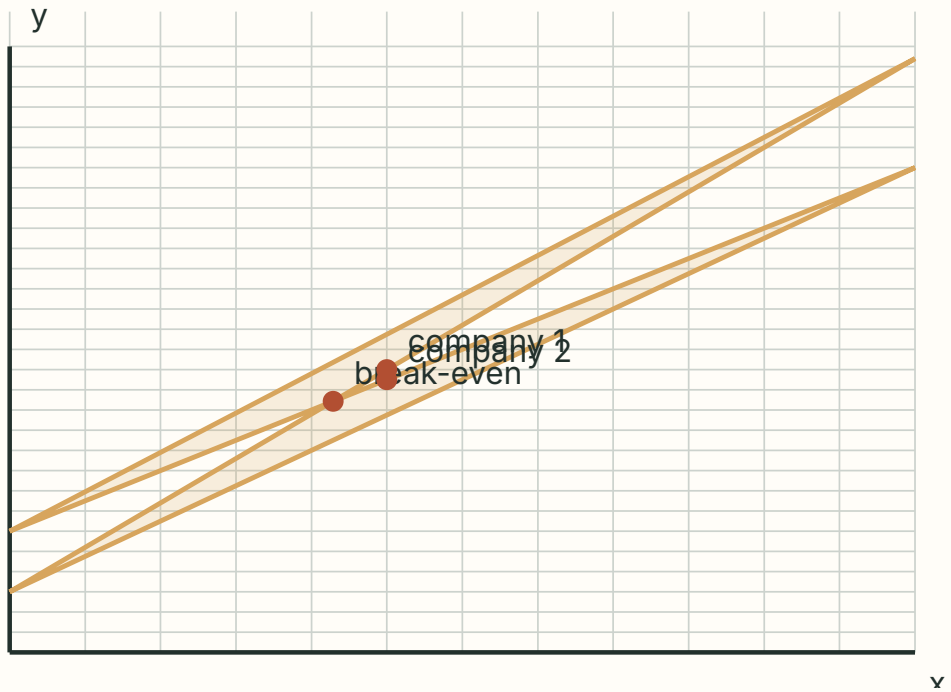
CHAPTER 4 · 4.6 · WORKED EXAMPLE 2

Investigation: modelling a fair fare

WORKED EXAMPLE 4.6.2

Problem. A second company charges 1.50 dollars per kilometre with a flag-fall of 6 dollars. Find the distance at which the two companies charge the same, and say which is fairer for a 5 km trip.

the two fare lines cross at about 4.3 km



Mathematical reasoning. Set the two models equal and solve, then compare at the stated distance.

1. The first company charges 3 + 2.2d and the second 6 + 1.5d.
2. Setting them equal gives 2.2d − 1.5d = 6 − 3.
3. So 0.7d = 3 and d ≈ 4.29 km.
4. For a 5 km trip the first charges 14 dollars and the second 13.50 dollars.
5. So the second is fairer for 5 km, and the first for longer trips.

Final answer. the fares are equal at about 4.29 km; the second company is cheaper for 5 km

Interpretation. Which fare is fairer depends on the distance, so a recommendation must state the journey it assumes.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

4.A1 FLUENCY

Find the midpoint of (2, 5) and (8, −1).

4.A2 FLUENCY

Find the gradient of the line through (1, 3) and (5, 11).

4.A3 FLUENCY

Find the gradient of the line $y = -3x + 7$.

4.A4 FLUENCY

Find $f(4)$ for $f(x) = 2x - 5$.

4.A5 FLUENCY

Find the y-intercept of $2x + 5y = 20$.

4.A6 FLUENCY

A line has gradient 3 and passes through (0, −2). Write its equation.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

4.A7 FLUENCY

Find the gradient of a line perpendicular to $y = 2x + 1$.

4.A8 FLUENCY

Find x when $f(x) = 9$ for $f(x) = 4x - 7$.

4.A9 FLUENCY

Find the gradient of the line through $(-2, 4)$ and $(4, -8)$.

4.A10 FLUENCY

A car travels 150 km in 2 hours. Find its average speed.

4.A11 FLUENCY

Find the x-intercept of $y = 5x - 15$.

4.A12 FLUENCY

Write the equation of the horizontal line through $(3, 4)$.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

4.B1 REASONING

The number of tiles in the nth figure of a pattern is $3n + 2$. Find the first four terms and test the rule on the tenth figure.

4.B2 REASONING

Investigate the pattern in the gradients of the lines joining (0, 0) to (1, 1), (2, 4), (3, 9) and (4, 16).

4.B3 REASONING

Conjecture a rule for the midpoint of the points (n, 0) and (0, n) and test it.

4.B4 REASONING

Investigate the pattern in the number of unit squares a diagonal of an n by n grid crosses.

4.B5 REASONING

Build a table of the first differences of $f(n) = n^2$ and describe the pattern.

4.B6 REASONING

Conjecture a rule for the distance from (0, 0) to (n, n) and justify it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

4.B7 REASONING

Investigate the gradients of the segments joining consecutive points of (0, 1), (1, 3), (2, 6), (3, 10).

4.B8 CHALLENGE

A pattern of lines is drawn from (0, 0) to (1, 1), (2, 1), (3, 1) and so on. Investigate the gradients and find the rule.

4.B9 REASONING

Find the pattern in the intercepts of the lines $y = nx + (3 - n)$ for $n = 1$ to 5 and justify it.

4.B10 REASONING

Investigate the point of intersection of $y = 2x + 1$ and $y = -x + 4$ as the coefficients change.

4.B11 REASONING

Conjecture a rule for the number of regions made by n lines in general position.

CRITERION C Communicating

Criterion C practice · Communicating

4.C1 REASONING
Explain in your own words what a function is.

4.C2 FLUENCY
Write a complete solution to: find the equation of the line through (1, 4) with gradient −3.

4.C3 REASONING
Choose between a graph and an equation to answer a question about speed and justify.

4.C4 FLUENCY
Set out a full solution to: find the gradient and intercepts of $4x - 2y = 8$.

4.C5 REASONING
Explain why the gradient is the same everywhere on a straight line.

4.C6 REASONING
Describe, with an example, the difference between interpolation and extrapolation.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

4.C7 REASONING
Explain why a vertical line is not the graph of a function of x.

4.C8 REASONING
Write the reasoning that shows the product of perpendicular gradients is −1.

4.C9 REASONING
Set out the argument that two quantities are in direct proportion exactly when their graph passes through the origin.

4.C10 REASONING
Explain what the intercept of a distance-time graph represents.

4.C11 REASONING
Describe the relationship between the gradient of a distance-time graph and speed.

4.C12 REASONING
Explain, with an example, when a linear model is a poor description of a real rate.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

4.D1 APPLICATION

A taxi charges 3 dollars plus 2.20 dollars per km. Model the fare, find the fare for 15 km, and find the distance affordable for 30 dollars.

4.D2 APPLICATION

A worker is paid 12 dollars per hour plus a 20 dollar bonus. Model the pay, find the pay for 35 hours, and find the hours needed for 400 dollars.

4.D3 APPLICATION

A tank holds 500 litres and is filled at 12 litres per minute. Model the volume, find when it is full, and find the volume after 20 minutes.

4.D4 APPLICATION

A candle 25 cm long burns 1.5 cm per hour. Model its height, find when it burns out, and find the height after 6 hours.

4.D5 APPLICATION

A train leaves at 60 km/h and a second leaves the same station 1 hour later at 90 km/h. Find when the second catches the first.

4.D6 APPLICATION

A school's roll falls by 15 students per year from 480. Model the roll, find when it reaches 300, and comment on the model's limits.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

4.D7 APPLICATION

A recipe serves 6 with 400 g of flour. Model the flour needed, find the amount for 11 people, and find the people served by 1.5 kg.

4.D8 APPLICATION

A cyclist covers 12 km in 40 minutes. Find the speed, model the distance, and find the distance covered in 2 hours 30 minutes.

4.D9 REASONING

A phone battery falls from 100% to 68% in 4 hours. Model the charge, find when it reaches zero, and comment on the model.

4.D10 APPLICATION

A plant grows from 15 cm to 47 cm in 8 weeks. Model the height, find the weekly growth, and predict the height after 20 weeks with a comment.

4.D11 APPLICATION

Two cyclists start 30 km apart and ride towards each other at 14 and 16 km/h. Find when and where they meet.

4.D12 REASONING

Explain, with a model of your own, when reporting a prediction to a false precision is misleading.

CHAPTER 4 · APPLICATION AND ASSESSMENT

Applying chapter 4

Each task asks you to move between representations. Say which representation you are using and why it answers the question.

A1 · The delivery route

A courier starts 3 km from the depot and rides away at 15 km/h for 40 minutes, rests for 15 minutes, then returns at 12 km/h. Model the distance from the depot as a function of time, draw the graph, and find the total time until the courier is back at the depot.

Criteria A and C · state the model for each part of the journey and read the graph.

A2 · The two rates

A printer charges 80 dollars plus 0.15 dollars per page. Another charges no fee but 0.40 dollars per page. Model both, find the number of pages at which the costs are equal, and find the number of pages for which the first printer is cheaper by at least 20 dollars.

Criteria A and D · solve the equation for equality and the inequality for the saving.

CHAPTER 4 · REVIEW

What you should now be able to do

Key ideas

- A coordinate pair fixes a point on a plane.
- The midpoint averages the coordinates; the gradient divides the differences.
- A function gives exactly one output for each permitted input.
- $y = mx + c$ displays the gradient and the intercept.
- Parallel lines have equal gradients; perpendicular gradients multiply to -1.
- A proportional relationship has a constant ratio and a graph through the origin.
- A rate is only meaningful with its units stated.
- Interpolation is inside the data; extrapolation assumes the rate continues.

Vocabulary

coordinate · midpoint · gradient · intercept · function · domain · range · linear model · rate · proportion · interpolation · extrapolation

CHAPTER 4 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

4.R1 Find the gradient of the line through (−1, 5) and (3, −7).

4.R2 Write the equation of the line with gradient −2 through (0, 6).

4.R3 Find f(3) and f(−2) for f(x) = 5 − 2x.

4.R4 Find the intercepts of 3x − 2y = 12.

4.R5 A car covers 210 km in 3 hours. Find the speed and the time for 350 km.

4.R6 Explain why y = mx + c is not proportional unless c = 0.

CHAPTER 4 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A function is a rule that gives exactly one output for each permitted input, and the form of that rule decides what can be read from it — a graph shows change, a table shows values, and an equation shows structure — so the same relationship can be represented in several ways, each of which answers a different question.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of representation, change and pattern appear in your work?
- How is the global context of orientation in space and time, and its exploration of maps, routes and rates, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a linear model of a journey or a rate, and what would you still like to ask?

CHAPTER 4 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A route, a rate or a map is a function in disguise. Model one and show which representation answers the question you care about.

Service as Action

COMMUNITY NEED	Journeys, fares and rates are quoted without the model behind them.
MATHEMATICAL CONNECTION	Linear functions, gradients and rates.
PROPOSED ACTION	Model one local journey or fare and publish the model with its assumptions.
EXPECTED IMPACT	A reader can compare two fares or routes on the same terms.
STUDENT REFLECTION	Record the model, the graph and what you found.

Community Project

PROJECT IDEA	A route audit for a school or neighbourhood.
NEED	Distances, times and costs are estimated without a shared method.
RESEARCH QUESTION	Which route is fastest for which purpose, and what does the model assume?
MATHEMATICS	Coordinates, gradients and linear models.
ACTION OR OUTCOME	A one-page map with a distance-time model for each route.
BENEFICIARIES	Everyone who travels the route.
EVIDENCE	The map, the models and the comparison.

Personal Project connection · MYP 5

PERSONAL INTEREST	A journey, a sport or a hobby with a rate.
LEARNING GOAL	To model a rate you care about and state what the model assumes.
PRODUCT OR OUTCOME	A linear model with a graph and a prediction.
MATHEMATICS	Gradients, intercepts and proportional reasoning.
PROCESS	Collect two data points, build the model, and test a prediction.
EVIDENCE	The model, the graph and the test of the prediction.

CHAPTER 5 · PART II · THINKING WITH MODELS

05

Quadratic Expressions and Equations

Every quadratic can be solved, and the method chosen decides how much work is needed and what the answer reveals about the equation.

ENGAGE

A ball is thrown upwards from a platform 5 m high. Its height after t seconds is $h = 20t - 5t^2$. When is it 15 m high, and when does it hit the ground?

Framing the inquiry

Key concept: Logic
Related concepts: Equivalence · Generalisation · Justification
Global context: Scientific and technical innovation
Global context exploration: Design, motion and optimisation

Statement of inquiry: A quadratic equation is solved by transforming it into an equivalent form whose solutions are visible, and the justification for each transformation is a reversible step; the discriminant then decides, before any solving, how many solutions exist, which turns a calculation into a general claim.

Questions to guide us

- **Factual:** What are the four methods of solving a quadratic equation, and what does the discriminant tell you?
- **Conceptual:** Why does completing the square work for every quadratic, and how does the discriminant justify a claim about the number of solutions?
- **Debatable:** Is the quadratic formula a tool for understanding or a tool for avoiding understanding, and does it matter which method a student uses?

ATL · REFLECTION. Students consider which method to use for a given quadratic, justify the choice, and reflect on the accuracy and the efficiency of the method they chose.

CHAPTER 5 · INVESTIGATION 5.1

From expressions to equations

INQUIRY QUESTION

How does a quadratic equation arise from a quadratic expression, and why must the equation be rearranged before it can be solved?

YOUR TASK

Take five equations that are not in standard form: one with terms on both sides, one with brackets, one with fractions, one with the variable on both sides, and one of your own. Rearrange each into the form $ax^2 + bx + c = 0$, state a, b and c, then check one solution by substitution into the original equation.

equation	standard form	a, b, c	check
$x^2 = 7x - 12$	$x^2 - 7x + 12 = 0$	1, −7, 12	$9 - 63 + 12 = 0$
$(x - 2)(x + 3) = 0$	$x^2 + x - 6 = 0$	1, 1, −6	$-3 - 3 - 6 = 0$
$2x(x - 5) = 0$	$2x^2 - 10x = 0$	2, −10, 0	$0 - 0 = 0$
$x^2 + 4 = 4x$	$x^2 - 4x + 4 = 0$	1, −4, 4	$4 - 8 + 4 = 0$
$3x^2 = 12$	$3x^2 - 12 = 0$	3, 0, −12	$12 - 12 = 0$

CHAPTER 5 · INVESTIGATION 5.1

From expressions to equations

What do you notice?

- Every equation became a quadratic once it was rearranged.
- The equation with no linear term gave symmetric roots.
- The equation $(x - 2)(x + 3) = 0$ was already factorised, so its roots came from the factors directly.
- The check into the original equation was as important as the rearrangement.

> **Observation.** Every quadratic equation can be rearranged into the standard form, and the standard form is what the solving methods need.

MATHEMATICAL CONCLUSION

Rearranging to $ax^2 + bx + c = 0$ is the first step of every quadratic method, and the zero-product property is the fact that turns a factorisation into solutions.

Your own words

CHAPTER 5 · 5.1

From expressions to equations

An expression is a rule for computing a value, and an equation is a claim that two expressions are equal. Solving an equation means finding every input that makes the claim true.

A quadratic equation is one that can be written in the standard form $ax^2 + bx + c = 0$, and the first step is always to rearrange it into that form, because every solving method assumes it.

The zero-product property says that if a product is zero then at least one factor is zero, and this is the single fact that turns factorisation into a method of solving.

Expanding and factorising are the two directions of the same equivalence, and a generalisation stated with letters is justified by the identity that proves it for every case at once.

A solution that has been found must be checked by substitution, because rearranging can introduce a value that does not satisfy the original claim.

> FORMAL DEVELOPMENT
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.1 · WORKED EXAMPLE 1

From expressions to equations

WORKED EXAMPLE 5.1.1

Problem. Evaluate $(2x - 3)(x + 4)$ when $x = 1$, first by substitution and then by expanding.

method	working	value
substitute first	$(2 - 3)(1 + 4)$	-5
expand first	$2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$	-5

Both routes give the same value, as equivalence requires.

Mathematical reasoning. Substitution gives the value directly; expansion gives a form that works for every x.

1. Substituting $x = 1$: $(2(1) - 3)(1 + 4) = (-1)(5) = -5$.
2. Expanding: $(2x - 3)(x + 4) = 2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$.
3. Substituting $x = 1$ into the expansion: $2 + 5 - 12 = -5$.
4. The two answers agree, which they must because expansion is an equivalence.
5. The expanded form would also give the value at any other x without re-multiplying.

Final answer. −5, by either route

Interpretation. Expanding is worth the effort only when more than one value is needed, which is the usual case in modelling.

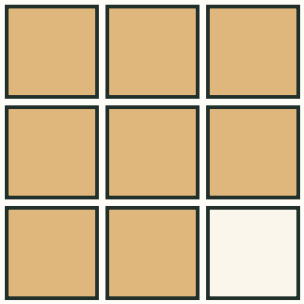
CHAPTER 5 · 5.1 · WORKED EXAMPLE 2

From expressions to equations

WORKED EXAMPLE 5.1.2

Problem. Expand $(3x + 2)^2$ and explain why it is not $9x^2 + 4$.

a 3 by 3 square of side 3x + 2



$9x^2$, 12x and 4

Mathematical reasoning. Use the perfect square identity and identify the missing cross term.

1. $(3x + 2)^2 = (3x)^2 + 2(3x)(2) + 2^2 = 9x^2 + 12x + 4$.
2. The term 12x is the cross term, and it is what $9x^2 + 4$ leaves out.
3. A check at x = 1 gives 25 against 13, so the claim fails.
4. The missing term is twice the product of the two parts.
5. This is the commonest error in expanding, and it comes from treating a square as distributive over addition.

Final answer. $9x^2 + 12x + 4$

Interpretation. A single substitution refutes a false identity, which is why checking with a number is a reliable habit.

CHAPTER 5 · INVESTIGATION 5.2

Solving by factorising

INQUIRY QUESTION

Why does factorisation solve a quadratic equation, and when does it fail?

YOUR TASK

Take eight quadratics of your own: four monic and four with a leading coefficient other than 1. Solve each by factorising, and record which factorised at the first attempt and which needed a different method.

quadratic	factorisation	roots	method
$x^2 - 9$	$(x - 3)(x + 3)$	3, −3	difference of squares
$x^2 + 10x + 25$	$(x + 5)^2$	−5 (repeated)	perfect square
$x^2 - 5x + 6$	$(x - 2)(x - 3)$	2, 3	monic
$3x^2 - 12$	$3(x - 2)(x + 2)$	2, −2	common factor first
$2x^2 - 5x - 3$	$(2x + 1)(x - 3)$	−½, 3	grouping

CHAPTER 5 · INVESTIGATION 5.2

Solving by factorising

What do you notice?

- Every quadratic that factorised gave its roots immediately.
- The difference of squares gave roots symmetric about zero.
- The perfect square gave one repeated root.
- The common factor had to be taken out before factorising.
- A quadratic that did not factorise would need the formula instead.

Observation. Factorising solves a quadratic by turning the equation into a product equal to zero, and the two factors give the two roots.

MATHEMATICAL CONCLUSION

The zero-product property is what makes factorisation a solving method, and the method applies whenever the quadratic factorises over the integers.

Your own words

CHAPTER 5 · 5.2

Solving by factorising

To solve a factorised quadratic, set each factor equal to zero. This is the zero-product property, and it works because a product is zero only when one of its factors is zero.

The method needs a quadratic that factorises over the integers, and it fails silently when the factors are not found, so a check by substitution is essential.

Factorising the difference of two squares $x^2 - a^2$ gives two roots that are symmetric about zero, and factorising a perfect square gives one repeated root.

When the leading coefficient is not 1, factorise by grouping or by inspection, and the two factors then give the two roots directly.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

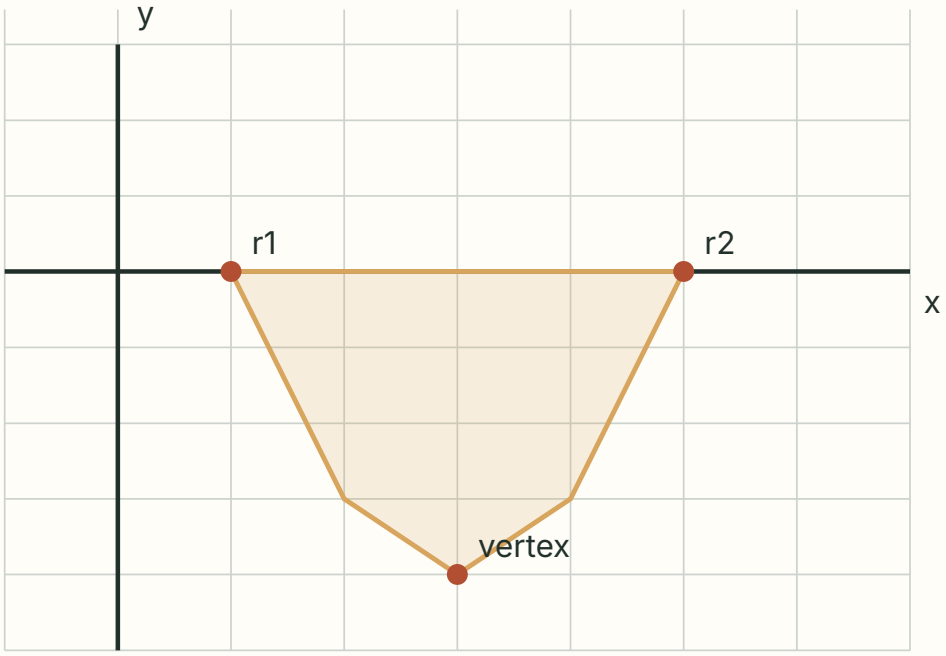
CHAPTER 5 · 5.2 · WORKED EXAMPLE 1

Solving by factorising

WORKED EXAMPLE 5.2.1

Problem. Solve $x^2 - 6x + 5 = 0$ by factorising, and check each root.

the roots are where the parabola meets the axis



Mathematical reasoning. Factorise the quadratic, then set each factor to zero.

1. Find two numbers that multiply to 5 and add to −6: −1 and −5.
2. So $x^2 - 6x + 5 = (x - 1)(x - 5)$.
3. Setting each factor to zero gives $x - 1 = 0$ or $x - 5 = 0$.
4. Hence $x = 1$ or $x = 5$.
5. Check: $1 - 6 + 5 = 0$ and $25 - 30 + 5 = 0$, so both roots work.

Final answer. $x = 1$ or $x = 5$

Interpretation. The two factors give the two roots, and each root is a value that makes the original equation true.

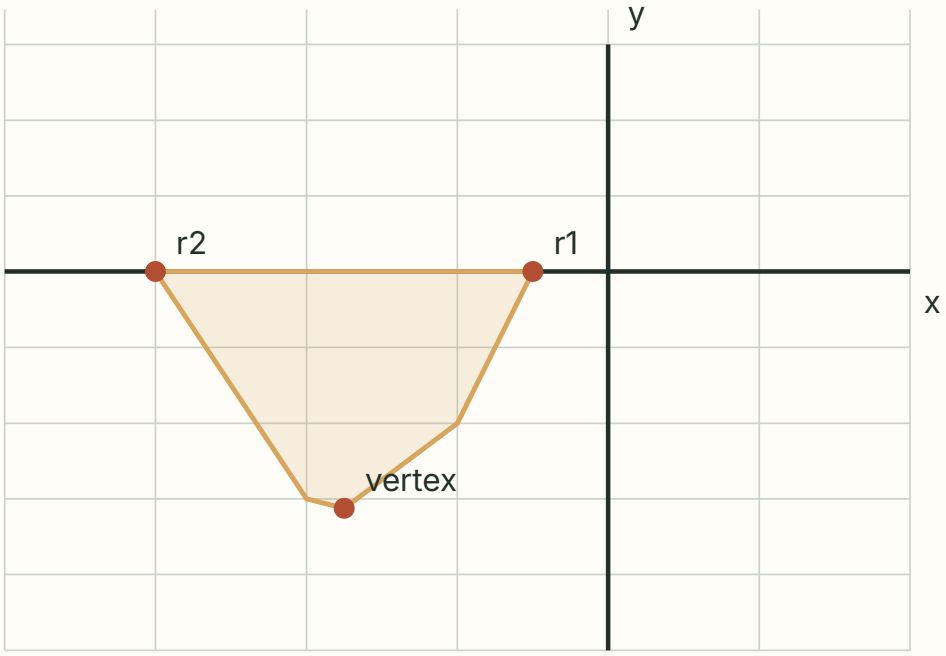
CHAPTER 5 · 5.2 · WORKED EXAMPLE 2

Solving by factorising

WORKED EXAMPLE 5.2.2

Problem. Solve $2x^2 + 7x + 3 = 0$ by factorising.

two roots, one of them fractional



Mathematical reasoning. The leading coefficient is not 1, so factorise by inspection or by grouping.

1. $2x^2 + 7x + 3 = (2x + 1)(x + 3)$, because $2x^2 + 6x + x + 3$ regroups.
2. Setting each factor to zero: $2x + 1 = 0$ or $x + 3 = 0$.
3. Hence $x = -½$ or $x = -3$.
4. Check: $2(\frac{1}{4}) + 7(-½) + 3 = 0.5 - 3.5 + 3 = 0$.
5. The roots are rational but not integers, so factorising still works.

Final answer. $x = -½$ or $x = -3$

Interpretation. A non-monic quadratic factorises just as well, and the roots can be fractions without spoiling the method.

CHAPTER 5 · INVESTIGATION 5.3

Completing the square

INQUIRY QUESTION

Why does completing the square work for every quadratic, and what does it reveal?

YOUR TASK

Take five quadratics: one monic with an even coefficient of x, one monic with an odd coefficient, one with a leading coefficient of 2, one with a negative leading coefficient, and one of your own. Complete the square for each, and use the completed form to find the vertex and the axis of symmetry.

quadratic	completed square	vertex
$x^2 + 4x + 1$	$(x + 2)^2 - 3$	$(-2, -3)$
$x^2 + 5x + 2$	$(x + 2.5)^2 - 4.25$	$(-2.5, -4.25)$
$2x^2 - 8x + 3$	$2(x - 2)^2 - 5$	$(2, -5)$
$-x^2 + 6x - 5$	$-(x - 3)^2 + 4$	$(3, 4)$
$3x^2 + 12x - 2$	$3(x + 2)^2 - 14$	$(-2, -14)$

CHAPTER 5 · INVESTIGATION 5.3

Completing the square

What do you notice?

- The x-coordinate of the vertex was always $-b/(2a)$.
- The y-coordinate was the value of the expression at that x.
- A negative leading coefficient turned the minimum into a maximum.
- The odd coefficient of x gave a half-integer in the vertex.

Observation. The completed square always has the form $a(x + p)^2 + q$, and the vertex is read from it immediately.

MATHEMATICAL CONCLUSION

Completing the square works for every quadratic, and it is the method that reveals the vertex, which is why the other methods are derived from it.

Your own words

CHAPTER 5 · 5.3

Completing the square

Completing the square rewrites $ax^2 + bx + c$ as $a(x + p)^2 + q$. It works for every quadratic because the square can always absorb the linear term.

The method is to factor out the leading coefficient, add and subtract the square of half the coefficient of x, and regroup. The value is unchanged because the same quantity is added and subtracted.

The completed form shows the vertex of the parabola at $(-p, q)$ and the axis of symmetry at $x = -p$, which is the information the expanded form hides.

The quadratic formula is derived by completing the square on $ax^2 + bx + c = 0$, so completing the square is the more fundamental of the two methods.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

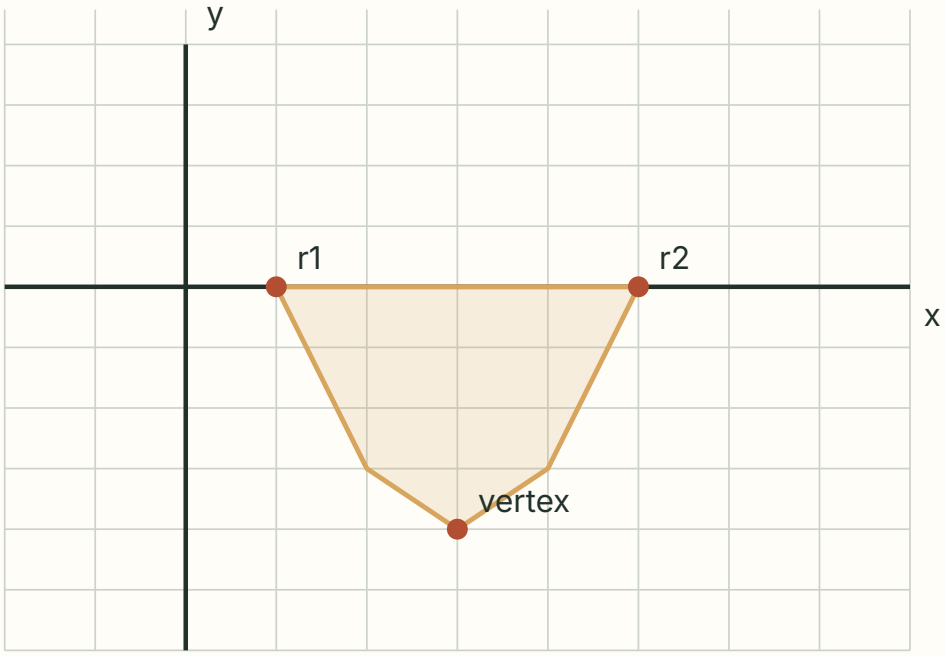
CHAPTER 5 · 5.3 · WORKED EXAMPLE 1

Completing the square

WORKED EXAMPLE 5.3.1

Problem. Solve $x^2 - 6x + 5 = 0$ by completing the square.

the vertex is at x = 3, halfway between the roots



Mathematical reasoning. Add and subtract the square of half the coefficient of x, then solve the resulting square.

1. Half of −6 is −3, and its square is 9.
2. So $x^2 - 6x + 5 = (x^2 - 6x + 9) - 9 + 5 = (x - 3)^2 - 4$.
3. The equation becomes $(x - 3)^2 = 4$.
4. Taking square roots gives $x - 3 = \pm 2$, so $x = 5$ or $x = 1$.
5. The vertex is (3, −4) and the axis of symmetry is $x = 3$, which is the average of the two roots.

Final answer. x = 5 or x = 1

Interpretation. The completed square gives the roots and the vertex from the same rewrite, which is why it is the more informative method.

CHAPTER 5 · INVESTIGATION 5.4

The quadratic formula and discriminant

INQUIRY QUESTION

How does the quadratic formula solve every quadratic, and what does its discriminant reveal about the roots?

YOUR TASK

Take five quadratics, including one that does not factorise over the integers. Solve each by the formula and by at least one other method, record which was quicker, and check one root by substitution in every case.

quadratic	formula root	other method	quicker
$x^2 - 5x + 6$	2 or 3	factorising	factorising
$x^2 + 4x + 1$	$-2 \pm \sqrt{3}$	completing the square	equal
$2x^2 + 5x - 3$	0.5 or −3	factorising	factorising
$x^2 + x + 1$	complex	discriminant only	discriminant
$3x^2 - 2x - 7$	$(1 \pm 22)/3$	formula	formula

CHAPTER 5 · INVESTIGATION 5.4

The quadratic formula and discriminant

What do you notice?

- Factorising was quickest whenever the discriminant was a perfect square.
- The formula always worked, even when factorising failed.
- One quadratic had a negative discriminant, so it had no real roots and no method could produce any.
- The check by substitution passed in every case where real roots existed.

> **Observation.** The choice of method is a question of efficiency, and the formula is the method that is never wrong but is not always the fastest.

MATHEMATICAL CONCLUSION

A quadratic is solved by the method that suits its coefficients, and the formula is the fallback that always succeeds.

Your own words

CHAPTER 5 · 5.4

The quadratic formula and discriminant

The quadratic formula $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ solves every quadratic equation, and it is obtained by completing the square on $ax^2 + bx + c = 0$.

The formula is used when factorising is not quick, and it always works, so it is the method of last resort that never fails.

The expression $b^2 - 4ac$ under the root is called the discriminant, and its sign decides the nature of the roots: positive gives two distinct real roots, zero gives one repeated root, and negative gives none.

The discriminant also justifies a claim about every quadratic of a given shape, because a single calculation proves what would otherwise need infinitely many substitutions.

The graph and the algebra must agree: the discriminant says whether the parabola crosses the x-axis twice, touches it once, or misses it entirely.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.4 · WORKED EXAMPLE 1

The quadratic formula and discriminant

WORKED EXAMPLE 5.4.1

Problem. Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula.

step	value
a, b, c	3, −5, −2
$b^2 - 4ac$	25 + 24 = 49
√49	7
x = (5 ± 7)/6	2 or −1/3

The discriminant is a perfect square, so the roots are rational.

Mathematical reasoning. Identify a, b and c, evaluate the discriminant, and simplify both roots.

1. Here a = 3, b = −5 and c = −2.
2. $b^2 - 4ac = 25 - 4(3)(-2) = 25 + 24 = 49$.
3. So $x = (5 \pm 7)/6$.
4. The roots are $x = 12/6 = 2$ and $x = -2/6 = -1/3$.
5. Check: $3(4) - 5(2) - 2 = 0$ and $3(1/9) - 5(-1/3) - 2 = 0$.

Final answer. x = 2 or x = −1/3

Interpretation. A perfect-square discriminant is the signal that the quadratic would also have factorised, and here it gives $(3x + 1)(x - 2)$.

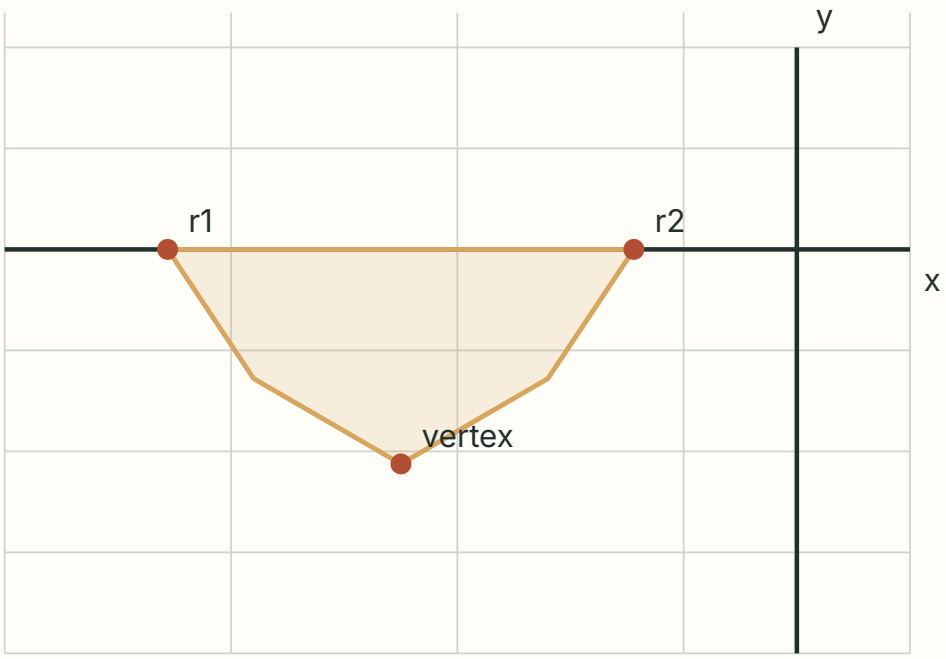
CHAPTER 5 · 5.4 · WORKED EXAMPLE 2

The quadratic formula and discriminant

WORKED EXAMPLE 5.4.2

Problem. Solve $2x^2 + 7x + 4 = 0$, giving the roots to three significant figures.

the two roots are close together



Mathematical reasoning. Apply the formula and round only at the end.

1. $a = 2$, $b = 7$, $c = 4$, so $b^2 - 4ac = 49 - 32 = 17$.
2. $x = (-7 \pm \sqrt{17})/4$.
3. $\sqrt{17} \approx 4.123$, so $x \approx (-7 + 4.123)/4 = -0.719$ and $x \approx (-7 - 4.123)/4 = -2.78$.
4. To three significant figures the roots are -0.719 and -2.78.
5. The exact form is $(-7 \pm \sqrt{17})/4$, which is preferable if the calculation continues.

Final answer. $x \approx -0.719$ or $x \approx -2.78$

Interpretation. Rounding was done once, at the end, so the answers carry the precision the inputs justify and no more.

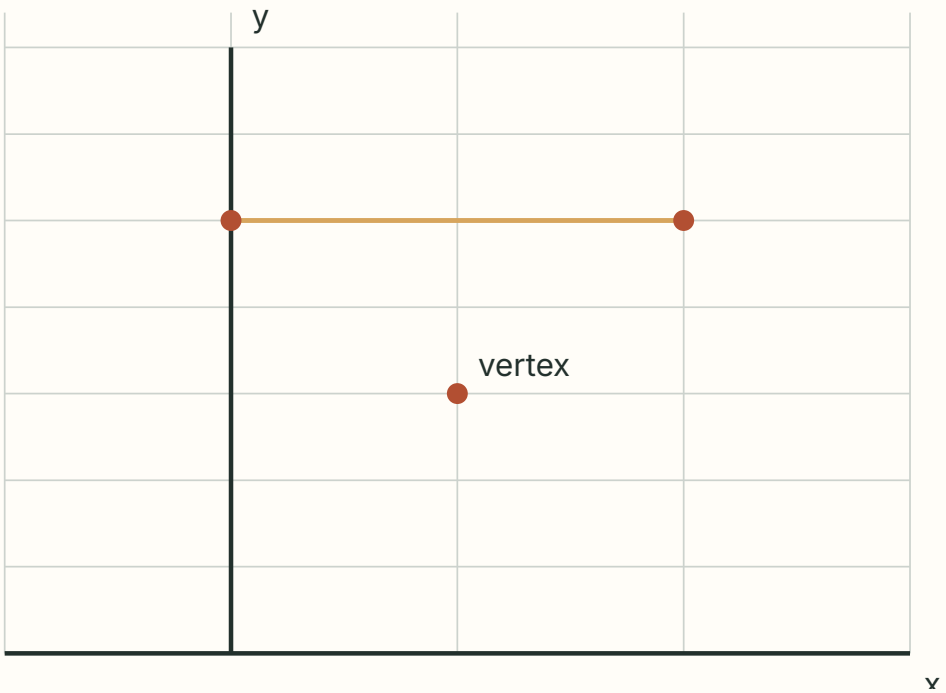
CHAPTER 5 · 5.4 · WORKED EXAMPLE 3

The quadratic formula and discriminant

WORKED EXAMPLE 5.4.3

Problem. Find the discriminant of $2x^2 - 4x + 5$ and state the nature of its roots, with a reason.

the parabola misses the x-axis entirely



Mathematical reasoning. Evaluate $b^2 - 4ac$ and read the sign.

1. $a = 2$, $b = -4$, $c = 5$.
2. $b^2 - 4ac = 16 - 40 = -24$.
3. The discriminant is negative, so there are no real roots.
4. The graph is a parabola opening upwards with its vertex above the x-axis, so it never crosses.
5. The statement 'no real solutions' is therefore justified, not merely observed.

Final answer. the discriminant is −24, so there are no real roots

Interpretation. The discriminant proves a general claim about every equation of this form, which testing values could never do.

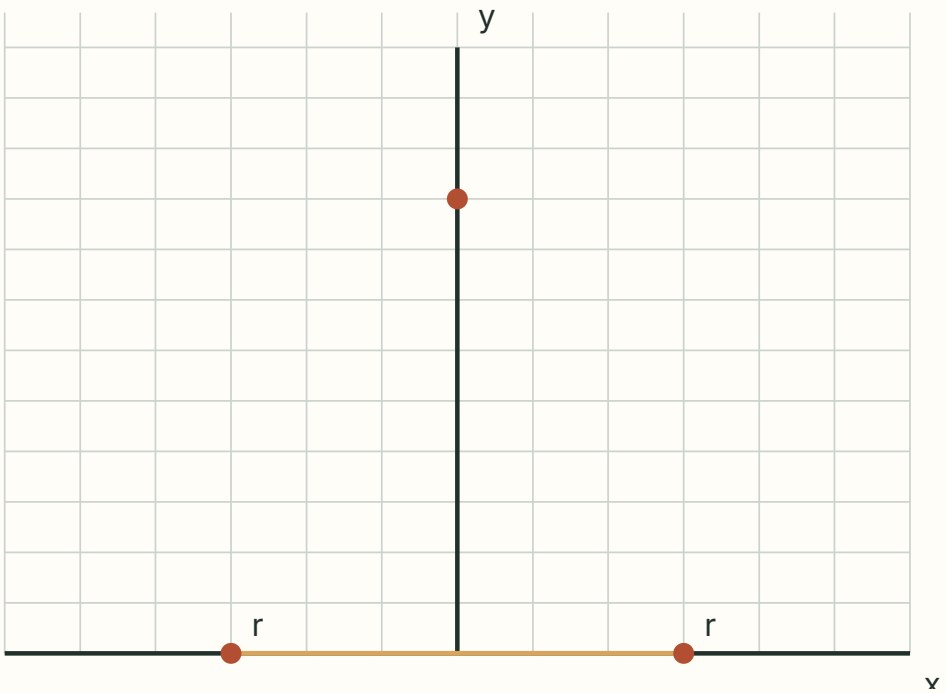
CHAPTER 5 · 5.4 · WORKED EXAMPLE 4

The quadratic formula and discriminant

WORKED EXAMPLE 5.4.4

Problem. Find the values of k for which $x^2 + kx + 9 = 0$ has exactly one real root.

a repeated root means the parabola touches the axis



Mathematical reasoning. A repeated root occurs when the discriminant is zero, so set $b^2 - 4ac = 0$ and solve for k.

1. Here $a = 1$, $b = k$ and $c = 9$.
2. The discriminant is $k^2 - 36$.
3. Setting it to zero gives $k^2 = 36$, so $k = \pm 6$.
4. For $k = 6$ the equation is $x^2 + 6x + 9 = (x + 3)^2$, with root -3.
5. For $k = -6$ the equation is $x^2 - 6x + 9 = (x - 3)^2$, with root 3.

Final answer. $k = 6$ or $k = -6$

Interpretation. A condition on the number of roots becomes an equation in the parameter, and the discriminant is what converts the one into the other.

CHAPTER 5 · INVESTIGATION 5.5

Forming and applying equations

INQUIRY QUESTION

How does a quadratic model describe motion, area and optimisation?

YOUR TASK

Find two real situations that produce a quadratic: one about motion and one about area or profit. Model each, solve it, and check that every root you keep is physically meaningful.

situation	model	roots	meaningful?
ball from a cliff	$h = 25 + 30t - 5t^2$	$t = 7$ and $t = -1$	only $t = 7$
rectangular pen	$A = 40x - 2x^2$	$x = 0$ and $x = 20$	the maximum at $x = 10$
ticket profit	$P = -2x^2 + 60x - 200$	$x = 4.2$ and $x = 25.8$	the maximum at $x = 15$
thrown stone	$h = 1.5 + 14t - 5t^2$	$t = 3$ and $t = -0.1$	only $t = 3$

CHAPTER 5 · INVESTIGATION 5.5

Forming and applying equations

What do you notice?

- Every motion model gave one positive and one negative root, and only the positive one described the future.
- Every area model gave a root at zero, which described the degenerate case.
- The maximum always lay halfway between the two roots.
- The negative roots were solutions of the equation but not of the situation.

Observation. A quadratic model of a physical situation returns roots that must be tested against the domain, and the negative or degenerate roots are rejected.

MATHEMATICAL CONCLUSION

A quadratic model is solved like any quadratic, and its roots are accepted only when they lie inside the domain of the situation.

Your own words

CHAPTER 5 · 5.5

Forming and applying equations

A projectile's height is a quadratic in time, because gravity gives a constant acceleration and the height is the integral of a linear velocity.

The vertex of the model gives the maximum height and the time at which it occurs, and the roots give when the object is at ground level.

Area problems often produce a quadratic when a fixed perimeter or a fixed area is rearranged, and the optimum is at the vertex.

A quadratic model is only valid within its domain, and a root outside that domain is a mathematical solution with no physical meaning.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

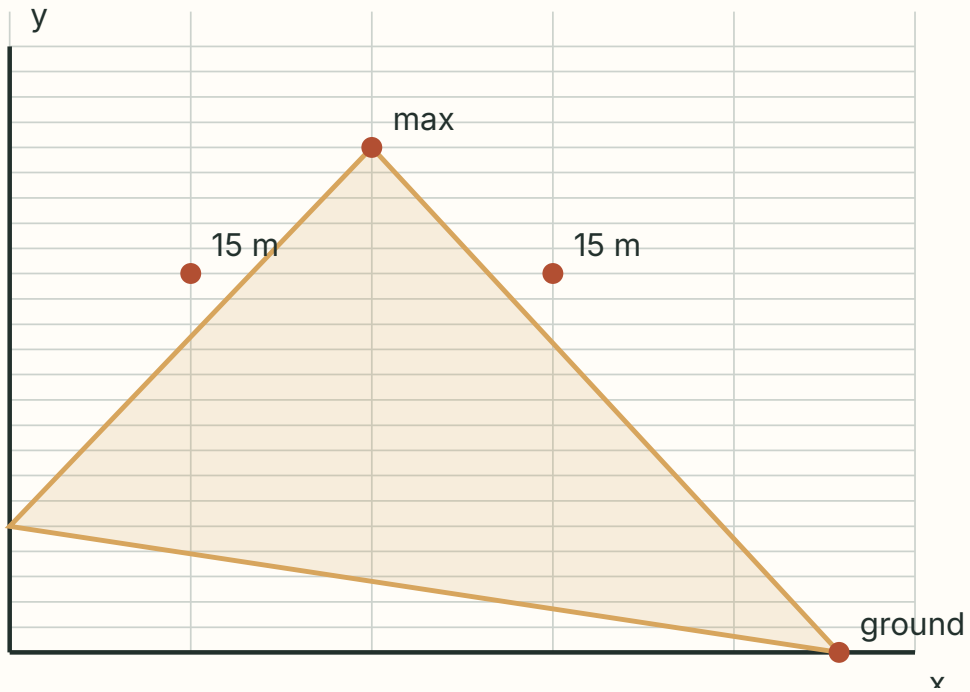
CHAPTER 5 · 5.5 · WORKED EXAMPLE 1

Forming and applying equations

WORKED EXAMPLE 5.5.1

Problem. A ball is thrown upwards from a platform 5 m high with height $h = 20t - 5t^2$ metres after t seconds. Find when it is 15 m high and when it hits the ground.

the height of the ball against time



Mathematical reasoning. Set the model equal to the required height and solve the resulting quadratic, keeping only the meaningful roots.

1. For $h = 15$: $20t - 5t^2 = 15$, so $5t^2 - 20t + 15 = 0$.
2. Dividing by 5 gives $t^2 - 4t + 3 = 0$, so $(t - 1)(t - 3) = 0$.
3. Hence $t = 1$ or $t = 3$ seconds, on the way up and on the way down.
4. For $h = 0$: $20t - 5t^2 = 0$, so $5t(4 - t) = 0$.
5. Hence $t = 0$ or $t = 4$ seconds; the answer is 4 seconds.

Final answer. 15 m at 1 s and 3 s; the ground at 4 s

Interpretation. Both roots of the height equation were meaningful, but the root $t = 0$ for the ground was the launch, not the landing.

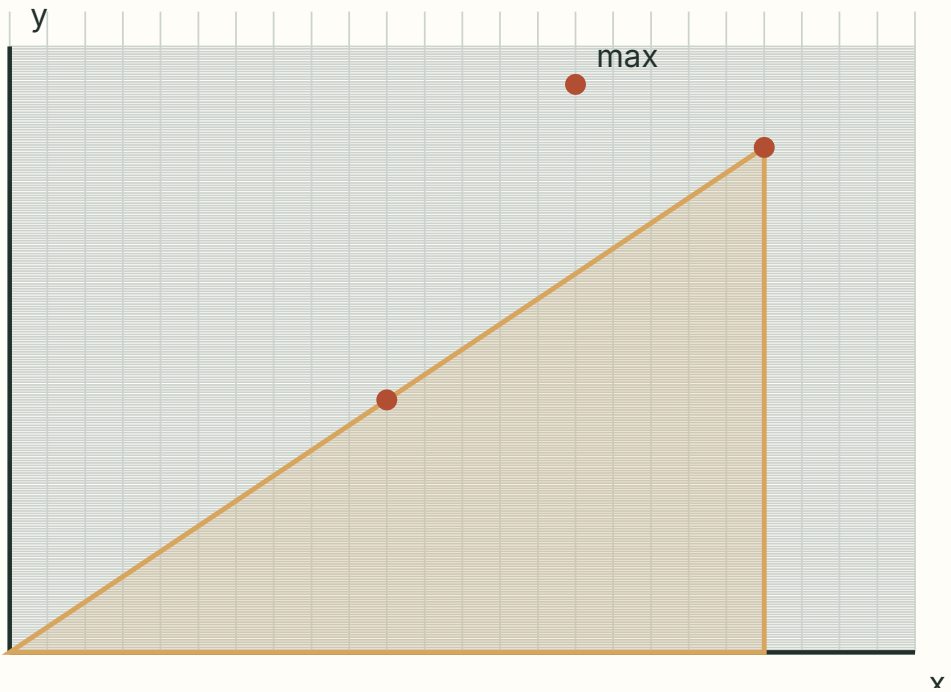
CHAPTER 5 · 5.5 · WORKED EXAMPLE 2

Forming and applying equations

WORKED EXAMPLE 5.5.2

Problem. A farmer has 40 m of fencing for a rectangular pen against a wall. Find the dimensions of greatest area.

area against width, with a maximum at 10 m



Mathematical reasoning. Write the area as a quadratic in one variable and find its vertex.

1. Let x be the width and y the depth; the fence gives $2x + y = 40$.
2. So $y = 40 - 2x$ and the area is $A = x(40 - 2x) = 40x - 2x^2$.
3. This is a downward parabola with roots at $x = 0$ and $x = 20$.
4. The maximum lies halfway between the roots, at $x = 10$.
5. Then $y = 20$ and the greatest area is $200\ m^2$.

Final answer. 10 m by 20 m, giving 200 m²

Interpretation. The symmetry of a quadratic with known roots gives the maximum without any calculus, which is why the factorised form is useful here.

CHAPTER 5 · INVESTIGATION 5.6

Investigation: the discriminant detective

INQUIRY QUESTION

What can the discriminant $b^2 - 4ac$ tell us about a quadratic before we solve it?

YOUR TASK

Take ten quadratics of your own. Compute $b^2 - 4ac$ for each before solving, predict the number and the type of roots, then solve and check whether the prediction held. Finally find the value of k for which $x^2 + kx + 9$ has a repeated root.

quadratic	discriminant	prediction	roots
$x^2 - 5x + 6$	1	two rational	2, 3
$x^2 + 4x + 4$	0	one repeated	−2
$x^2 + x + 1$	−3	no real roots	none
$2x^2 - 3x + 1$	1	two rational	½, 1
$x^2 - 2$	8	two irrational	$\pm\sqrt{2}$

CHAPTER 5 · INVESTIGATION 5.6

Investigation: the discriminant detective

What do you notice?

- A perfect-square discriminant always gave rational roots.
- A zero discriminant gave exactly one repeated root.
- A negative discriminant gave no real roots, and the parabola missed the axis.
- The repeated-root condition $b^2 = 4ac$ gave $k = \pm 6$ for $x^2 + kx + 9$.

Observation. The discriminant reports the number and the type of the real roots, and it agrees with the graph in every case.

MATHEMATICAL CONCLUSION

The discriminant is the quickest test of a quadratic's roots, and setting it to zero is how a parameter is chosen for a repeated root.

Your own words

CHAPTER 5 · 5.6

Investigation: the discriminant detective

The discriminant is a detective's tool: a single calculation reports the number of real roots, and it does so without solving anything.

A discriminant that is a perfect square tells us the roots are rational, so factorising will work. A positive non-square discriminant gives irrational roots, and a negative one gives none at all.

Setting the discriminant to zero is how a parameter is chosen so that a quadratic has a repeated root, which is the condition for a parabola to touch a line or an axis.

The discriminant must agree with the graph. If the calculation says two roots and the graph shows none, one of the two is wrong, and the check is what makes the tool trustworthy.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 5 · 5.6 · WORKED EXAMPLE 1

Investigation: the discriminant detective

WORKED EXAMPLE 5.6.1

Problem. Solve $x^2 + 10x + 21 = 0$ by two methods and say which you would recommend.

method	working	roots
factorising	$(x + 3)(x + 7) = 0$	$-3, -7$
completing the square	$(x + 5)^2 - 4 = 0$	$-3, -7$
formula	$x = (-10 \pm 4)/2$	$-3, -7$

Three methods, the same two roots.

Mathematical reasoning. All three methods are valid, and the choice is a question of speed and of what is revealed.

1. Factorising: two numbers with sum 10 and product 21 are 3 and 7, so $(x + 3)(x + 7) = 0$.
2. The roots are $x = -3$ and $x = -7$.
3. Completing the square gives $(x + 5)^2 = 4$, so $x = -5 \pm 2$, the same roots.
4. The formula gives $x = (-10 \pm 4)/2$, again the same.
5. Factorising is quickest here, but completing the square also shows the vertex at $(-5, -4)$.

Final answer. $x = -3$ or $x = -7$; factorising is quickest, completing the square is most informative

Interpretation. The fastest method and the most informative method are not always the same, and the choice depends on the purpose.

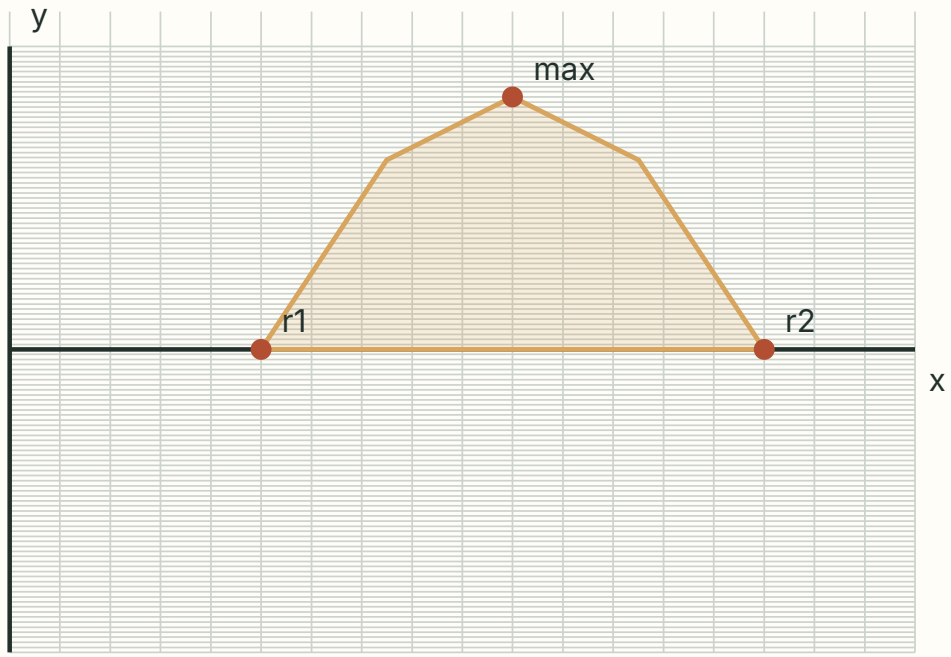
CHAPTER 5 · 5.6 · WORKED EXAMPLE 2

Investigation: the discriminant detective

WORKED EXAMPLE 5.6.2

Problem. A model of a company's profit is $P = -2x^2 + 40x - 150$. Find the maximum profit and the number of items that gives it, using the method that shows the most.

the profit parabola with its maximum at x = 10



Mathematical reasoning. The factorised or vertex form shows the maximum immediately.

1. $P = -2x^2 + 40x - 150 = -2(x^2 - 20x + 75)$.
2. Now $x^2 - 20x + 75 = (x - 5)(x - 15)$, so $P = -2(x - 5)(x - 15)$.
3. The roots are 5 and 15, so the maximum is at $x = 10$.
4. At $x = 10$, $P = -2(25) + 400 - 150 = 50$.
5. The maximum profit is 50 dollars at 10 items, and the company loses money below 5 or above 15 items.

Final answer. maximum profit 50 dollars at 10 items; break-even at 5 and 15 items

Interpretation. The factorised form gave the break-even points and the maximum in one step, which the expanded form did not.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

5.A1 FLUENCY

Expand $(x + 5)(x - 2)$.

5.A2 FLUENCY

Expand $(2x + 3)^2$.

5.A3 FLUENCY

Solve $x^2 - 9x + 20 = 0$ by factorising.

5.A4 FLUENCY

Solve $x^2 - 7x = 0$.

5.A5 FLUENCY

Solve $2x^2 - 50 = 0$.

5.A6 FLUENCY

Complete the square for $x^2 + 8x + 3$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

5.A7 FLUENCY

Find the discriminant of $x^2 + 4x + 5$.

5.A8 FLUENCY

Solve $x^2 + 6x + 5 = 0$.

5.A9 FLUENCY

Solve $3x^2 - 12 = 0$.

5.A10 FLUENCY

Find the vertex of $y = x^2 - 6x + 1$.

5.A11 FLUENCY

Solve $x^2 + 2x - 8 = 0$ using the formula.

5.A12 FLUENCY

Solve $(x - 2)^2 = 9$.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

5.B1 REASONING

The areas of a sequence of squares are 1, 4, 9, 16. Find the rule for the nth area and test it on the seventh square.

5.B2 CHALLENGE

Investigate the pattern in the sums of the first n square numbers and justify the general rule.

5.B3 REASONING

Conjecture a rule for the difference between consecutive squares and test it.

5.B4 CHALLENGE

Investigate the pattern in the number of rectangles in an n by n grid and justify it.

5.B5 REASONING

Build a table of the roots of $x^2 + bx + 1 = 0$ for $b = 2$ to 6 and describe the pattern.

5.B6 REASONING

Conjecture a rule for the maximum product of two numbers with a fixed sum and justify it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

5.B7 REASONING

Investigate the pattern in the discriminants of $x^2 + nx + (n + 1)$ and state when real roots exist.

5.B8 CHALLENGE

A pattern of nested squares has side lengths following the rule $s = 1 + 1/n$. Investigate the areas and their limit.

5.B9 REASONING

Find the pattern in the heights $h = 20t - 5t^2$ at $t = 0, 1, 2, 3, 4$ and justify the shape.

5.B10 REASONING

Investigate the pattern in the roots of $x^2 - 2x - k = 0$ as k grows.

5.B11 REASONING

Conjecture a rule for the number of regions into which n lines divide a plane when no two are parallel and no three meet.

CRITERION C Communicating

Criterion C practice · Communicating

5.C1 REASONING

Explain in your own words what the discriminant is and what it decides.

5.C2 FLUENCY

Write a complete solution to $x^2 + 4x - 5 = 0$ by completing the square.

5.C3 REASONING

Choose between factorising and the formula for $2x^2 + 7x + 3$ and justify.

5.C4 FLUENCY

Set out a full solution to $3x^2 + 2x - 4 = 0$, giving roots to three significant figures.

5.C5 REASONING

Explain why a negative discriminant means there are no real roots.

5.C6 REASONING

Describe, with an example, the difference between a repeated root and two equal-looking roots.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

5.C7 REASONING

Explain why completing the square works for every quadratic.

5.C8 REASONING

Write the reasoning that shows the vertex of $y = ax^2 + bx + c$ is at $x = -b/(2a)$.

5.C9 REASONING

Set out the argument that the maximum product of two numbers with a fixed sum occurs when they are equal.

5.C10 REASONING

Explain what a root of a model means in a real situation.

5.C11 REASONING

Describe the relationship between the roots of a quadratic and the x-intercepts of its graph.

5.C12 REASONING

Explain, with an example, why a root of an equation may not be an acceptable answer.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

5.D1 APPLICATION

A ball is thrown from a 20 m cliff with height $h = 20 + 18t - 5t^2$. Find when it hits the ground and its greatest height.

5.D2 APPLICATION

A rectangular garden has a perimeter of 60 m. Find the dimensions of greatest area and the area.

5.D3 APPLICATION

A company's profit is $P = -3x^2 + 120x - 800$. Find the break-even values and the maximum profit.

5.D4 APPLICATION

A bridge arch is modelled by $y = -0.02x^2 + 0.8x$. Find its greatest height and the width at the base.

5.D5 APPLICATION

A rectangle has area 48 m^2 and one side 2 m longer than the other. Find the dimensions.

5.D6 APPLICATION

A rocket's height is $h = 40t - 8t^2$. Find when it returns to the ground and its greatest height.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

5.D7 REASONING

A hall's seating revenue is $R = x(30 - x)$ for a ticket price of x dollars. Investigate whether a revenue of 200 dollars is possible and justify.

5.D8 APPLICATION

A photograph is 20 cm by 15 cm and is surrounded by a frame of uniform width with area 136 cm^2. Find the width.

5.D9 REASONING

A ball is thrown so that $h = 1.5 + 14t - 5t^2$. Investigate whether it clears a 12 m wall and justify.

5.D10 CHALLENGE

A field is divided into two rectangles with a total perimeter of 80 m. Investigate the division that maximises the total area and justify.

5.D11 APPLICATION

A company finds that its cost is $C = 2x^2 + 60x + 500$ and its revenue is $R = 100x$. Find the production level that maximises profit and the profit.

5.D12 REASONING

Explain, with a model of your own, when a quadratic model is a poor description of a situation.

CHAPTER 5 · APPLICATION AND ASSESSMENT

Applying chapter 5

Each task asks for a solution and a justification. State the method you chose and why it suits the equation.

A1 · The water fountain

A fountain shoots water upwards from ground level with height $h = 24t - 6t^2$ metres after t seconds. Find the greatest height reached, the time at which it occurs, the times at which the water is 18 m high, and the time at which it returns to the ground. Justify each answer with the method you used.

Criteria A and C · solve each quadratic and state the domain that makes the answers meaningful.

A2 · The parameter problem

The equation $kx^2 - 4x + 1 = 0$ has real roots for some values of k and not for others. Find the values of k for which it has two distinct real roots, one repeated root and no real roots, and explain why $k = 0$ must be excluded from your answer.

Criteria B and D · use the discriminant to justify each case and state the condition on k.

CHAPTER 5 · REVIEW

What you should now be able to do

Key ideas

- $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$.
- $a^2 - b^2 = (a + b)(a - b)$ is the difference of two squares.
- Every quadratic can be written as $a(x + p)^2 + q$.
- The quadratic formula solves $ax^2 + bx + c = 0$ for any $a \neq 0$.
- $b^2 - 4ac > 0$ gives two roots, $= 0$ gives one, < 0 gives none.
- The vertex of a parabola is at $x = -b/(2a)$.
- The roots of a model must lie inside the domain of the situation.
- The best method depends on the coefficients and on the question.

Vocabulary

perfect square · cross term · complete the square · vertex · axis of symmetry · quadratic formula · discriminant · repeated root · parabola · domain · projectile · optimisation

CHAPTER 5 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

5.R1 Expand $(3x - 4)(x + 2)$.

5.R2 Solve $x^2 - 8x + 12 = 0$.

5.R3 Write $x^2 - 10x + 7$ in completed square form.

5.R4 Find the discriminant of $2x^2 + 3x + 5$ and state the nature of the roots.

5.R5 Solve $2x^2 + 5x - 3 = 0$ using the formula.

5.R6 Find the maximum value of $12x - 3x^2$ and the x that gives it.

CHAPTER 5 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A quadratic equation is solved by transforming it into an equivalent form whose solutions are visible, and the justification for each transformation is a reversible step; the discriminant then decides, before any solving, how many solutions exist, which turns a calculation into a general claim.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of equivalence, generalisation and justification appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of design, motion and optimisation, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a quadratic model for motion or profit, and what would you still like to ask?

CHAPTER 5 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Optimisation and motion are quadratic. Model one and say which method answers the question most clearly.

Service as Action

COMMUNITY NEED	Optimisation decisions in design and sport are made without showing the model.
MATHEMATICAL CONNECTION	Quadratic models, vertices and discriminants.
PROPOSED ACTION	Model one real optimisation and publish the model with its answer.
EXPECTED IMPACT	A reader can see why the best choice is the best.
STUDENT REFLECTION	Record the model, the solution and the check.

Community Project

PROJECT IDEA	An optimisation audit of the school's own spaces.
NEED	Room layouts, timetables and budgets are decided by guesswork.
RESEARCH QUESTION	Which layout or allocation maximises a stated quantity, and what does the model assume?
MATHEMATICS	Quadratic models, vertices and the discriminant.
ACTION OR OUTCOME	A one-page report with the model and the optimum.
BENEFICIARIES	Everyone who uses the space.
EVIDENCE	The report with the model, the vertex and the check.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, gaming or a design you enjoy.
LEARNING GOAL	To model an optimisation you care about and justify the answer.
PRODUCT OR OUTCOME	A quadratic model with a maximum or minimum found and checked.
MATHEMATICS	Completing the square, the formula and the discriminant.
PROCESS	Build the model, solve it, and check the root against the situation.
EVIDENCE	The model, the solution and the check.

CHAPTER 6 · PART II · THINKING WITH MODELS

06

Quadratic Functions and Graphical Reasoning

A parabola is the graph of a quadratic function. Its shape is fixed, and every other parabola is obtained from it by a shift, a stretch or a reflection.

ENGAGE

A garden arch follows the curve $y = -0.2x^2 + 2$. How wide is it at ground level, how high is it in the middle, and how would you draw it without a calculator?

Framing the inquiry

Key concept: Form
Related concepts: Representation · Change · Pattern
Global context: Orientation in space and time
Global context exploration: Arches, paths and trajectories

Statement of inquiry: The graph of a function is a representation that shows change at a glance — where a quantity rises and falls, where it is greatest, and how it behaves far from the origin — and reading that representation is a form of reasoning that the equation alone does not provide.

Questions to guide us

- **Factual:** What are the key features of the graph of a quadratic function, and how do the coefficients a, b and c affect its shape?
- **Conceptual:** Why does every parabola have the same shape, and how does a graph show change that an equation hides?
- **Debatable:** Is a sketch drawn from reasoning more trustworthy than one drawn by plotting points, or does precision always require computation?

ATL · COMMUNICATION. Students read a graph, a table and an equation of the same function, and explain in words what each shows about the change it describes.

CHAPTER 6 · INVESTIGATION 6.1

Parabolas and key features

INQUIRY QUESTION

What are the key features of a parabola, and how are they found from the equation?

YOUR TASK

Take five quadratic functions with different signs and sizes of a, b and c. For each, find the vertex, the axis of symmetry and the intercepts, sketch the graph, and record whether the parabola opens up or down.

function	vertex	axis	opens
$y = x^2 - 4$	$(0, -4)$	$x = 0$	up
$y = -x^2 + 9$	$(0, 9)$	$x = 0$	down
$y = x^2 + 4x + 3$	$(-2, -1)$	$x = -2$	up
$y = -2x^2 + 8x - 6$	$(2, 2)$	$x = 2$	down
$y = 3x^2 + 6x - 2$	$(-1, -5)$	$x = -1$	up

CHAPTER 6 · INVESTIGATION 6.1

Parabolas and key features

What do you notice?

- The sign of a decided the direction of opening in every case.
- The axis of symmetry was always $x = -b/(2a)$.
- The vertex's y-coordinate was the value of the function at the axis.
- The y-intercept was always c, and it moved the graph without changing its shape.

> **Observation.** The three coefficients decide the direction, the position and the width of the parabola, and none of them changes its fundamental shape.

MATHEMATICAL CONCLUSION

Every quadratic graph is a parabola whose direction, position and width are read from a, b and c, so a sketch can be made from the equation without plotting any points.

Your own words

CHAPTER 6 · 6.1

Parabolas and key features

The graph of $y = ax^2 + bx + c$ is a parabola. It is symmetric about the vertical line through its vertex, and it opens upwards when $a > 0$ and downwards when $a < 0$.

A pattern in a table of values becomes a shape on a graph, and the representation as a picture is what makes the pattern of change visible.

The vertex is the maximum or minimum point, and its x-coordinate is $-b/(2a)$. The axis of symmetry is the line $x = -b/(2a)$.

The y-intercept is c, because setting $x = 0$ leaves $y = c$. The x-intercepts, if they exist, are the roots of $ax^2 + bx + c = 0$.

The value of a controls the width: a larger $|a|$ gives a narrower parabola. The value of c moves the whole graph up or down without changing its shape.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

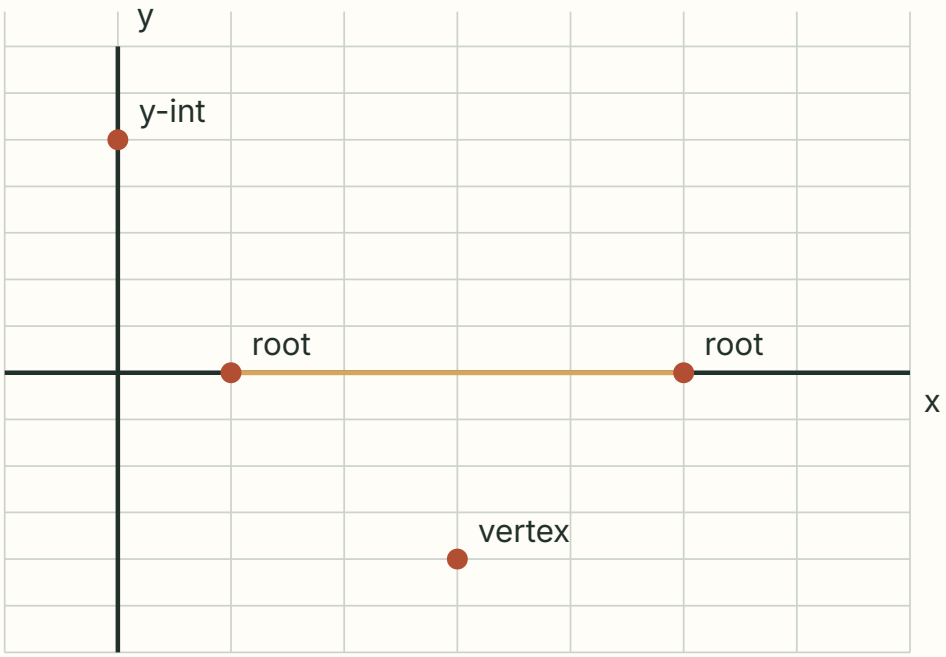
CHAPTER 6 · 6.1 · WORKED EXAMPLE 1

Parabolas and key features

WORKED EXAMPLE 6.1.1

Problem. For $y = x^2 - 6x + 5$, find the vertex, the axis of symmetry, the intercepts, and sketch the graph.

the parabola $y = x^2 - 6x + 5$



Mathematical reasoning. Read the intercepts directly, complete the square for the vertex, and factorise for the roots.

1. The y-intercept is 5, at (0, 5).
2. The axis of symmetry is $x = -(-6)/2 = 3$.
3. The vertex is at $x = 3$, where $y = 9 - 18 + 5 = -4$, so $(3, -4)$.
4. Factorising gives $x^2 - 6x + 5 = (x - 1)(x - 5)$, so the roots are 1 and 5.
5. The parabola opens upwards, so $(3, -4)$ is a minimum.

Final answer. vertex (3, −4), axis x = 3, intercepts (0, 5), (1, 0), (5, 0)

Interpretation. Four features and a sketch come from one equation, and each feature answers a different question about the function.

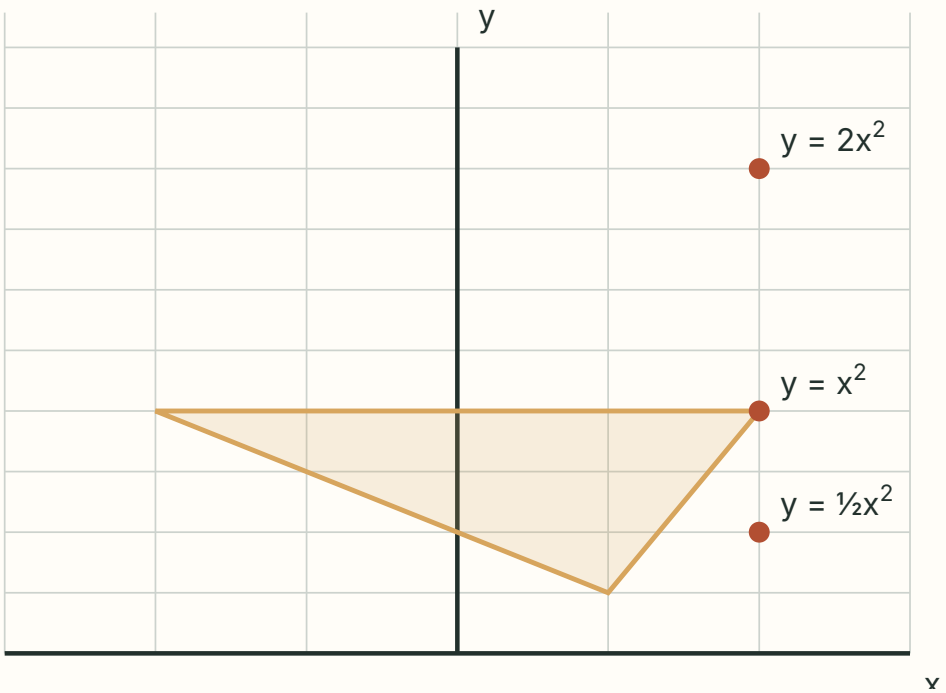
CHAPTER 6 · 6.1 · WORKED EXAMPLE 2

Parabolas and key features

WORKED EXAMPLE 6.1.2

Problem. Compare the graphs of $y = x^2$, $y = 2x^2$ and $y = \frac{1}{2}x^2$ on the same axes.

the same shape at three widths



Mathematical reasoning. All three pass through the origin and are symmetric about the y-axis; only the width differs.

1. At $x = 2$ the values are 4, 8 and 2 respectively.
2. So $y = 2x^2$ rises twice as fast as $y = x^2$, and $y = \frac{1}{2}x^2$ half as fast.
3. All three pass through (0, 0) and are symmetric about the y-axis.
4. The shape is the same in every case; only the scale of the y-axis differs.
5. A larger leading coefficient means a narrower parabola.

Final answer. the three graphs have the same shape and vertex but different widths

Interpretation. The leading coefficient changes the width without changing the form, which is why every parabola is the same curve at a different scale.

CHAPTER 6 · INVESTIGATION 6.2

Vertex form and transformations

INQUIRY QUESTION

How is one graph obtained from another, and what does each transformation do?

YOUR TASK

Take the graph of $y = x^2$ and apply six transformations of your own choice: two translations, two stretches and two reflections. For each, write the equation, sketch the graph, and record the new vertex.

transformation	equation	vertex
up 3	$y = x^2 + 3$	(0, 3)
right 4	$y = (x - 4)^2$	(4, 0)
stretch 3	$y = 3x^2$	(0, 0)
reflect in x-axis	$y = -x^2$	(0, 0)
left 2, up 5	$y = (x + 2)^2 + 5$	(−2, 5)
reflect, stretch 2, up 6	$y = -2x^2 + 6$	(0, 6)

CHAPTER 6 · INVESTIGATION 6.2

Vertex form and transformations

What do you notice?

- Every translation moved the vertex without changing the shape.
- The stretches changed the width but not the vertex.
- The reflection in the x-axis turned a minimum into a maximum.
- A reflection in the y-axis left $y = x^2$ unchanged, because it is symmetric.

Observation. Translations move the vertex, stretches change the width, and reflections reverse the direction, and each transformation is visible in the equation.

MATHEMATICAL CONCLUSION

Any parabola can be obtained from $y = x^2$ by a sequence of translations, stretches and reflections, and the sequence can be read from the equation.

Your own words

CHAPTER 6 · 6.2

Vertex form and transformations

A translation moves every point by the same amount. Replacing y by y − k shifts the graph up by k, and replacing x by x − h shifts it right by h.

A stretch multiplies distances from the axis. Replacing y by y/a stretches vertically by a, and replacing x by x/a stretches horizontally by a.

A reflection in the x-axis replaces y by −y, and a reflection in the y-axis replaces x by −x. Both change the direction of the curve.

Transformations compose, and the order matters for stretches combined with translations. Describing a transformation is a precise use of language that a diagram alone cannot give.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

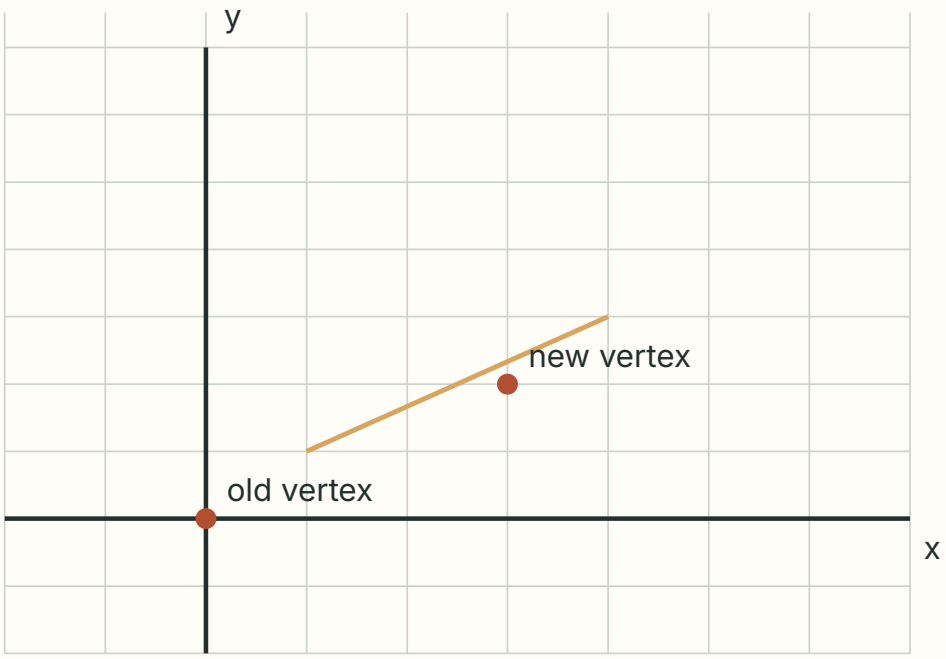
CHAPTER 6 · 6.2 · WORKED EXAMPLE 1

Vertex form and transformations

WORKED EXAMPLE 6.2.1

Problem. Describe the transformation that maps $y = x^2$ to $y = (x - 3)^2 + 2$, and find the new vertex.

a translation by (3, 2)



Mathematical reasoning. Compare the two equations term by term.

1. The term $(x - 3)^2$ is x^2 shifted 3 units to the right.
2. The $+ 2$ outside the bracket shifts the graph 2 units up.
3. So the transformation is a translation by the vector (3, 2).
4. The vertex moves from (0, 0) to (3, 2).
5. The shape is unchanged, because no stretch or reflection is involved.

Final answer. a translation by (3, 2); the new vertex is (3, 2)

Interpretation. Reading the transformation from the equation is faster than plotting points, and it gives the vertex immediately.

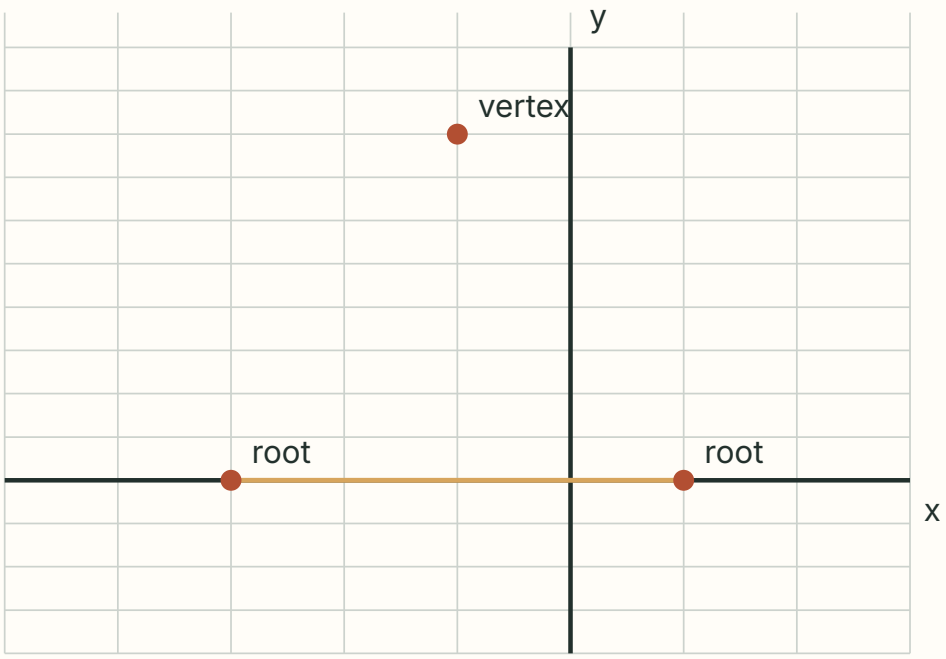
CHAPTER 6 · 6.2 · WORKED EXAMPLE 2

Vertex form and transformations

WORKED EXAMPLE 6.2.2

Problem. Sketch $y = -2(x + 1)^2 + 8$, stating each transformation from $y = x^2$.

a reflected, stretched and translated parabola



Mathematical reasoning. Apply the transformations in order from the inside out.

1. The bracket $(x + 1)^2$ shifts the graph 1 unit to the left.
2. The -2 reflects it in the x-axis and stretches it vertically by 2.
3. The $+ 8$ then shifts the whole graph 8 units up.
4. The vertex is at $(-1, 8)$, and the parabola opens downwards.
5. Setting $y = 0$ gives $(x + 1)^2 = 4$, so the roots are 1 and -3.

Final answer. left 1, reflect in the x-axis, stretch by 2, up 8; vertex (−1, 8)

Interpretation. The order of the transformations is visible in the equation, and reading it from the inside out avoids mistakes.

CHAPTER 6 · INVESTIGATION 6.3

Quadratic inequalities

INQUIRY QUESTION

How is the solution of a quadratic inequality found, and why is it a region rather than a number?

YOUR TASK

Take six quadratic inequalities of your own, including one with a negative leading coefficient, one with no real roots, one with a repeated root, and one whose solution is a single interval. Solve each, draw the solution on a number line, and sketch the parabola to confirm the region.

inequality	roots	solution	region
$x^2 - 4 < 0$	$-2, 2$	$-2 < x < 2$	one interval
$x^2 + 1 > 0$	none	all x	whole line
$x^2 - 4x + 4 \geq 0$	2	all x	whole line
$-x^2 + x + 6 > 0$	$-2, 3$	$-2 < x < 3$	one interval
$x^2 - x - 6 < 0$	$-2, 3$	$-2 < x < 3$	one interval

CHAPTER 6 · INVESTIGATION 6.3

Quadratic inequalities

What do you notice?

- The roots always formed the boundary of the solution.
- A negative leading coefficient reversed the region.
- A quadratic with no real roots was either always positive or always negative, so the answer was every value or none.
- The repeated root gave a parabola that touched the axis, so the region was the whole line.

Observation. A quadratic inequality is solved by finding the roots and testing the sign of one interval, and the answer is a region of the number line.

MATHEMATICAL CONCLUSION

The roots of the equation divide the number line into intervals of constant sign, and the solution of an inequality is the set of intervals whose sign the inequality asks for.

Your own words

CHAPTER 6 · 6.3

Quadratic inequalities

A quadratic inequality is solved by first solving the equation, because the roots divide the number line into the intervals where the sign of the quadratic cannot change.

The sign of the quadratic is then tested in one interval, and the sign of the other intervals follows from the shape of the parabola: it has the same sign throughout each interval between and beyond the roots.

The solution is a set of values, usually a union of intervals, and it is drawn on a number line as a region. A single value is almost never the answer.

On the graph the solution is where the parabola lies above or below the x-axis, so the algebra and the picture must give the same region. This is the test that catches a sign error.

> FORMAL DEVELOPMENT
>
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In your own words

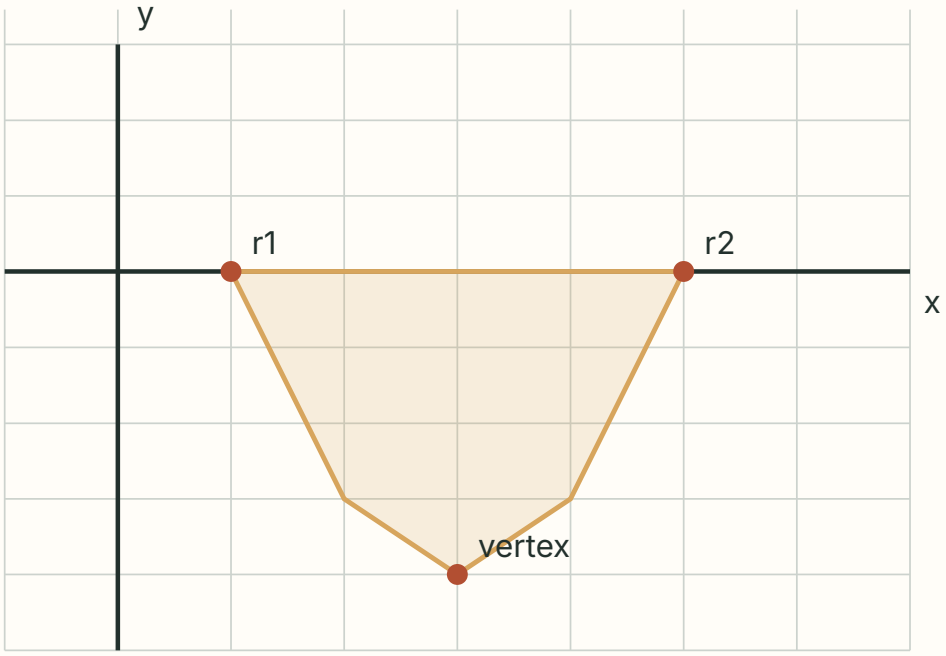
CHAPTER 6 · 6.3 · WORKED EXAMPLE 1

Quadratic inequalities

WORKED EXAMPLE 6.3.1

Problem. Solve $x^2 - 6x + 5 < 0$, and show the solution on a number line.

the parabola is below the axis between the roots



Mathematical reasoning. Solve the equation first, then test the sign in one interval.

1. $x^2 - 6x + 5 = 0$ gives $(x - 1)(x - 5) = 0$, so $x = 1$ or $x = 5$.
2. The roots divide the line into three intervals: $x < 1$, $1 < x < 5$ and $x > 5$.
3. Test $x = 3$: $9 - 18 + 5 = -4$, which is negative, so the middle interval satisfies the inequality.
4. The parabola opens upwards, so it is positive outside the roots and negative between them.
5. Hence the solution is $1 < x < 5$, drawn as the open interval between 1 and 5.

Final answer. $1 < x < 5$

Interpretation. The roots are the boundary of the solution, and the sign between them is the sign that the inequality asks for.

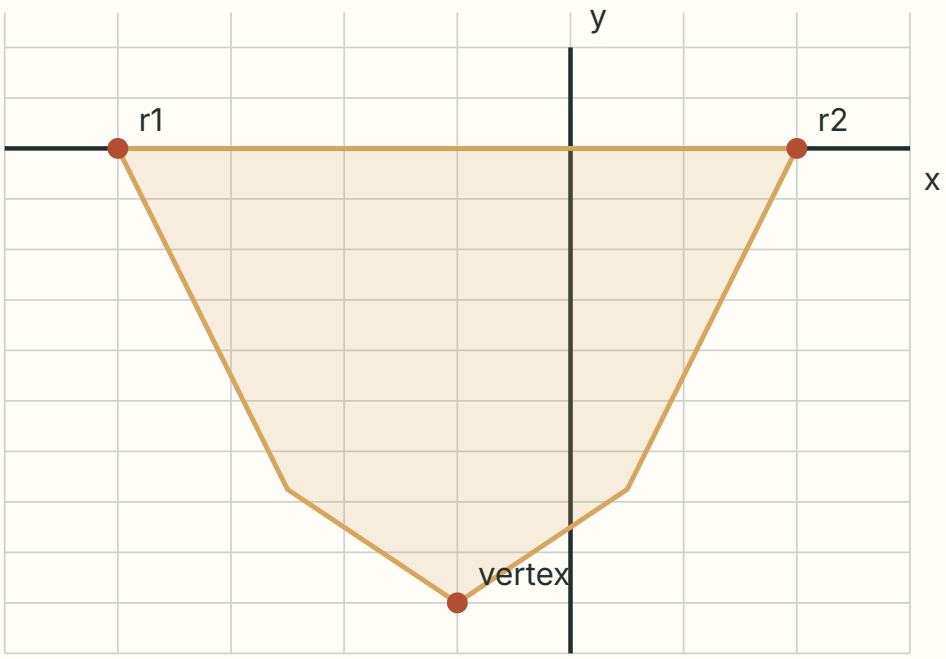
CHAPTER 6 · 6.3 · WORKED EXAMPLE 2

Quadratic inequalities

WORKED EXAMPLE 6.3.2

Problem. Solve $x^2 + 2x - 8 \geq 0$.

the parabola is above the axis outside the roots



Mathematical reasoning. Factorise, find the roots, then decide which intervals are positive.

1. $x^2 + 2x - 8 = (x + 4)(x - 2) = 0$, so $x = -4$ or $x = 2$.
2. Test $x = 0$: $0 + 0 - 8 = -8$, which is negative.
3. So the middle interval is negative and the two outer intervals are positive.
4. The inequality allows equality, so the roots are included.
5. Hence the solution is $x \leq -4$ or $x \geq 2$.

Final answer. $x \leq -4$ or $x \geq 2$

Interpretation. An inequality that allows equality includes the roots, and the solution is a union of two intervals rather than one.

CHAPTER 6 · INVESTIGATION 6.4

Optimisation and modelling

INQUIRY QUESTION

How is a graph used to answer a question that the equation asks badly?

YOUR TASK

Choose two real quantities that can be modelled on the same axes — income and spending, height and weight, distance and time. Draw both graphs, find where they cross if they do, and state what the crossing means and what the graphs do not show.

pair	crossing	meaning	not shown
revenue and cost	break-even	no profit, no loss	fixed costs detail
distance and time	none	different quantities	the speed
height and age	several	growth spurts	the cause

CHAPTER 6 · INVESTIGATION 6.4

Optimisation and modelling

What do you notice?

- Two graphs on the same axes answer questions neither answers alone.
- The crossing point had a clear meaning in one case and none in another.
- Different quantities on the same axes can mislead if the scales differ.
- The graphs showed correlation but never cause.

Observation. Graphs are compared by placing them on the same axes, and the crossing point is the solution of the equation that sets the two models equal.

MATHEMATICAL CONCLUSION

A graph is used to answer questions about comparison and trend, and its answers are limited by the assumptions of the model it was drawn from.

Your own words

CHAPTER 6 · 6.4

Optimisation and modelling

A graph answers questions about comparison, about trend and about limits, and it answers them faster than an equation.

Two graphs on the same axes answer questions about which is larger, where they cross, and how the gap changes.

The intersection of two graphs is the solution of the equation obtained by setting them equal, so the graphical and algebraic methods are the same.

A graph drawn from a model carries the model's assumptions. Drawing a smooth curve through scattered data asserts a relationship that the data may not support.

> FORMAL DEVELOPMENT
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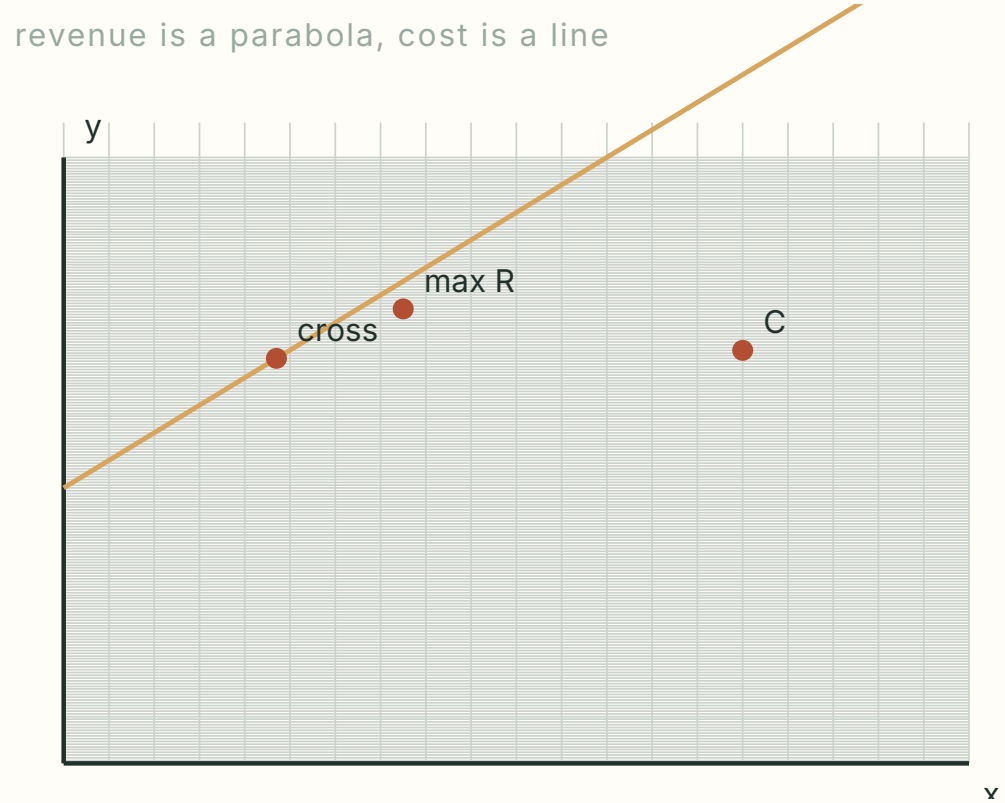
In your own words

CHAPTER 6 · 6.4 · WORKED EXAMPLE 1

Optimisation and modelling

WORKED EXAMPLE 6.4.1

Problem. A company's revenue is $R = 40x - 2x^2$ and its cost is $C = 10x + 100$. Draw both graphs on the same axes and find the values of x for which the company makes a profit.



Mathematical reasoning. Set the two models equal to find the crossing points, and read the profit region from the graphs.

1. Setting $40x - 2x^2 = 10x + 100$ gives $2x^2 - 30x + 100 = 0$.
2. Dividing by 2 gives $x^2 - 15x + 50 = 0$, so $(x - 5)(x - 10) = 0$.
3. The graphs cross at $x = 5$ and $x = 10$.
4. Between these values the parabola is above the line, so the company makes a profit.
5. The maximum profit is at $x = 7.5$, halfway between the two crossings.

Final answer. a profit is made for $5 < x < 10$ items, with the maximum at $x = 7.5$

Interpretation. The graphs show the region of profit at a glance, and the equation gives its exact boundaries.

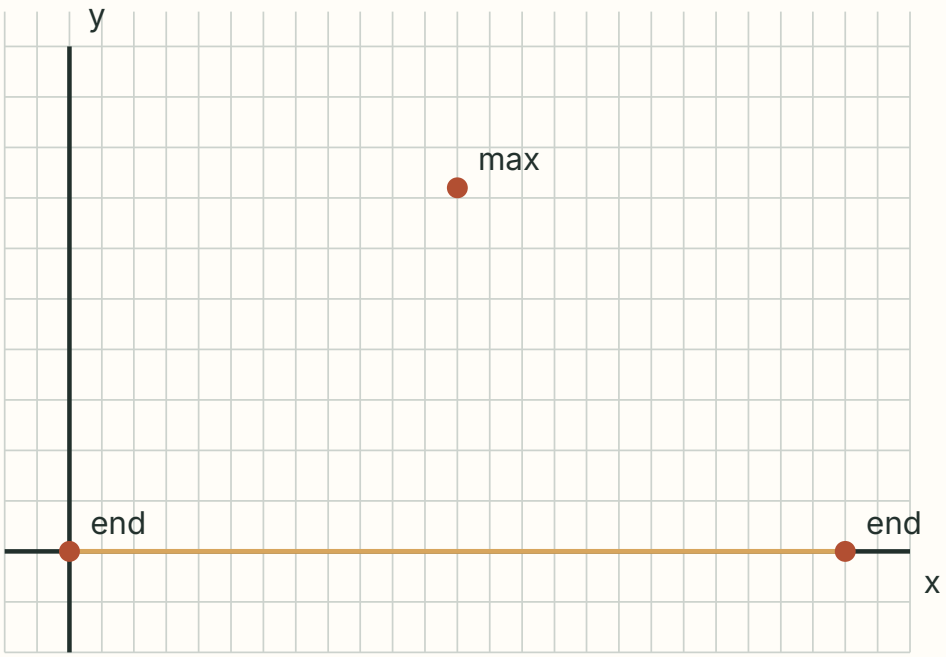
CHAPTER 6 · 6.4 · WORKED EXAMPLE 2

Optimisation and modelling

WORKED EXAMPLE 6.4.2

Problem. A bridge arch is modelled by $y = -0.05x^2 + 1.2x$. Find the span of the arch and its greatest height, and say what the model assumes.

the arch as a parabola



Mathematical reasoning. The roots give the span and the vertex gives the height.

1. Setting $y = 0$ gives $x(-0.05x + 1.2) = 0$, so $x = 0$ or $x = 24$.
2. The span is therefore 24 m.
3. The vertex is at $x = 12$, where $y = -0.05(144) + 14.4 = 7.2$ m.
4. The greatest height is 7.2 m at the middle of the span.
5. The model assumes a parabolic arch, which is an idealisation of any real structure.

Final answer. span 24 m, greatest height 7.2 m

Interpretation. The model is an idealisation, and the graph makes the idealisation visible so that its limits can be discussed.

CHAPTER 6 · INVESTIGATION 6.5

Comparing models

INQUIRY QUESTION

How does a graph show growth that is not linear?

YOUR TASK

Take three quantities that change by a fixed percentage each period: one growing, one decaying, and one of your own. Model each exponentially, tabulate six values, sketch the graph, and compare with the linear model that starts at the same point.

period	exponential	linear comparison
0	1	1
1	1.5	1.5
2	2.25	2
3	3.375	2.5
4	5.06	3
5	7.59	3.5

CHAPTER 6 · INVESTIGATION 6.5

Comparing models

What do you notice?

- The two models agreed at the first period and then diverged.
- The exponential grew faster than the linear model at every later period.
- The decaying model fell quickly at first and then slowly.
- The multiplier was the same at every step, which is the defining property.

Observation. An exponential model changes by a constant multiplier, so its graph curves away from any linear model that starts at the same point.

MATHEMATICAL CONCLUSION

Exponential growth and decay are modelled by repeated multiplication, and their graphs separate from linear models without limit.

Your own words

CHAPTER 6 · 6.5

Comparing models

An exponential function has the form $y = a \cdot b^x$ with $b > 0$ and $b \neq 1$. Its graph rises or falls at a rate proportional to its own value.

Exponential growth is faster than any linear growth in the long run, because the rate itself grows. This is the difference between adding and multiplying.

The y-intercept is a, because $b^0 = 1$. The graph never crosses the x-axis, so the x-axis is a horizontal asymptote.

Halving and doubling are the two most common exponential models, and their graphs are mirror images in the y-axis.

> FORMAL DEVELOPMENT
>
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In your own words

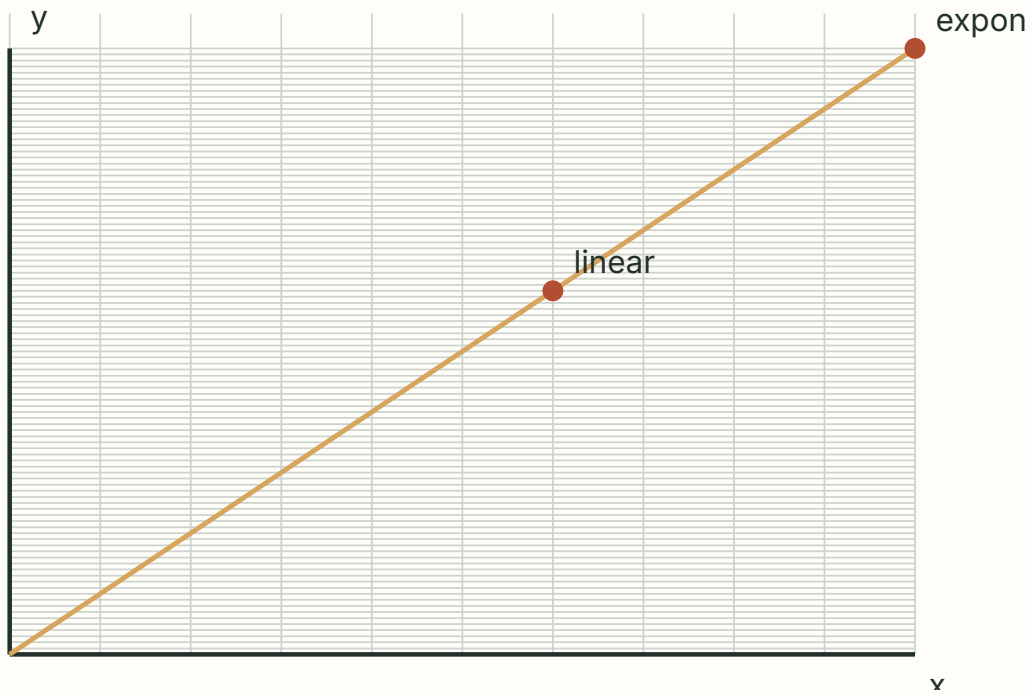
CHAPTER 6 · 6.5 · WORKED EXAMPLE 1

Comparing models

WORKED EXAMPLE 6.5.1

Problem. Compare the values of y = 10x and y = 2^x for x = 0 to 10, and say which grows faster and where they cross.

the line is overtaken by the curve



Mathematical reasoning. Compute both values and compare, then find the crossing point by solving.

1. At x = 0 the values are 0 and 1, so the line is lower.
2. At x = 5 the values are 50 and 32, so the line is higher.
3. At x = 10 the values are 100 and 1024, so the curve is far higher.
4. Solving 2^x = 10x gives x ≈ 1.4 and x ≈ 6.6.
5. Beyond x = 6.6 the exponential is always larger, and the gap grows without limit.

Final answer. the exponential overtakes the line at about x = 6.6 and then grows far faster

Interpretation. A linear model and an exponential model can agree for a while and then diverge completely, which is why the choice of model matters.

CHAPTER 6 · 6.5 · WORKED EXAMPLE 2

Comparing models

WORKED EXAMPLE 6.5.2

Problem. A car loses 15% of its value each year from 20 000 dollars. Model the value and find it after 5 years and after 10 years.

year	value
0	20000
1	17000
2	14450
5	8874
10	3932

Each year multiplies the previous value by 0.85.

Mathematical reasoning. Repeated multiplication by a constant factor is an exponential model.

1. The multiplier is $1 - 0.15 = 0.85$.
2. So the model is $V = 20000 \times 0.85^{n}$.
3. After 5 years, $V = 20000 \times 0.85^{5} \approx 8874$ dollars.
4. After 10 years, $V = 20000 \times 0.85^{10} \approx 3932$ dollars.
5. The value falls quickly at first and then more slowly, which the graph shows as a curve flattening out.

Final answer. $V = 20000 \times 0.85^{n}$; about 8874 dollars after 5 years and 3932 dollars after 10

Interpretation. The model never reaches zero, so a car always has some value in the model even though it does not in reality.

CHAPTER 6 · INVESTIGATION 6.6

Investigation: arches and trajectories

INQUIRY QUESTION

What does the graph of a trajectory show, and what does it leave out?

YOUR TASK

Model the flight of a ball, a stone or a water jet with a quadratic. Find the greatest height, the range and the symmetry, then compare with a real throw or a video if you can, and list the differences.

quantity	model	measured	difference
greatest height	6 m	about 5.5 m	air resistance
range	20 m	about 18 m	air resistance
time of flight	1.4 s	about 1.3 s	air resistance
shape	symmetric	slightly asymmetric	spin and wind

CHAPTER 6 · INVESTIGATION 6.6

Investigation: arches and trajectories

What do you notice?

- The model overestimated both the height and the range.
- The measured path was not quite symmetric.
- The differences were small for a heavy ball and large for a light one.
- The model was still useful, because it predicted the right order of magnitude.

Observation. A parabolic model of a trajectory is a good first approximation, and its errors come from the forces the model leaves out.

MATHEMATICAL CONCLUSION

The graph of a trajectory shows the height, the range and the symmetry of a flight under gravity alone, and its usefulness depends on how small the neglected forces are.

Your own words

CHAPTER 6 · 6.6

Investigation: arches and trajectories

A trajectory is a parabola under gravity alone, and its graph shows the highest point, the range and the symmetry of the flight.

Air resistance, spin and wind all change a real trajectory, and a parabolic model is a first approximation that is often good enough.

The graph shows position but not velocity, and the two are related by the gradient: the velocity is zero at the highest point.

Reading a trajectory means stating what the model assumes, and a reader who does not know the assumptions can be misled by the picture.

> FORMAL DEVELOPMENT
>
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In your own words

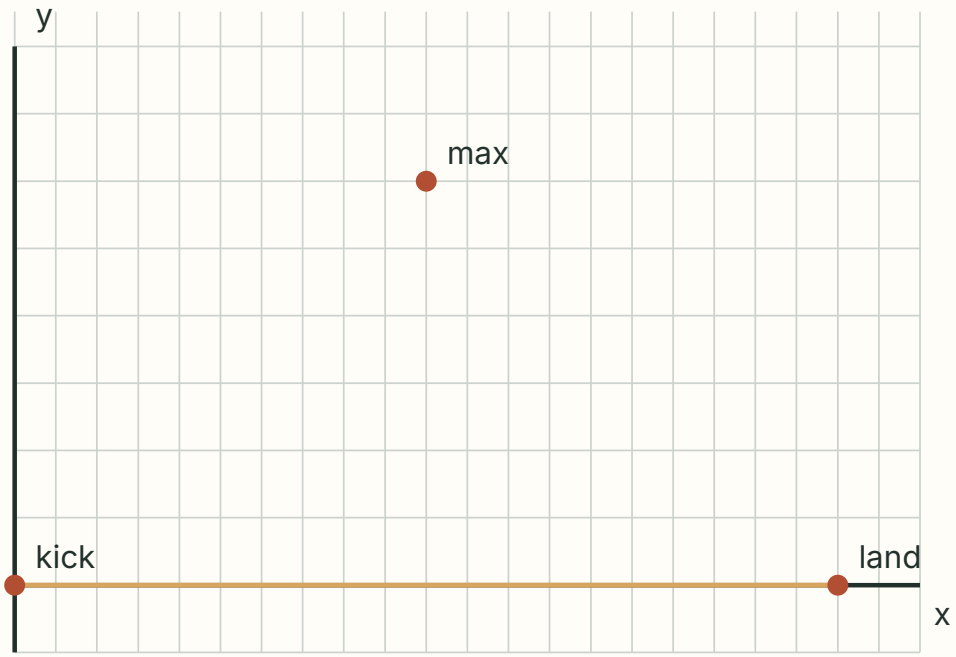
CHAPTER 6 · 6.6 · WORKED EXAMPLE 1

Investigation: arches and trajectories

WORKED EXAMPLE 6.6.1

Problem. A ball is kicked from the ground at an angle, and its height is $h = 1.2x - 0.06x^2$ metres when it is x metres from the kicker. Find the greatest height and the horizontal distance when it lands.

the trajectory as a parabola



Mathematical reasoning. The vertex gives the greatest height and the second root gives the landing distance.

1. The parabola opens downwards, so it has a maximum.
2. The axis of symmetry is $x = -1.2/(2 \times -0.06) = 10$.
3. At $x = 10$ the height is $12 - 6 = 6$ m, the greatest height.
4. Setting $h = 0$ gives $x(1.2 - 0.06x) = 0$, so $x = 0$ or $x = 20$.
5. The ball lands 20 m from the kicker, twice the distance of the highest point.

Final answer. greatest height 6 m at 10 m; it lands at 20 m

Interpretation. The symmetry of the parabola means the landing distance is twice the distance of the highest point, which is a fact about parabolas rather than about football.

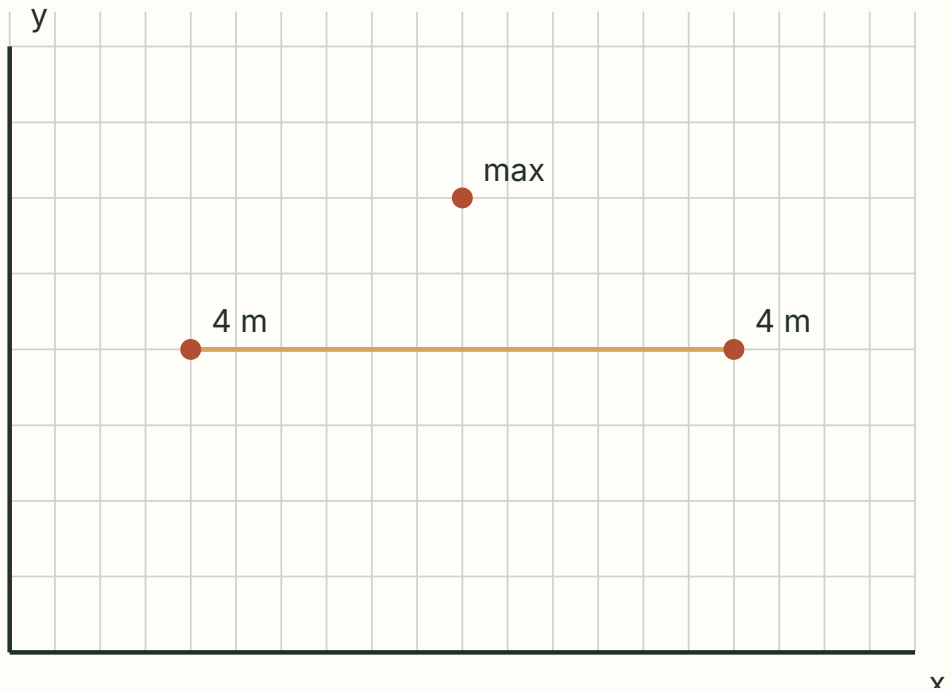
CHAPTER 6 · 6.6 · WORKED EXAMPLE 2

Investigation: arches and trajectories

WORKED EXAMPLE 6.6.2

Problem. A graph of a trajectory is symmetric about x = 10. The ball is at height 4 m at x = 4. Find the other point at which it is 4 m high, and explain why this works.

symmetry gives the second point



Mathematical reasoning. Symmetry about the axis means points at equal height are equally spaced from it.

1. The axis of symmetry is x = 10.
2. The point (4, 4) is 6 units to the left of the axis.
3. By symmetry there is a point 6 units to the right, at x = 16.
4. So the ball is also 4 m high at x = 16.
5. This works for any height, which is what symmetry means.

Final answer. the ball is 4 m high again at x = 16

Interpretation. Symmetry is a property of the graph that can be used instead of solving the equation, and it is a check on any calculation.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

6.A1 FLUENCY

Find the vertex of $y = x^2 - 4x + 7$.

6.A2 FLUENCY

Find the y-intercept of $y = 2x^2 - 5x + 3$.

6.A3 FLUENCY

State whether $y = -3x^2 + 6$ opens up or down.

6.A4 FLUENCY

Find the axis of symmetry of $y = x^2 + 8x + 3$.

6.A5 FLUENCY

Find the roots of $y = x^2 - 9$.

6.A6 FLUENCY

Describe the transformation from $y = x^2$ to $y = x^2 - 5$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

6.A7 FLUENCY

Describe the transformation from $y = x^2$ to $y = (x + 4)^2$.

6.A8 FLUENCY

Find the vertex of $y = (x - 2)^2 + 6$.

6.A9 FLUENCY

Find the y-intercept of $y = 3 \times 2^x$.

6.A10 FLUENCY

Evaluate 5×0.8^3.

6.A11 FLUENCY

Find the roots of $y = x^2 + x - 12$.

6.A12 FLUENCY

State the greatest value of $y = -(x - 1)^2 + 4$.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

6.B1 REASONING

The heights of a bouncing ball are 2, 1.6, 1.28, 1.024 m. Find the rule for the nth height and test it on the sixth bounce.

6.B2 REASONING

Investigate the pattern in the number of regions made by n lines through a single point.

6.B3 REASONING

Conjecture a rule for the sum of the first n powers of 2 and justify it.

6.B4 CHALLENGE

Investigate the pattern in the areas under the parabola $y = x^2$ between $x = 0$ and $x = 1, 2, 3$.

6.B5 REASONING

Build a table of the values of $y = 2^x$ for $x = -3$ to 3 and describe the pattern.

6.B6 REASONING

Conjecture a rule for the width of the parabola $y = kx^2$ at a height of 1 unit.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

6.B7 REASONING

Investigate the pattern in the number of intersection points of $y = x^2$ and $y = nx$ for $n = 1$ to 5.

6.B8 CHALLENGE

A ball's rebound heights follow a geometric sequence. Investigate the total distance travelled before it comes to rest.

6.B9 REASONING

Find the pattern in the gradients of $y = x^2$ at $x = 1, 2, 3$ and justify the rule.

6.B10 REASONING

Investigate the pattern in the maximum height of $y = -x^2 + 2nx$ for $n = 1$ to 5.

6.B11 REASONING

Conjecture a rule for the number of points with integer coordinates inside $y = x^2$ between $x = -n$ and n.

CRITERION C Communicating

Criterion C practice · Communicating

6.C1 REASONING

Explain in your own words what the vertex of a parabola is.

6.C2 FLUENCY

Write a complete solution to: sketch $y = x^2 - 2x - 8$, marking the vertex, the intercepts and the axis of symmetry.

6.C3 REASONING

Choose between a graph and an equation to answer a question about the greatest value of a quadratic and justify.

6.C4 FLUENCY

Set out a full solution to: describe the transformation from $y = x^2$ to $y = -(x + 3)^2 - 1$.

6.C5 REASONING

Explain why every parabola has the same shape.

6.C6 REASONING

Describe, with an example, the difference between a stretch and a translation.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

6.C7 REASONING

Explain why the graph of $y = 2^x$ never crosses the x-axis.

6.C8 REASONING

Write the reasoning that shows the axis of symmetry of $y = ax^2 + bx + c$ is $x = -b/(2a)$.

6.C9 REASONING

Set out the argument that exponential growth eventually overtakes any linear growth.

6.C10 REASONING

Explain what the crossing point of two graphs means.

6.C11 REASONING

Describe the relationship between a sketch and a plotted graph.

6.C12 REASONING

Explain, with an example, when a graph can be misleading.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

6.D1 APPLICATION

A ball is thrown so that its height is $h = 12t - 5t^2$. Find the greatest height, the time at which it occurs, and the time it returns to the ground.

6.D2 APPLICATION

A bridge arch is modelled by $y = -0.04x^2 + 0.8x$. Find the span and the greatest height.

6.D3 APPLICATION

A shop's revenue is $R = 30x - x^2$ and its cost is $C = 4x + 96$. Find the range of x for which a profit is made and the maximum profit.

6.D4 APPLICATION

A population of 400 grows by 12% per year. Model it, find the population after 5 years, and find when it doubles.

6.D5 APPLICATION

A radioactive sample decays from 80 g at 20% per year. Model it, find the mass after 4 years, and find the half-life.

6.D6 APPLICATION

A fountain's water follows $y = 6x - 0.3x^2$. Find the greatest height and the horizontal distance when it lands.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

6.D7 REASONING

A phone's value falls from 900 dollars by 25% per year. Investigate whether it is worth more than 100 dollars after 6 years and justify.

6.D8 APPLICATION

A field's yield is $Y = -2x^2 + 60x$ for x kg of fertiliser. Find the amount giving the greatest yield and the yield.

6.D9 REASONING

A ball is thrown from a window 10 m high with $h = 10 + 20t - 5t^2$. Investigate whether it reaches a height of 30 m and justify.

6.D10 CHALLENGE

Two savings accounts pay 5% simple and 5% compound interest on 1000 dollars. Investigate when the compound account first gives more, and by how much after 10 years.

6.D11 REASONING

A ball is kicked with $h = 0.6x - 0.02x^2$. Investigate whether it clears a 4 m wall at $x = 12$ and justify.

6.D12 REASONING

Explain, with a model of your own, when a graph drawn from a model is more useful than the equation.

CHAPTER 6 · APPLICATION AND ASSESSMENT

Applying chapter 6

Each task asks you to read a graph or to sketch one. State the feature you are reading and the evidence for it.

A1 · The garden arch

A garden arch is modelled by $y = -0.08x^2 + 1.6x$ metres, where x is the horizontal distance from one end. Find the span of the arch, its greatest height, and the height at the quarter points. Then sketch the arch and state two features of the model that a real arch would not have.

Criteria A and C · find the roots and the vertex, and comment on the model's assumptions.

A2 · The two savings plans

Plan A pays 6% simple interest and plan B pays 5% compound interest on a deposit of 2000 dollars. Model both, draw the graphs on the same axes, find the year in which plan B overtakes plan A, and say which you would choose for a 10-year saving.

Criteria A and D · model both plans, solve for the crossing point, and justify the recommendation.

CHAPTER 6 · REVIEW

What you should now be able to do

Key ideas

- The graph of $y = ax^2 + bx + c$ is a parabola.
- The vertex is at $x = -b/(2a)$ and is the maximum or minimum.
- The y-intercept is c and the x-intercepts are the roots.
- A larger $|a|$ gives a narrower parabola.
- Translations move the vertex; stretches change the width; reflections reverse the direction.
- An exponential function has the form $y = a \cdot b^x$.
- The y-axis is an asymptote of an exponential graph.
- The crossing point of two graphs solves the equation that sets them equal.

Vocabulary

parabola · vertex · axis of symmetry · intercept · translation · stretch · reflection · asymptote · exponential · trajectory · sketch · range

CHAPTER 6 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

6.R1 Find the vertex and axis of symmetry of $y = x^2 + 6x + 1$.

6.R2 Describe the transformation from $y = x^2$ to $y = (x - 5)^2 + 3$.

6.R3 Find the greatest value of $y = -x^2 + 8x - 3$.

6.R4 Evaluate 200×1.05^4.

6.R5 Find the roots of $y = 2x^2 + 5x - 3$.

6.R6 Explain why $y = 2^x$ has no x-intercept.

CHAPTER 6 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The graph of a function is a representation that shows change at a glance — where a quantity rises and falls, where it is greatest, and how it behaves far from the origin — and reading that representation is a form of reasoning that the equation alone does not provide.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of representation, change and pattern appear in your work?
- How is the global context of orientation in space and time, and its exploration of arches, paths and trajectories, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a parabolic model of a real path, and what would you still like to ask?

CHAPTER 6 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Arches, paths and trajectories are parabolas. Model one and show what its graph tells you that its equation does not.

Service as Action

COMMUNITY NEED	Arches, paths and trajectories are designed without showing the mathematics.
MATHEMATICAL CONNECTION	Quadratic functions, vertices and graphs.
PROPOSED ACTION	Model one real arch or path and publish the model with its graph.
EXPECTED IMPACT	A reader sees why the shape works and where it would fail.
STUDENT REFLECTION	Record the model, the graph and the check.

Community Project

PROJECT IDEA	A shape audit of the school's own architecture.
NEED	Arches, roofs and paths are chosen by appearance alone.
RESEARCH QUESTION	Which quadratic models the shapes we use, and what do they reveal?
MATHEMATICS	Quadratic functions, transformations and graphs.
ACTION OR OUTCOME	A one-page report with models and sketches of three shapes.
BENEFICIARIES	Everyone who uses the buildings.
EVIDENCE	The report with the models, the sketches and the checks.

Personal Project connection · MYP 5

PERSONAL INTEREST	Sport, gaming or a path you travel.
LEARNING GOAL	To model a trajectory you care about and read its graph.
PRODUCT OR OUTCOME	A quadratic model with a graph and its key features.
MATHEMATICS	Parabolas, vertices and symmetry.
PROCESS	Model the path, sketch the graph, and compare with a measurement.
EVIDENCE	The model, the graph and the comparison.

CHAPTER 7 · PART II · THINKING WITH MODELS

07

Patterns, Sequences and Generalisation

Arithmetic sequences add a constant; geometric sequences multiply by one. Quadratic sequences have constant second differences, and each kind has its own signature.

ENGAGE

A savings account starts at 100 dollars and grows by 20 dollars a month. Another grows by 20% a month. After how many months does the second overtake the first?

Framing the inquiry

Key concept: Relationships
Related concepts: Pattern · Generalisation · Representation
Global context: Orientation in space and time
Global context exploration: Cycles, growth and prediction

Statement of inquiry: A sequence is a pattern in time, and generalising it means finding a rule that connects the position of a term to its value; the rule is a representation of the pattern that allows prediction beyond the terms observed, and its justification is what separates a general claim from a guess.

Questions to guide us

- **Factual:** What are arithmetic, geometric and quadratic sequences, and how is the nth term of each found?
- **Conceptual:** Why does a constant second difference signal a quadratic rule, and how is a general rule justified?
- **Debatable:** If a rule fits every term observed, is it reasonable to use it to predict, or is prediction always a leap beyond the evidence?

ATL · CRITICAL THINKING. Students test a conjectured rule against terms they have not yet used, and revise the rule when the test fails rather than defending it.

CHAPTER 7 · INVESTIGATION 7.1

Describing and extending patterns

INQUIRY QUESTION

How is a pattern described so that another person can continue it, and what does extending it predict?

YOUR TASK

Take four patterns of your own: one you can describe by adding, one by multiplying, one from a picture and one from a real situation. For each, write the term-to-term rule, the position-to-term rule, and the next two terms. Then check your position-to-term rule against the term-to-term rule for a term you did not use to find it.

pattern	term-to-term	position-to-term	next two terms
4, 9, 14, 19, …	add 5	$5n - 1$	24, 29
3, 6, 12, 24, …	double	$3 \times 2^{n-1}$	48, 96
1, 4, 9, 16, …	add 3, 5, 7, …	n^2	25, 36
squares of side 1, 2, 3, …	add $2n + 1$	n^2	25, 36

CHAPTER 7 · INVESTIGATION 7.1

Describing and extending patterns

What do you notice?

- Every pattern had both a term-to-term and a position-to-term rule.
- The multiplicative pattern grew far faster than the additive ones.
- The picture pattern and the numeric pattern gave the same rule.
- The check against an unused term held for every rule.

> **Observation.** A pattern is described by a term-to-term rule and a position-to-term rule, and the two descriptions of the same pattern must agree.

MATHEMATICAL CONCLUSION

Describing a pattern well means giving a rule another person can continue, and extending it is a prediction that the rule alone justifies.

Your own words

CHAPTER 7 · 7.1

Describing and extending patterns

A pattern can be described in words, in pictures, in a table of values or as a rule, and each description is a different representation of the same pattern. A good description is one another person can continue without asking a question.

A term-to-term rule says how to get from one term to the next, and it is the easiest rule to see. A position-to-term rule connects the position of a term to its value, and it is the rule that predicts a distant term without listing the ones before it.

Extending a pattern is a prediction, and a prediction is only as safe as the rule behind it. Two different rules can fit the same first few terms, so a pattern is never fixed by its first terms alone.

A generalisation is a claim about every term, and it needs an argument, not just agreement with the terms already listed. A single term that disagrees is enough to refute it.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 7 · 7.1 · WORKED EXAMPLE 1

Describing and extending patterns

WORKED EXAMPLE 7.1.1

Problem. Describe the pattern 2, 5, 8, 11, … in two ways, and use each to find the 20th term.

position	1	2	3	4	5	20
term	2	5	8	11	14	59

Mathematical reasoning. Read the term-to-term rule from the differences, then turn it into a position-to-term rule.

1. The differences are 3, 3, 3, so the term-to-term rule is add 3.
2. Term 1 is 2, term 2 is $2 + 3$, term 3 is $2 + 2 \times 3$.
3. So term n is $2 + (n - 1) \times 3$, which is the position-to-term rule.
4. Term 20 is $2 + 19 \times 3 = 59$.
5. The two descriptions agree, and the second reaches term 20 without listing the terms between.

Final answer. add 3 each time; term n = 3n − 1; the 20th term is 59

Interpretation. A term-to-term rule is quick to see and a position-to-term rule is quick to use, and the same pattern has both.

CHAPTER 7 · 7.1 · WORKED EXAMPLE 2

Describing and extending patterns

WORKED EXAMPLE 7.1.2

Problem. The first three terms of a pattern are 1, 2, 4. Give two different rules that fit, and say what each predicts for the fourth term.

position	1	2	3	4 (rule 1)	4 (rule 2)
term	1	2	4	8	7

Mathematical reasoning. Look for a multiplicative rule and an additive rule.

1. Multiplicative: each term is double the previous, so the next is 8.
2. Additive: the differences are 1 then 2, so the next difference might be 3, giving 7.
3. Both rules fit 1, 2, 4 exactly.
4. They predict different fourth terms, so the three terms given do not fix the pattern.
5. A fourth term of 8 would support the multiplicative rule and a fourth term of 7 the additive one.

Final answer. double each time gives 8; add the next integer gives 7

Interpretation. A pattern is never determined by its first few terms, so a prediction is a conjecture until more terms confirm it.

CHAPTER 7 · INVESTIGATION 7.2

Linear sequences and the nth term

INQUIRY QUESTION

What does a constant first difference tell you, and how is the nth term found?

YOUR TASK

Build three arithmetic sequences of your own: one with a positive difference, one with a negative difference, and one with a fractional difference. For each, find the rule, the 15th term and the sum of the first 15 terms, and check the 15th term by listing.

sequence	rule	15th term	sum of 15
3, 7, 11, …	$4n - 1$	59	465
20, 17, 14, …	$-3n + 23$	−22	−15
1, 1.5, 2, …	$0.5n + 0.5$	8	71.25
−5, −1, 3, …	$4n - 9$	51	345

CHAPTER 7 · INVESTIGATION 7.2

Linear sequences and the nth term

What do you notice?

- Every sequence with a constant difference gave a linear rule.
- The negative difference made the terms fall, and the 15th term became negative.
- The fractional difference still gave a whole-number sum here, by chance.
- Listing the terms confirmed every rule that was found algebraically.

> **Observation.** A constant first difference is the signature of a linear rule, and the rule can be recovered from any two terms of the sequence.

MATHEMATICAL CONCLUSION

An arithmetic sequence is described by $a + (n - 1)d$, and the constant difference is both the gradient of its graph and the test that identifies it.

Your own words

CHAPTER 7 · 7.2

Linear sequences and the nth term

An arithmetic sequence has a constant difference between consecutive terms, called the common difference d. The nth term is a + (n − 1)d, where a is the first term.

The constant difference is the signature of a linear rule, and the graph of the terms against their position is a straight line.

The sum of the first n terms is n/2 × (first + last), which comes from pairing the first with the last, the second with the second-last, and so on.

A sequence given by a rule such as 3n + 2 starts at 5, not 2, because the first term corresponds to n = 1. Substituting n = 0 gives the value before the sequence begins.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

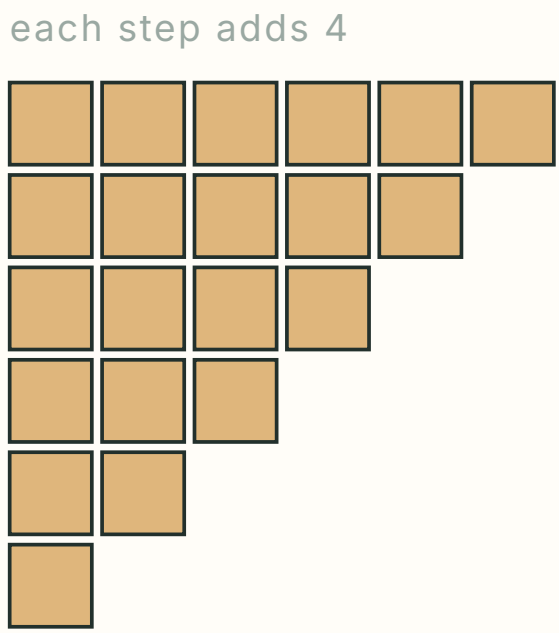
In your own words

CHAPTER 7 · 7.2 · WORKED EXAMPLE 1

Linear sequences and the nth term

WORKED EXAMPLE 7.2.1

Problem. Find the 20th term and the sum of the first 20 terms of 7, 11, 15, 19, …



Mathematical reasoning. Use the formula for the nth term and the pairing formula for the sum.

1. The first term is 7 and the common difference is 4.
2. The 20th term is $7 + 19 \times 4 = 7 + 76 = 83$.
3. The sum is $20/2 \times (7 + 83) = 10 \times 90 = 900$.
4. The terms lie on the line whose rule is $4n + 3$.
5. The sum formula works because the terms pair to a constant total.

Final answer. the 20th term is 83 and the sum is 900

Interpretation. One formula gives any term and another gives the total, and neither requires listing the terms.

CHAPTER 7 · 7.2 · WORKED EXAMPLE 2

Linear sequences and the nth term

WORKED EXAMPLE 7.2.2

Problem. The 5th term of an arithmetic sequence is 14 and the 12th is 42. Find the first term and the common difference.

term	position	value
5th	5	14
12th	12	42

Two terms fix the whole sequence.

Mathematical reasoning. The difference between the values equals the difference between the positions times d.

1. The value rises by 42 − 14 = 28 over 12 − 5 = 7 positions.
2. So 7d = 28 and d = 4.
3. The 5th term is a + 4d = a + 16 = 14, so a = −2.
4. The rule is therefore 4n − 6.
5. Check: the 12th term is 48 − 6 = 42.

Final answer. first term −2, common difference 4

Interpretation. Two terms are enough to fix an arithmetic sequence, because a linear rule has only two unknowns.

CHAPTER 7 · INVESTIGATION 7.3

Quadratic sequences and differences

INQUIRY QUESTION

Why does a constant second difference signal a quadratic rule?

YOUR TASK

Take four sequences: two with constant first differences, one with constant second differences, and one that is neither. For each, build a difference table, identify the kind of rule, fit it, and test it on a term you did not use.

sequence	first diff	second diff	rule
5, 8, 11, 14	3, 3, 3	0	$3n + 2$
1, 4, 9, 16	3, 5, 7	2, 2	n^2
2, 6, 12, 20	4, 6, 8	2, 2	$n^2 + n$
1, 2, 4, 8	1, 2, 4	1, 2	2^{n-1}

CHAPTER 7 · INVESTIGATION 7.3

Quadratic sequences and differences

What do you notice?

- Constant first differences always gave a linear rule.
- Constant second differences always gave a quadratic.
- The doubling sequence had growing differences and was exponential.
- Each rule was tested on a term not used to find it, and every test passed.

Observation. The difference table identifies the kind of rule, and the number of terms needed to fix it equals one more than the degree of the rule.

MATHEMATICAL CONCLUSION

A sequence with constant kth differences is generated by a polynomial of degree k, and the difference table is the tool that identifies which polynomial to fit.

Your own words

CHAPTER 7 · 7.3

Quadratic sequences and differences

The first differences of a quadratic sequence form an arithmetic sequence, so the second differences are constant.

A constant second difference means the rule is a quadratic of the form $an^2 + bn + c$, and the second difference is $2a$.

Finding the rule means solving three equations from the first three terms, or using the standard form with the differences read directly.

Quadratic sequences arise from counting in two dimensions, such as the number of dots in a growing pattern or the number of diagonals of a polygon.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

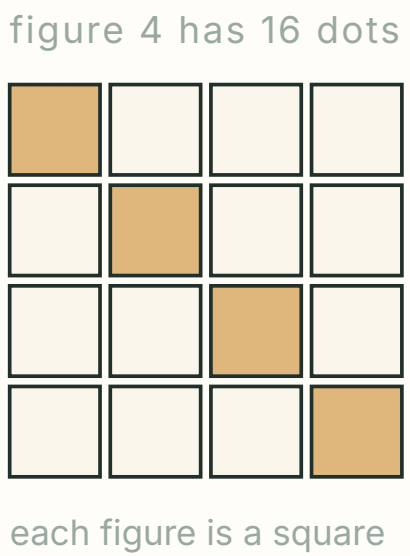
In your own words

CHAPTER 7 · 7.3 · WORKED EXAMPLE 1

Quadratic sequences and differences

WORKED EXAMPLE 7.3.1

Problem. The numbers of matches needed for 1, 2, 3 and 4 squares in a row are 4, 7, 10, 13. A different pattern gives 1, 4, 9, 16. Find the rule for the second and test it on the 6th figure.



Mathematical reasoning. Take differences to identify the kind of rule, then fit the quadratic.

1. The sequence is 1, 4, 9, 16, so the first differences are 3, 5, 7.
2. The second differences are 2, 2, which are constant.
3. So the rule is quadratic with $2a = 2$, hence $a = 1$.
4. Fitting gives n^2, and the 6th figure has 36 dots.
5. The first pattern had constant first differences, so it was linear with rule $3n + 1$.

Final answer. the rule is n^2; the 6th figure has 36 dots

Interpretation. The differences identify the kind of rule before any fitting is attempted, which is the whole value of the difference table.

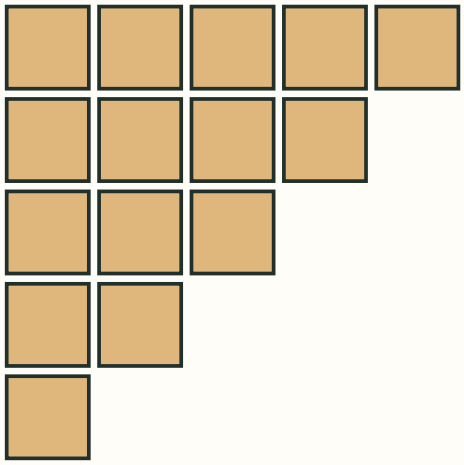
CHAPTER 7 · 7.3 · WORKED EXAMPLE 2

Quadratic sequences and differences

WORKED EXAMPLE 7.3.2

Problem. Find the rule for the sequence 3, 8, 15, 24, 35 and use it to find the 12th term.

first differences 5, 7, 9, 11



Mathematical reasoning. Take differences to find the coefficient of n^2, then solve for the rest.

1. The first differences are 5, 7, 9, 11, so the second difference is 2.
2. Hence the rule is $n^2 + bn + c$.
3. Using the first term: $1 + b + c = 3$, so $b + c = 2$.
4. Using the second: $4 + 2b + c = 8$, so $2b + c = 4$.
5. Subtracting gives $b = 2$ and $c = 0$, so the rule is $n^2 + 2n$.
6. The 12th term is $144 + 24 = 168$.

Final answer. the rule is $n^2 + 2n$, and the 12th term is 168

Interpretation. Three terms are enough to fix a quadratic, exactly as two fix a line, and the difference table says how many are needed.

CHAPTER 7 · INVESTIGATION 7.4

Recursive rules and spreadsheets

INQUIRY QUESTION

What does a constant ratio tell you, and why does growth outrun addition?

YOUR TASK

Take three geometric sequences: one growing, one decaying and one with a negative ratio. For each, find the rule, the 8th term and the sum of the first 8 terms. Then find the limiting sum for the decaying one and check it by adding many terms.

sequence	rule	8th term	sum of 8
2, 6, 18, …	$2 \times 3^{n-1}$	4374	6560
64, 32, 16, …	$64 \times 0.5^{n-1}$	0.5	127.5
1, −2, 4, …	$(-2)^{n-1}$	−128	85
100, 90, 81, …	$100 \times 0.9^{n-1}$	47.83	471.6

CHAPTER 7 · INVESTIGATION 7.4

Recursive rules and spreadsheets

What do you notice?

- The growing sequence's last term was almost as large as the whole sum.
- The decaying sequence's sum approached 128, its limit.
- The negative ratio made the terms alternate in sign.
- Adding many terms of the decaying sequence confirmed the limiting sum.

> **Observation.** A constant ratio gives an exponential rule, and the behaviour of the sum depends on whether the ratio is greater or less than one in size.

MATHEMATICAL CONCLUSION

A geometric sequence is described by ar^{n-1}, and when $|r| < 1$ its terms shrink towards zero while its total approaches the finite limit $a/(1 - r)$.

Your own words

CHAPTER 7 · 7.4

Recursive rules and spreadsheets

A recursive rule gives each term from the one before it: term n = term n−1 + d for an arithmetic pattern, or term n = r × term n−1 for a geometric one. It is the rule a spreadsheet uses, because a cell can refer to the cell above it.

A spreadsheet makes recursion visible: type the first term, then fill the formula down, and the whole sequence appears. Changing the first term or the step changes every term at once, which is why a spreadsheet is a model and not just a list.

A position-to-term rule gives any term directly, and a recursive rule gives only the next one. To reach term 100 a recursive rule needs 99 steps, and a position-to-term rule needs one substitution.

A geometric sequence is the multiplicative recursive rule, and its nth term is ar^{n-1}. Growth by a constant ratio eventually overtakes growth by a constant amount, because the increase itself grows.

A ratio between −1 and 1 makes the terms shrink towards zero, and the sum of infinitely many such terms is a finite number. This is the first infinite process in the course, and it is what makes a bouncing ball eventually stop.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

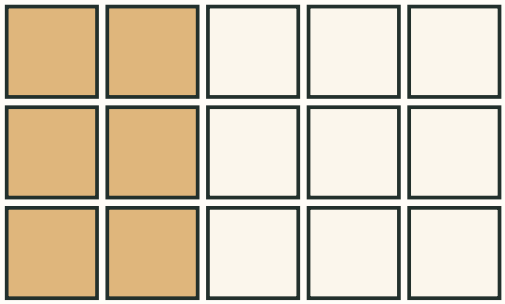
CHAPTER 7 · 7.4 · WORKED EXAMPLE 1

Recursive rules and spreadsheets

WORKED EXAMPLE 7.4.1

Problem. A pattern of squares grows so that figure 1 has 1 square, figure 2 has 3, figure 3 has 9 and figure 4 has 27. Find the number of squares in figure 10 and the total for the first 10 figures.

figures 1 to 4 of the pattern



each figure has three times the previous

Mathematical reasoning. The constant ratio is 3, so the rule is geometric.

1. The ratio between consecutive figures is 3.
2. So the rule is 3^{n-1}, and figure 10 has $3^9 = 19683$ squares.
3. The total is $(3^{10} - 1)/(3 - 1) = (59049 - 1)/2 = 29524$.
4. The 10th figure alone has twice as many squares as all the others together.
5. That is the defining property of exponential growth.

Final answer. figure 10 has 19683 squares; the total is 29524

Interpretation. The last term dominates the sum, which never happens in an arithmetic sequence.

CHAPTER 7 · 7.4 · WORKED EXAMPLE 2

Recursive rules and spreadsheets

WORKED EXAMPLE 7.4.2

Problem. A ball is dropped from 2 m and rebounds to 80% of its previous height. Find the total distance it travels before coming to rest.

bounce	height	distance
drop	2	2
1	1.6	3.2
2	1.28	2.56
3	1.024	2.048

Each bounce adds two journeys of 80% of the previous height.

Mathematical reasoning. The rebound heights form a geometric sequence, and each bounce contributes twice the height.

1. The rebound heights are 2×0.8, 2×0.8^2, 2×0.8^3, ...
2. Each bounce adds the upward and the downward journey, so the total is $2 + 2(2 \times 0.8 + 2 \times 0.8^2 + \ldots)$.
3. The sum inside is $1.6/(1 - 0.8) = 8$.
4. So the total distance is $2 + 2 \times 8 = 18$ m.
5. The number of bounces is infinite but the total distance is finite.

Final answer. 18 m

Interpretation. An infinite number of bounces covers a finite distance, which is the first hint that infinite processes can have finite answers.

CHAPTER 7 · INVESTIGATION 7.5

Modelling with sequences

INQUIRY QUESTION

How is a sequence used to predict, and how far can the prediction be trusted?

YOUR TASK

Choose a quantity that changes over time: a population, a saving, a skill level or a phone bill. Model it with a sequence, predict three terms ahead, and then check one prediction against reality if you can.

quantity	model	prediction	reality
savings	arithmetic, +50 a month	600 after 10 months	matched
reading speed	arithmetic, +2 words a minute	120 after 10 weeks	slower
plant height	geometric, ×1.1 a week	2.6 times after 10 weeks	slower

CHAPTER 7 · INVESTIGATION 7.5

Modelling with sequences

What do you notice?

- The arithmetic savings model matched reality because the rule was exact.
- The reading model overestimated, because improvement slows.
- The plant model overestimated, because growth is not a constant multiplier.
- Every failure pointed to the assumption, not the arithmetic.

Observation. A sequence model predicts well when its rule matches the mechanism of change, and badly when the mechanism changes.

MATHEMATICAL CONCLUSION

A sequence is a model whose predictions are trustworthy only while the rule that generates it continues to describe the situation.

Your own words

CHAPTER 7 · 7.5

Modelling with sequences

A sequence models a situation when its rule matches the way the situation changes, and the model is used to predict terms beyond those observed.

Prediction inside the range of the data is interpolation and is usually safe; prediction beyond it is extrapolation and depends on the rule continuing to hold.

Recursive and explicit rules describe the same sequence. A recursive rule gives each term from the one before, and an explicit rule gives any term from its position.

A model's failure is informative. When a prediction is wrong, the rule or the assumption behind it is wrong, and finding which is the next task.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 7 · 7.5 · WORKED EXAMPLE 1

Modelling with sequences

WORKED EXAMPLE 7.5.1

Problem. A population of rabbits grows from 40 by a factor of 1.3 each month. Model it, predict the population after 12 months, and comment on the model.

month	population
0	40
3	88
6	193
12	928

the population more than doubles every six months

Mathematical reasoning. Repeated multiplication by a constant factor is a geometric model.

1. The model is $P = 40 \times 1.3^{n}$.
2. After 12 months, $P = 40 \times 1.3^{12} \approx 928$.
3. That is more than twenty times the starting population.
4. The model assumes no limit on food or space, which is false.
5. So the prediction is reliable for a few months and unreliable beyond a year.

Final answer. $P = 40 \times 1.3^{n}$; about 928 after 12 months; unreliable in the long term

Interpretation. An exponential model is a local description, and extrapolating it far enough always produces an impossible answer.

CHAPTER 7 · 7.5 · WORKED EXAMPLE 2

Modelling with sequences

WORKED EXAMPLE 7.5.2

Problem. A phone contract costs 20 dollars a month plus 5 dollars for each gigabyte. Model the monthly cost, and find the cost in month 12 and the total over 12 months if usage rises by 1 gigabyte each month.

month	usage	cost
1	1	25
2	2	30
3	3	35
12	12	80

the cost is an arithmetic sequence

Mathematical reasoning. The cost is a linear function of the usage, and the usage is an arithmetic sequence.

1. The cost in month n is 20 + 5n dollars.
2. In month 12 the cost is 20 + 60 = 80 dollars.
3. The costs form an arithmetic sequence 25, 30, 35, … with difference 5.
4. The total over 12 months is 12/2 × (25 + 80) = 6 × 105 = 630 dollars.
5. The model assumes the usage really does rise by exactly 1 gigabyte each month.

Final answer. 80 dollars in month 12; 630 dollars over 12 months

Interpretation. Two models combine: an arithmetic sequence of usage and a linear cost, and the total uses the arithmetic sum formula.

CHAPTER 7 · INVESTIGATION 7.6

Investigation: figures that grow

INQUIRY QUESTION

How is a repeating pattern predicted, and what does a cycle hide?

YOUR TASK

Find three periodic patterns: a traffic light, a weekly timetable, and one of your own. For each, state the period, predict the state at a large position, and check by counting if the period is short.

pattern	period	prediction	check
traffic light	4	time 47 is red	confirmed
weekly timetable	7	day 100 is the second day	confirmed
last digit of 7^n	4	7^{50} ends in 9	confirmed

CHAPTER 7 · INVESTIGATION 7.6

Investigation: figures that grow

What do you notice?

- Every prediction came from a division with a remainder.
- The remainder identified the position in the cycle exactly.
- A short cycle made counting a practical check.
- The convention for a zero remainder had to be decided each time.

> **Observation.** A periodic pattern is predicted by dividing the position by the period and reading the remainder, and the remainder convention must be stated.

MATHEMATICAL CONCLUSION

A repeating pattern is predicted from the remainder of a division, and the cycle is a representation that hides the long-term behaviour by making a short sample look complete.

Your own words

CHAPTER 7 · 7.6

Investigation: figures that grow

A periodic sequence repeats after a fixed number of terms, and the position within the cycle is found from the remainder when the position is divided by the period.

Predicting a periodic pattern is therefore a division with a remainder, and the remainder identifies the point in the cycle.

A cycle hides the long-term behaviour, because a short sample of a cycle looks like a rule that will continue, and it does.

Some sequences are eventually periodic, and some are not, and telling them apart requires more terms than a short observation provides.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

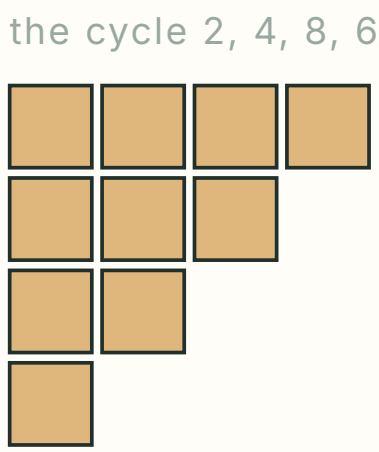
In your own words

CHAPTER 7 · 7.6 · WORKED EXAMPLE 1

Investigation: figures that grow

WORKED EXAMPLE 7.6.1

Problem. The last digits of the powers of 2 cycle 2, 4, 8, 6. Find the last digit of 2^{100}.



Mathematical reasoning. Divide the exponent by the period and use the remainder to identify the point in the cycle.

1. The cycle has period 4.
2. Dividing 100 by 4 gives a remainder of 0.
3. A remainder of 0 means the last position in the cycle, which is 6.
4. So the last digit of 2^{100} is 6.
5. Check: the last digit of 2^4 is 6 and of 2^8 is 6, confirming the pattern.

Final answer. 6

Interpretation. A remainder of zero points to the end of the cycle rather than the beginning, which is the one detail that catches people out.

CHAPTER 7 · 7.6 · WORKED EXAMPLE 2

Investigation: figures that grow

WORKED EXAMPLE 7.6.2

Problem. A traffic light cycles green, amber, red, red, green. If it is green at time 0, what colour is it at time 47?

time	colour
0	green
1	amber
2	red
3	red
4	green
5	amber

the cycle has period 4

Mathematical reasoning. Divide the time by the period and read the position in the cycle.

1. The cycle is green, amber, red, red, so the period is 4.
2. Dividing 47 by 4 gives 11 remainder 3.
3. A remainder of 3 is the fourth position, which is red.
4. So the light is red at time 47.
5. A remainder of 0 would mean green, the first position.

Final answer. red

Interpretation. The remainder identifies the position in the cycle, and the convention for a zero remainder must be stated.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

7.A1 FLUENCY

Find the 10th term of 3, 7, 11, 15, ...

7.A2 FLUENCY

Find the 8th term of 20, 17, 14, ...

7.A3 FLUENCY

Find the rule for 5, 9, 13, 17, ...

7.A4 FLUENCY

Find the sum of the first 12 terms of 2, 5, 8, ...

7.A5 FLUENCY

Find the 6th term of 3, 6, 12, 24, ...

7.A6 FLUENCY

Find the rule for 81, 27, 9, 3, ...

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

7.A7 FLUENCY

Find the next two terms of 1, 4, 9, 16, ...

7.A8 FLUENCY

Find the next term of 2, 6, 12, 20, ...

7.A9 FLUENCY

Find the sum of the first 6 terms of 1, 2, 4, 8, ...

7.A10 FLUENCY

Find the 5th term of 50, 45, 40.5, ...

7.A11 FLUENCY

Find the rule for 7, 12, 17, 22, ...

7.A12 FLUENCY

Find the next two terms of 1, −3, 9, −27, ...

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

7.B1 REASONING

The number of dots in figure n is $n^2 + n$. Find the first four terms and test the rule on figure 8.

7.B2 REASONING

Investigate the pattern in the sums 1, 1 + 2, 1 + 2 + 3 and justify the general rule.

7.B3 REASONING

Conjecture a rule for the number of diagonals of an n-sided polygon and test it for n = 5, 6, 7.

7.B4 REASONING

Investigate the pattern in the number of cubes needed to build a staircase of n steps.

7.B5 REASONING

Build a difference table for 2, 5, 10, 17, 26 and identify the rule.

7.B6 REASONING

Conjecture a rule for the number of matches in a row of n hexagons and justify it.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

7.B7 REASONING

Investigate the pattern in the sums $1/2$, $1/2 + 1/4$, $1/2 + 1/4 + 1/8$ and justify the rule.

7.B8 CHALLENGE

The heights of a bouncing ball are 2, 1.6, 1.28 m. Investigate the total distance before it comes to rest.

7.B9 REASONING

Find the pattern in the last digits of 3^n for n = 1 to 8 and use it to predict 3^{50}.

7.B10 CHALLENGE

Investigate the pattern in the number of squares of all sizes in an n by n grid.

7.B11 CHALLENGE

Conjecture a rule for the number of triangles formed by n lines in general position.

CRITERION C Communicating

Criterion C practice · Communicating

7.C1 REASONING

Explain in your own words what an arithmetic sequence is.

7.C2 FLUENCY

Write a complete solution to: find the rule for 4, 11, 18, 25 and the 30th term.

7.C3 REASONING

Choose between a recursive and an explicit rule for finding the 100th term and justify.

7.C4 FLUENCY

Set out a full solution to: find the sum of the first 10 terms of 5, 10, 20, 40, …

7.C5 REASONING

Explain why a constant second difference signals a quadratic rule.

7.C6 REASONING

Describe, with an example, the difference between an arithmetic and a geometric sequence.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

7.C7 REASONING

Explain why the sum of infinitely many terms can be finite.

7.C8 REASONING

Write the reasoning that shows the sum of the first n odd numbers is n^2.

7.C9 REASONING

Set out the argument that the number of diagonals of an n-sided polygon is $n(n - 3)/2$.

7.C10 REASONING

Explain what the common ratio of a geometric sequence represents in a real situation.

7.C11 REASONING

Describe the relationship between a difference table and the degree of a rule.

7.C12 REASONING

Explain, with an example, when a pattern that fits every observed term is still wrong.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

7.D1 APPLICATION

A population of 500 grows by 8% a year. Model it, find the population after 10 years, and find when it reaches 1000.

7.D2 APPLICATION

A car loses 12% of its value a year from 15000 dollars. Model it, find the value after 5 years, and find when it falls below 5000 dollars.

7.D3 APPLICATION

A contractor charges 200 dollars plus 40 dollars per hour for the first 5 hours and 60 dollars per hour after. Model the cost and find it for an 8-hour job.

7.D4 APPLICATION

A student saves 20 dollars in month 1 and 5 dollars more each month. Model the saving, find the saving in month 12, and find the total over 12 months.

7.D5 APPLICATION

A ball is dropped from 10 m and rebounds to 60% of its previous height. Find the total distance travelled before it comes to rest.

7.D6 APPLICATION

A phone plan costs 15 dollars a month, rising by 2 dollars a month each month. Model the cost, find it in month 12, and find the total over 12 months.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

7.D7 REASONING

A bank pays 4% compound interest on 5000 dollars. Investigate how long it takes to double and justify.

7.D8 APPLICATION

A colony of bacteria doubles every 3 hours from 200. Model it, find the population after a day, and comment on the model.

7.D9 REASONING

A staircase of n steps uses $n(n + 1)/2$ blocks. Investigate the number of steps possible with 1000 blocks and justify.

7.D10 CHALLENGE

A savings account pays 50 dollars a month plus 0.5% interest monthly. Investigate the balance after 12 months and justify the method.

7.D11 APPLICATION

A tree grows 0.4 m a year for the first 5 years and 0.2 m a year after. Model the height and find it after 10 years from 1.2 m.

7.D12 REASONING

Explain, with a model of your own, when a sequence model is a poor description of a situation.

CHAPTER 7 · APPLICATION AND ASSESSMENT

Applying chapter 7

Each task asks for a rule and a justification. Show the differences or ratios you used, and test your rule on a term you did not use to find it.

A1 · The growing savings club

A savings club starts with 5 members and the number of members doubles every month. Each member contributes 10 dollars a month. Model the total contribution in month n, find the contribution in month 12, and find the total contributed over the first 12 months. Then comment on how long the model can continue.

Criteria A and D · build the geometric model, use the series formula, and justify the limits of the model.

A2 · The bouncing ball

A ball is dropped from 8 m and rebounds to 70% of its previous height each time. Find the height of the third rebound, the total distance travelled before it comes to rest, and the number of bounces after which the rebound is less than 10 cm. Justify the use of an infinite series.

Criteria B and C · use a geometric model, justify the limiting sum, and solve the inequality.

CHAPTER 7 · REVIEW

What you should now be able to do

Key ideas

- An arithmetic sequence has a constant difference and a linear rule.
- A geometric sequence has a constant ratio and an exponential rule.
- Constant second differences signal a quadratic rule.
- The sum of an arithmetic sequence is n/2 × (first + last).
- The sum of a geometric sequence is $a(r^n - 1)/(r - 1)$.
- When $|r| < 1$ the terms shrink and the total approaches $a/(1 - r)$.
- A periodic pattern is predicted from the remainder of a division.
- A general rule needs an argument, not a list of checked cases.

Vocabulary

arithmetic sequence · geometric sequence · common difference · common ratio · nth term · difference table · periodic · recursive rule · explicit rule · generalisation · counterexample · limiting sum

CHAPTER 7 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

7.R1 Find the 15th term of 4, 9, 14, ...

7.R2 Find the rule for 2, 6, 18, 54, ...

7.R3 Find the next term of 1, 8, 27, 64, ...

7.R4 Find the sum of the first 8 terms of 3, 6, 12, ...

7.R5 Find the rule for 0, 3, 8, 15, ...

7.R6 Explain why the sum 1 + 1/2 + 1/4 + ... is finite.

CHAPTER 7 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> A sequence is a pattern in time, and generalising it means finding a rule that connects the position of a term to its value; the rule is a representation of the pattern that allows prediction beyond the terms observed, and its justification is what separates a general claim from a guess.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of pattern, generalisation and representation appear in your work?
- How is the global context of orientation in space and time, and its exploration of cycles, growth and prediction, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a sequence model for prediction, and what would you still like to ask?

CHAPTER 7 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Cycles and growth are sequences in disguise. Model one and say how far your rule can be trusted.

Service as Action

COMMUNITY NEED	Growth, cycles and savings are predicted without stating the model.
MATHEMATICAL CONNECTION	Arithmetic and geometric sequences and series.
PROPOSED ACTION	Model one community growth or savings plan and publish the model with its limits.
EXPECTED IMPACT	A reader can see how long a plan can last.
STUDENT REFLECTION	Record the model, the prediction and the check.

Community Project

PROJECT IDEA	A growth audit of a school club or team.
NEED	Membership and budgets are projected by guesswork.
RESEARCH QUESTION	Which sequence models the growth, and how far can it be trusted?
MATHEMATICS	Sequences, series and exponential models.
ACTION OR OUTCOME	A one-page projection with the model and its limits.
BENEFICIARIES	Everyone who plans for the club.
EVIDENCE	The projection with the model and a test.

Personal Project connection · MYP 5

PERSONAL INTEREST	A game, a sport or a savings goal.
LEARNING GOAL	To model something that grows or cycles and predict it.
PRODUCT OR OUTCOME	A sequence model with a prediction and a check.
MATHEMATICS	Arithmetic and geometric sequences.
PROCESS	Identify the mechanism, build the model, and test a prediction.
EVIDENCE	The model, the prediction and the test.

CHAPTER 8 · PART II · THINKING WITH MODELS

08

Growth and Financial Mathematics

A percentage rise followed by the same percentage fall does not return to the start, because the two percentages are taken on different quantities.

ENGAGE

A shop raises a price by 15% and then reduces it by 15%. Is the final price the same as the original, higher, or lower?

Framing the inquiry

Key concept: Relationships
Related concepts: Change · Quantity · Model
Global context: Fairness and development
Global context exploration: Access, saving and long-term planning

Statement of inquiry: Repeated percentage change is modelled by multiplication rather than addition, so the quantity after n periods is the initial quantity times a multiplier to the power n; recognising this changes how a percentage rise and a percentage fall are understood, and the model makes the long-term effect of a rate visible.

Questions to guide us

- **Factual:** What is the difference between simple and compound interest, and how is a repeated percentage change modelled?
- **Conceptual:** Why does a percentage rise followed by an equal percentage fall not restore the original quantity, and what does a multiplier reveal that a percentage does not?
- **Debatable:** Is a loan that is affordable in the short term a fair one, and who is best placed to judge the long-term cost of borrowing?

ATL · SELF-MANAGEMENT. Students organise a multi-step financial calculation, check each step against the model, and manage the time needed to complete a comparison of several options.

CHAPTER 8 · INVESTIGATION 8.1

Repeated percentage change

INQUIRY QUESTION

Why is a percentage change modelled by multiplication rather than addition?

YOUR TASK

Take five pairs of percentage changes: two rises, two falls, and one of each. For each, compute the combined multiplier in both orders, check that the order does not matter, and find the single percentage change that gives the same result.

changes	combined multiplier	single change
+10%, +10%	1.21	+21%
−10%, −10%	0.81	−19%
+20%, −20%	0.96	−4%
+50%, −50%	0.75	−25%
+5%, −3%	1.0235	+2.35%

CHAPTER 8 · INVESTIGATION 8.1

Repeated percentage change

What do you notice?

- The combined multiplier was the same in both orders every time.
- A rise and a fall of the same size always gave a multiplier below one.
- The single percentage change was always the multiplier minus one.
- The loss from a rise and a fall was the square of the rate.

Observation. Percentage changes combine by multiplying their multipliers, and the combined multiplier is independent of the order of the changes.

MATHEMATICAL CONCLUSION

A percentage change is modelled by a multiplier, and repeated changes combine by multiplication, which is why equal rises and falls do not cancel.

Your own words

CHAPTER 8 · 8.1

Repeated percentage change

A percentage is a fraction with denominator 100, and a percentage change is a change relative to the quantity it is applied to. Because the reference quantity changes, the same percentage is not the same amount.

A multiplier is 1 plus the rate as a decimal. A 15% rise is a multiplier of 1.15 and a 15% fall is a multiplier of 0.85, and their product is 0.9775, which is a fall.

A rise of r followed by a fall of r gives a multiplier of $(1 + r)(1 - r) = 1 - r^2$, which is always less than 1 for any non-zero r. This is why the two never cancel.

Working with multipliers makes repeated changes a single power, and it also makes the order of the changes irrelevant, because multiplication is commutative.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.1 · WORKED EXAMPLE 1

Repeated percentage change

WORKED EXAMPLE 8.1.1

Problem. A price of 80 dollars is increased by 15% and then decreased by 15%. Find the final price and the overall percentage change.

step	multiplier	price
start	—	80
rise 15%	1.15	92
fall 15%	0.85	78.20
combined	0.9775	78.20

the combined multiplier is less than 1

Mathematical reasoning. Apply each multiplier in turn, then compare the final price with the original.

1. After the rise the price is $80 \times 1.15 = 92$ dollars.
2. After the fall it is $92 \times 0.85 = 78.20$ dollars.
3. The combined multiplier is $1.15 \times 0.85 = 0.9775$.
4. The overall change is a fall of 2.25%.
5. The final price is lower than the start, even though the two percentages were equal.

Final answer. 78.20 dollars, a fall of 2.25%

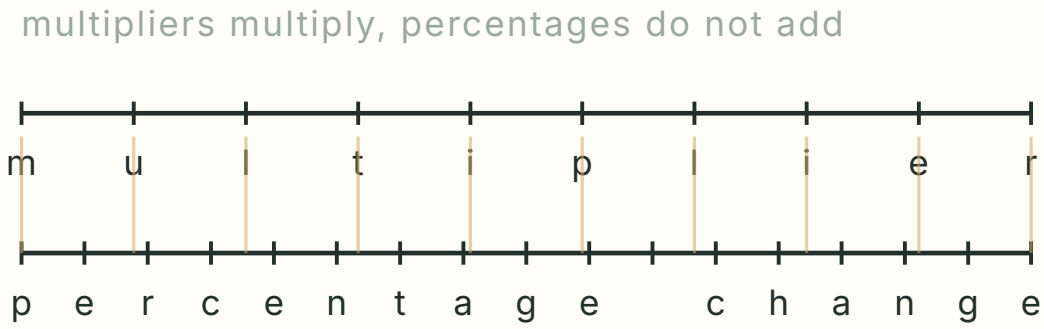
Interpretation. The two percentages were equal but were taken on different quantities, so they did not cancel, and the fall is $r^2 = 2.25\%$.

CHAPTER 8 · 8.1 · WORKED EXAMPLE 2

Repeated percentage change

WORKED EXAMPLE 8.1.2

Problem. A salary rises by 5% and then by 8%. Find the single multiplier and the overall percentage rise, and compare it with a single rise of 13%.



Mathematical reasoning. Multiply the two multipliers and compare with the sum of the percentages.

1. The multipliers are 1.05 and 1.08.
2. Their product is 1.134, so the overall rise is 13.4%.
3. A single rise of 13% would give a multiplier of 1.13.
4. The difference is 0.4 percentage points, which is $1.05 \times 1.08 - 1.13$.
5. The extra comes from the cross term $0.05 \times 0.08 = 0.004$.

Final answer. multiplier 1.134, a rise of 13.4%, which is 0.4 points more than 13%

Interpretation. Percentage rises add only approximately, and the error grows with the product of the rates, which is why multipliers are safer.

CHAPTER 8 · INVESTIGATION 8.2

Compound interest and multipliers

INQUIRY QUESTION

What is the difference between interest on the principal and interest on the interest?

YOUR TASK

Take three loans or savings plans of your own: one short, one long and one with a high rate. Model each with simple and compound interest, tabulate five periods, and find the period at which the difference exceeds 10% of the principal.

plan	rate	simple at 5y	compound at 5y	difference
savings	3%	1150	1159.27	9.27
loan	8%	1400	1469.33	69.33
long loan	12%	1600	1762.34	162.34

CHAPTER 8 · INVESTIGATION 8.2

Compound interest and multipliers

What do you notice?

- The difference grew with both the rate and the time.
- The long loan at 12% differed by more than 10% of the principal.
- The short low-rate plan differed by less than 1%.
- Compound interest always gave the larger amount, whether it was a debt or a saving.

Observation. Simple interest grows by a constant amount and compound interest grows by a constant multiplier, and the two separate faster at higher rates and over longer times.

MATHEMATICAL CONCLUSION

A financial quantity is modelled by $P(1 + rn)$ for simple interest and $P(1 + r)^n$ for compound interest, and the choice between them changes the answer by an amount that grows with the rate and the time.

Your own words

CHAPTER 8 · 8.2

Compound interest and multipliers

Simple interest pays a fixed percentage of the original principal each period, so the amount grows by the same amount every period and the model is linear.

Compound interest pays a percentage of the current amount, so the interest itself earns interest and the model is exponential.

The compound formula is $A = P(1 + r)^n$, where P is the principal, r the rate per period and n the number of periods. The simple formula is $A = P(1 + rn)$.

The two agree after one period and then diverge, and the gap grows with the rate and the number of periods, which is why the length of a loan matters so much.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.2 · WORKED EXAMPLE 1

Compound interest and multipliers

WORKED EXAMPLE 8.2.1

Problem. Compare 2000 dollars at 6% simple interest and 6% compound interest for 5 years.

year	simple	compound
0	2000	2000
1	2120	2120
3	2360	2382.03
5	2600	2676.45

the two agree at year 1 and then separate

Mathematical reasoning. Apply each formula and compare the amounts.

1. Simple: $A = 2000(1 + 0.06 \times 5) = 2000 \times 1.30 = 2600$ dollars.
2. Compound: $A = 2000 \times 1.06^5 = 2000 \times 1.3382 = 2676.45$ dollars.
3. The difference is 76.45 dollars, which is the interest on the interest.
4. After one year both give 2120 dollars, so the models agree there.
5. The gap grows each year, because the compound interest is charged on a larger amount.

Final answer. simple 2600 dollars, compound 2676.45 dollars, a difference of 76.45 dollars

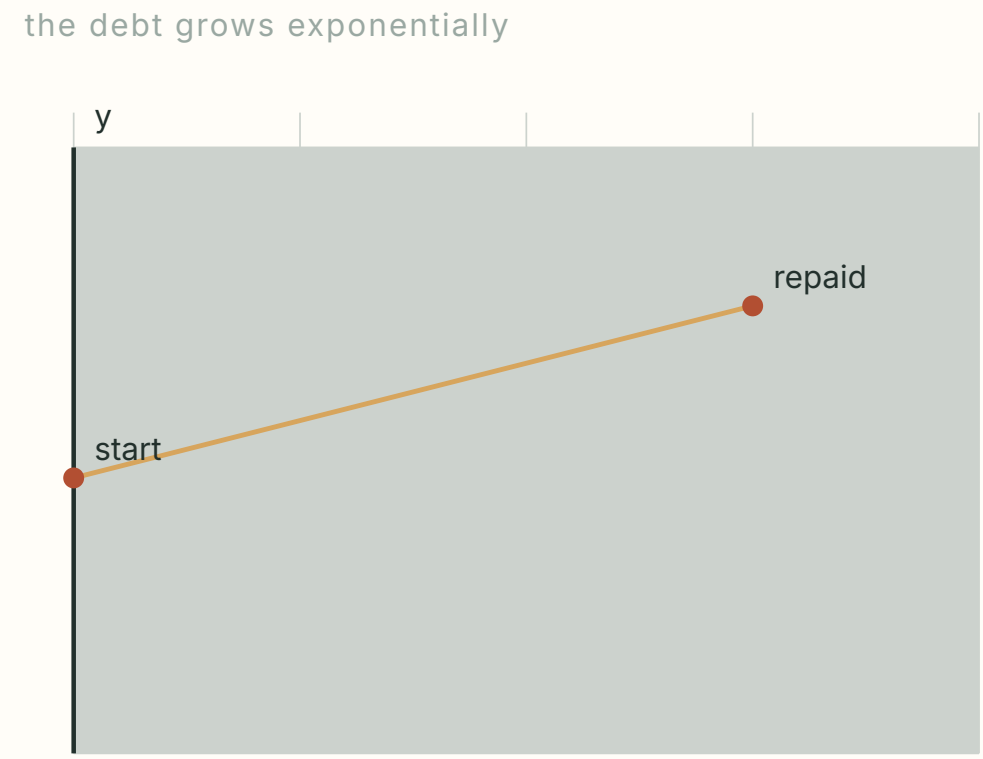
Interpretation. The difference is entirely the interest earned on interest, and it grows with time, which is why a long loan is expensive.

CHAPTER 8 · 8.2 · WORKED EXAMPLE 2

Compound interest and multipliers

WORKED EXAMPLE 8.2.2

Problem. A loan of 5000 dollars is repaid at 4% compound interest after 3 years. Find the amount repaid and the interest paid.



Mathematical reasoning. Apply the compound formula and subtract the principal.

1. $A = 5000 \times 1.04^3$.
2. $1.04^3 = 1.124864$, so $A = 5624.32$ dollars.
3. The interest paid is $5624.32 - 5000 = 624.32$ dollars.
4. That is 12.5% of the principal over three years.
5. At simple interest the interest would be 600 dollars, so compounding added 24.32 dollars.

Final answer. 5624.32 dollars repaid, with 624.32 dollars of interest

Interpretation. The interest is a substantial fraction of the loan, and the compounding is only part of the reason; the rate and the time matter more.

CHAPTER 8 · INVESTIGATION 8.3

Depreciation and reverse percentages

INQUIRY QUESTION

How is a falling value modelled, and how are two offers compared?

YOUR TASK

Find three real best-value comparisons: two appliances, two phone plans, or two transport options. Model the total cost of each over a stated period, find the break-even point, and recommend one with a stated assumption about how long it will be used.

comparison	model A	model B	break-even
two phones	300 + 10n	150 + 25n	10 months
two bus passes	40 + 2n	90 + 0.5n	33.3 rides
two printers	200 + 40n	120 + 60n	4 years

CHAPTER 8 · INVESTIGATION 8.3

Depreciation and reverse percentages

What do you notice?

- Every comparison had a break-even point.
- The recommendation changed with the assumed period of use.
- The cheaper purchase was better only for short use.
- Stating the assumption was essential to the recommendation.

Observation. A best-value comparison is a model of total cost over a stated period, and the recommendation depends on the period assumed.

MATHEMATICAL CONCLUSION

Two offers are compared by modelling their total cost over the same period, and the recommendation is only as good as the assumption about how long the item will be used.

Your own words

CHAPTER 8 · 8.3

Depreciation and reverse percentages

Depreciation is a repeated percentage decrease, so it is modelled by a multiplier below 1 and the value falls exponentially towards zero without reaching it.

The value after n periods is $V_0(1 - r)^n$, and the rate can be found from two values by taking the nth root of their ratio.

Comparing two offers means comparing the same quantity: the total cost over the same period, or the value at the same future date.

A lower price with a higher running cost can be worse than a higher price with a lower running cost, and only a model over a stated period can decide.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.3 · WORKED EXAMPLE 1

Depreciation and reverse percentages

WORKED EXAMPLE 8.3.1

Problem. A machine costs 8000 dollars and loses 20% of its value each year. Find its value after 4 years and the annual depreciation in the fourth year.

year	value	loss that year
0	8000	—
1	6400	1600
2	5120	1280
3	4096	1024
4	3276.80	819.20

the annual loss falls each year

Mathematical reasoning. Apply the depreciation multiplier and compare consecutive values.

1. The multiplier is 0.80, so the value is 8000×0.8^{n}.
2. After 4 years the value is $8000 \times 0.4096 = 3276.80$ dollars.
3. The loss in the fourth year is $4096 - 3276.80 = 819.20$ dollars.
4. The loss falls each year, because it is a percentage of a smaller amount.
5. The value never reaches zero in the model, though it becomes negligible.

Final answer. 3276.80 dollars after 4 years, with a fourth-year loss of 819.20 dollars

Interpretation. A percentage depreciation gives a falling amount each year, which is the opposite of the rising amount a percentage rise gives.

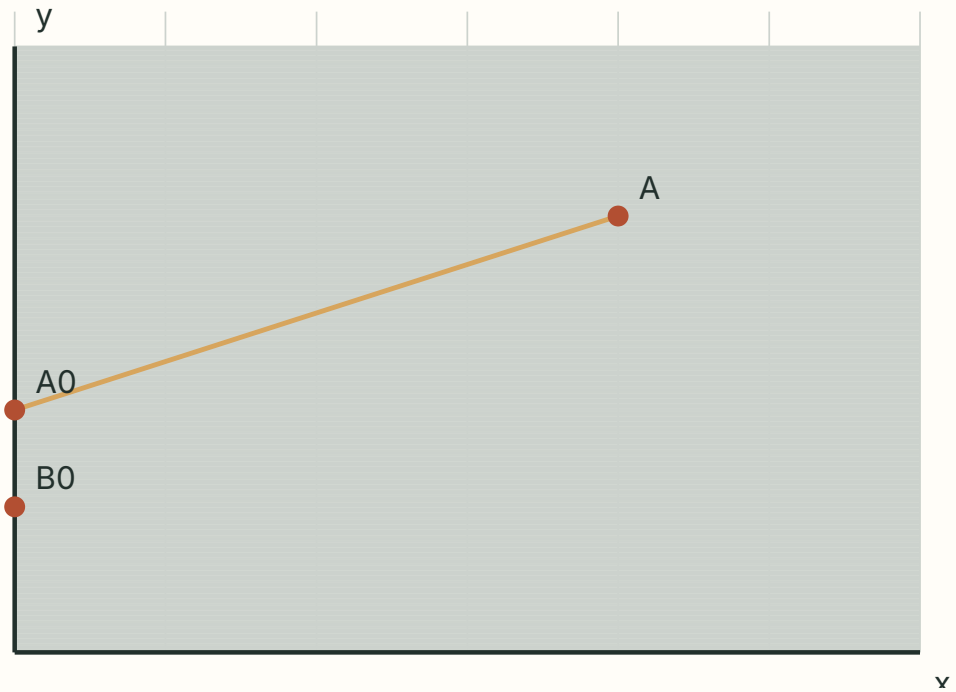
CHAPTER 8 · 8.3 · WORKED EXAMPLE 2

Depreciation and reverse percentages

WORKED EXAMPLE 8.3.2

Problem. Printer A costs 200 dollars with 40 dollars of ink a year; printer B costs 120 dollars with 60 dollars a year. Find the number of years after which B is more expensive.

the two cost lines cross at 4 years



Mathematical reasoning. Model both total costs and find where they are equal.

1. Printer A costs 200 + 40n dollars after n years.
2. Printer B costs 120 + 60n dollars after n years.
3. Setting them equal gives 80 = 20n, so n = 4.
4. For fewer than 4 years B is cheaper; for more, A is cheaper.
5. A buyer who keeps a printer for 6 years should choose A, saving 40 dollars.

Final answer. B is more expensive after 4 years

Interpretation. The cheaper purchase is not always the cheaper option, and only a model over a stated period can decide.

CHAPTER 8 · INVESTIGATION 8.4

Comparing financial offers

INQUIRY QUESTION

How are two financial offers compared fairly, and what does a single figure hide?

YOUR TASK

Collect three real financial offers: two savings accounts, two loan advertisements, or two phone plans. Model each over the same period, convert any rate to the same compounding period, include every fee you can find, and rank them. Then write the limitation of your ranking.

offer	rate	period	fees	total after 1 year	rank
account A	0.5% monthly	12 months	0	2123.36	1
account B	6% yearly	12 months	5	2115	2
account C	5.5% yearly	12 months	0	2110	3

CHAPTER 8 · INVESTIGATION 8.4

Comparing financial offers

What do you notice?

- Rates quoted in different periods had to be converted before comparing.
- A small fee changed the ranking.
- The offer with the highest headline rate was not the best.
- The ranking held only for the period modelled.

Observation. Offers are compared by modelling the total over the same period, with every fee included, and the ranking is valid only for that period.

MATHEMATICAL CONCLUSION

A fair comparison states the period, the compounding and the fees, and the model's limitation is part of the answer.

Your own words

CHAPTER 8 · 8.4

Comparing financial offers

Two offers are compared by modelling the total amount paid or received over the same period, because a headline rate says nothing about the length of the agreement or the fees attached to it.

An annual percentage rate is not the same as a monthly one multiplied by twelve, because compounding happens inside the year. Offers quoted in different periods must be converted before they are compared.

A single figure such as a monthly payment hides the total, and the total is what the customer pays. The honest comparison states the period, the fees and the total together.

A model that ignores a fee, a penalty or a change of rate is optimistic, and the limitation must be stated. Comparing offers is a modelling task before it is an arithmetic one.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.4 · WORKED EXAMPLE 1

Comparing financial offers

WORKED EXAMPLE 8.4.1

Problem. Offer A is 1000 dollars at 12% a year simple interest for 3 years. Offer B is 1000 dollars at 10% a year compound interest for 3 years. Which costs more in total?

year	A total	B total
1	1120	1100
2	1240	1210
3	1360	1331

Mathematical reasoning. Model each offer over the same period and compare the totals.

1. Offer A adds 12% of 1000 each year, which is 120 a year.
2. After 3 years A is $1000 + 3 \times 120 = 1360$ dollars.
3. Offer B multiplies by 1.10 each year: $1000 \times 1.10^3 = 1331$.
4. So A costs 29 dollars more in total over the 3 years.
5. The lower rate on B wins here, but only because the period is short; over 10 years B would be dearer.

Final answer. offer A costs more: 1360 against 1331 dollars

Interpretation. The cheaper rate is not always the cheaper offer, because the period and the compounding both matter.

CHAPTER 8 · 8.4 · WORKED EXAMPLE 2

Comparing financial offers

WORKED EXAMPLE 8.4.2

Problem. A savings account pays 0.5% a month with no fees. Another pays 6% a year with a 5 dollar annual fee. Compare them on a balance of 2000 dollars over one year.

account	interest	fee	balance after a year
monthly	123.36	0	2123.36
annual	120	5	2115

Mathematical reasoning. Convert both to the same period, then subtract the fees.

1. The monthly account gives $2000 \times 1.005^{12} = 2123.36$ dollars.
2. The annual account gives $2000 \times 1.06 = 2120$ dollars, minus the 5 dollar fee.
3. So the annual account leaves 2115 dollars.
4. The monthly account wins by 8.36 dollars.
5. The fee of 5 dollars was the difference between the two rates, and leaving it out would have reversed the answer.

Final answer. the monthly account wins: 2123.36 against 2115 dollars

Interpretation. A fee can decide a comparison, so a model that omits it is not a model of the offer.

CHAPTER 8 · INVESTIGATION 8.5

Modelling decisions and limitations

INQUIRY QUESTION

Who does a rate favour, and how can a model show that?

YOUR TASK

Choose one real lending or saving situation, such as a micro-loan, a phone contract or a school fee plan. Model the total cost, find the effective annual rate, and write a short account of who the arrangement favours and who it does not.

arrangement	total cost	effective annual rate	favours
micro-loan at 3% a month	about 1430 for 1200	about 43%	the lender
phone contract over 24 months	about 1200 for a 600 phone	about 25%	the provider
savings at 5% a year	1260 after 5 years	5%	the saver

CHAPTER 8 · INVESTIGATION 8.5

Modelling decisions and limitations

What do you notice?

- Every lending arrangement cost far more than the headline figure suggested.
- The effective annual rate was much larger than the monthly rate.
- The savings arrangement was the only one that paid the customer.
- The arrangements favoured whoever already had money.

> **Observation.** A monthly rate compounded over a year gives an effective annual rate much larger than the monthly rate suggests, and the model makes that visible.

MATHEMATICAL CONCLUSION

A model of a financial arrangement shows its total cost and its effective rate, and the question of who it favours is answered by comparing that cost with the alternative of saving.

Your own words

CHAPTER 8 · 8.5

Modelling decisions and limitations

A rate of interest is the price of access to money now, and it is paid by those who do not have the money and received by those who do.

A model can show the total cost of a loan over its life, and that total is often several times the amount borrowed, which the monthly figure hides.

Comparing a loan with saving for the same amount shows how much cheaper patience is, and the difference is the cost of access.

Fairness in lending is a question about the model's assumptions as much as about its arithmetic, because a rate that is affordable in one situation is crushing in another.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.5 · WORKED EXAMPLE 1

Modelling decisions and limitations

WORKED EXAMPLE 8.5.1

Problem. A loan of 1000 dollars at 3% a month is repaid over 12 months. Model the total repaid and compare it with saving 84 dollars a month for 12 months.

month	balance	interest that month
1	916	30
6	500	15
12	0	2.5

the interest falls as the balance falls

Mathematical reasoning. Model the loan recursively and compare the total with the amount saved.

1. Each month the balance grows by 3% and falls by 84 dollars.
2. The total repaid is $12 \times 84 = 1008$ dollars plus the interest already included in the schedule.
3. Modelling month by month gives a total interest of about 197 dollars.
4. So the loan costs about 1197 dollars in total for 1000 borrowed.
5. Saving 84 dollars a month for 12 months raises 1008 dollars with no interest at all.

Final answer. the loan costs about 1197 dollars in total; saving raises 1008 dollars

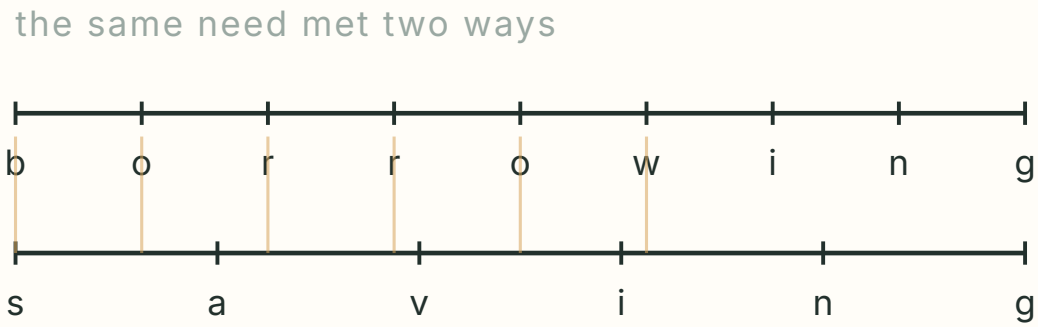
Interpretation. Borrowing to get the money now costs about 20% of the amount, which is the price of access, and the monthly figure of 84 dollars hides it.

CHAPTER 8 · 8.5 · WORKED EXAMPLE 2

Modelling decisions and limitations

WORKED EXAMPLE 8.5.2

Problem. Two families each need 1200 dollars. One borrows at 3% a month and one saves 100 dollars a month. Model both and compare the cost.



Mathematical reasoning. Model the loan recursively and the saving linearly, and compare the totals.

1. The borrowing family repays about 1430 dollars in total, including interest.
2. The saving family needs 12 months to reach 1200 dollars and pays nothing extra.
3. The difference of about 230 dollars is the cost of access.
4. The saving family waits a year; the borrowing family has the money now.
5. Which is fairer depends on why the money is needed, which the model cannot say.

Final answer. borrowing costs about 1430 dollars; saving costs 1200 dollars after 12 months

Interpretation. The model quantifies the cost of access, and the judgement about whether it is worth paying is not a mathematical one.

CHAPTER 8 · INVESTIGATION 8.6

Investigation: doubling and halving

INQUIRY QUESTION

How far ahead can a financial model be trusted, and what breaks it?

YOUR TASK

Choose a long-term financial plan: a mortgage, a pension, a school fees plan or a business loan. Model it over its full term, find the total paid and the total interest, and then test the model with a rate 2 percentage points higher to see how much the answer changes.

plan	total paid at 5%	total at 7%	change
mortgage 200000 over 25 years	353700	424200	70500
pension saving 200 a month for 30 years	149000	n/a	rate matters less
loan 20000 over 5 years	23000	23800	800

CHAPTER 8 · INVESTIGATION 8.6

Investigation: doubling and halving

What do you notice?

- A 2-point rise in the rate changed the total by tens of thousands.
- The longer the term, the larger the effect of the rate change.
- The short loan was barely affected.
- Every model depended on the rate holding for the whole term.

> **Observation.** The sensitivity of a financial model to its rate grows with the length of the term, so a long plan is a statement about the future as much as a calculation.

MATHEMATICAL CONCLUSION

A long-term financial model is reliable only as a statement of what happens if its rates hold, and its sensitivity to those rates is itself an important result.

Your own words

CHAPTER 8 · 8.6

Investigation: doubling and halving

A financial model is trusted while its rates hold, and rates change with inflation, policy and personal circumstance.

The longer the horizon, the larger the effect of a small error in the rate, because the model compounds the error.

A model that is accurate for one year can be badly wrong for twenty, and the difference is not arithmetic error but assumption error.

A responsible model states its horizon, its assumptions and the point at which it should be reviewed.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 8 · 8.6 · WORKED EXAMPLE 1

Investigation: doubling and halving

WORKED EXAMPLE 8.6.1

Problem. A family saves 200 dollars a month at 4% a year compound interest. Model the balance after 10 and after 20 years, and comment on the assumptions.

year	balance
0	0
5	13206
10	29401
20	73700

the balance grows faster than the payments

Mathematical reasoning. Model the payments as a series with compound growth.

1. Each year the balance grows by 4% and 2400 dollars are added.
2. After 10 years the balance is about 29401 dollars, on payments of 24000.
3. After 20 years it is about 73700 dollars, on payments of 48000.
4. The interest exceeds the payments after about 13 years.
5. The model assumes the rate stays at 4% for twenty years, which no one can promise.

Final answer. about 29401 dollars after 10 years and 73700 after 20

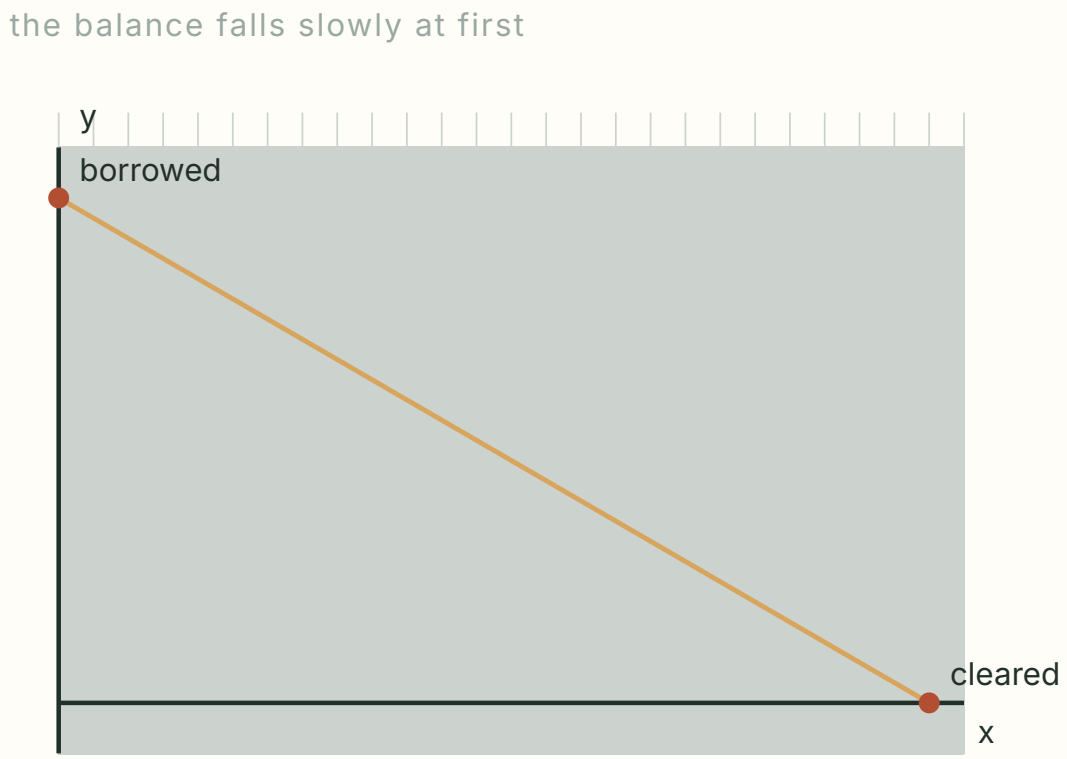
Interpretation. The interest eventually exceeds the contributions, which is the power of compounding over a long horizon and also the source of the model's fragility.

CHAPTER 8 · 8.6 · WORKED EXAMPLE 2

Investigation: doubling and halving

WORKED EXAMPLE 8.6.2

Problem. A loan of 200000 dollars at 5% a year is repaid over 25 years. Model the total repaid and compare it with the amount borrowed.



Mathematical reasoning. Model the repayments and compare the total with the principal.

1. The annual repayment is about 14150 dollars.
2. Over 25 years the total repaid is about 353700 dollars.
3. The interest is about 153700 dollars, which is 77% of the amount borrowed.
4. In the first years most of each payment is interest.
5. The model assumes the rate is fixed for the whole term.

Final answer. about 353700 dollars repaid in total, of which about 153700 dollars is interest

Interpretation. The total cost of a long loan can approach twice the amount borrowed, and the monthly figure of about 1179 dollars gives no hint of it.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

8.A1 FLUENCY

Increase 80 by 15% and then decrease the result by 15%.

8.A2 FLUENCY

Find the single multiplier for a rise of 10% followed by a fall of 10%.

8.A3 FLUENCY

Find the multiplier for a fall of 25%.

8.A4 FLUENCY

Find 2000×1.06^5 to the nearest cent.

8.A5 FLUENCY

Find the simple interest on 3000 dollars at 5% for 4 years.

8.A6 FLUENCY

Find the value after 3 years of 5000 dollars depreciating at 20% a year.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

8.A7 FLUENCY

Find the annual multiplier for a 4% rise.

8.A8 FLUENCY

A price of 50 dollars rises by 20% and then by 10%. Find the final price.

8.A9 FLUENCY

Find the compound interest on 1000 dollars at 3% for 2 years.

8.A10 FLUENCY

A car worth 12000 dollars falls to 9000 dollars in a year. Find the percentage fall.

8.A11 FLUENCY

Find the value after 2 years of 800 dollars growing at 5% a year.

8.A12 FLUENCY

Find the overall percentage change for multipliers 1.2 and 0.9.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

8.B1 REASONING

A population of 1000 falls by 10% a year. Find the rule and test it on the fifth year.

8.B2 REASONING

Investigate the pattern in the balances of a loan of 1000 at 10% a month with 200 dollar repayments.

8.B3 CHALLENGE

Conjecture a rule for the total repaid on a loan of P at rate r over n months with fixed repayments, and test it.

8.B4 REASONING

Investigate the pattern in the difference between simple and compound interest on 1000 at 5% over 1 to 10 years.

8.B5 REASONING

Build a table of the value of 1000 dollars at 3% a month for 1 to 12 months and describe the pattern.

8.B6 CHALLENGE

Conjecture a rule for the number of years for an investment to double at r% compound interest.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

8.B7 REASONING

Investigate the pattern in the total cost of a 1000 dollar loan at different monthly rates over 12 months.

8.B8 REASONING

A savings account pays interest monthly. Investigate the effective annual rate for a monthly rate of 1%.

8.B9 REASONING

Find the pattern in the values of a car that loses 15% a year and predict the value after 10 years from 20000 dollars.

8.B10 REASONING

Investigate the pattern in the interest paid each year on a fixed mortgage.

8.B11 CHALLENGE

Conjecture a rule for the total interest paid on a loan of P at rate r over n years with annual repayments, for small r.

CRITERION C Communicating

Criterion C practice · Communicating

8.C1 REASONING

Explain in your own words the difference between simple and compound interest.

8.C2 FLUENCY

Write a complete solution to: find the amount after 3 years on 1500 dollars at 4% compound interest.

8.C3 REASONING

Choose between simple and compound interest to model a 6-month saving and justify.

8.C4 FLUENCY

Set out a full solution to: find the value after 4 years of 9000 dollars depreciating at 30% a year.

8.C5 REASONING

Explain why a rise of 20% followed by a fall of 20% is a net fall.

8.C6 REASONING

Describe, with an example, the difference between a multiplier and a percentage.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

8.C7 REASONING
Explain why the effective annual rate is larger than twelve monthly rates added together.

8.C8 REASONING
Write the reasoning that shows the interest portion of a fixed loan repayment falls over time.

8.C9 REASONING
Set out the argument that depreciation never reaches zero in the model.

8.C10 REASONING
Explain what the break-even point means in a best-value comparison.

8.C11 REASONING
Describe the relationship between the rate of a loan and its total cost.

8.C12 REASONING
Explain, with an example, when a financial model is misleading.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

8.D1 APPLICATION

A worker earns 3000 dollars a month and spends 2550. Find the monthly surplus and the time to save 9000 dollars.

8.D2 APPLICATION

A family borrows 1500 dollars at 4% a month and repays 200 dollars a month. Model the balance after 3 months.

8.D3 APPLICATION

A school fee plan charges 120 dollars a month for 10 months for a 900 dollar fee. Find the total paid and the effective annual rate.

8.D4 APPLICATION

A shop's takings rise by 8% a year from 40000 dollars. Find the takings after 5 years and the total over 5 years.

8.D5 APPLICATION

A loan of 8000 dollars at 6% a year is repaid over 4 years. Model the total repaid and compare it with the principal.

8.D6 APPLICATION

A bicycle worth 600 dollars loses 18% of its value a year. Find its value after 3 years and the total loss.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

8.D7 REASONING

Two bus passes are offered: 40 dollars plus 2 dollars a ride, or 90 dollars plus 0.50 dollars a ride. Investigate which is better for 20 rides a month and justify.

8.D8 APPLICATION

A microfinance loan of 200 dollars at 5% a month is repaid over 6 months. Model the total repaid and find the effective annual rate.

8.D9 REASONING

A savings account pays 0.5% a month. Investigate whether it beats 6% a year simple interest over 3 years and justify.

8.D10 REASONING

A company's revenue falls by 5% a year from 500000 dollars. Investigate when it falls below 300000 dollars and justify.

8.D11 APPLICATION

A farmer's yield rises by 3% a year from 20 tonnes. Find the yield after 15 years and comment on the model's usefulness.

8.D12 REASONING

Explain, with a model of your own, when a monthly figure is a misleading way to report a cost.

CHAPTER 8 · APPLICATION AND ASSESSMENT

Applying chapter 8

Each task involves a rate and a time. State the multiplier before you calculate, and say whether the change is simple or compound.

A1 · The two loans

A family needs 1200 dollars. Option A is a loan at 3% a month repaid over 12 months. Option B is a savings plan of 100 dollars a month. Model both, find the total cost of each, find the effective annual rate of the loan, and write a short recommendation stating the assumption on which it depends.

Criteria A and D · model both options, compare the totals, and justify the recommendation.

A2 · The long view

A deposit of 5000 dollars is invested at 5% compound interest a year, and 200 dollars a month is added. Model the balance after 5, 10 and 20 years, find the year in which the accumulated interest first exceeds the total paid in, and test the model with a rate of 3% to see how much the 20-year balance changes.

Criteria B and C · build the recursive model, justify the series, and comment on the sensitivity to the rate.

CHAPTER 8 · REVIEW

What you should now be able to do

Key ideas

- A percentage change is modelled by a multiplier.
- A rise of r then a fall of r gives a multiplier of $1 - r^2$.
- Multipliers combine by multiplication, so the order does not matter.
- Simple interest is $P(1 + rn)$; compound interest is $P(1 + r)^n$.
- Depreciation is a repeated percentage decrease, modelled by a multiplier below 1.
- The effective annual rate compounds the monthly rate.
- The interest portion of a fixed repayment falls as the balance falls.
- A long-term model is sensitive to its assumed rate.

Vocabulary

percentage change · multiplier · simple interest · compound interest · principal · depreciation · effective annual rate · break-even · surplus · recursive model · horizon · sensitivity

CHAPTER 8 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

8.R1 Find the multiplier for a rise of 12%.

8.R2 Find 1500×1.04^3 to the nearest cent.

8.R3 Find the value after 5 years of 9000 dollars at 20% depreciation.

8.R4 Find the simple interest on 2500 dollars at 6% for 3 years.

8.R5 A price rises 25% then falls 25%. Find the overall change.

8.R6 Explain why a monthly rate of 1% is more than 12% a year.

CHAPTER 8 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

Repeated percentage change is modelled by multiplication rather than addition, so the quantity after n periods is the initial quantity times a multiplier to the power n; recognising this changes how a percentage rise and a percentage fall are understood, and the model makes the long-term effect of a rate visible.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of change, quantity and model appear in your work?
- How is the global context of fairness and development, and its exploration of access, saving and long-term planning, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a financial model for a decision about borrowing, and what would you still like to ask?

CHAPTER 8 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Access to money is a mathematical question with a social answer. Model one real financial decision and say who it favours.

Service as Action

COMMUNITY NEED	Borrowing and saving decisions are made from monthly figures that hide the total.
MATHEMATICAL CONNECTION	Multipliers, compound models and effective rates.
PROPOSED ACTION	Model one real loan or savings plan and publish its total cost and effective rate.
EXPECTED IMPACT	A reader can compare two offers on the same terms.
STUDENT REFLECTION	Record the model, the totals and what you found.

Community Project

PROJECT IDEA	A transparency audit of the costs the school publishes.
NEED	Fees, trips and contracts are quoted per month or per term without totals.
RESEARCH QUESTION	What is the true total cost of each arrangement, and who does it favour?
MATHEMATICS	Multipliers, compound interest and effective rates.
ACTION OR OUTCOME	A one-page table of totals and effective rates.
BENEFICIARIES	Every family paying the school's costs.
EVIDENCE	The table with the models and the totals.

Personal Project connection · MYP 5

PERSONAL INTEREST	Saving, spending or a small business idea.
LEARNING GOAL	To model a financial decision and state its total cost.
PRODUCT OR OUTCOME	A model with a total, a rate and a recommendation.
MATHEMATICS	Multipliers and compound models.
PROCESS	State the rate, model the periods, and compare the total with the alternative.
EVIDENCE	The model, the total and the comparison.

CHAPTER 9 · PART III · SPATIAL REASONING

09

Congruence, Similarity and Transformations

A translation, a reflection and a rotation preserve shape and size. An enlargement preserves shape alone, and the scale factor connects every length on the model to the object.

ENGAGE

A scale model of a bridge is 40 cm long and the real bridge is 200 m long. What is the scale, and how heavy is the real bridge if the model weighs 300 g?

Framing the inquiry

Key concept: Form
Related concepts: Space · Equivalence · Representation
Global context: Scientific and technical innovation
Global context exploration: Scale models, design and measurement at a distance

Statement of inquiry: Two shapes are congruent when one can be moved onto the other by a rigid transformation, and similar when one can be obtained from the other by an enlargement; the transformations are therefore the test of equivalence, and the scale factor is what connects a model to the object it represents.

Questions to guide us

- **Factual:** What are the four transformations, and what are the four tests for congruent triangles?
- **Conceptual:** Why is a transformation a better test of congruence than measurement, and how does the scale factor connect lengths, areas and volumes?
- **Debatable:** Is a scale model a representation of the real thing or a different object, and does the answer change when the model is used for a decision?

ATL · COMMUNICATION. Students describe a transformation precisely enough for another person to reproduce it, and justify why two shapes are or are not congruent.

CHAPTER 9 · INVESTIGATION 9.1

Transformations and congruence

INQUIRY QUESTION

What does each transformation preserve, and how is it described exactly?

YOUR TASK

Take one shape on squared paper. Apply a translation, a reflection, a rotation and an enlargement of your own choice. For each, record the image, the invariant quantities and the quantities that change.

transformation	preserved	changed
translation	length, angle, area, orientation	position
reflection	length, angle, area	orientation, position
rotation	length, angle, area	position, orientation
enlargement k = 2	angle	length, area, position

CHAPTER 9 · INVESTIGATION 9.1

Transformations and congruence

What do you notice?

- The three rigid transformations preserved every measurement.
- The enlargement preserved angles but doubled lengths and quadrupled areas.
- Only the reflection and the rotation changed the orientation.
- The enlargement was the only transformation that was not an equivalence of measurement.

Observation. The rigid transformations preserve every measurement, and the enlargement preserves angles alone while scaling lengths, areas and volumes by k, k^2 and k^3.

MATHEMATICAL CONCLUSION

A transformation is a rule that moves a shape, and what it preserves decides whether the image is congruent or merely similar to the original.

Your own words

CHAPTER 9 · 9.1

Transformations and congruence

A translation moves every point by the same vector. It preserves length, angle, area and orientation, and it is described by a column vector.

A reflection in a mirror line turns the shape over. It preserves length and angle but reverses orientation, and the mirror line is the perpendicular bisector of every segment joining a point to its image.

A rotation about a centre through an angle preserves length and angle. It is described by the centre, the angle and the direction, and all three are needed.

An enlargement about a centre by a scale factor k multiplies every distance from the centre by k. Lengths scale by k, areas by k^2 and volumes by k^3, so an enlargement is not a rigid transformation when $k \neq 1$.

A transformation is a rule in space: it takes each point of a shape to exactly one image point, and the whole shape is carried with it.

Two shapes are congruent when one can be carried onto the other by translations, reflections and rotations alone, so congruence is the equivalence that a rigid transformation preserves.

Four tests decide congruence of triangles from three pieces of information: SSS, SAS, ASA and RHS. Each is sufficient because three measurements fix a triangle, so any two triangles with them must be identical.

ASS is not a test, because two sides and a non-included angle allow two different triangles. This ambiguous case is the reason the angle in SAS must be the included one.

A representation of a shape as a set of coordinates makes each transformation a calculation on those coordinates, which is how a design is checked exactly rather than by eye.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

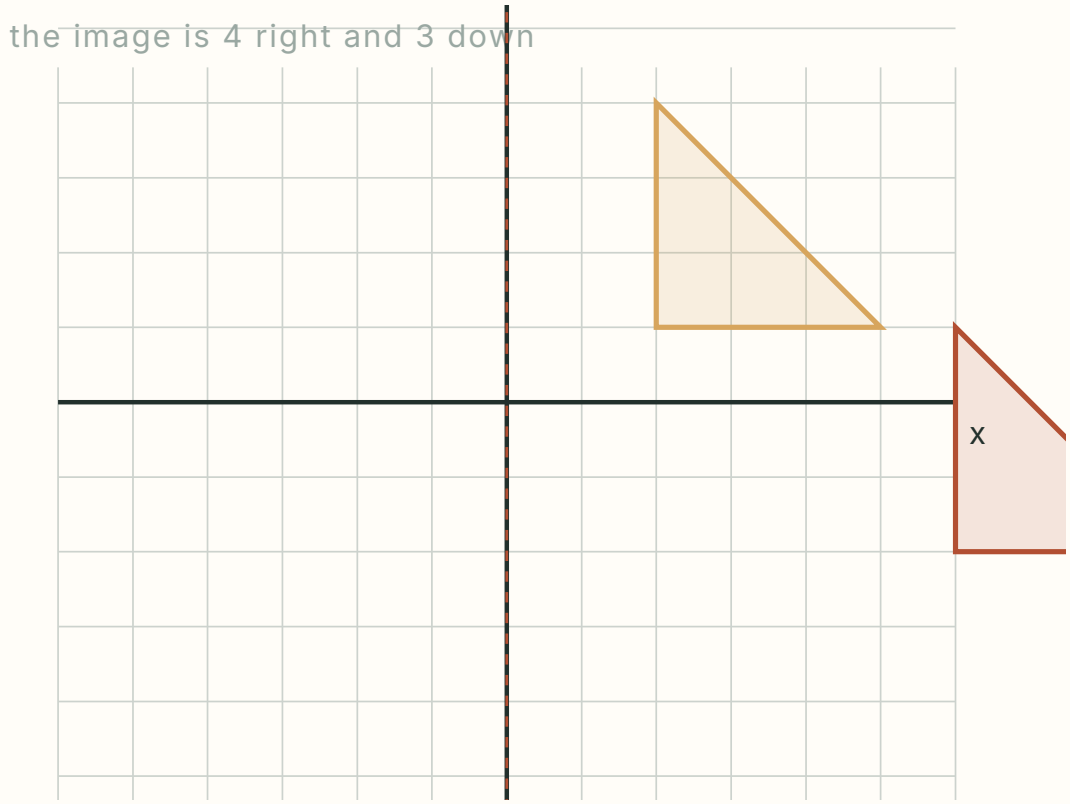
In your own words

CHAPTER 9 · 9.1 · WORKED EXAMPLE 1

Transformations and congruence

WORKED EXAMPLE 9.1.1

Problem. Translate the triangle with vertices (2, 1), (5, 1) and (2, 4) by the vector (4, −3), and describe what is preserved.



Mathematical reasoning. Add the vector to each vertex.

1. (2, 1) maps to (2 + 4, 1 − 3) = (6, −2).
2. (5, 1) maps to (5 + 4, 1 − 3) = (9, −2).
3. (2, 4) maps to (2 + 4, 4 − 3) = (6, 1).
4. Every side keeps its length and every angle keeps its size, so the image is congruent to the original.
5. The orientation is unchanged, because no reflection is involved.

Final answer. the image has vertices (6, −2), (9, −2) and (6, 1)

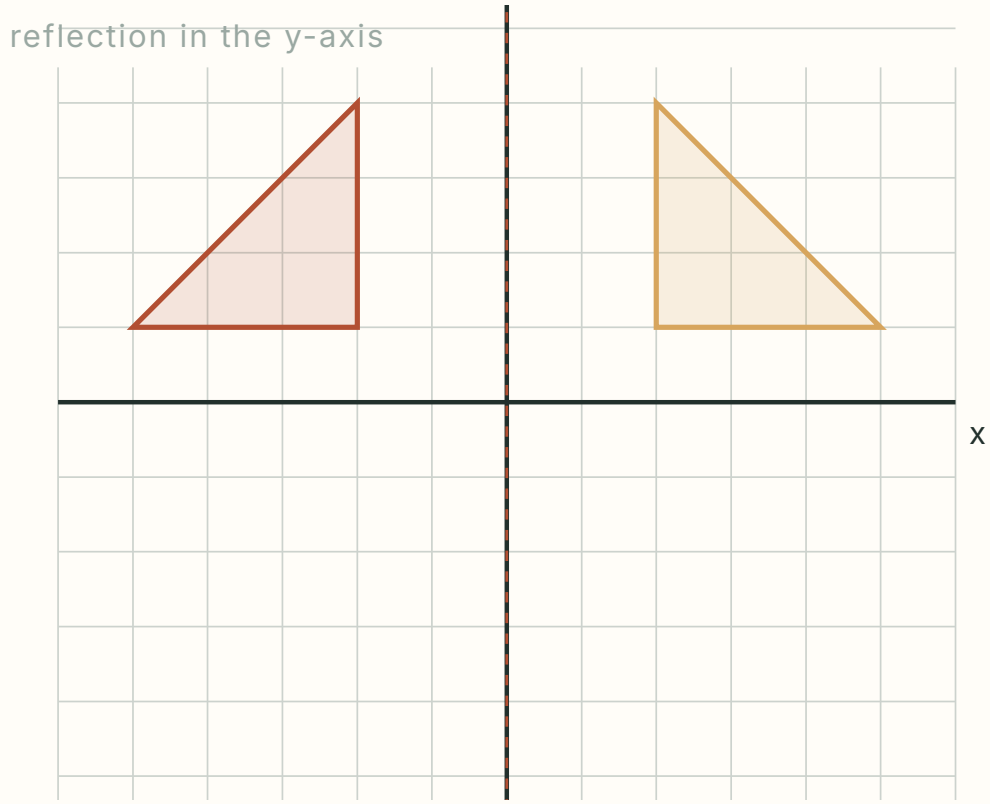
Interpretation. A translation is fully described by one vector, and it preserves everything except position.

CHAPTER 9 · 9.1 · WORKED EXAMPLE 2

Transformations and congruence

WORKED EXAMPLE 9.1.2

Problem. Reflect the triangle with vertices (2, 1), (5, 1) and (2, 4) in the y-axis, and state what changes.



Mathematical reasoning. Change the sign of each x-coordinate.

1. Reflecting in the y-axis maps (x, y) to (−x, y).
2. So the vertices become (−2, 1), (−5, 1) and (−2, 4).
3. The side lengths and the angles are unchanged, so the image is congruent.
4. The orientation is reversed: a clockwise order becomes anticlockwise.
5. The y-axis is the perpendicular bisector of each joining segment.

Final answer. the image has vertices (−2, 1), (−5, 1) and (−2, 4)

Interpretation. A reflection preserves shape and size but reverses orientation, which is the one thing a translation cannot do.

CHAPTER 9 · INVESTIGATION 9.2

Similarity and similar triangles

INQUIRY QUESTION

How is it shown that two triangles are similar, and what does the scale factor decide?

YOUR TASK

Find three pairs of similar triangles in the room or in a picture: one pair from shadows, one from a photograph, and one you construct by enlarging a drawn triangle by a factor of 2. For each, name the equal angles, list the ratios of corresponding sides, and check that the ratios are all the same. Then measure one area in each pair and check that it scales by the square of the factor.

pair	equal angles	scale factor	area ratio
stick and tree	sun angle, right angle	3.75	14.06
photo and object	corresponding angles	0.2	0.04
drawn and enlarged	all three	2	4

CHAPTER 9 · INVESTIGATION 9.2

Similarity and similar triangles

What do you notice?

- Every similar pair had equal corresponding angles.
- The ratios of corresponding sides were all the same number.
- The area ratio was always the square of the length ratio.
- The shadow pair gave a measurement that could not be taken directly.

Observation. Similar triangles have equal angles and proportional sides, and the scale factor multiplies lengths while its square multiplies areas.

MATHEMATICAL CONCLUSION

Similarity is an equivalence of shape, and the scale factor is the single number that converts every length, every area and every volume between two similar figures.

Your own words

CHAPTER 9 · 9.2

Similarity and similar triangles

Two shapes are similar when one is an enlargement of the other, so corresponding angles are equal and corresponding sides are in the same ratio. The ratio is the scale factor.

Two tests decide similarity of triangles from angles alone: AA, because two equal angles force the third, and the equiangular test, because three equal angles force the sides into proportion.

The scale factor k multiplies every length, so a pair of corresponding sides gives k. Areas then scale by k^2 and volumes by k^3, which is why the factor for an area is the square of the factor for a length.

Similarity is the equivalence that lets an inaccessible measurement be found from an accessible one, and the argument is complete only when the equal angles are named.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

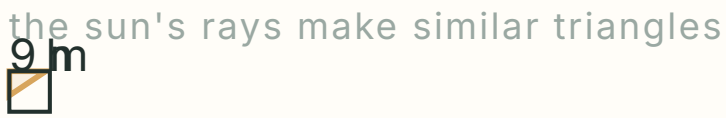
In your own words

CHAPTER 9 · 9.2 · WORKED EXAMPLE 1

Similarity and similar triangles

WORKED EXAMPLE 9.2.1

Problem. A tree and a 1.8 m stick stand vertically. The stick's shadow is 2.4 m and the tree's is 9 m. Find the height of the tree.



Mathematical reasoning. The two triangles are similar, so the ratios of corresponding sides are equal.

1. The stick and the tree are both vertical, and the sun's rays make the same angle with the ground.
2. So the two right-angled triangles are similar by AA.
3. The scale factor is 9/2.4 = 3.75.
4. The tree is therefore 3.75 × 1.8 = 6.75 m tall.
5. The height was found without measuring the tree, which is the point of the method.

Final answer. the tree is 6.75 m tall

Interpretation. Similarity converts one measurement into another, and the equal angles are what make the conversion valid.

CHAPTER 9 · 9.2 · WORKED EXAMPLE 2

Similarity and similar triangles

WORKED EXAMPLE 9.2.2

Problem. Two triangles have angles 50°, 60° and 70°, and 50°, 60° and 70°. Are they similar, and what else is needed to find the scale factor?



Mathematical reasoning. Equal corresponding angles give similarity, and one pair of corresponding sides gives the factor.

1. All three angles match, so the triangles are similar by the equiangular test.
2. Similarity fixes the ratios but not the sizes.
3. One pair of corresponding sides is needed to find k.
4. If a side of 4 cm corresponds to a side of 10 cm, then k = 2.5.
5. Every other length in the second triangle is 2.5 times the matching length in the first.

Final answer. yes, similar by the equiangular test; one pair of corresponding sides gives the scale factor

Interpretation. Similarity is about shape, and the scale factor is about size, so the two questions need different information.

CHAPTER 9 · INVESTIGATION 9.3

Enlargement and scale factors

INQUIRY QUESTION

How does a scale factor connect lengths, areas and volumes?

YOUR TASK

Take four pairs of similar objects: two photographs, two bottles, two maps and one pair of your own. For each, find the length scale factor, then use it to predict the area and the volume ratios, and check one prediction by measurement.

pair	length ratio	area ratio	volume ratio
two photos	2 : 1	4 : 1	—
two bottles	3 : 1	9 : 1	27 : 1
two maps	1 : 2	1 : 4	—
two cubes	5 : 1	25 : 1	125 : 1

CHAPTER 9 · INVESTIGATION 9.3

Enlargement and scale factors

What do you notice?

- Every area ratio was the square of the length ratio.
- Every volume ratio was the cube of the length ratio.
- The maps were smaller than the objects, so the ratio was less than one.
- A measurement of one pair confirmed the prediction exactly.

Observation. The scale factor connects lengths directly, areas through its square and volumes through its cube, and the connection holds for every pair of similar objects.

MATHEMATICAL CONCLUSION

Two similar objects are connected by one scale factor, and that factor raised to the second and third powers connects their areas and their volumes.

Your own words

CHAPTER 9 · 9.3

Enlargement and scale factors

Two shapes are similar when corresponding angles are equal and corresponding sides are in the same ratio. The ratio is the scale factor.

For triangles, equal angles are enough, because the sides are then forced into proportion. This is the AA test for similarity.

Lengths scale by k, areas by k^2 and volumes by k^3. This is why a model twice as long has four times the surface and eight times the volume.

The scale factor can be found from any pair of corresponding lengths, and it is the same for all of them. A single measurement is enough to connect a model to the object.

> FORMAL DEVELOPMENT
>
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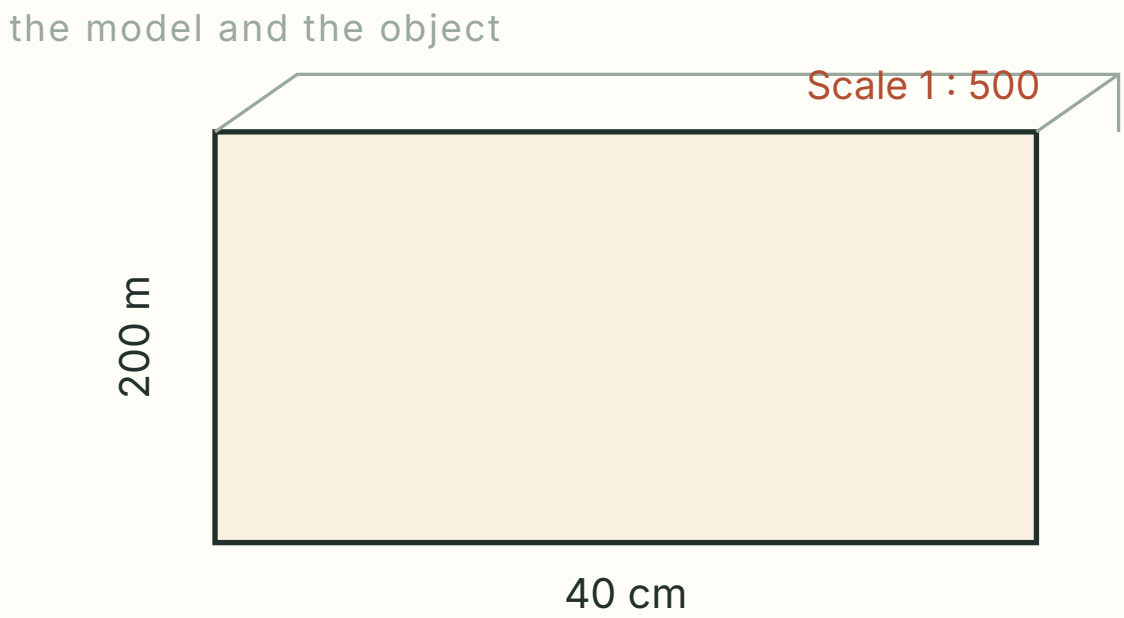
In your own words

CHAPTER 9 · 9.3 · WORKED EXAMPLE 1

Enlargement and scale factors

WORKED EXAMPLE 9.3.1

Problem. A model bridge is 40 cm long and the real bridge is 200 m long. Find the scale, and find the real length of a part that is 3 cm on the model.



Mathematical reasoning. Convert both lengths to the same unit and form the ratio.

1. 200 m = 20000 cm.
2. The scale is 40 : 20000, which is 1 : 500.
3. So every centimetre on the model is 500 cm, or 5 m, in reality.
4. A part of 3 cm on the model is 3 × 500 = 1500 cm, or 15 m, long.
5. The scale is stated as a ratio of model to object, and the order must be said.

Final answer. the scale is 1 : 500, and a 3 cm part is 15 m long

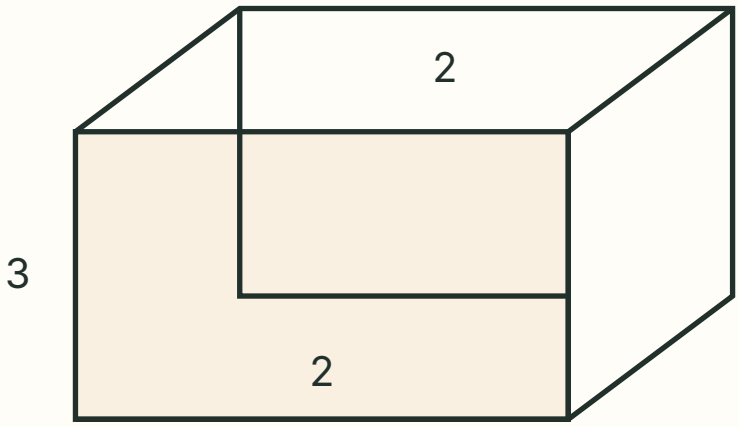
Interpretation. One ratio connects every length on the model to the object, which is what makes a scale drawing useful.

CHAPTER 9 · 9.3 · WORKED EXAMPLE 2

Enlargement and scale factors

WORKED EXAMPLE 9.3.2

Problem. Two similar bottles have heights in the ratio 1 : 3. Find the ratio of their surface areas and of their capacities.



the smaller bottle

Mathematical reasoning. Areas scale by k^2 and volumes by k^3.

1. The length scale factor is 3.
2. The surface area ratio is $3^2 : 1 = 9 : 1$.
3. The capacity ratio is $3^3 : 1 = 27 : 1$.
4. So the taller bottle holds twenty-seven times as much and needs nine times the material.
5. That is why large containers are cheaper per unit volume than small ones.

Final answer. surface areas 9 : 1, capacities 27 : 1

Interpretation. The scale factor connects lengths directly, areas through its square and volumes through its cube, and the three are often confused.

CHAPTER 9 · INVESTIGATION 9.4

Transformations on the coordinate plane

INQUIRY QUESTION

How is each transformation written as a rule on coordinates, and what does the rule preserve?

YOUR TASK

Take a shape of your own with three or four vertices. Apply four transformations in turn: a translation, a reflection in the y-axis, a rotation of 90° anticlockwise about the origin, and an enlargement of factor 2 about the origin. Record the coordinates after each step, and for each transformation state whether lengths, angles, area and orientation are preserved. Then try the same four in a different order and compare the final images.

step	rule	A(2, 1)	B(5, 1)	lengths	area
start	—	(2, 1)	(5, 1)	—	—
translate (4, −3)	(x, y) maps to (x + 4, y − 3)	(6, −2)	(9, −2)	yes	same
reflect in y-axis	(x, y) maps to (−x, y)	(−6, −2)	(−9, −2)	yes	same
rotate 90° anticlockwise	(x, y) maps to (−y, x)	(2, −6)	(2, −9)	yes	same
enlarge factor 2	(x, y) maps to (2x, 2y)	(4, −12)	(4, −18)	no	×4

CHAPTER 9 · INVESTIGATION 9.4

Transformations on the coordinate plane

What do you notice?

- The three rigid transformations preserved every length and angle.
- The enlargement preserved angles but doubled lengths and quadrupled area.
- The reflection reversed the orientation.
- Doing the four steps in a different order gave a different final image.

Observation. Each transformation is a rule on coordinates, and the rule shows immediately which properties are preserved.

MATHEMATICAL CONCLUSION

Transformations on the coordinate plane are arithmetic rules on pairs, and the rigid ones preserve length while an enlargement preserves only angle and ratio.

Your own words

CHAPTER 9 · 9.4

Transformations on the coordinate plane

On the coordinate plane a transformation is a rule on pairs of numbers, and the rule is what makes the image exact rather than approximate.

A translation by the vector (a, b) sends (x, y) to (x + a, y + b). A reflection in the y-axis sends (x, y) to (−x, y), and a reflection in the x-axis sends it to (x, −y).

A rotation of 90° anticlockwise about the origin sends (x, y) to (−y, x), and a rotation of 180° sends it to (−x, −y). The order of the coordinates is what changes, so the rule is easy to check on a single point.

An enlargement by scale factor k about the origin sends (x, y) to (kx, ky), so the coordinates are multiplied rather than added, and the shape grows away from the centre.

A transformation is an equivalence of shape when it is a translation, reflection or rotation, and the coordinate rule shows which property is preserved: lengths for the rigid ones, ratios for the enlargement.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

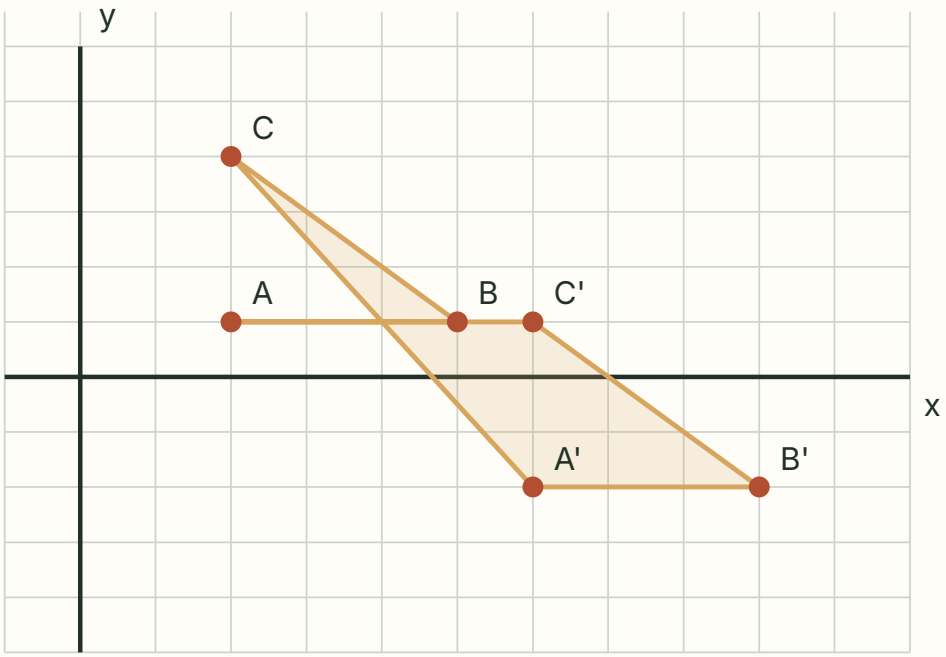
CHAPTER 9 · 9.4 · WORKED EXAMPLE 1

Transformations on the coordinate plane

WORKED EXAMPLE 9.4.1

Problem. The triangle has vertices A(2, 1), B(5, 1) and C(2, 4). Find its image under the translation (4, −3), and state what the transformation preserves.

the triangle and its image under the translation



Mathematical reasoning. Add the vector to each vertex.

1. A(2, 1) maps to (2 + 4, 1 − 3) = (6, −2).
2. B(5, 1) maps to (9, −2).
3. C(2, 4) maps to (6, 1).
4. The image is congruent to the original, so every length and angle is preserved.
5. The rule (x, y) maps to (x + 4, y − 3) describes the whole transformation at once.

Final answer. A′(6, −2), B′(9, −2), C′(6, 1); lengths, angles and area are preserved

Interpretation. One rule applied to three vertices gives the whole image, which is why coordinates are used for exact transformations.

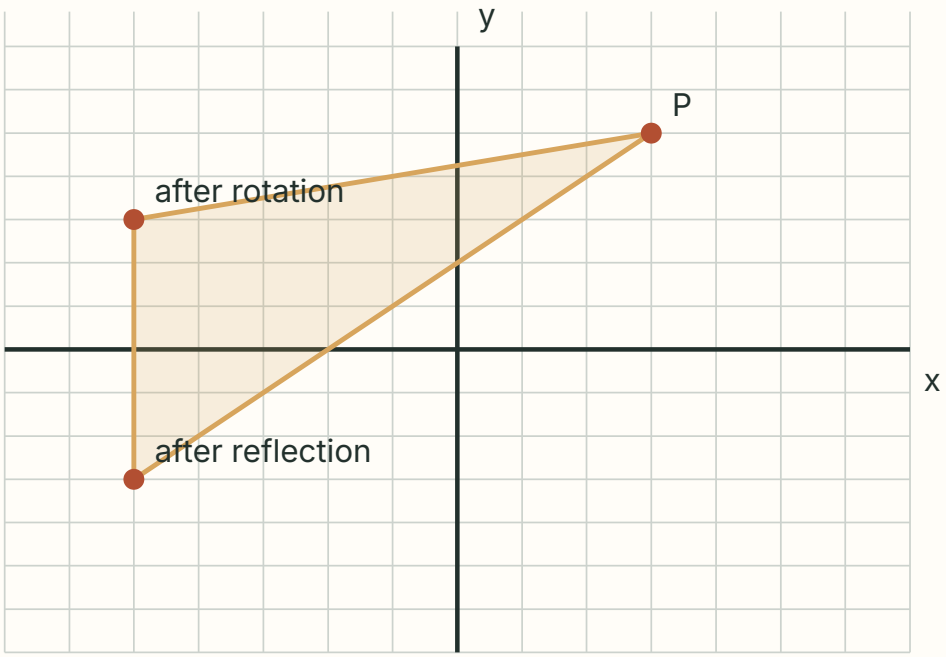
CHAPTER 9 · 9.4 · WORKED EXAMPLE 2

Transformations on the coordinate plane

WORKED EXAMPLE 9.4.2

Problem. The point P(3, 5) is rotated 90° anticlockwise about the origin, then reflected in the x-axis. Find the final image.

a rotation followed by a reflection



Mathematical reasoning. Apply each rule in turn, keeping track of the order.

1. A 90° anticlockwise rotation sends (x, y) to (−y, x).
2. So P(3, 5) maps to (−5, 3).
3. A reflection in the x-axis sends (x, y) to (x, −y).
4. So (−5, 3) maps to (−5, −3).
5. The order matters: reflecting first and rotating second gives a different point.

Final answer. the final image is (−5, −3)

Interpretation. Transformations compose, and the composition is not the same in both orders, so the sequence must be stated.

CHAPTER 9 · INVESTIGATION 9.5

Similarity in area and volume

INQUIRY QUESTION

How do area and volume scale when a figure is enlarged, and why is the factor squared or cubed?

YOUR TASK

Take three pairs of similar objects: two photographs at different scales, two maps of the same area at different scales, and two containers of the same shape but different sizes. For each pair, find the length factor, then measure or calculate the area factor and the volume factor, and check that they are the square and the cube of the length factor.

pair	length factor	area factor	volume factor
photo 1 : photo 2	2	4	—
map 1 : 25 000	4	16	—
cup A : cup B	1.5	2.25	3.375

CHAPTER 9 · INVESTIGATION 9.5

Similarity in area and volume

What do you notice?

- Every area factor was the square of the length factor.
- The volume factor was the cube, where a volume could be measured.
- A map factor of 4 in length meant 16 times the ground area.
- The two cups held very different amounts for a small difference in height.

Observation. Lengths scale by k, areas by k^2 and volumes by k^3, and each factor is the length factor raised to the number of dimensions.

MATHEMATICAL CONCLUSION

The exponent in a scale factor counts dimensions, so similarity converts a length ratio into an area ratio or a volume ratio by squaring or cubing it.

Your own words

CHAPTER 9 · 9.5

Similarity in area and volume

When a figure is enlarged by scale factor k, every length is multiplied by k, every area by k^2 and every volume by k^3. The exponents come from the number of dimensions the quantity measures.

The reason is dimensional: an area is a product of two lengths and a volume a product of three, so the factor appears twice or three times. This is why a model twice as long has four times the surface and eight times the volume.

A quantity per unit area, such as paint per square metre, is divided by k^2 under an enlargement, and a quantity per unit volume, such as the mass of a solid, is multiplied by k^3. Forgetting which way the factor acts is the commonest error.

Scaling is a representation of a real object at a different size, and it is valid only for the quantities that scale with the dimensions named. Weight, strength and cost need their own models.

> FORMAL DEVELOPMENT
>
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In your own words

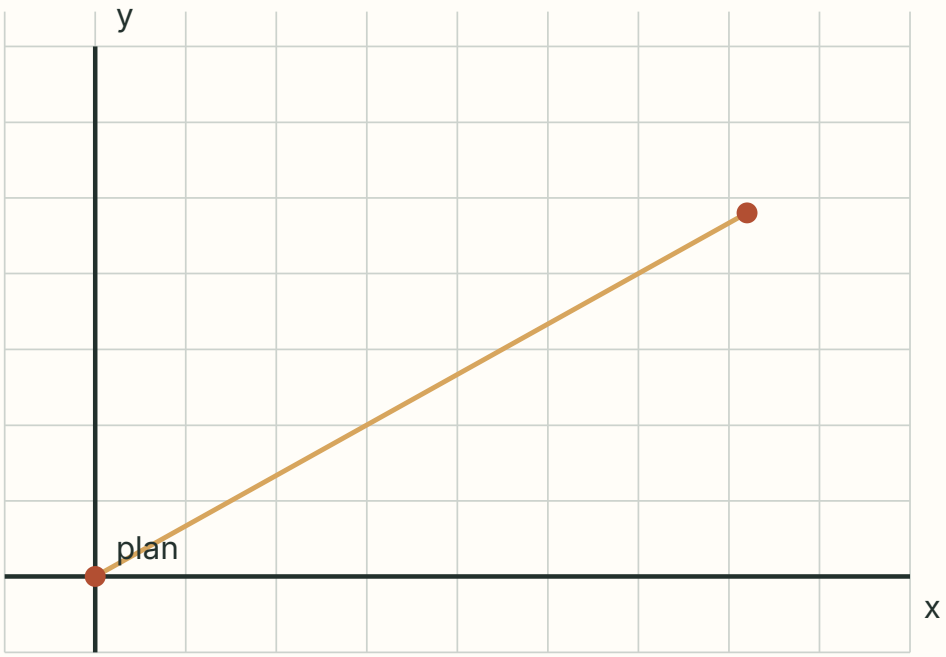
CHAPTER 9 · 9.5 · WORKED EXAMPLE 1

Similarity in area and volume

WORKED EXAMPLE 9.5.1

Problem. A scale model of a courtyard is drawn at 1 : 50. The plan shows a rectangular lawn 7.2 cm by 4.8 cm. Find the real area and the real perimeter.

the plan at 1 : 50



Mathematical reasoning. Lengths scale by 50, so the real dimensions are 50 times the drawn ones.

1. The real length is $7.2 \times 50 = 360$ cm $= 3.6$ m.
2. The real width is $4.8 \times 50 = 240$ cm $= 2.4$ m.
3. The real area is $3.6 \times 2.4 = 8.64$ m^2.
4. The real perimeter is $2(3.6 + 2.4) = 12$ m.
5. The drawn area is $7.2 \times 4.8 = 34.56$ cm^2, and $50^2 \times 34.56$ cm$^2 = 8.64$ m^2, so the area scales by the square of the factor.

Final answer. real area 8.64 m², real perimeter 12 m

Interpretation. A length scale factor of 50 becomes an area scale factor of 2500, and the two must not be confused.

CHAPTER 9 · 9.5 · WORKED EXAMPLE 2

Similarity in area and volume

WORKED EXAMPLE 9.5.2

Problem. Two similar bottles have heights in the ratio 1 : 3. The small bottle holds 0.5 litres. How much does the large one hold, and what is the ratio of their surface areas?

quantity	small	large	factor
height	h	3h	3
surface area	S	9S	9
volume	0.5 L	13.5 L	27

Mathematical reasoning. The volume scales by the cube of the length factor and the area by its square.

1. The length factor is 3, so the area factor is $3^2 = 9$.
2. The volume factor is $3^3 = 27$.
3. The large bottle holds $27 \times 0.5 = 13.5$ litres.
4. The surface area of the large bottle is 9 times that of the small one.
5. So trebling the height multiplies the capacity by 27, which is why a taller bottle looks far bigger than it is.

Final answer. 13.5 litres; surface areas in the ratio 1 : 9

Interpretation. Volume grows faster than area, and area faster than length, so a small change in size changes capacity a great deal.

CHAPTER 9 · INVESTIGATION 9.6

Investigation: shadows and scale

INQUIRY QUESTION

How can the size of something inaccessible be found from a model or a photograph?

YOUR TASK

Measure the height of something you cannot reach: a tree, a building, a flagpole or a lamp post. Use either the shadow method or the photograph method with a reference object, state your assumptions, and compare with a direct measurement if one is possible.

method	reference	estimate	assumption
shadow	a person 1.7 m, shadow 2.4 m	12.75 m	level ground, parallel rays
photograph	a door 2 m tall	18 m	no lens distortion
direct	tape measure	12.8 m	the tape reaches

CHAPTER 9 · INVESTIGATION 9.6

Investigation: shadows and scale

What do you notice?

- The shadow estimate was within a few centimetres of the direct measurement.
- The photograph estimate was less accurate, because of the lens.
- Both indirect methods needed a reference of known size.
- Stating the assumptions showed where each method could fail.

Observation. An inaccessible length is estimated by setting up a pair of similar triangles with a reference of known size, and the estimate carries the assumptions of the method.

MATHEMATICAL CONCLUSION

Similarity allows a measurement at a distance, and the honesty of the result depends on stating the assumptions that make the two triangles similar.

Your own words

CHAPTER 9 · 9.6

Investigation: shadows and scale

Similarity allows a measurement of an inaccessible object to be found from a photograph or a shadow, provided something of known size is in the same picture.

A shadow method uses the fact that the sun's rays are parallel, so the object and its shadow form a pair of similar triangles with a reference object and its shadow.

The method assumes the ground is level, the object is vertical and the light source is effectively at infinity. Each assumption is a place where the method can fail.

A measurement obtained this way is an estimate, and stating the assumptions is part of reporting it.

> FORMAL DEVELOPMENT
>
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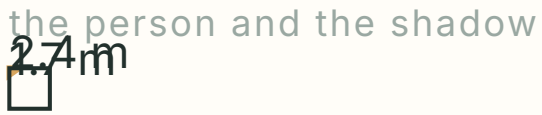
In your own words

CHAPTER 9 · 9.6 · WORKED EXAMPLE 1

Investigation: shadows and scale

WORKED EXAMPLE 9.6.1

Problem. A person 1.7 m tall casts a shadow 2.4 m long. A tree casts a shadow 18 m long at the same time. Find the height of the tree.



Mathematical reasoning. The two triangles are similar, so the ratios of corresponding sides are equal.

1. The person and the shadow form a right triangle with legs 1.7 m and 2.4 m.
2. The tree and its shadow form a similar triangle with legs h and 18 m.
3. So $h/18 = 1.7/2.4$.
4. Hence $h = 18 \times 1.7/2.4 = 12.75$ m.
5. The method assumes level ground, a vertical tree and parallel rays.

Final answer. the tree is 12.75 m tall

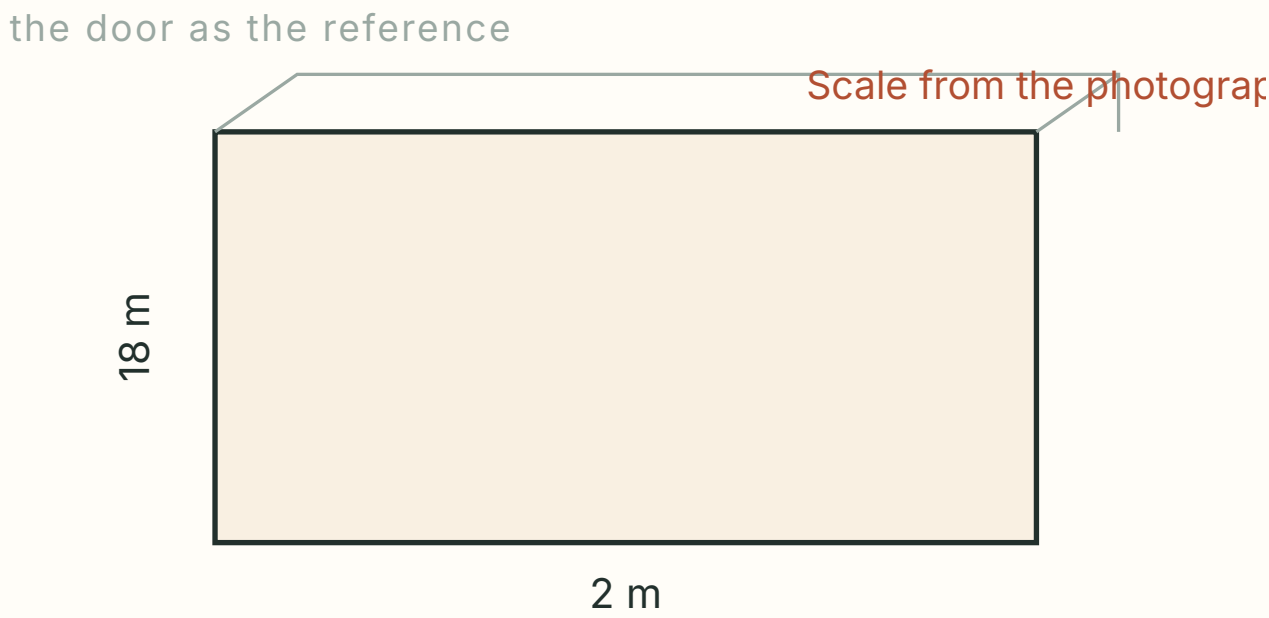
Interpretation. One ratio of similar triangles gives the height of something that cannot be measured directly, and the assumptions must be stated for the answer to be honest.

CHAPTER 9 · 9.6 · WORKED EXAMPLE 2

Investigation: shadows and scale

WORKED EXAMPLE 9.6.2

Problem. A photograph shows a door 2 m tall and a building 9 times as tall in the picture. Estimate the building's height and comment on the method.



Mathematical reasoning. Use the ratio of the heights in the picture as the scale factor.

1. The door is 2 m tall and appears 9 times smaller than the building.
2. So the building is about $9 \times 2 = 18$ m tall.
3. The estimate assumes the photograph is not distorted and the camera was level.
4. A wide-angle lens would make the building look smaller near the edges.
5. The estimate is therefore approximate, and its error comes from the photograph.

Final answer. about 18 m, subject to the distortion of the photograph

Interpretation. A reference object of known size makes an inaccessible measurement possible, and the reliability of the answer depends on the assumptions about the picture.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

9.A1 FLUENCY

Translate (3, 2) by the vector (−2, 5).

9.A2 FLUENCY

Reflect (4, −1) in the x-axis.

9.A3 FLUENCY

Reflect (4, −1) in the y-axis.

9.A4 FLUENCY

State whether two triangles with sides 3, 4, 5 and 3, 4, 6 are congruent.

9.A5 FLUENCY

Two triangles have sides 5 and 7 with an included angle of 60°. Are they congruent?

9.A6 FLUENCY

Find the ratio of the areas of two similar shapes with length ratio 3 : 5.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

9.A7 FLUENCY

Find the ratio of the volumes of two similar solids with length ratio 2 : 3.

9.A8 FLUENCY

A map has scale 1 : 25000. Find the real distance for 4 cm on the map.

9.A9 FLUENCY

Enlarge (2, 3) by scale factor 3 about the origin.

9.A10 FLUENCY

Rotate (1, 0) through 90° anticlockwise about the origin.

9.A11 FLUENCY

A model is 15 cm long at a scale of 1 : 200. Find the real length.

9.A12 FLUENCY

State whether two right-angled triangles with hypotenuse 10 and one leg 6 are congruent.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

9.B1 REASONING

The perimeters of similar figures are in the ratio 2 : 7. Find the ratio of their areas and justify.

9.B2 REASONING

Investigate the pattern in the areas of the images of a shape under enlargements of factor 1, 2, 3, 4.

9.B3 REASONING

Conjecture a rule for the ratio of the volumes of two similar cylinders with radii in the ratio 1 : 4.

9.B4 REASONING

Investigate whether two triangles with angles 50°, 60° and 70° are congruent, and justify.

9.B5 REASONING

Build a table of the images of a triangle under reflections in the two axes and describe the combined effect.

9.B6 REASONING

Conjecture a rule for the scale factor that doubles the area of a shape.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

9.B7 REASONING

Investigate the pattern in the number of congruent copies of a triangle that tile a rectangle.

9.B8 REASONING

A shape is enlarged by factor k and then by factor m. Investigate the combined effect and justify.

9.B9 REASONING

Find the pattern in the side lengths of the images of a square under enlargements of factor 1, ½, $\frac{1}{4}$.

9.B10 REASONING

Investigate the ratio of the surface areas of two similar spheres with radii in the ratio 3 : 1.

9.B11 REASONING

Conjecture a rule for the number of lines of symmetry of a regular n-sided polygon.

CRITERION C Communicating

Criterion C practice · Communicating

9.C1 REASONING

Explain in your own words what it means for two shapes to be congruent.

9.C2 FLUENCY

Write a complete solution to: enlarge the triangle (1, 1), (4, 1), (1, 3) by factor 2 about the origin.

9.C3 REASONING

Choose between measuring and transforming to decide congruence and justify.

9.C4 FLUENCY

Set out a full solution to: find the scale factor between two similar rectangles with areas 12 cm^2 and 108 cm^2.

9.C5 REASONING

Explain why ASS is not a congruence test.

9.C6 REASONING

Describe, with an example, the difference between congruence and similarity.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

9.C7 REASONING
Explain why the area scale factor is the square of the length scale factor.

9.C8 REASONING
Write the reasoning that shows reflecting in both axes is a half turn.

9.C9 REASONING
Set out the argument that RHS is a congruence test for right-angled triangles.

9.C10 REASONING
Explain what a tolerance is and why it is an interval.

9.C11 REASONING
Describe the relationship between a scale drawing and the object it represents.

9.C12 REASONING
Explain, with an example, when a model is a misleading representation of an object.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

9.D1 APPLICATION

A map has a scale of 1 : 50000. A field measures 6 cm by 4 cm on the map. Find its real area in hectares.

9.D2 APPLICATION

Two similar bottles have heights 12 cm and 18 cm. The smaller holds 500 ml. Find the capacity of the larger.

9.D3 APPLICATION

A model car is 20 cm long at a scale of 1 : 20. Find the real length and the ratio of the real and model surface areas.

9.D4 APPLICATION

A photograph is enlarged from 10 cm by 15 cm to 25 cm by 37.5 cm. Find the scale factor and the ratio of the areas.

9.D5 APPLICATION

A part is specified as 25.0 mm with a tolerance of ±0.1 mm. Investigate whether measurements of 24.95 and 25.12 mm are acceptable.

9.D6 APPLICATION

A tree casts a shadow 15 m long while a 1.6 m post casts a shadow 2 m long. Find the height of the tree and state one assumption.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

9.D7 REASONING

Two similar solids have volumes in the ratio 1 : 64. Find the ratio of their surface areas and justify.

9.D8 APPLICATION

A scale model of a room is built at 1 : 25. The model has a floor area of 0.32 m^2. Find the real floor area and comment on the model's limits.

9.D9 REASONING

A photograph shows a 1.8 m door and a wall 6.5 times as tall. Investigate the wall's height and comment on the reliability.

9.D10 CHALLENGE

A company makes a cup in two sizes with heights in the ratio 1 : 1.5. Investate whether the larger uses 2.25 times the material and justify.

9.D11 APPLICATION

A map has a scale of 1 : 200000. A lake measures 3.5 cm by 2 cm on the map. Find its real area in square kilometres and state one limitation of the method.

9.D12 REASONING

Explain, with an example, when using a scale factor directly on an area gives a wrong answer.

CHAPTER 9 · APPLICATION AND ASSESSMENT

Applying chapter 9

Each task asks you to describe a transformation or to use a scale factor. State the transformation in full, and say what it preserves.

A1 · The model bridge

A scale model of a footbridge is built at 1 : 75. The model is 48 cm long and its deck area is 0.031 m^2. Find the real length and the real deck area, estimate the real volume of concrete if the model used 0.0004 m^3, and comment on two properties of the real bridge that the model cannot establish.

Criteria A and C · use the scale factor, its square and its cube, and justify the limits of the model.

A2 · The two triangles

Two triangles are described. Triangle P has sides 7 cm and 9 cm with an included angle of 50°. Triangle Q has sides 7 cm and 9 cm with an angle of 50° opposite the 7 cm side. Decide, with a full justification, whether each pair of triangles with these descriptions must be congruent, and explain the difference between the two descriptions.

Criteria B and D · apply the congruence tests and justify why one description decides the shape and the other does not.

CHAPTER 9 · REVIEW

What you should now be able to do

Key ideas

- A translation, a reflection and a rotation preserve length and angle.
- An enlargement by k scales lengths by k, areas by k^2 and volumes by k^3.
- SSS, SAS, ASA and RHS are tests for congruent triangles.
- ASS and AAA are not congruence tests.
- Similar shapes have equal angles and sides in proportion.
- The scale factor connects a model to the object it represents.
- A tolerance defines an interval of acceptable measurements.
- An inaccessible length can be estimated from similar triangles.

Vocabulary

translation · vector · reflection · rotation · enlargement · scale factor · congruent · similar · tolerance · invariant · orientation · ambiguous case

CHAPTER 9 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

9.R1 Reflect (3, −5) in the y-axis.

9.R2 Enlarge (2, −4) by factor 3 about the origin.

9.R3 Find the ratio of the areas of similar shapes with length ratio 4 : 7.

9.R4 A map has scale 1 : 100000. Find the real length of 5 cm.

9.R5 Two triangles have sides 6, 8 and 10. Are they congruent to a 3, 4, 5 triangle?

9.R6 Explain why a scale factor of 2 quadruples the area.

CHAPTER 9 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> Two shapes are congruent when one can be moved onto the other by a rigid transformation, and similar when one can be obtained from the other by an enlargement; the transformations are therefore the test of equivalence, and the scale factor is what connects a model to the object it represents.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of space, equivalence and representation appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of scale models, design and measurement at a distance, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a scale model for a decision about a real object, and what would you still like to ask?

CHAPTER 9 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Scale models let us build what we cannot yet measure. Model one and say what the scale factor does and does not preserve.

Service as Action

COMMUNITY NEED	Scale models and maps are used for decisions without stating what they cannot show.
MATHEMATICAL CONNECTION	Transformations, similarity and scale factors.
PROPOSED ACTION	Build or analyse one scale model and publish what it establishes and what it does not.
EXPECTED IMPACT	A reader knows which questions a model can answer.
STUDENT REFLECTION	Record the model, the scale factor and the limits.

Community Project

PROJECT IDEA	A scale model of one community space.
NEED	Plans for a garden, a market or a classroom are discussed without a shared picture.
RESEARCH QUESTION	What does a scale model establish about the space, and what does it leave out?
MATHEMATICS	Transformations, similarity and area scaling.
ACTION OR OUTCOME	A scale model or drawing with a stated scale and a list of its limits.
BENEFICIARIES	Everyone who will use the space.
EVIDENCE	The model, the scale and the list of limits.

Personal Project connection · MYP 5

PERSONAL INTEREST	A model, a map or a design you enjoy.
LEARNING GOAL	To build or analyse a scale model and state its limits.
PRODUCT OR OUTCOME	A scale model with a stated factor and a justification.
MATHEMATICS	Transformations and similarity.
PROCESS	Choose a scale, build or analyse the model, and test one prediction.
EVIDENCE	The model, the scale and the test.

CHAPTER 10 · PART III · SPATIAL REASONING

10

Pythagoras, Trigonometry and Measurement

Pythagoras connects the three sides. The sine, cosine and tangent connect the angles to the sides. Together they decide every right-angled triangle from two pieces of information.

ENGAGE

A ladder leans against a wall. Its foot is 1.5 m from the wall and its top is 3.6 m up. How long is the ladder, and what angle does it make with the ground?

Framing the inquiry

Key concept: Relationships
Related concepts: Space · Measurement · Equivalence
Global context: Scientific and technical innovation
Global context exploration: Surveying, navigation and safe design

Statement of inquiry: The sides and angles of a right-angled triangle are related by Pythagoras' theorem and by the trigonometric ratios, so knowing two of them decides the rest; these relations turn an inaccessible length or angle into a measurement, and the equivalence between the theorem and the ratios is what makes surveying possible.

Questions to guide us

- **Factual:** What does Pythagoras' theorem state, and what are the three trigonometric ratios?
- **Conceptual:** Why is the choice of ratio decided by which sides are known, and how do the theorem and the ratios relate to each other?
- **Debatable:** When a measurement is used to decide whether a structure is safe, is a calculation enough, or must the measurement itself be justified?

ATL · THINKING. Students select the appropriate relation for a given right-angled triangle, justify the choice, and check the answer by a second route.

CHAPTER 10 · INVESTIGATION 10.1

Pythagoras' theorem and proof

INQUIRY QUESTION

How does the theorem connect the three sides of a right-angled triangle?

YOUR TASK

Measure the diagonal and the two sides of five rectangles, including a sheet of paper, a book, a door and a screen. Check Pythagoras' theorem on each, and then use it to predict the diagonal of a rectangle you cannot measure.

object	sides	diagonal	check
sheet of paper	21.0, 29.7	36.3	$21^2 + 29.7^2 \approx 36.3^2$
book	15.0, 21.0	25.8	$15^2 + 21^2 \approx 25.8^2$
door	0.8, 2.0	2.15	$0.8^2 + 2^2 \approx 2.15^2$
screen	0.45, 0.80	0.92	$0.45^2 + 0.8^2 \approx 0.92^2$

CHAPTER 10 · INVESTIGATION 10.1

Pythagoras' theorem and proof

What do you notice?

- The theorem held on every rectangle, within the precision of the ruler.
- The predicted diagonal of the unmeasured rectangle was within a centimetre.
- The diagonal was always longer than either side and shorter than their sum.
- The small disagreements came from rounding the measurements, not from the theorem.

> **Observation.** The square on the diagonal equals the sum of the squares on the sides, and the agreement is as close as the measurements allow.

MATHEMATICAL CONCLUSION

Pythagoras' theorem connects the three sides of a right-angled triangle, and it works in both directions: it finds a missing side and it tests whether an angle is right.

Your own words

CHAPTER 10 · 10.1

Pythagoras' theorem and proof

In a right-angled triangle the square on the hypotenuse equals the sum of the squares on the other two sides: $a^2 + b^2 = c^2$, where c is the hypotenuse.

The hypotenuse is the side opposite the right angle and is always the longest side. The other two sides are the legs.

A right-angled triangle is the simplest figure in space that has a fixed shape once two measurements are known, which is why it is used to measure distances that cannot be reached.

The theorem is used forwards to find a missing side and backwards to test whether a triangle is right-angled: if $a^2 + b^2 = c^2$ then the angle opposite c is a right angle.

The converse is a test, and it is the reason a builder can check a corner is square by measuring 3, 4 and 5 units along the two edges and the diagonal.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

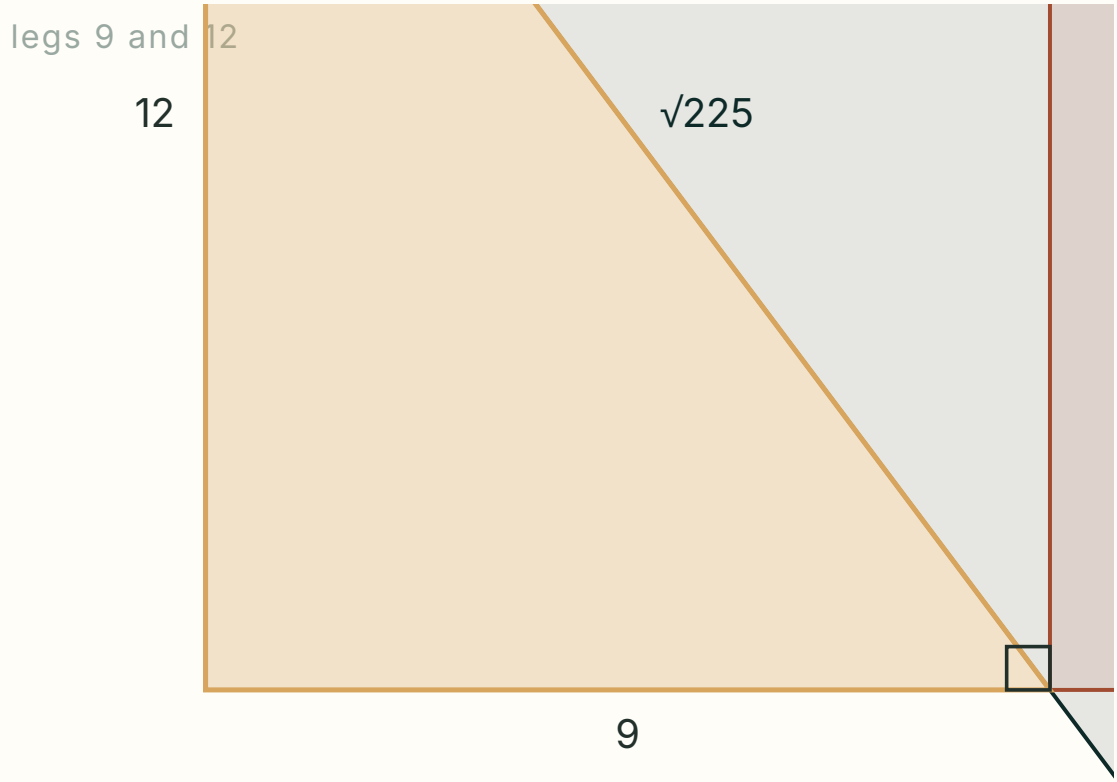
In your own words

CHAPTER 10 · 10.1 · WORKED EXAMPLE 1

Pythagoras' theorem and proof

WORKED EXAMPLE 10.1.1

Problem. A right-angled triangle has legs 9 cm and 12 cm. Find the hypotenuse.



Mathematical reasoning. Apply $a^2 + b^2 = c^2$ and take the square root.

1. Let c be the hypotenuse. Then $c^2 = 9^2 + 12^2$.
2. So $c^2 = 81 + 144 = 225$.
3. Hence $c = \sqrt{225} = 15$ cm.
4. The hypotenuse is longer than either leg, as expected.
5. This is the 3-4-5 triangle scaled by 3, which is why the answer is a whole number.

Final answer. 15 cm

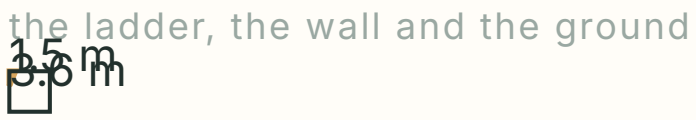
Interpretation. The theorem converts two known sides into the third, and the answer is checked by the fact that it exceeds both legs.

CHAPTER 10 · 10.1 · WORKED EXAMPLE 2

Pythagoras' theorem and proof

WORKED EXAMPLE 10.1.2

Problem. A ladder reaches 3.6 m up a wall with its foot 1.5 m from the wall. Find the length of the ladder.



Mathematical reasoning. The ladder is the hypotenuse of a right-angled triangle whose legs are known.

1. The ladder is the hypotenuse, and the legs are 1.5 m and 3.6 m.
2. So $L^2 = 1.5^2 + 3.6^2 = 2.25 + 12.96 = 15.21$.
3. Hence $L = \sqrt{15.21} = 3.9$ m.
4. The ladder is longer than the height it reaches, as it must be.
5. The angle with the ground is about 67.4°, found in section 10.2.

Final answer. the ladder is 3.9 m long

Interpretation. A length that cannot be measured directly is found from two measurements that can, which is the practical value of the theorem.

CHAPTER 10 · INVESTIGATION 10.2

Pythagoras in three dimensions

INQUIRY QUESTION

How is Pythagoras' theorem used inside a three-dimensional solid?

YOUR TASK

Take four solids: a cuboid, a cone, a cylinder and a pyramid. For each, draw or describe the cross-section that contains a right angle, and use Pythagoras to find the internal length named: the space diagonal of the cuboid, the slant height of the cone and the pyramid, and the longest rod that fits inside the cylinder lying flat.

solid	cross-section	Pythagoras on	result
cuboid 3, 4, 12	face then body	$\sqrt{(3^2+4^2+12^2)}$	13
cone r = 6, h = 8	through the axis	$\sqrt{(6^2 + 8^2)}$	10
cylinder r = 5, h = 12	through the axis	$\sqrt{(5^2 + 12^2)}$	13
pyramid square base 6, height 4	through apex and mid-edge	$\sqrt{(3^2 + 4^2)}$	5

CHAPTER 10 · INVESTIGATION 10.2

Pythagoras in three dimensions

What do you notice?

- Every solid gave a right-angled triangle in some cross-section.
- The cuboid needed Pythagoras twice, the others once.
- The slant heights were all whole numbers because the triangles were Pythagorean triples.
- The cylinder's longest flat rod was its space diagonal, not its height.

> **Observation.** A three-dimensional length is found by choosing the cross-section that contains a right angle, and Pythagoras then applies unchanged.

MATHEMATICAL CONCLUSION

Pythagoras' theorem is not limited to a flat page: it applies to every right-angled triangle, including those hidden inside a solid.

Your own words

CHAPTER 10 · 10.2

Pythagoras in three dimensions

Pythagoras' theorem applies to any right-angled triangle, including one drawn inside a solid, so a three-dimensional problem is solved by finding the right triangle first.

The diagonal of a rectangle of sides a and b is $\sqrt{(a^2 + b^2)}$, and the space diagonal of a cuboid of sides a, b and c is $\sqrt{(a^2 + b^2 + c^2)}$, which is Pythagoras applied twice.

The longest rod that fits in a box is the space diagonal, so the theorem decides what can be carried through a doorway or stored in a container.

A two-dimensional cross-section is a representation of a three-dimensional problem, and choosing the cross-section that contains the right angle is the whole skill.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

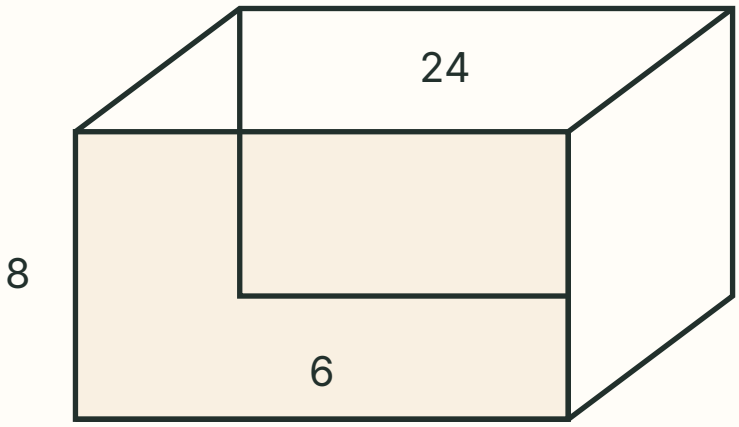
In your own words

CHAPTER 10 · 10.2 · WORKED EXAMPLE 1

Pythagoras in three dimensions

WORKED EXAMPLE 10.2.1

Problem. A cuboid measures 6 cm by 8 cm by 24 cm. Find the length of its space diagonal.



a cuboid of 6 cm, 8 cm and 24 cm

Mathematical reasoning. Apply Pythagoras twice: first to a face, then to the cross-section.

1. The face diagonal is $\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$ cm.
2. This diagonal and the third side form a right-angled triangle whose hypotenuse is the space diagonal.
3. So the space diagonal is $\sqrt{(10^2 + 24^2)} = \sqrt{676} = 26$ cm.
4. Equivalently $\sqrt{(6^2 + 8^2 + 24^2)} = \sqrt{676} = 26$ cm.
5. The two routes agree, which is the check.

Final answer. 26 cm

Interpretation. Pythagoras used twice turns a three-dimensional measurement into two two-dimensional ones.

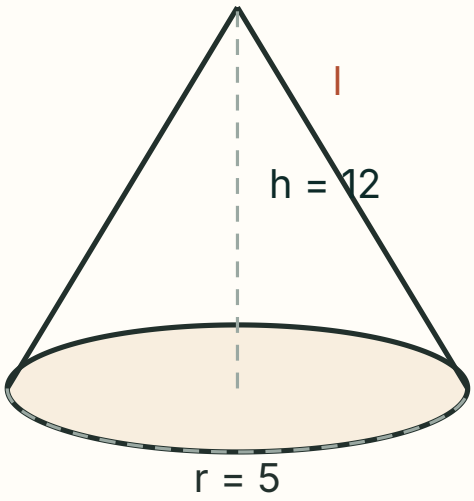
CHAPTER 10 · 10.2 · WORKED EXAMPLE 2

Pythagoras in three dimensions

WORKED EXAMPLE 10.2.2

Problem. A cone has height 12 cm and base radius 5 cm. Find its slant height.

radius 5 cm and height 12 cm



Mathematical reasoning. The vertical cross-section through the axis is an isosceles triangle, and half of it is right-angled.

1. The cross-section through the axis contains the height, the radius and the slant height.
2. The height is perpendicular to the radius at the centre of the base.
3. So the slant height is the hypotenuse: $\sqrt{(5^2 + 12^2)} = \sqrt{169}$.
4. Hence the slant height is 13 cm.
5. The slant height is what is needed for the curved surface area.

Final answer. 13 cm

Interpretation. A solid is measured by choosing the cross-section that contains the right angle, and the cone's cross-section is a right triangle.

CHAPTER 10 · INVESTIGATION 10.3

Trigonometric ratios

INQUIRY QUESTION

How do the sine, cosine and tangent connect an angle to the sides?

YOUR TASK

Measure five right-angled triangles of your own: a set square, a leaning ladder, a roof pitch, a ramp and one of your own. For each, compute the three ratios for one acute angle, then use them to find the other sides and check with Pythagoras.

triangle	angle	sin	cos	tan
set square	30°	0.5	0.866	0.577
ladder	67.4°	0.923	0.385	2.4
roof pitch	22.6°	0.385	0.923	0.417
ramp	11.3°	0.196	0.981	0.2

CHAPTER 10 · INVESTIGATION 10.3

Trigonometric ratios

What do you notice?

- The three ratios for the same angle agreed across triangles of different sizes.
- sin θ and cos θ swapped when the two acute angles were exchanged.
- The tangent was the ratio of the other two, in every case.
- Pythagoras was satisfied by every triangle, within measurement error.

> **Observation.** The three ratios depend on the angle alone, so a table of values serves every right-angled triangle of the same shape.

MATHEMATICAL CONCLUSION

The sine, cosine and tangent of an angle are the three ratios of the sides of a right-angled triangle, and any one of them connects a known side to an unknown one.

Your own words

CHAPTER 10 · 10.3

Trigonometric ratios

In a right-angled triangle, sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse and tan θ = opposite/adjacent, where the sides are named relative to the angle θ.

The ratios depend only on the angle, not on the size of the triangle, which is why similar triangles give the same values and a table of ratios serves every triangle.

Choosing a ratio is decided by which sides are known and which is wanted. If the hypotenuse is involved, use sine or cosine; if not, use tangent.

The inverse ratios recover the angle: if sin θ = 0.6 then θ = 37°, and the two questions — finding a side and finding an angle — use the same three relations.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.3 · WORKED EXAMPLE 1

Trigonometric ratios

WORKED EXAMPLE 10.3.1

Problem. A right-angled triangle has hypotenuse 10 cm and an angle of 35°. Find the side opposite and the side adjacent to the angle.



Mathematical reasoning. Use sine for the opposite side and cosine for the adjacent side.

1. $\sin 35° = \text{opposite}/10$, so the opposite side is $10 \times \sin 35°$.
2. $\sin 35° \approx 0.5736$, so the opposite side is about 5.74 cm.
3. $\cos 35° = \text{adjacent}/10$, so the adjacent side is $10 \times \cos 35°$.
4. $\cos 35° \approx 0.8192$, so the adjacent side is about 8.19 cm.
5. Check: $5.74^2 + 8.19^2 \approx 100$, as Pythagoras requires.

Final answer. opposite 5.74 cm, adjacent 8.19 cm

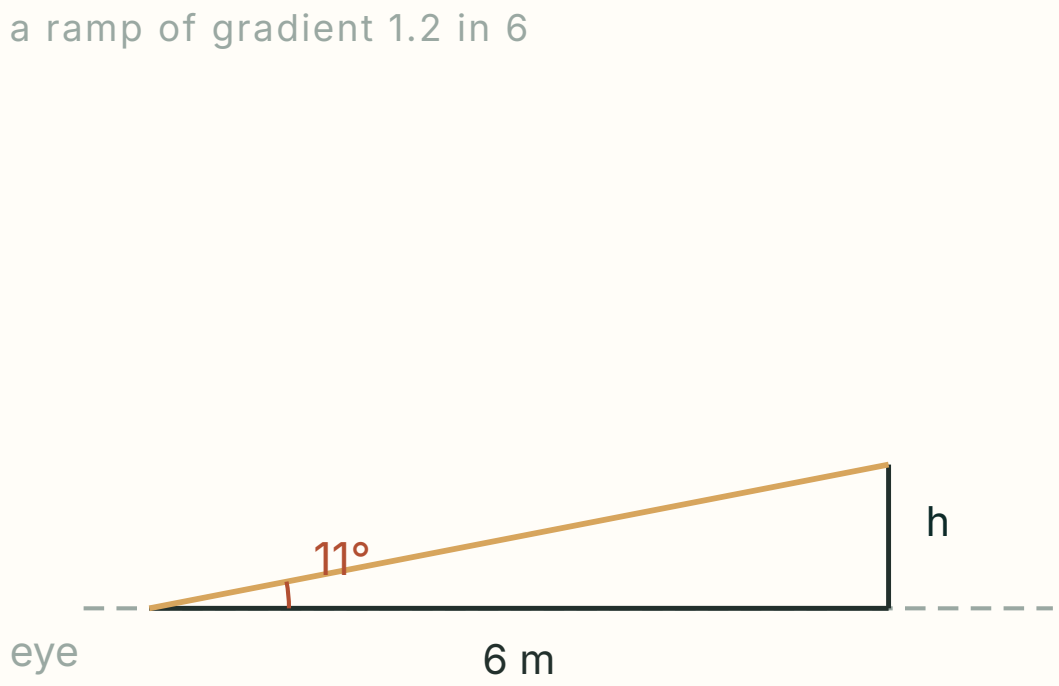
Interpretation. One angle and the hypotenuse fix the whole triangle, and the two ratios give the two legs.

CHAPTER 10 · 10.3 · WORKED EXAMPLE 2

Trigonometric ratios

WORKED EXAMPLE 10.3.2

Problem. A ramp rises 1.2 m over a horizontal distance of 6 m. Find the angle the ramp makes with the ground.



Mathematical reasoning. The tangent relates the rise to the run, so use the inverse tangent.

1. The rise is opposite the angle and the run is adjacent.
2. So $\tan\theta = 1.2/6 = 0.2$.
3. Hence $\theta = \arctan(0.2) \approx 11.3°$.
4. A gradient of 1 in 5 is a gentle slope, as the angle shows.
5. Building regulations often limit ramps to about 1 in 12, which is about 4.8°.

Final answer. about 11.3°

Interpretation. The tangent of the angle of elevation is the gradient, which connects trigonometry to the linear models of chapter 4.

CHAPTER 10 · INVESTIGATION 10.4

Solving right-angled triangles

INQUIRY QUESTION

How is a right-angled triangle solved when two pieces of information are given?

YOUR TASK

Solve six right-angled triangles of your own: two given two legs, two given a leg and the hypotenuse, and two given an angle and a side. For each, state which ratio you used and why, find every unknown, and check by a second route such as Pythagoras or the sum of the angles.

given	ratio used	found	check
legs 5 and 12	Pythagoras then tan	hypotenuse 13, angles 22.6°, 67.4°	$5^2 + 12^2 = 13^2$
leg 7, hypotenuse 25	sin then Pythagoras	angles 16.3°, 73.7°	$7^2 + 24^2 = 25^2$
angle 40°, hypotenuse 10	sin and cos	legs 6.43 and 7.66	$6.43^2 + 7.66^2 \approx 100$
angle 55°, leg 6	tan and cos	hypotenuse 10.49	$6/10.49 \approx \cos 55°$

CHAPTER 10 · INVESTIGATION 10.4

Solving right-angled triangles

What do you notice?

- The ratio used was always the one containing the two known quantities.
- Finding a side multiplied; finding an angle used an inverse ratio.
- Pythagoras checked every side and the angle sum checked every angle.
- The two acute angles always added to 90°.

Observation. A right-angled triangle is solved by choosing the ratio that uses the two known quantities, then checking by a second route.

MATHEMATICAL CONCLUSION

Two pieces of information fix a right-angled triangle, and the choice of ratio depends on which two they are.

Your own words

CHAPTER 10 · 10.4

Solving right-angled triangles

Solving a right-angled triangle means finding every unknown side and angle from the information given, and two pieces of information are enough because the right angle is already known.

The choice of ratio depends on what is given and what is wanted: the tangent uses the two legs, the sine uses the opposite and the hypotenuse, and the cosine uses the adjacent and the hypotenuse.

Finding a side uses multiplication by a ratio, and finding an angle uses the inverse ratio. Writing the equation before substituting is what stops the two being confused.

The two acute angles of a right-angled triangle add to 90°, so one angle can be found from the other, and this is a useful check on a calculation.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 10 · 10.4 · WORKED EXAMPLE 1

Solving right-angled triangles

WORKED EXAMPLE 10.4.1

Problem. A right-angled triangle has legs 9 cm and 12 cm. Find the hypotenuse and the two acute angles.



Mathematical reasoning. Pythagoras gives the hypotenuse, then a ratio gives an angle.

1. The hypotenuse is $\sqrt{(9^2 + 12^2)} = \sqrt{225} = 15$ cm.
2. For the angle opposite the 9 cm side, $\tan A = 9/12$.
3. So $A = \tan^{-1}(0.75) \approx 36.9°$.
4. The other angle is $90° - 36.9° = 53.1°$.
5. Check: $36.9 + 53.1 = 90$, as the two acute angles must.

Final answer. hypotenuse 15 cm; angles 36.9° and 53.1°

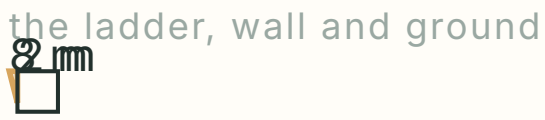
Interpretation. Pythagoras finds the sides and the ratios find the angles, and the two acute angles check each other.

CHAPTER 10 · 10.4 · WORKED EXAMPLE 2

Solving right-angled triangles

WORKED EXAMPLE 10.4.2

Problem. A ladder 8 m long leans against a wall with its foot 2 m from the wall. Find the angle the ladder makes with the ground.



Mathematical reasoning. The hypotenuse and the adjacent side are known, so use the cosine.

1. The hypotenuse is 8 m and the side adjacent to the angle is 2 m.
2. So $\cos\theta = 2/8 = 0.25$.
3. Hence $\theta = \cos^{-1}(0.25) \approx 75.5°$.
4. The height reached is $\sqrt{(8^2 - 2^2)} = \sqrt{60} \approx 7.75$ m.
5. A steeper angle would give a greater height, so the answer is sensible.

Final answer. about 75.5° to the ground

Interpretation. Choosing the ratio that uses the two known quantities is the first step, and it is decided by what is given, not by what is wanted.

CHAPTER 10 · INVESTIGATION 10.5

Angles of elevation, depression and bearings

INQUIRY QUESTION

How is an inaccessible height found from an angle and a distance?

YOUR TASK

Measure the height of three objects you cannot reach: a tree, a building and a flagpole. For each, measure the horizontal distance and the angle of elevation with a clinometer or a phone, compute the height, and add the observer's eye level.

object	distance	angle	height
tree	18 m	32°	13.1 m
building	25 m	41°	23.3 m
flagpole	12 m	55°	18.7 m

CHAPTER 10 · INVESTIGATION 10.5

Angles of elevation, depression and bearings

What do you notice?

- Every height needed the observer's eye level added.
- A small error in the angle produced a large error in the height, because the tangent is steep.
- The ground was not perfectly level, which added a further error.
- The method worked because the distance along the ground could be measured.

Observation. A height is found from the tangent of the angle of elevation and the horizontal distance, and the accuracy of the answer is limited by the accuracy of the angle.

MATHEMATICAL CONCLUSION

An inaccessible height is measured by combining an angle of elevation with a measured horizontal distance, and the observer's own height must be included.

Your own words

CHAPTER 10 · 10.5

Angles of elevation, depression and bearings

The angle of elevation is the angle above the horizontal at which an object is seen, and the angle of depression is the angle below the horizontal from which it is seen.

The two angles are equal when the lines of sight are parallel, which is why a person on a cliff and a person on the ground see each other at the same angle.

A height is found from the tangent of the angle of elevation and the horizontal distance, and the observer's own height must be added if the angle was measured at eye level.

A clinometer or a phone app measures the angle, and the distance along the ground is measured separately. The two measurements combine in the tangent ratio.

A bearing is the angle measured clockwise from north, written with three figures, and it is the convention that makes a direction unambiguous. Without it, two people can mean different things by the same words.

A back bearing differs from the forward bearing by 180°, because reversing a direction turns it through a half turn, and a journey of two legs is combined by Pythagoras or trigonometry on the triangle the legs form.

A measured angle carries an error interval, and a height calculated from it carries one too. A small error in an angle can produce a large error in a height, so an answer near a safety limit needs a margin.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

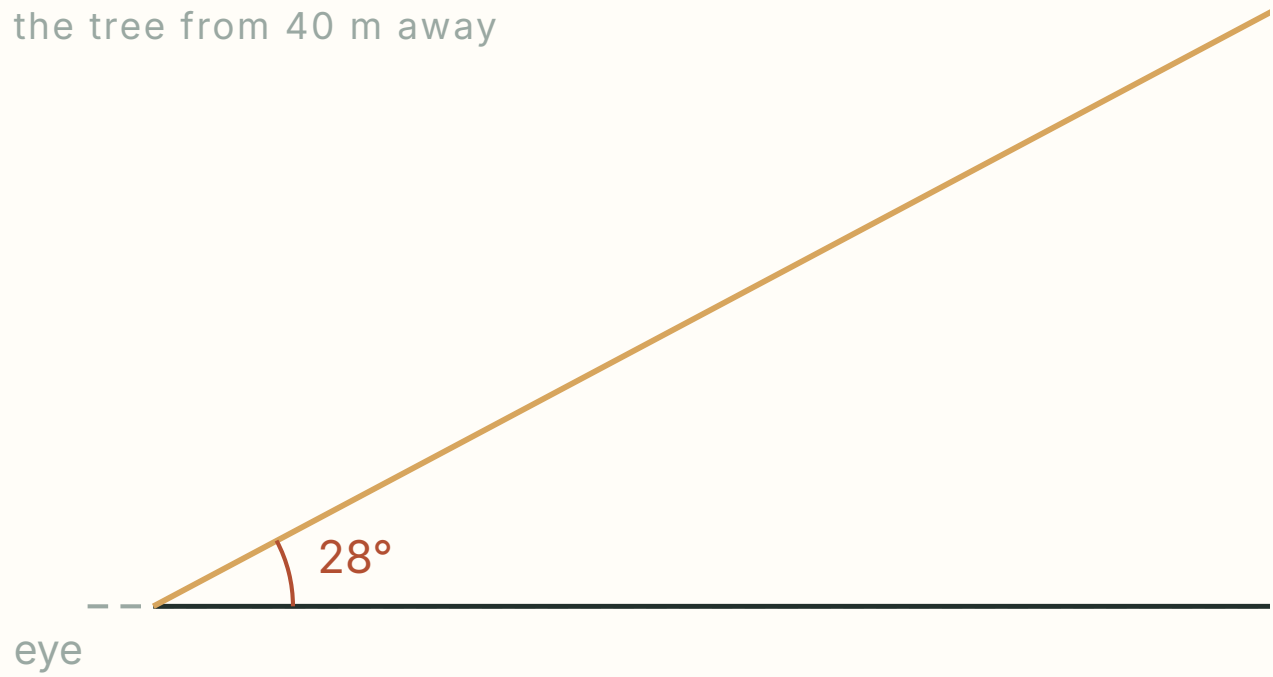
In your own words

CHAPTER 10 · 10.5 · WORKED EXAMPLE 1

Angles of elevation, depression and bearings

WORKED EXAMPLE 10.5.1

Problem. From 40 m away, the angle of elevation to the top of a tree is 28°. The observer's eye is 1.6 m above the ground. Find the height of the tree.



Mathematical reasoning. Use the tangent of the angle of elevation, then add the observer's height.

1. tan 28° = height above eye/40.
2. So the height above eye level is 40 × tan 28°.
3. tan 28° ≈ 0.5317, so the height above eye level is about 21.3 m.
4. Adding the observer's height gives 21.3 + 1.6 = 22.9 m.
5. The tree is about 22.9 m tall.

Final answer. about 22.9 m

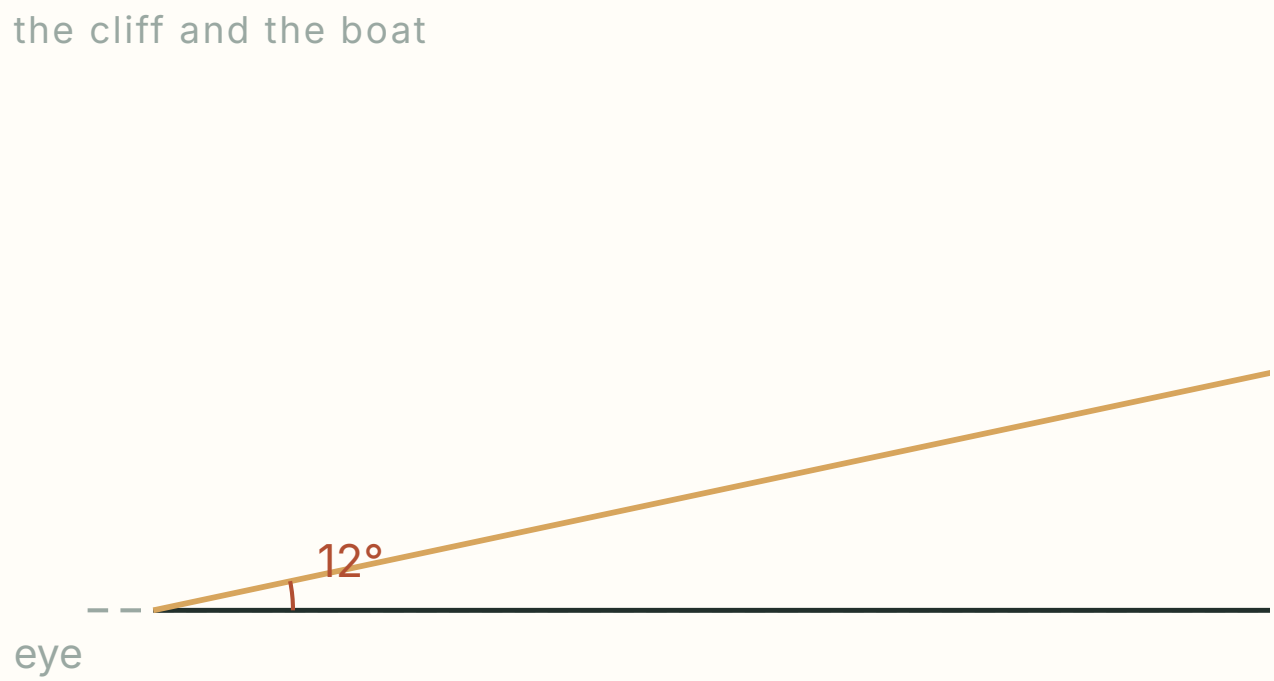
Interpretation. The measurement gives the height above the observer's eye, and forgetting to add the eye level is the commonest error in this calculation.

CHAPTER 10 · 10.5 · WORKED EXAMPLE 2

Angles of elevation, depression and bearings

WORKED EXAMPLE 10.5.2

Problem. From the top of a 30 m cliff, the angle of depression to a boat is 12°. Find the distance of the boat from the base of the cliff.



Mathematical reasoning. The angle of depression equals the angle of elevation from the boat, so the same triangle is used.

1. The angle of elevation from the boat is also 12°.
2. tan 12° = 30/d, where d is the distance from the base.
3. So d = 30/tan 12°.
4. tan 12° ≈ 0.2126, so d ≈ 141 m.
5. The boat is about 141 m from the base of the cliff.

Final answer. about 141 m

Interpretation. The equality of the two angles lets one measurement serve both observers, which is why the method is used in navigation.

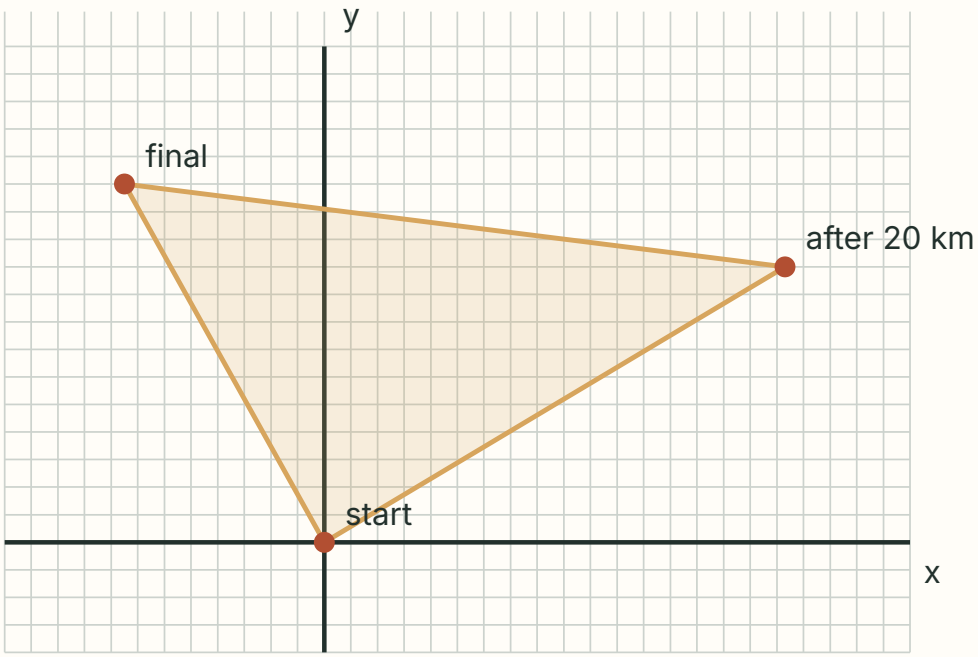
CHAPTER 10 · 10.5 · WORKED EXAMPLE 3

Angles of elevation, depression and bearings

WORKED EXAMPLE 10.5.3

Problem. A ship sails 20 km on a bearing of 030° and then 15 km on a bearing of 120°. Find its distance and bearing from the start.

two legs and the resultant



Mathematical reasoning. Draw the two displacements, find the resultant by components, then convert back to distance and bearing.

1. The first leg has components $20 \sin 30° = 10$ km east and $20 \cos 30° = 17.3$ km north.
2. The second leg has components $15 \sin 120° = 13$ km east and $15 \cos 120° = -7.5$ km north.
3. Adding gives 23 km east and 9.8 km north.
4. The distance is $\sqrt{(23^2 + 9.8^2)} \approx 25$ km.
5. The bearing is $\arctan(23/9.8) \approx 067°$.

Final answer. about 25 km on a bearing of 067°

Interpretation. Two legs combine into one resultant, and the components make the combination a matter of addition.

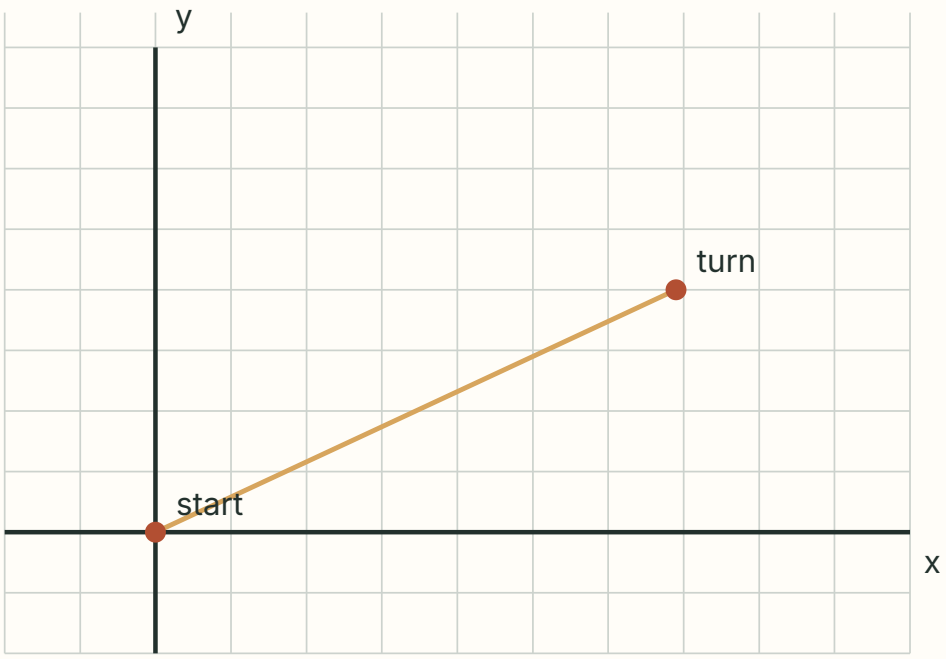
CHAPTER 10 · 10.5 · WORKED EXAMPLE 4

Angles of elevation, depression and bearings

WORKED EXAMPLE 10.5.4

Problem. A walker goes 8 km on a bearing of 060°. Find the bearing of the return journey.

the return bearing is 240°



Mathematical reasoning. A back bearing differs by 180°.

1. The forward bearing is 060°.
2. Adding 180° gives 240°.
3. So the return bearing is 240°.
4. This is the same line travelled in the opposite direction.
5. If the sum exceeds 360°, subtract 360° instead.

Final answer. 240°

Interpretation. A back bearing is found by adding or subtracting 180°, and the three-figure form keeps 240° distinct from 024°.

CHAPTER 10 · INVESTIGATION 10.6

Investigation: measuring the inaccessible

INQUIRY QUESTION

How is a real site measured and drawn, and what does the drawing decide?

YOUR TASK

Survey a small area: a classroom, a garden or a playground. Measure four distances and four bearings, plot the points, check that the traverse closes, and draw the plan to a stated scale. Then find the area and comment on the largest source of error.

point	distance	bearing	east	north
A to B	6.0 m	090°	6.00	0.00
B to C	4.0 m	180°	0.00	−4.00
C to D	6.0 m	270°	−6.00	0.00
D to A	4.0 m	000°	0.00	4.00
closure	—	—	0.00	0.00

CHAPTER 10 · INVESTIGATION 10.6

Investigation: measuring the inaccessible

What do you notice?

- The traverse closed exactly, because the measurements were chosen to do so.
- The plot was a rectangle, which the bearings showed directly.
- The area was 24 m^2, found from the two side lengths.
- In a real survey the closure would not be exact, and the discrepancy would estimate the error.

Observation. A survey is a set of distances and bearings whose components must close back on the start, and the closure error estimates the accuracy of the whole survey.

MATHEMATICAL CONCLUSION

A site is measured by distances and bearings, plotted by components, and checked by closure, and the plan drawn from it carries the survey's accuracy.

Your own words

CHAPTER 10 · 10.6

Investigation: measuring the inaccessible

A site is surveyed by measuring distances and bearings between points, and the points are then plotted by resolving each measurement into components.

A survey is a representation that fixes the shape and the size of a site, and it is the basis of every decision made from a plan.

The survey's accuracy is limited by the instruments, and the errors accumulate along a chain of measurements, which is why a survey closes back on its starting point.

A drawing made from a survey is a model, and a decision taken from it inherits the survey's errors as well as its assumptions.

> FORMAL DEVELOPMENT
>
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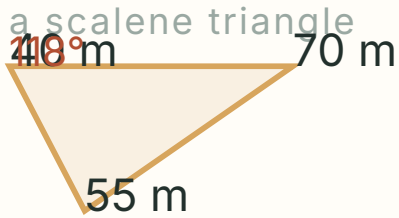
In your own words

CHAPTER 10 · 10.6 · WORKED EXAMPLE 1

Investigation: measuring the inaccessible

WORKED EXAMPLE 10.6.1

Problem. A triangular plot has sides 40 m, 55 m and 70 m. Show that it is not right-angled, and find its largest angle.



Mathematical reasoning. Test Pythagoras first, then use the cosine rule for the largest angle.

1. The longest side is 70 m, and $40^2 + 55^2 = 1600 + 3025 = 4625$.
2. But $70^2 = 4900$, which is not 4625, so the triangle is not right-angled.
3. The largest angle is opposite the 70 m side.
4. $\cos C = (40^2 + 55^2 - 70^2)/(2 \times 40 \times 55) = (4625 - 4900)/4400 = -0.0625$.
5. So $C \approx 93.6°$, an obtuse angle.

Final answer. not right-angled; the largest angle is about 93.6°

Interpretation. The converse of Pythagoras rules out a right angle, and the cosine rule then finds the actual angle.

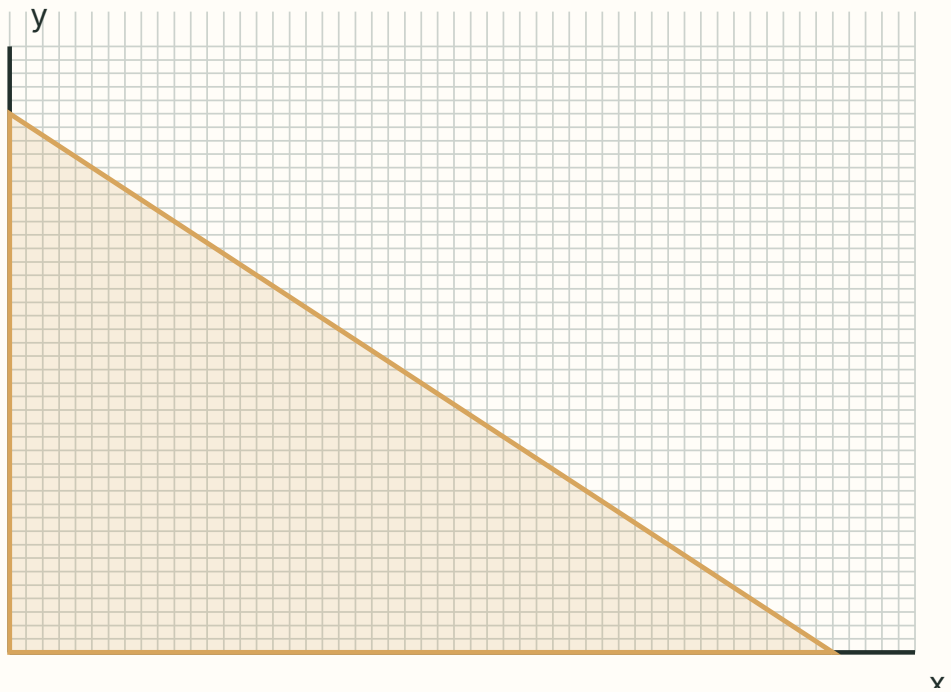
CHAPTER 10 · 10.6 · WORKED EXAMPLE 2

Investigation: measuring the inaccessible

WORKED EXAMPLE 10.6.2

Problem. A surveyor measures three sides of a plot as 30 m, 40 m and 50 m and closes the traverse. Find the area of the plot.

a right-angled plot



Mathematical reasoning. The sides satisfy Pythagoras, so the plot is right-angled and the area is half the product of the legs.

1. $30^2 + 40^2 = 900 + 1600 = 2500 = 50^2$.
2. So the plot is right-angled with legs 30 m and 40 m.
3. The area is $\frac{1}{2} \times 30 \times 40 = 600$ m^2.
4. The hypotenuse of 50 m is the diagonal of the plot.
5. The right angle means no angle measurement was needed.

Final answer. 600 m²

Interpretation. A side measurement that satisfies Pythagoras also establishes a right angle, which is why a builder's 3-4-5 check works.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

10.A1 FLUENCY

Find the hypotenuse of a right-angled triangle with legs 6 cm and 8 cm.

10.A2 FLUENCY

Find the hypotenuse when the legs are 9 cm and 12 cm.

10.A3 FLUENCY

A right-angled triangle has hypotenuse 13 cm and one leg 5 cm. Find the other leg.

10.A4 FLUENCY

Find $\sin 30°$.

10.A5 FLUENCY

Find $\cos 60°$.

10.A6 FLUENCY

Find $\tan 45°$.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

10.A7 FLUENCY

A right-angled triangle has hypotenuse 10 and an angle of 30°. Find the opposite side.

10.A8 FLUENCY

A right-angled triangle has an adjacent side of 8 and an angle of 40°. Find the opposite side.

10.A9 FLUENCY

Find the angle whose sine is 0.5.

10.A10 FLUENCY

A ladder 5 m long leans with its foot 3 m from a wall. Find the height it reaches.

10.A11 FLUENCY

Find the back bearing of 137°.

10.A12 FLUENCY

A triangle has sides 5, 12 and 13. Show that it is right-angled.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

10.B1 REASONING

The diagonals of a rectangle are 13 cm and the sides are in the ratio 5 : 12. Find the sides and justify.

10.B2 REASONING

Investigate the pattern in the values of $\sin\theta$ and $\cos\theta$ for $\theta = 0°, 30°, 45°, 60°, 90°$.

10.B3 REASONING

Conjecture a rule connecting $\sin\theta$ and $\cos\theta$ and justify it.

10.B4 REASONING

Investigate the pattern in the height reached by a 5 m ladder as its foot moves from 0 to 5 m from a wall.

10.B5 REASONING

Build a table of the gradients of lines making angles of $0°$ to $90°$ with the horizontal and describe the pattern.

10.B6 REASONING

Conjecture a rule for the angle of elevation that doubles the height seen at a fixed distance.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

10.B7 CHALLENGE

Investigate the pattern in the areas of right-angled triangles with a fixed hypotenuse of 10 cm.

10.B8 REASONING

A ship sails 30 km east and then 40 km north. Investigate the direct distance and bearing home.

10.B9 REASONING

Find the pattern in the back bearings of 000°, 045°, 090°, 135°, 180°.

10.B10 REASONING

Investigate the error in a height computed from an angle of 89° and a distance of 10 m when the angle is out by 1°.

10.B11 CHALLENGE

Conjecture a rule for the distance to the horizon from a height h.

CRITERION C Communicating

Criterion C practice · Communicating

10.C1 REASONING
Explain in your own words what Pythagoras' theorem states.

10.C2 FLUENCY
Write a complete solution to: find the hypotenuse of a triangle with legs 7 cm and 24 cm.

10.C3 REASONING
Choose between Pythagoras and a trigonometric ratio to find a missing side and justify.

10.C4 FLUENCY
Set out a full solution to: find the angle of elevation of a slope that rises 2 m over 5 m.

10.C5 REASONING
Explain why the choice of trigonometric ratio depends on which sides are known.

10.C6 REASONING
Describe, with an example, the difference between an angle of elevation and an angle of depression.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

10.C7 REASONING

Explain why the angle of elevation equals the angle of depression for two observers.

10.C8 REASONING

Write the reasoning that shows $\sin^2\theta + \cos^2\theta = 1$.

10.C9 REASONING

Set out the argument that a triangle with sides satisfying $a^2 + b^2 = c^2$ is right-angled.

10.C10 REASONING

Explain what a bearing is and why it is written in three figures.

10.C11 REASONING

Describe the relationship between a back bearing and a forward bearing.

10.C12 REASONING

Explain, with an example, when a trigonometric calculation is not enough for a decision.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

10.D1 APPLICATION

A tree is 18 m from an observer whose eye is 1.5 m above the ground. The angle of elevation to the top is 31°. Find the height of the tree.

10.D2 APPLICATION

From a cliff 45 m high, the angle of depression to a boat is 9°. Find the distance of the boat from the base.

10.D3 APPLICATION

A ramp must rise no more than 1 m in 12. Find the greatest angle and the length of a ramp that rises 0.9 m.

10.D4 APPLICATION

A ship sails 25 km on a bearing of 040° and then 20 km on a bearing of 130°. Find the distance and bearing from the start.

10.D5 APPLICATION

A television mast is supported by a wire from a point 30 m from its base. The wire makes an angle of 62° with the ground. Find the height of the mast and the length of the wire.

10.D6 APPLICATION

A plane flies 300 km north and then 400 km east. Find the direct distance and bearing from the start.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

10.D7 REASONING
A roof has a rise of 2.4 m and a run of 6 m. Investigate whether it meets a rule limiting the pitch to 25° and justify.

10.D8 REASONING
A surveyor measures a plot as 25 m, 39 m and 56 m on a traverse that closes. Investigate whether the plot contains a right angle and justify.

10.D9 REASONING
A ladder must reach a window 5 m up with its foot no more than 1.8 m from the wall. Investigate whether a 5.5 m ladder is suitable and justify.

10.D10 CHALLENGE
A drone flies 200 m on a bearing of 315° and then 150 m on a bearing of 045°. Investigate the final position and justify.

10.D11 REASONING
A height is measured from 50 m with an angle of 20°. Investigate the effect on the height of an error of 1° in the angle.

10.D12 REASONING
Explain, with a measurement of your own, when a measurement is too imprecise for the decision being made.

CHAPTER 10 · APPLICATION AND ASSESSMENT

Applying chapter 10

Each task involves a right-angled triangle. Draw it, label the sides relative to the angle you are using, and state which relation you apply.

A1 · The flagpole

A flagpole stands on level ground. From a point 22 m from its base the angle of elevation to the top is 34°, measured with a clinometer held 1.5 m above the ground. Find the height of the flagpole, find the error interval if the distance is 22 ± 0.3 m, and state whether the flagpole is certainly taller than 16 m.

Criteria A and C · use the tangent ratio, carry the bounds through, and justify the conclusion.

A2 · The two bearings

A fishing boat sails 18 km on a bearing of 065° and then 24 km on a bearing of 155°. Find its distance and bearing from the harbour, find the bearing of the direct route home, and explain how the errors in the two bearings would affect the final position.

Criteria B and D · resolve into components, find the resultant, and comment on the accumulation of error.

CHAPTER 10 · REVIEW

What you should now be able to do

Key ideas

- $a^2 + b^2 = c^2$ connects the sides of a right-angled triangle.
- The hypotenuse is opposite the right angle and is the longest side.
- $\sin\theta$ = opposite/hypotenuse, $\cos\theta$ = adjacent/hypotenuse, $\tan\theta$ = opposite/adjacent.
- The ratios depend on the angle alone, so one table serves every triangle.
- $\sin^2\theta + \cos^2\theta = 1$, from dividing Pythagoras by c^2.
- A bearing is measured clockwise from north in three figures.
- A back bearing differs by 180°.
- Errors in the inputs combine in the answer, and relative errors add.

Vocabulary

hypotenuse · leg · converse · sine · cosine · tangent · inverse ratio · angle of elevation · angle of depression · bearing · back bearing · traverse

CHAPTER 10 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

10.R1 Find the hypotenuse when the legs are 8 cm and 15 cm.

10.R2 Find tan 30° to three significant figures.

10.R3 A triangle has hypotenuse 20 and an angle of 50°. Find the opposite side.

10.R4 Find the angle whose tangent is 0.75.

10.R5 Find the back bearing of 246°.

10.R6 Explain why the sine and cosine of complementary angles swap.

CHAPTER 10 · REVIEW

Returning to the statement of inquiry

> **STATEMENT OF INQUIRY**
>
> The sides and angles of a right-angled triangle are related by Pythagoras' theorem and by the trigonometric ratios, so knowing two of them decides the rest; these relations turn an inaccessible length or angle into a measurement, and the equivalence between the theorem and the ratios is what makes surveying possible.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of space, measurement and equivalence appear in your work?
- How is the global context of scientific and technical innovation, and its exploration of surveying, navigation and safe design, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a measurement-based calculation for a safety decision, and what would you still like to ask?

CHAPTER 10 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Surveying and navigation turn inaccessible distances into measurements. Model one and say how accurate your answer is.

Service as Action

COMMUNITY NEED	Heights, slopes and routes are estimated without stating the accuracy.
MATHEMATICAL CONNECTION	Pythagoras, trigonometry and error intervals.
PROPOSED ACTION	Measure one inaccessible height or slope and publish the result with its error interval.
EXPECTED IMPACT	A reader knows how much to trust the figure.
STUDENT REFLECTION	Record the measurements, the calculation and the interval.

Community Project

PROJECT IDEA	An accessibility survey of one route or building.
NEED	Ramps, steps and paths are used without a check against the rules.
RESEARCH QUESTION	Which slopes and heights on our site meet the accessibility rules, and how sure are we?
MATHEMATICS	Trigonometric ratios and error intervals.
ACTION OR OUTCOME	A one-page survey with each slope, its angle and a verdict.
BENEFICIARIES	Everyone who uses the route or building.
EVIDENCE	The survey with the measurements and the intervals.

Personal Project connection · MYP 5

PERSONAL INTEREST	A climb, a flight path or a sports kick.
LEARNING GOAL	To measure something inaccessible and state how accurate the answer is.
PRODUCT OR OUTCOME	A measurement report with an error interval.
MATHEMATICS	Pythagoras and the trigonometric ratios.
PROCESS	Measure a distance and an angle, calculate, and bound the error.
EVIDENCE	The measurements, the calculation and the interval.

CHAPTER 11 · PART III · SPATIAL REASONING

11

Surface Area and Volume

Surface area measures the material; volume measures the space inside. For a fixed volume, a cube wastes less material than a long thin box, and a sphere wastes least of all.

ENGAGE

A drinks company must ship 1 litre of juice. Which container uses least cardboard: a cube, a tall cylinder, or a flat box?

Framing the inquiry

Key concept: Form
Related concepts: Space · Measurement · Quantity
Global context: Globalisation and sustainability
Global context exploration: Packaging, water storage and material efficiency

Statement of inquiry: The surface area of a solid is the quantity of material needed to make it and its volume is the quantity it can hold, and for a fixed volume the shape that minimises surface area is the one that wastes least; the ratio of the two therefore decides how efficient a container is.

Questions to guide us

- **Factual:** What are the formulae for the surface area and volume of prisms, cylinders, pyramids, cones and spheres?
- **Conceptual:** Why does the surface-area-to-volume ratio change with size, and what does that mean for the efficiency of a container?
- **Debatable:** Is the container that uses least material always the one that should be used, or do handling, transport and recycling change the answer?

ATL · TRANSFER. Students transfer the formulae for area and volume between solids, and apply them to a real packaging or storage problem of their own choice.

CHAPTER 11 · INVESTIGATION 11.1

Prisms and cylinders

INQUIRY QUESTION

How are the surface area and volume of a prism found, and why does one formula serve them all?

YOUR TASK

Take five prisms and cylinders of your own, including one with a triangular section, one with a rectangular section and one cylinder. For each, compute the volume and the surface area, then compute the ratio of surface area to volume and rank them.

solid	volume	surface area	area/volume
cube, side 4	64 cm^3	96 cm^2	1.50
cuboid 2 × 4 × 8	64 cm^3	112 cm^2	1.75
cylinder r = 2, h = 5	62.8 cm^3	87.9 cm^2	1.40
cylinder r = 4, h = 1.25	62.8 cm^3	125.7 cm^2	2.00
triangular prism 3-4-5, l = 10	60 cm^3	132 cm^2	2.20

CHAPTER 11 · INVESTIGATION 11.1

Prisms and cylinders

What do you notice?

- All five had about the same volume but very different surface areas.
- The cube had the smallest ratio, and the long thin prism the largest.
- The short wide cylinder was worse than the tall narrow one.
- The ratio fell as the solid became more compact.

Observation. For a fixed volume the surface area depends on the shape, and the compact shapes use least material.

MATHEMATICAL CONCLUSION

A prism's volume is the area of its cross-section times its length and its surface area is twice that area plus the perimeter times the length, and the ratio of the two measures how much material a given capacity requires.

Your own words

CHAPTER 11 · 11.1

Prisms and cylinders

A prism has a constant cross-section, and its volume is the area of the cross-section times the length. One formula serves every prism, whatever the shape of the section.

Every formula in this chapter is a rule for measuring space: the volume measures the space enclosed and the surface area measures the material that encloses it.

The surface area of a prism is twice the area of the cross-section plus the perimeter of the cross-section times the length, because the curved part unrolls into a rectangle.

A cylinder is a prism with a circular cross-section, so its volume is $\pi r^2 h$ and its curved surface area is $2\pi rh$.

Units must be tracked: an area is in square units and a volume in cubic units, and 1 litre is 1000 cm^3, which is the conversion that connects the two.

Density is mass per unit volume, so mass = density × volume, and a capacity in litres is a volume in cubic decimetres because 1 litre = 1000 cm^3.

> FORMAL DEVELOPMENT
>
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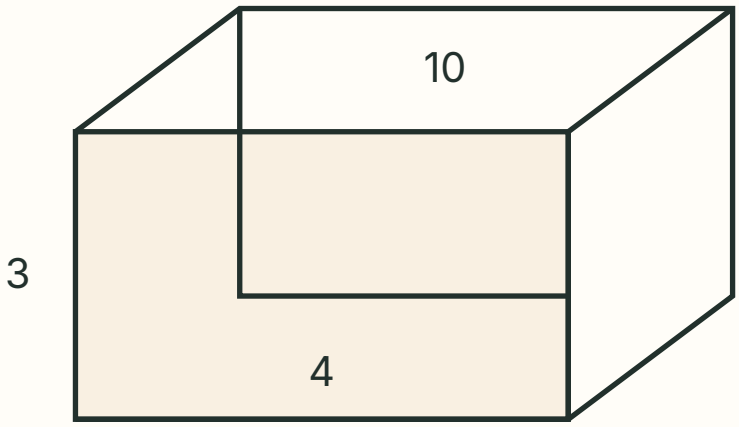
In your own words

CHAPTER 11 · 11.1 · WORKED EXAMPLE 1

Prisms and cylinders

WORKED EXAMPLE 11.1.1

Problem. A triangular prism has a right-angled cross-section with legs 3 cm and 4 cm, and a length of 10 cm. Find its volume and surface area.



a triangular prism of length 10 cm

Mathematical reasoning. Find the cross-section area and perimeter, then apply the two prism formulae.

1. The cross-section is a right triangle with legs 3 and 4, so its area is $\frac{1}{2} \times 3 \times 4 = 6$ cm^2.
2. Its hypotenuse is 5 cm, so the perimeter is $3 + 4 + 5 = 12$ cm.
3. The volume is $6 \times 10 = 60$ cm^3.
4. The surface area is $2 \times 6 + 12 \times 10 = 12 + 120 = 132$ cm^2.
5. The two triangular faces contribute only 12 cm^2 of the 132 cm^2.

Final answer. volume 60 cm^3, surface area 132 cm^2

Interpretation. The prism formulae reduce a three-dimensional problem to a two-dimensional one, which is why they work for every prism.

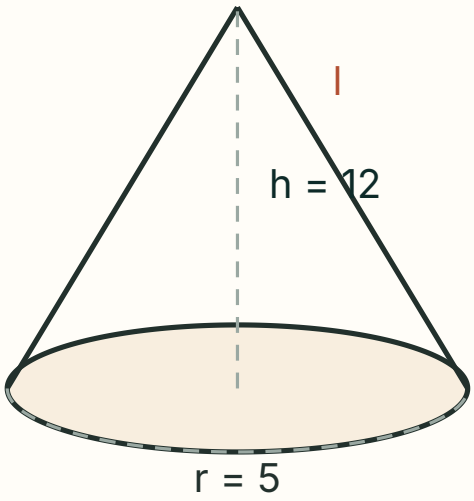
CHAPTER 11 · 11.1 · WORKED EXAMPLE 2

Prisms and cylinders

WORKED EXAMPLE 11.1.2

Problem. A cylinder has radius 5 cm and height 12 cm. Find its volume and total surface area.

a cylinder of radius 5 and height 12



Mathematical reasoning. Apply the cylinder formulae.

1. The volume is $\pi r^2 h = \pi \times 25 \times 12 = 300\pi \approx 942$ cm^3.
2. The curved surface area is $2\pi rh = 2\pi \times 5 \times 12 = 120\pi \approx 377$ cm^2.
3. The two circular ends contribute $2 \times \pi r^2 = 50\pi \approx 157$ cm^2.
4. The total surface area is $170\pi \approx 534$ cm^2.
5. In litres the volume is about 0.942 litres.

Final answer. volume about 942 cm³, surface area about 534 cm²

Interpretation. The two ends are a third of the surface area, which is why a short wide cylinder uses relatively more material than a tall one.

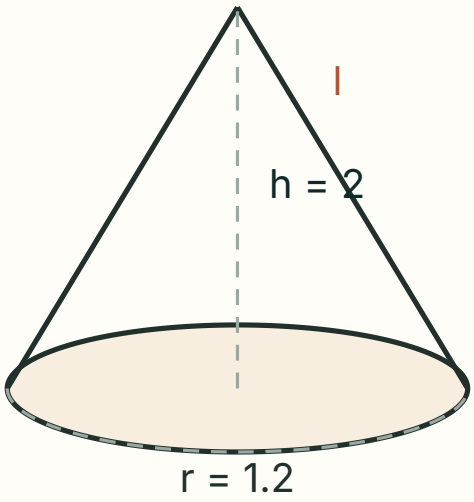
CHAPTER 11 · 11.1 · WORKED EXAMPLE 3

Prisms and cylinders

WORKED EXAMPLE 11.1.3

Problem. A steel tank is a cylinder of radius 1.2 m and height 2 m, made from sheet steel of mass 8 kg per m^2. Find the mass of the empty tank and the mass when full of water.

a cylindrical tank



Mathematical reasoning. Find the surface area for the mass of the steel, and the volume for the mass of the water.

1. The surface area is $2\pi rh + 2\pi r^2 = 2\pi \times 1.2 \times 2 + 2\pi \times 1.44 = 4.8\pi + 2.88\pi = 7.68\pi \approx 24.1\ m^2$.
2. The mass of the steel is $24.1 \times 8 \approx 193$ kg.
3. The volume is $\pi r^2 h = \pi \times 1.44 \times 2 = 2.88\pi \approx 9.05\ m^3$.
4. Water has density 1000 kg/m^3, so the water has mass about 9050 kg.
5. The full tank has mass about 9240 kg, which needs a strong support.

Final answer. about 193 kg empty and about 9240 kg full

Interpretation. The water outweighs the tank by a factor of about 47, which is why the support matters more than the material.

CHAPTER 11 · INVESTIGATION 11.2

Pyramids and cones

INQUIRY QUESTION

How are the formulae for pyramids, cones and spheres obtained, and what do they have in common?

YOUR TASK

Take four solids: a cone, a pyramid, a sphere and a cylinder. For each, find the volume and the surface area, then arrange them so that all have the same volume and compare their surface areas.

solid	volume	surface area	same volume comparison
cone r = 6, h = 8	301 cm^3	301 cm^2	—
sphere of volume 301	301 cm^3	252 cm^2	least material
cylinder of volume 301	301 cm^3	318 cm^2	—
cube of volume 301	301 cm^3	381 cm^2	most material

CHAPTER 11 · INVESTIGATION 11.2

Pyramids and cones

What do you notice?

- The sphere had the smallest surface area for the same volume.
- The cube had the largest.
- The cone was close to the cylinder, because a cone is a third of one.
- The ranking of the shapes was the same for every volume.

> **Observation.** For a fixed volume the sphere has the least surface area and the cube the most among these shapes, and the ordering does not depend on the volume chosen.

MATHEMATICAL CONCLUSION

The formulae for pyramids, cones and spheres all reduce volume and surface area to the base, the height and the radius, and the sphere is the shape that encloses a given volume with least surface.

Your own words

CHAPTER 11 · 11.2

Pyramids and cones

A pyramid has volume one third of the prism with the same base and height, and a cone has volume one third of the cylinder with the same base and height, so V = (1/3) × base area × height for both.

The one-third rule can be checked by pouring: a cone holds exactly a third of what a cylinder of the same base and height holds, which is the experimental reason for the factor.

The curved surface area of a cone is πrl, where l is the slant height, and the slant height, the radius and the height form a right-angled triangle, so Pythagoras finds l from the other two.

A pyramid's surface area is the base area plus the sum of the triangular faces, and each triangular face is found from the slant height of that face rather than from the vertical height.

The one-third rule and the slant-height triangle are the two facts that turn a drawing of a solid into a calculation, and both are consequences of a cross-section.

> FORMAL DEVELOPMENT
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

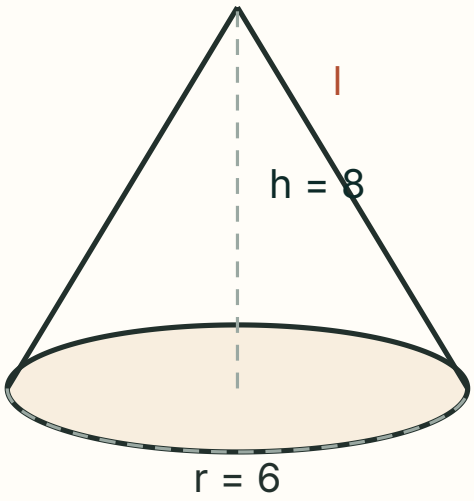
CHAPTER 11 · 11.2 · WORKED EXAMPLE 1

Pyramids and cones

WORKED EXAMPLE 11.2.1

Problem. A cone has radius 6 cm and height 8 cm. Find its volume and curved surface area.

a cone of radius 6 and height 8



Mathematical reasoning. Find the slant height by Pythagoras, then apply the cone formulae.

1. The slant height is $l = \sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$ cm.
2. The volume is $(1/3)\pi r^2 h = (1/3)\pi \times 36 \times 8 = 96\pi \approx 301$ cm^3.
3. The curved surface area is $\pi r l = \pi \times 6 \times 10 = 60\pi \approx 188$ cm^2.
4. The base area is $36\pi \approx 113$ cm^2, so the total is $96\pi \approx 301$ cm^2.
5. A cylinder with the same base and height would hold $288\pi \approx 905$ cm^3, three times as much.

Final answer. volume $96\pi \approx 301$ cm^3, curved surface area $60\pi \approx 188$ cm^2

Interpretation. The slant height is not the height, and using the height in the surface area formula is the commonest error with cones.

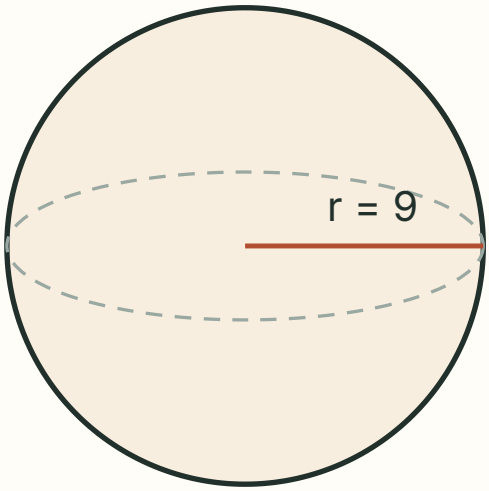
CHAPTER 11 · 11.2 · WORKED EXAMPLE 2

Pyramids and cones

WORKED EXAMPLE 11.2.2

Problem. A sphere has radius 9 cm. Find its volume and surface area, and compare with a cube of the same volume.

a sphere of radius 9



Mathematical reasoning. Apply the sphere formulae, then find the cube's side.

1. The volume is $(4/3)\pi r^3 = (4/3)\pi \times 729 = 972\pi \approx 3054$ cm^3.
2. The surface area is $4\pi r^2 = 4\pi \times 81 = 324\pi \approx 1018$ cm^2.
3. A cube of the same volume has side $\sqrt[3]{3054} \approx 14.5$ cm.
4. Its surface area is $6 \times 14.5^2 \approx 1262$ cm^2.
5. So the sphere uses about 19% less material for the same capacity.

Final answer. volume about 3054 cm³, surface area about 1018 cm², against about 1262 cm² for the cube

Interpretation. The sphere is the most efficient shape for a given volume, which is why balls, tanks and drops are spherical.

CHAPTER 11 · INVESTIGATION 11.3

Spheres and composite solids

INQUIRY QUESTION

How is the volume of a sphere found, and how is a solid made of several parts measured?

YOUR TASK

Take four composite solids: a cylinder with a hemisphere on top, a cuboid with a pyramid on top, a cone with a hemisphere, and a shape of your own. For each, find the total volume and the total surface area, stating clearly which faces are shared and therefore not counted. Then check one volume by displacement in a measuring cylinder.

solid	parts	volume	shared faces
cylinder r = 3, h = 10 + hemisphere r = 3	90π + 18π	108π	one circle
cuboid 4 × 4 × 6 + pyramid base 4 × 4, h = 3	96 + 16	112	one square
cone r = 3, h = 4 + hemisphere r = 3	12π + 18π	30π	one circle

CHAPTER 11 · INVESTIGATION 11.3

Spheres and composite solids

What do you notice?

- Every composite volume was the sum of the parts.
- The shared faces were left out of the surface area but made no difference to the volume.
- The displacement check agreed with the calculation to within the precision of the measuring cylinder.
- A hemisphere is half a sphere, which is why its formula has the factor of one half.

Observation. A composite solid is measured by adding its parts, and the surface area counts only the faces that are actually exposed.

MATHEMATICAL CONCLUSION

Volumes add and exposed faces add, so a composite solid is measured by decomposing it into parts whose formulae are known.

Your own words

CHAPTER 11 · 11.3

Spheres and composite solids

A sphere of radius r has volume $(4/3)\pi r^3$ and surface area $4\pi r^2$, and the two formulae are related because a small increase in the radius adds a shell whose volume is approximately the surface area times the thickness.

The formulae can be checked by experiment: a sphere displaces its own volume of water, and the displacement can be measured in a cylinder, which is the historical reason for the formula.

A composite solid is measured by adding the volumes of its parts and, for the surface area, adding the exposed faces only. A face where two parts meet is not on the surface and must not be counted.

Choosing how to cut a solid into parts is a representation of the problem, and a different cutting can make the same calculation much shorter.

Every quantity in this chapter is a measurement of space: a volume measures the space a solid encloses and a surface area measures the material that encloses it, and the two are reported in different units.

> FORMAL DEVELOPMENT
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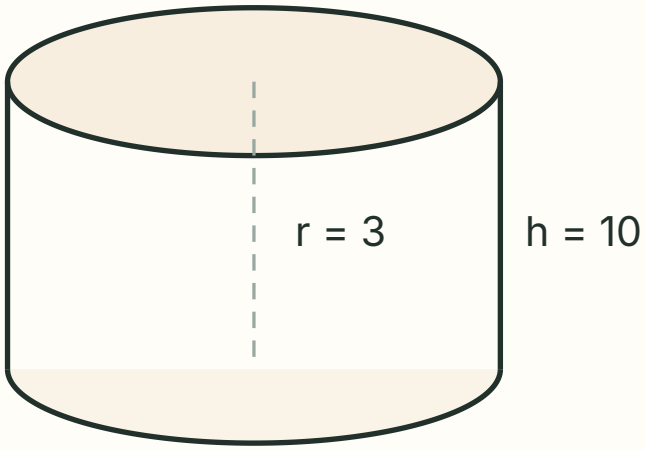
In your own words

CHAPTER 11 · 11.3 · WORKED EXAMPLE 1

Spheres and composite solids

WORKED EXAMPLE 11.3.1

Problem. A solid is a cylinder of radius 3 cm and height 10 cm with a hemisphere of radius 3 cm on top. Find its total volume.



a cylinder of radius 3 cm and height 10 cm

Mathematical reasoning. Add the volume of the cylinder to half the volume of a sphere.

1. The cylinder has volume $\pi \times 3^2 \times 10 = 90\pi$ cm^3.
2. A hemisphere of radius 3 has volume half of $(4/3)\pi \times 3^3 = 18\pi$ cm^3.
3. The total volume is $90\pi + 18\pi = 108\pi \approx 339.3$ cm^3.
4. The two parts share a circular face, which does not affect the volume.
5. For the surface area the shared face would be left out, which is the difference between the two measures.

Final answer. $108\pi \approx 339.3$ cm^3

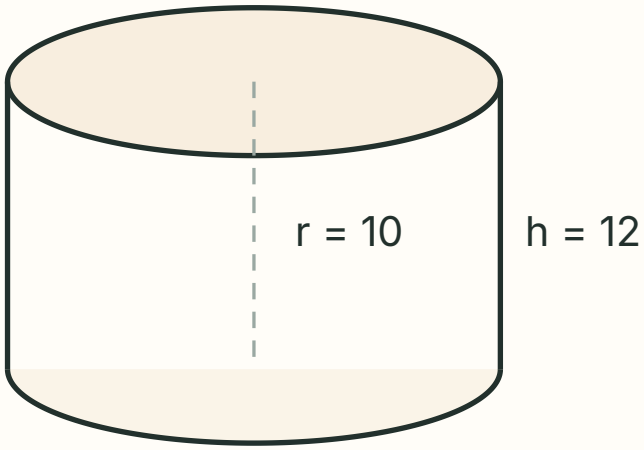
Interpretation. A composite volume is a sum, and the shared face matters only for the surface area.

CHAPTER 11 · 11.3 · WORKED EXAMPLE 2

Spheres and composite solids

WORKED EXAMPLE 11.3.2

Problem. A sphere of radius 5 cm is dropped into a full cylinder of radius 5 cm, and the water that overflows is collected. How much water overflows?



the sphere displaces its own volume

Mathematical reasoning. The overflow is the volume of the sphere.

1. The sphere has volume $(4/3)\pi \times 5^3 = (500/3)\pi \approx 523.6$ cm^3.
2. The displaced water has exactly that volume.
3. So about 523.6 cm^3, or 0.524 litres, overflows.
4. The radius of the cylinder does not matter, only that the sphere fits.
5. This displacement is the experimental check on the formula.

Final answer. about 523.6 cm³, or 0.524 litres

Interpretation. A volume can be measured by displacement, and the measurement is a check on the formula rather than a replacement for it.

CHAPTER 11 · INVESTIGATION 11.4

Scaling surface area and volume

INQUIRY QUESTION

What happens to the surface area and volume of a solid when every length is multiplied?

YOUR TASK

Take three pairs of similar solids: two cubes of different edge, two spheres of different radius, and two cylinders of the same shape but different size. For each pair, find the length factor, then measure or calculate the surface area factor and the volume factor, and check that they are the square and the cube of the length factor. Finally compute the surface-area-to-volume ratio for each member of one pair and see how it changes with size.

pair	length factor	area factor	volume factor
cube 1 cm : cube 3 cm	3	9	27
sphere r : sphere 2r	2	4	8
cylinder h : cylinder 2h	2	4	8
ratio for the cubes	3	6 : 2	1 : 27

CHAPTER 11 · INVESTIGATION 11.4

Scaling surface area and volume

What do you notice?

- Every area factor was the square of the length factor.
- Every volume factor was the cube.
- The surface-area-to-volume ratio fell as the solid grew.
- The sphere kept the smallest ratio of the three shapes at every size.

Observation. Lengths scale by k, areas by k^2 and volumes by k^3, so the surface-area-to-volume ratio scales by 1/k.

MATHEMATICAL CONCLUSION

Scaling multiplies each measure by the scale factor raised to the number of dimensions, and the falling surface-to-volume ratio is why size changes what is efficient.

Your own words

CHAPTER 11 · 11.4

Scaling surface area and volume

When every length of a solid is multiplied by k, the surface area is multiplied by k^2 and the volume by k^3, because an area counts two dimensions and a volume counts three.

The ratio of surface area to volume is therefore divided by k, so a larger solid of the same shape has proportionally less surface for each unit of volume. This is why large animals cool slowly and large packets are cheaper per litre.

The cube of the scale factor is the one that surprises: doubling a length multiplies the volume by eight, so a model twice as long as the original needs eight times the material to fill.

Scaling is a representation of an object at a different size, and it is valid only for the quantities that scale with length. Mass, strength and cost depend on density and price as well, so they need their own model.

> FORMAL DEVELOPMENT
>
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In your own words

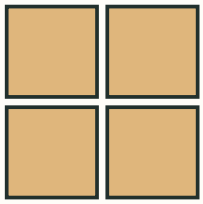
CHAPTER 11 · 11.4 · WORKED EXAMPLE 1

Scaling surface area and volume

WORKED EXAMPLE 11.4.1

Problem. A cube has edges of 2 cm. Find its surface area and volume. Then find both for a cube whose edges are three times as long.

a cube of edge 2 cm has 6 faces of 4 cm^2



surface 24 cm^2, volume 8 cm^3

Mathematical reasoning. Apply the scale factor to each measure with the right power.

1. The small cube has surface area $6 \times 2^2 = 24$ cm^2 and volume $2^3 = 8$ cm^3.
2. The large cube has edges of 6 cm.
3. Its surface area is $6 \times 6^2 = 216$ cm^2, which is 9 times the small one.
4. Its volume is $6^3 = 216$ cm^3, which is 27 times the small one.
5. So a length factor of 3 gave an area factor of 9 and a volume factor of 27.

Final answer. 24 cm^2 and 8 cm^3; 216 cm^2 and 216 cm^3

Interpretation. The powers of the scale factor are the number of dimensions in the quantity, and confusing them is the commonest error.

CHAPTER 11 · 11.4 · WORKED EXAMPLE 2

Scaling surface area and volume

WORKED EXAMPLE 11.4.2

Problem. Two similar bottles have heights in the ratio 1 : 4. The small bottle holds 250 ml. How much does the large one hold?

measure	small	large	factor
height	h	4h	4
surface area	S	16S	16
volume	250 ml	16 litres	64

Mathematical reasoning. The volume scales by the cube of the length factor.

1. The length factor is 4.
2. The volume factor is $4^3 = 64$.
3. The large bottle holds 64×250 ml = 16 000 ml.
4. That is 16 litres, which is 64 times the small bottle.
5. The surface area is only 16 times as large, so the large bottle uses proportionally less material per litre.

Final answer. 16 litres

Interpretation. Capacity grows far faster than height, and faster than the material needed to make the bottle.

CHAPTER 11 · INVESTIGATION 11.5

Packaging, cost and design

INQUIRY QUESTION

How is the efficiency of a container measured, and why does size matter?

YOUR TASK

Choose a real container: a drinks carton, a tin, a bottle or a water tank. Measure or find its dimensions, compute its volume and surface area, and compare with the most efficient shape of the same volume. Then list the reasons the real container is not the most efficient one.

container	volume	surface area	ratio	efficient shape ratio
drinks carton	1000 cm^3	630 cm^2	0.63	0.48
tin	330 cm^3	290 cm^2	0.88	0.56
bottle	500 cm^3	420 cm^2	0.84	0.53

CHAPTER 11 · INVESTIGATION 11.5

Packaging, cost and design

What do you notice?

- Every real container used more material than the efficient shape.
- The carton was closest, because it is nearly a cuboid of good proportions.
- The tin was worst, because a cylinder taller than it is wide is inefficient.
- Handling, stacking and printing all explain the differences.

Observation. Real containers use more material than the most efficient shape of the same volume, and the difference is explained by constraints other than material.

MATHEMATICAL CONCLUSION

The efficiency of a container is measured by its surface-area-to-volume ratio, and the real choice is decided by constraints that a material model does not include.

Your own words

CHAPTER 11 · 11.5

Packaging, cost and design

The efficiency of a container is measured by the ratio of its surface area to its volume, and a smaller ratio means less material for the same capacity.

For similar shapes the ratio is proportional to 1/length, so a container twice as large has half the ratio. This is why large containers are cheaper per unit volume.

A cube of side a has ratio 6/a, a cylinder of radius r and height 2r has ratio 3/r, and a sphere has ratio 3/r. The sphere is the best of the three.

Efficiency is only one criterion. Handling, stacking, transport and recycling all affect the real choice, and a model of material alone can recommend the wrong container.

> FORMAL DEVELOPMENT
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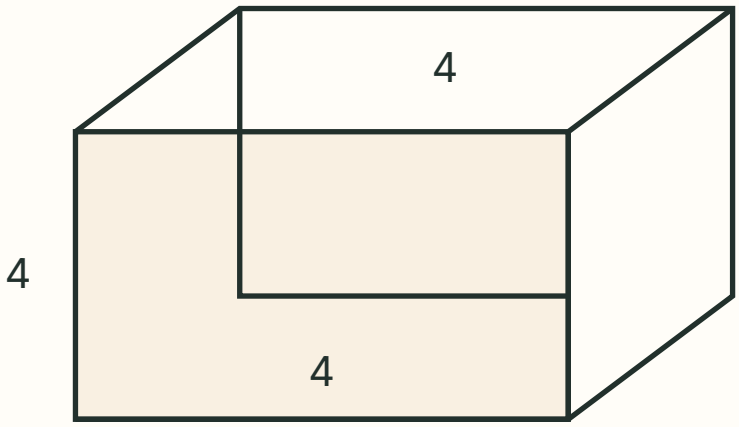
In your own words

CHAPTER 11 · 11.5 · WORKED EXAMPLE 1

Packaging, cost and design

WORKED EXAMPLE 11.5.1

Problem. Compare the efficiency of a cube of side 10 cm and a cylinder of radius 5 cm and height 10 cm, both holding about 1 litre.



a cube of side 10 cm

Mathematical reasoning. Compute both volumes and both surface areas, then the ratios.

1. The cube has volume 1000 cm^3 and surface area $6 \times 100 = 600$ cm^2, so the ratio is 0.60.
2. The cylinder has volume $\pi \times 25 \times 10 = 250\pi \approx 785$ cm^3.
3. Its surface area is $2\pi \times 5 \times 10 + 2 \times 25\pi = 150\pi \approx 471$ cm^2, so the ratio is 0.60.
4. The cylinder holds less, so the comparison is not yet fair.
5. Scaling the cylinder to 1000 cm^3 gives a height of about 12.7 cm and a ratio of about 0.54, so it is more efficient.

Final answer. the cube has ratio 0.60; the cylinder scaled to the same volume has about 0.54

Interpretation. Comparing ratios requires the same volume, and the cylinder wins because it is closer to a sphere.

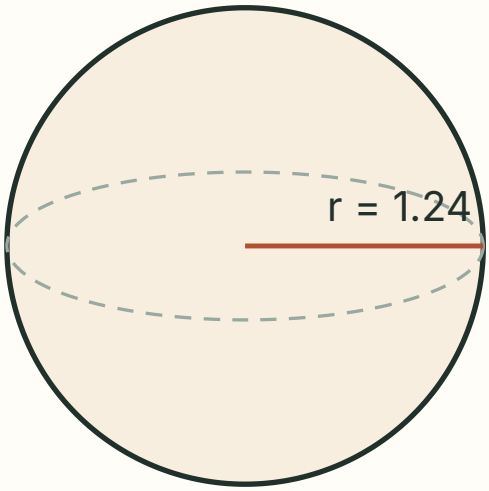
CHAPTER 11 · 11.5 · WORKED EXAMPLE 2

Packaging, cost and design

WORKED EXAMPLE 11.5.2

Problem. A spherical tank and a cubical tank each hold 8 m^3. Find the surface area of each and the material saved by the sphere.

a sphere holding 8 m^3



Mathematical reasoning. Find the radius from the volume, then the surface area.

1. For the sphere, $(4/3)\pi r^3 = 8$, so $r^3 = 6/\pi$ and $r \approx 1.24$ m.
2. The surface area is $4\pi r^2 \approx 4\pi \times 1.54 \approx 19.4$ m^2.
3. The cube has side 2 m and surface area $6 \times 4 = 24$ m^2.
4. So the sphere saves $24 - 19.4 = 4.6$ m^2 of material.
5. That is about 19% less material for the same capacity.

Final answer. sphere about 19.4 m^2, cube 24 m^2, saving about 4.6 m^2

Interpretation. The saving is a fifth of the material, which over many tanks is a substantial quantity of metal.

CHAPTER 11 · INVESTIGATION 11.6

Investigation: the most efficient package

INQUIRY QUESTION

Which container for a given capacity uses least material, and what does the answer ignore?

YOUR TASK

Design a container for a stated capacity and a stated constraint, such as standing on a shelf, fitting in a hand, or stacking in a lorry. Model the material used, find the mathematical optimum, then choose a design and justify the departure from the optimum.

design	capacity	surface area	departure from optimum
optimal cylinder	1000 cm^3	549 cm^2	none
short wide can	1000 cm^3	559 cm^2	2% more material
cuboid carton	1000 cm^3	600 cm^2	9% more material
sphere	1000 cm^3	484 cm^2	12% less, but rolls

CHAPTER 11 · INVESTIGATION 11.6

Investigation: the most efficient package

What do you notice?

- The sphere used least material but cannot stand or stack.
- The carton used most but stacks and prints well.
- Every real design departed from the optimum for a stated reason.
- The departure was always justified by a constraint, not by ignorance.

> **Observation.** The mathematical optimum minimises material alone, and every real design departs from it because of constraints the model does not contain.

MATHEMATICAL CONCLUSION

A container is designed by minimising surface area for a fixed volume and then relaxing the optimum to satisfy the constraints of use, and the report of a design states both.

Your own words

CHAPTER 11 · 11.6

Investigation: the most efficient package

For a given capacity the shape with least surface area is the sphere, and among cylinders the one whose height equals its diameter.

The optimal proportions can be found by writing the surface area in terms of one variable, using the volume constraint, and minimising.

A real container must stand, stack, be held and be filled, and each of these adds a constraint that the mathematical optimum ignores.

A model of material alone is a first step, and a good design report states what the model ignored as well as what it found.

> FORMAL DEVELOPMENT
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In your own words

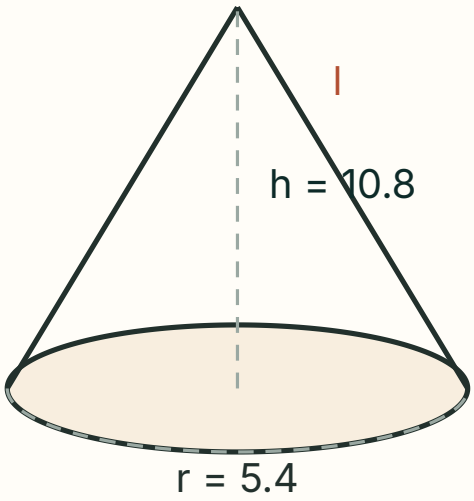
CHAPTER 11 · 11.6 · WORKED EXAMPLE 1

Investigation: the most efficient package

WORKED EXAMPLE 11.6.1

Problem. Find the dimensions of the cylinder of volume 1000 cm^3 with the least surface area.

the optimal cylinder has h = 2r



Mathematical reasoning. Write the surface area in terms of r alone using the volume constraint, then minimise.

1. Let the radius be r and the height h, with $\pi r^2 h = 1000$.
2. So $h = 1000/(\pi r^2)$.
3. The surface area is $2\pi r^2 + 2\pi rh = 2\pi r^2 + 2000/r$.
4. Differentiating and setting to zero gives $4\pi r = 2000/r^2$, so $r^3 = 500/\pi$.
5. Hence $r \approx 5.42$ cm and $h \approx 10.8$ cm, so $h = 2r$.

Final answer. radius about 5.42 cm and height about 10.8 cm, with h = 2r

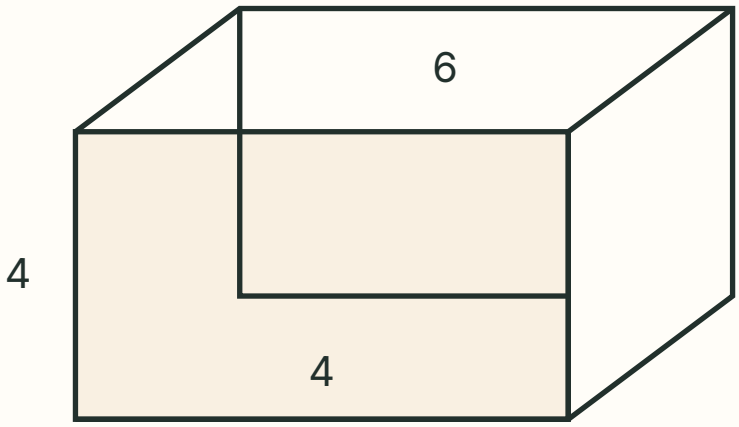
Interpretation. The optimal cylinder has height equal to its diameter, which is a result about shape rather than about size.

CHAPTER 11 · 11.6 · WORKED EXAMPLE 2

Investigation: the most efficient package

WORKED EXAMPLE 11.6.2

Problem. A drinks company must ship 1 litre in a cylinder that can stand on a shelf. Compare the optimal cylinder with a shorter, wider one and comment on the choice.



a shorter, wider cylinder

Mathematical reasoning. Compare the surface areas of two cylinders of the same volume.

1. The optimal cylinder has $r \approx 5.42$ cm, $h \approx 10.8$ cm and surface area about 549 cm^2.
2. A cylinder with $r = 6$ cm has $h = 1000/(36\pi) \approx 8.84$ cm and surface area about 559 cm^2.
3. The difference is only about 10 cm^2, or 2%.
4. The shorter can is easier to hold and to stack, which may be worth more than the 2% saving.
5. So the mathematical optimum is not automatically the best design.

Final answer. the optimal cylinder uses about 549 cm^2; the shorter one about 559 cm^2

Interpretation. A 2% material saving can be outweighed by handling and transport, which is why the optimum is a starting point rather than an answer.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

11.A1 FLUENCY

Find the volume of a cuboid 5 cm by 4 cm by 3 cm.

11.A2 FLUENCY

Find the surface area of a cube of side 6 cm.

11.A3 FLUENCY

Find the volume of a cylinder of radius 3 cm and height 7 cm.

11.A4 FLUENCY

Find the curved surface area of a cylinder of radius 4 cm and height 9 cm.

11.A5 FLUENCY

Find the volume of a cone of radius 3 cm and height 8 cm.

11.A6 FLUENCY

Find the volume of a sphere of radius 6 cm.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

11.A7 FLUENCY

Find the surface area of a sphere of radius 5 cm.

11.A8 FLUENCY

A triangular prism has cross-section area 12 cm^2 and length 8 cm. Find its volume.

11.A9 FLUENCY

Convert 2.5 litres to cubic centimetres.

11.A10 FLUENCY

Find the slant height of a cone of radius 5 cm and height 12 cm.

11.A11 FLUENCY

Find the volume of a pyramid with base area 48 cm^2 and height 9 cm.

11.A12 FLUENCY

A cuboid has volume 240 cm^3 and base area 40 cm^2. Find its height.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

11.B1 REASONING
The surface areas of similar cubes are in the ratio 4 : 9. Find the ratio of their volumes and justify.

11.B2 REASONING
Investigate the pattern in the surface-area-to-volume ratio of cubes of side 1 to 6 cm.

11.B3 CHALLENGE
Conjecture a rule for the dimensions of the cylinder of fixed volume with least surface area and justify.

11.B4 REASONING
Investigate the pattern in the volumes of cylinders of fixed surface area as the radius grows.

11.B5 REASONING
Build a table of the volumes of spheres of radius 1 to 5 and describe the pattern.

11.B6 REASONING
Conjecture a rule for the ratio of the surface area to the volume of a sphere of radius r.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

11.B7 REASONING

Investigate the pattern in the material needed for 1-litre containers of different shapes.

11.B8 REASONING

A can's height is twice its radius. Investigate the effect on its surface-area-to-volume ratio of doubling the radius.

11.B9 REASONING

Find the pattern in the number of 1 cm cubes needed to build a cube of side n cm.

11.B10 REASONING

Investigate the pattern in the surface area of a stack of n cubes of side 1 cm.

11.B11 REASONING

Conjecture a rule for the volume of a solid formed by rotating a right triangle about one leg.

CRITERION C Communicating

Criterion C practice · Communicating

11.C1 REASONING
Explain in your own words the difference between surface area and volume.

11.C2 FLUENCY
Write a complete solution to: find the volume and surface area of a cylinder of radius 4 cm and height 10 cm.

11.C3 REASONING
Choose between surface area and volume to answer a question about how much a container holds and justify.

11.C4 FLUENCY
Set out a full solution to: find the curved surface area of a cone of radius 5 cm and slant height 13 cm.

11.C5 REASONING
Explain why the surface-area-to-volume ratio falls as a container grows.

11.C6 REASONING
Describe, with an example, the difference between a prism and a cylinder.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

11.C7 REASONING
Explain why the volume of a cone is one third of the cylinder with the same base and height.

11.C8 REASONING
Write the reasoning that shows the slant height of a cone is not its height.

11.C9 CHALLENGE
Set out the argument that the sphere encloses a given volume with the least surface area.

11.C10 REASONING
Explain what density is and how it connects mass to volume.

11.C11 REASONING
Describe the relationship between the mass of a container and its surface area.

11.C12 REASONING
Explain, with an example, when the most efficient container is not the best choice.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

11.D1 APPLICATION

A tank is a cylinder of radius 0.8 m and height 1.5 m. Find its capacity in litres and the material needed for its curved surface.

11.D2 APPLICATION

A box measures 30 cm by 20 cm by 15 cm. Find its volume, its surface area, and the amount of cardboard for 100 boxes with 10% extra for joins.

11.D3 APPLICATION

A spherical tank holds 10 m^3. Find its radius and the material needed for its surface.

11.D4 APPLICATION

A cone-shaped funnel has radius 6 cm and height 10 cm. Find its capacity in millilitres and the area of tin needed for its curved surface.

11.D5 APPLICATION

A manufacturer must package 2 litres. Compare the material for a cube, an optimal cylinder and a sphere, and recommend one with a reason.

11.D6 APPLICATION

A glass jar holds 750 ml. Its surface area is 480 cm^2 and the glass has mass 0.5 g per cm^2. Find the mass of the empty jar and the fraction of the full mass that is glass.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

11.D7 REASONING

A concrete pipe is a cylinder of outer radius 0.5 m, inner radius 0.4 m and length 2 m. Investigate the volume of concrete and the mass at 2400 kg per m^3.

11.D8 APPLICATION

A grain silo is a cylinder of radius 3 m and height 8 m with a hemispherical roof. Find its capacity and its external surface area.

11.D9 REASONING

A company ships balls of radius 4 cm in cuboid boxes of 8 cm by 8 cm by 8 cm. Investigate the fraction of the box volume occupied and the material wasted.

11.D10 CHALLENGE

A water tank is a cuboid 2 m by 1.5 m by 1 m. Investigate the dimensions of a cylinder of the same capacity with the least surface area.

11.D11 CHALLENGE

A packet of biscuits is a cuboid 20 cm by 12 cm by 6 cm. Investigate the material saved by reducing the height to 5 cm and keeping the volume.

11.D12 REASONING

Explain, with a container of your own, when a material model is not enough to choose a design.

CHAPTER 11 · APPLICATION AND ASSESSMENT

Applying chapter 11

Each task involves a solid and a quantity. State the formula you use, the units of your answer, and whether you are measuring material or capacity.

A1 · The water tank

A community needs a tank holding 10 m^3 of water. Three designs are proposed: a cube, a cylinder whose height equals its diameter, and a sphere. Find the surface area of each, the mass of steel needed at 10 kg per m^2, and the mass of the water each holds. Recommend one design and justify the recommendation, stating at least one constraint your model ignores.

Criteria A and D · compute all three designs, compare them, and justify the recommendation.

A2 · The packaging audit

A product of volume 500 cm^3 is sold in a cuboid measuring 10 cm by 10 cm by 5 cm. Find the surface area of the box, the surface area of the most efficient cylinder of the same volume, and the material saved. Then investigate the effect on the saving of allowing the cylinder to be twice as tall as it is wide, and explain why a manufacturer might still choose the cuboid.

Criteria B and C · use the volume constraint, minimise the surface area, and justify the choice.

CHAPTER 11 · REVIEW

What you should now be able to do

Key ideas

- A prism's volume is the cross-section area times its length.
- A prism's surface area is twice the cross-section area plus the perimeter times the length.
- A cone or pyramid has one third the volume of the corresponding prism or cylinder.
- A cone's curved surface area is πrl, where l is the slant height.
- A sphere has volume $(4/3)\pi r^3$ and surface area $4\pi r^2$.
- Surface area scales as k^2 and volume as k^3, so the ratio scales as $1/k$.
- Mass = density × volume, and material mass = surface area × mass per area.
- The sphere encloses a given volume with least surface.

Vocabulary

prism · cross-section · cylinder · cone · pyramid · sphere · slant height · density · capacity · surface-area-to-volume ratio · isoperimetric · efficient shape

CHAPTER 11 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

11.R1 Find the volume and surface area of a cube of side 7 cm.

11.R2 Find the volume of a cone of radius 4 cm and height 9 cm.

11.R3 Find the surface area of a sphere of radius 3.5 cm.

11.R4 Convert 0.75 litres to cubic centimetres.

11.R5 Find the slant height of a cone of radius 8 cm and height 15 cm.

11.R6 Explain why a large tank uses less material per litre than a small one.

CHAPTER 11 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The surface area of a solid is the quantity of material needed to make it and its volume is the quantity it can hold, and for a fixed volume the shape that minimises surface area is the one that wastes least; the ratio of the two therefore decides how efficient a container is.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of space, measurement and quantity appear in your work?
- How is the global context of globalisation and sustainability, and its exploration of packaging, water storage and material efficiency, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a material model for a design decision, and what would you still like to ask?

CHAPTER 11 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Packaging is a mathematical optimisation with an environmental consequence. Model one container and say what your model ignores.

Service as Action

COMMUNITY NEED	Water storage and packaging decisions are made without the material model.
MATHEMATICAL CONNECTION	Surface area, volume and efficiency.
PROPOSED ACTION	Model the containers used for one community need and publish the material used by each.
EXPECTED IMPACT	A group can choose the design that wastes least.
STUDENT REFLECTION	Record the models, the comparison and the recommendation.

Community Project

PROJECT IDEA	A packaging audit of one product sold locally.
NEED	Packaging is chosen by appearance and cost per unit, with no account of material.
RESEARCH QUESTION	How much material does each design use, and what would the efficient shape use?
MATHEMATICS	Surface area, volume and ratios.
ACTION OR OUTCOME	A one-page comparison of three designs with the material used by each.
BENEFICIARIES	Everyone who buys and disposes of the packaging.
EVIDENCE	The comparison with the models and the recommendation.

Personal Project connection · MYP 5

PERSONAL INTEREST	A container, a box or a tank you use.
LEARNING GOAL	To model the material used by a container and find the efficient shape.
PRODUCT OR OUTCOME	A model with a comparison and a recommendation.
MATHEMATICS	Surface area, volume and ratios.
PROCESS	Measure a container, compute both quantities, and compare with the efficient shape.
EVIDENCE	The model, the comparison and the recommendation.

CHAPTER 12 · PART IV · REASONING WITH DATA

12

Statistics: Distributions and Data

An average summarises a group; a spread says how typical that average is. The same numbers can be made to look very different by the choice of display.

ENGAGE

A school reports that its average class size is 24. Half the classes have 18 students and half have 30. Is the school's figure honest, and is it useful?

Framing the inquiry

Key concept: Relationships
Related concepts: Statistics · Representation · Quantity
Global context: Identities and relationships
Global context exploration: How data about groups shape decisions about individuals

Statement of inquiry: A data set is a collection of quantities that describe a group, and its distribution is a representation of how those quantities are spread; the average and the spread are two different statements about the same group, and a decision about an individual made from a group statistic is an inference whose strength depends on how the data were collected.

Questions to guide us

- **Factual:** What are the mean, median, mode and range, and how are cumulative frequency and box plots constructed?
- **Conceptual:** Why does the choice of average matter, and how does a sampling method affect what a conclusion can claim?
- **Debatable:** Is it fair to make a decision about an individual from a statistic about a group, and when does it become discrimination?

ATL · INFORMATION LITERACY. Students evaluate the source, the sampling method and the presentation of a data set before using it, and state what the data can and cannot support.

CHAPTER 12 · INVESTIGATION 12.1

Collecting and classifying data

INQUIRY QUESTION

How does the way data are collected affect what a conclusion can claim?

YOUR TASK

Find two published surveys: one you trust and one you do not. For each, identify the population, the sampling method, the sample size and the claim made, and decide whether the claim is supported.

survey	population	method	supported?
national census	everyone	complete count	yes
gym exit survey	town	convenience	no
online poll	voters	self-selected	no
school register	students	complete count	yes

CHAPTER 12 · INVESTIGATION 12.1

Collecting and classifying data

What do you notice?

- The complete counts supported their claims exactly.
- The convenience and self-selected samples did not.
- Sample size was irrelevant where the method was biased.
- The population and the sampling frame had to match for the claim to hold.

Observation. A claim about a population is supported only when the sampling frame matches the population and the selection is unbiased.

MATHEMATICAL CONCLUSION

An inference from a sample to a population depends on the sampling method, and a biased sample supports no conclusion about the population however large it is.

Your own words

CHAPTER 12 · 12.1

Collecting and classifying data

Data are classified before they are displayed: a variable is categorical when its values are names and numerical when they are numbers, and a numerical variable is discrete when it is counted and continuous when it is measured.

Primary data are collected for the question in hand and secondary data were collected for another purpose, and the second kind often needs reclassifying before it can answer a new question.

A population is the whole group a claim is about, and a sample is the part that is measured. A conclusion about the population is an inference from the sample.

A random sample gives every member of the population the same chance of selection, and it is the only method that supports a claim about the population without bias.

A stratified sample takes each group in proportion to its share of the population, which guarantees that small groups are represented.

A biased sample supports no conclusion about the population, however large it is. Sample size reduces random error but cannot remove bias, and the two are different problems.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 12 · 12.1 · WORKED EXAMPLE 1

Collecting and classifying data

WORKED EXAMPLE 12.1.1

Problem. A school has 720 students in four year groups of 180, 200, 170 and 170. A stratified sample of 72 is to be taken. Find the number from each group.

group	population	share	sample
Year 6	180	25%	18
Year 7	200	27.8%	20
Year 8	170	23.6%	17
Year 9	170	23.6%	17

each group is sampled in proportion to its size

Mathematical reasoning. Take the same fraction of each group as the sample is of the population.

1. The sampling fraction is $72/720 = 0.1$.
2. Year 6 contributes $180 \times 0.1 = 18$ students.
3. Year 7 contributes $200 \times 0.1 = 20$ students.
4. Year 8 contributes $170 \times 0.1 = 17$ students.
5. Year 9 contributes $170 \times 0.1 = 17$ students, and the total is 72.

Final answer. 18, 20, 17 and 17 students

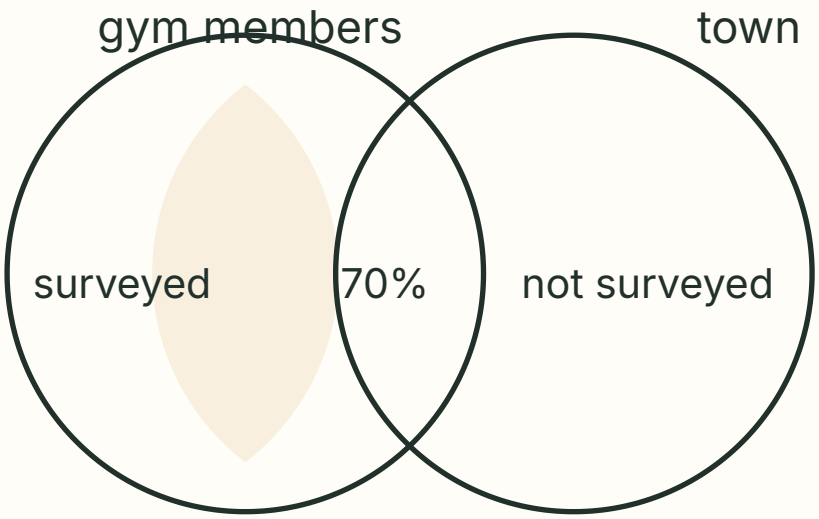
Interpretation. A stratified sample guarantees that every group appears in proportion, which a simple random sample does not.

CHAPTER 12 · 12.1 · WORKED EXAMPLE 2

Collecting and classifying data

WORKED EXAMPLE 12.1.2

Problem. A survey of 200 people leaving a gym finds that 70% exercise three times a week. Explain why this does not describe the town.



Mathematical reasoning. Identify the population and the sampling frame, and decide whether the sample represents the population.

1. The population is the town, but the sample is drawn only from people leaving a gym.
2. People leaving a gym are more likely to exercise than the general population.
3. So the sample is biased, and the figure of 70% describes the sample, not the town.
4. A larger sample of the same kind would not fix the bias.
5. A random sample of the town would be needed for a claim about the town.

Final answer. the sample is biased, so the figure describes gym users, not the town

Interpretation. Bias is a property of how the sample was chosen, and no amount of data from a biased sample removes it.

CHAPTER 12 · INVESTIGATION 12.2

Frequency distributions and histograms

INQUIRY QUESTION

How does the choice of display change what a reader sees?

YOUR TASK

Collect one set of continuous data with at least 30 values: heights, times or masses. Group it into classes of equal width, draw a frequency table with a frequency density column, and draw the histogram. Then regroup the same data into unequal classes and draw the histogram again, and say which display shows the shape more clearly.

class	frequency	width	frequency density
$10 \le t < 20$	6	10	0.6
$20 \le t < 30$	14	10	1.4
$30 \le t < 40$	9	10	0.9
$40 \le t < 60$	5	20	0.25
$60 \le t < 80$	2	20	0.1

CHAPTER 12 · INVESTIGATION 12.2

Frequency distributions and histograms

What do you notice?

- The frequency density column was needed because the last two classes were twice as wide.
- With equal classes the bar heights and the frequencies told the same story.
- With unequal classes the heights alone were misleading, and only the areas were correct.
- The narrower classes showed a bumpier shape than the wider ones.

Observation. A histogram draws frequency density rather than frequency, so the area of each bar is the frequency and unequal classes can be used.

MATHEMATICAL CONCLUSION

Grouping data into classes is the first step of a display, and the histogram's use of density is what allows classes of different widths to appear on one pair of axes.

Your own words

CHAPTER 12 · 12.2

Frequency distributions and histograms

A frequency table groups data into classes, and the class width is the difference between the class boundaries. Equal widths make the heights of the bars directly comparable.

A histogram has frequency density on the vertical axis, where frequency density is frequency divided by class width, so the area of each bar represents the frequency. This is what makes the histogram suitable for unequal class widths.

Grouping loses information: the exact values are replaced by classes, and a class width that is too wide hides the shape while one that is too narrow makes the display noisy.

The shape of a histogram is a representation of the distribution, and the words used to describe it — symmetric, skewed, bimodal — are claims about the population as well as about the sample.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 12 · 12.2 · WORKED EXAMPLE 1

Frequency distributions and histograms

WORKED EXAMPLE 12.2.1

Problem. Draw a histogram for data grouped into classes of width 5 with frequencies 4, 11, 20, 9, 3, and state the frequency density of the tallest class.



Mathematical reasoning. Divide each frequency by the class width to get the frequency density.

1. The class width is 5 for every class.
2. The frequency densities are 4/5, 11/5, 20/5, 9/5 and 3/5.
3. That is 0.8, 2.2, 4.0, 1.8 and 0.6.
4. The tallest class has density 4.0, and its area is $4.0 \times 5 = 20$, the frequency.
5. The total area is 47, the total frequency.

Final answer. frequency densities 0.8, 2.2, 4.0, 1.8, 0.6

Interpretation. With equal class widths the shape of the histogram is the same whether the axis is frequency or frequency density, and only the scale changes.

CHAPTER 12 · INVESTIGATION 12.3

Cumulative frequency and box plots

INQUIRY QUESTION

How is a cumulative frequency diagram built, and what does a box plot add to it?

YOUR TASK

Take the data set you grouped in the previous investigation. Draw a cumulative frequency diagram, read the median and the quartiles from it, and then draw a box plot on the same axis as a box plot for a second data set of your own. Compare the two distributions in terms of centre, spread and shape, and mark any value that lies more than 1.5 interquartile ranges beyond a quartile.

quantity	data set 1	data set 2
minimum	12	20
lower quartile	22	31
median	27	38
upper quartile	35	47
maximum	48	60
interquartile range	13	16

CHAPTER 12 · INVESTIGATION 12.3

Cumulative frequency and box plots

What do you notice?

- The quartiles read from the diagram matched those from the ordered data.
- The second data set was shifted upwards but had a similar spread.
- The box plots made the shift visible at a glance.
- No value lay far enough beyond a quartile to be marked as an outlier.

> **Observation.** A cumulative frequency diagram gives the quartiles directly, and a box plot is the compact picture of the same five values.

MATHEMATICAL CONCLUSION

The cumulative diagram and the box plot are two representations of the same five-number summary, and the box plot is the one that supports comparison.

Your own words

CHAPTER 12 · 12.3

Cumulative frequency and box plots

A cumulative frequency diagram plots the running total of frequencies against the upper class boundary, so the height at any value is the number of observations at or below it.

The gradient of a cumulative frequency diagram is the frequency density, so a steep part of the curve is a class with many observations and a flat part is a class with few.

The median, the lower quartile and the upper quartile are read from the diagram at the half, the quarter and the three-quarter points of the total frequency, which is why the total must be known before the diagram can be read.

A box plot shows the minimum, the lower quartile, the median, the upper quartile and the maximum in one compact picture, and it is the best display for comparing several distributions on the same axis.

A value more than 1.5 interquartile ranges beyond a quartile is usually marked as an outlier, and an outlier is a value that deserves a second look rather than automatic deletion.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

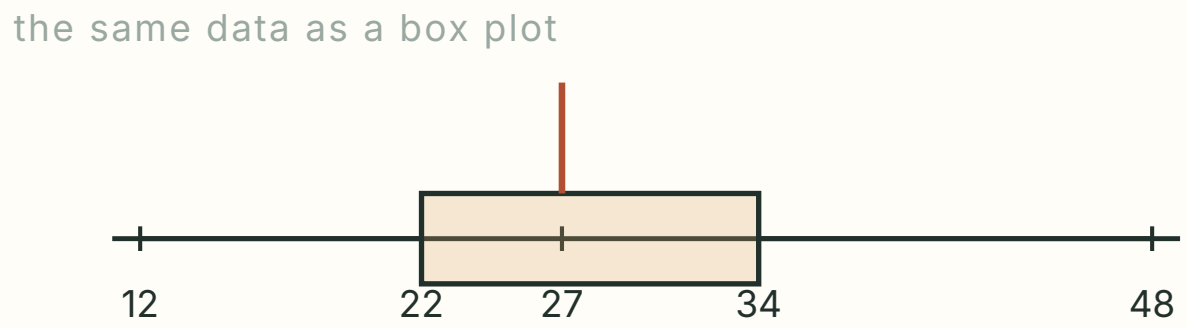
In your own words

CHAPTER 12 · 12.3 · WORKED EXAMPLE 1

Cumulative frequency and box plots

WORKED EXAMPLE 12.3.1

Problem. A cumulative frequency diagram gives 24 at 20, 58 at 30 and 74 at 40, for a total of 100 observations. Estimate the median and the interquartile range.



Mathematical reasoning. Read the diagram at the half, quarter and three-quarter points of the total.

1. The total frequency is 100, so the median is at 50.
2. Reading across from 50 gives about 27.
3. The lower quartile is at 25, which gives about 22.
4. The upper quartile is at 75, which gives about 35.
5. So the interquartile range is about 35 − 22 = 13.

Final answer. median about 27, interquartile range about 13

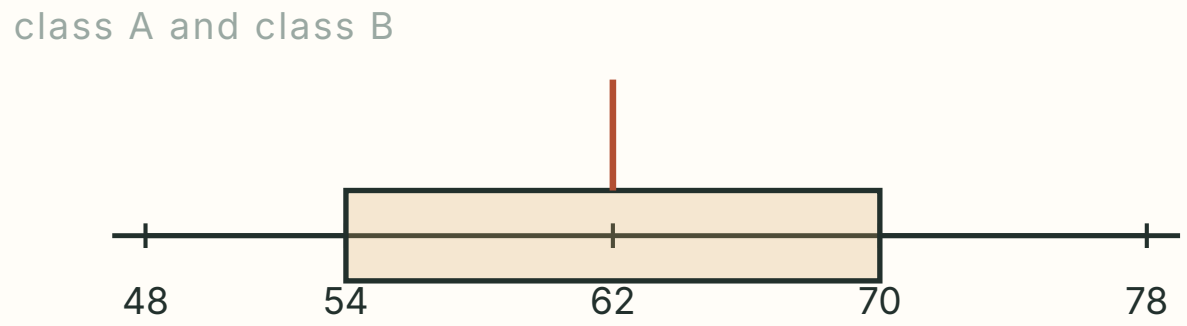
Interpretation. The quartiles are read from the cumulative total, so the total frequency must be known before the diagram can be interpreted.

CHAPTER 12 · 12.3 · WORKED EXAMPLE 2

Cumulative frequency and box plots

WORKED EXAMPLE 12.3.2

Problem. Two classes take the same test. Class A has median 62, lower quartile 54 and upper quartile 70; class B has median 60, lower quartile 48 and upper quartile 72. Compare the two distributions.



Mathematical reasoning. Compare the centres and the spreads separately, and say which is larger.

1. The medians are 62 and 60, so the centres are close.
2. The interquartile ranges are 16 and 24, so class B is more spread.
3. Class B's lower quartile is lower and its upper quartile higher.
4. So class B contains both weaker and stronger students.
5. A comparison must mention both the centre and the spread, because either alone is incomplete.

Final answer. the centres are similar; class B is more spread out

Interpretation. A box plot makes a comparison of two distributions a matter of reading two pictures, which is why it is preferred for comparison.

CHAPTER 12 · INVESTIGATION 12.4

Spread, outliers and robust measures

INQUIRY QUESTION

Why does the choice of average matter, and what does spread add?

YOUR TASK

Collect two data sets from your own life: one with an extreme value and one without. For each, compute the mean, the median, the mode, the range and the interquartile range, and say which average you would report and why.

data set	mean	median	range	interquartile range
weekly spending	42.50	35.00	80	20
daily steps	8200	8100	3000	1800

CHAPTER 12 · INVESTIGATION 12.4

Spread, outliers and robust measures

What do you notice?

- The spending data had one large value that raised the mean above the median.
- The steps data was nearly symmetric, so the mean and median were close.
- The interquartile range was much smaller than the range in both cases.
- The choice of average followed the shape of the distribution.

Observation. The mean and the median agree for a symmetric distribution and diverge for a skewed one, and the spread decides how typical either average is.

MATHEMATICAL CONCLUSION

An average is a summary of a group whose usefulness depends on the spread, and the choice between the mean and the median is decided by the shape of the distribution.

Your own words

CHAPTER 12 · 12.4

Spread, outliers and robust measures

The mean is the sum of the values divided by how many there are, and it uses every value. The median is the middle value when they are ordered, and the mode is the most frequent.

The mean is pulled by extreme values and the median is not, so a skewed distribution has a mean above or below its median. Reporting only one of them can mislead.

The range is the difference between the largest and smallest values, and the interquartile range is the difference between the upper and lower quartiles. The range is sensitive to one extreme value; the interquartile range is not.

A measure of spread is what makes an average meaningful. Two classes with the same mean can be very different, and the spread is what says so.

A relationship between two variables is what a scatter diagram looks for, and the strength of that relationship is a quantity that can be measured.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

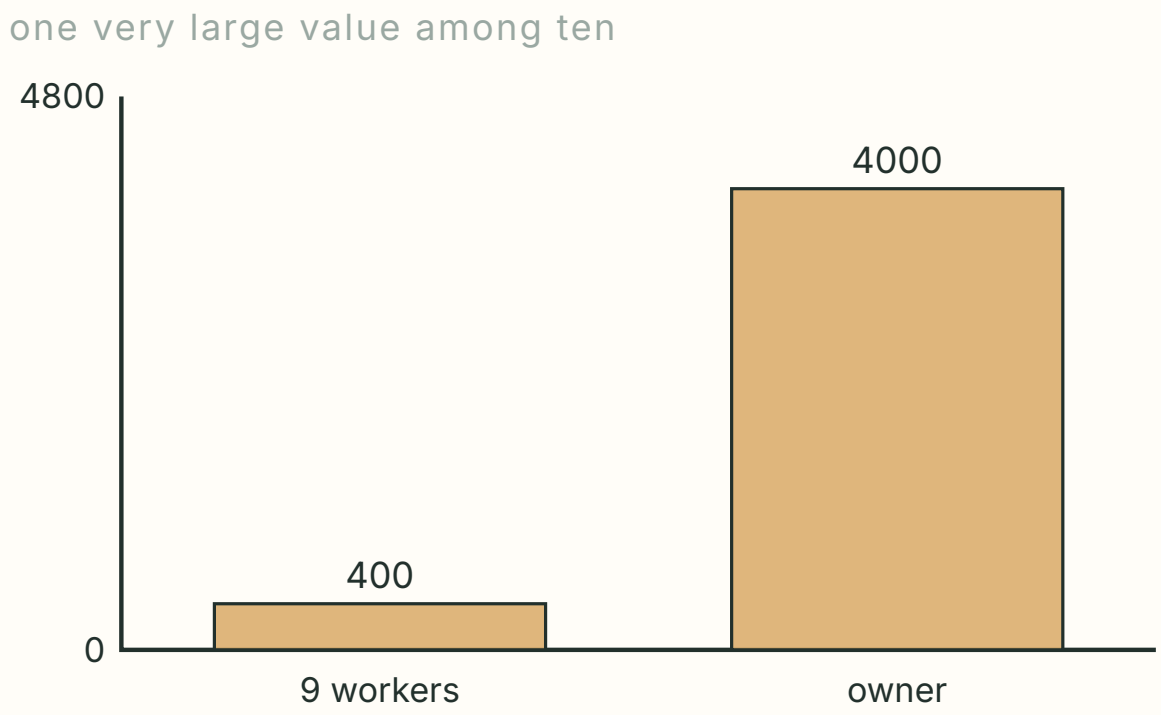
In your own words

CHAPTER 12 · 12.4 · WORKED EXAMPLE 1

Spread, outliers and robust measures

WORKED EXAMPLE 12.4.1

Problem. A factory pays 9 workers 400 dollars a week and the owner 4000. Find the mean and the median, and say which better describes a typical worker's pay.



Mathematical reasoning. Compute both averages and compare their resistance to the extreme value.

1. The total pay is $9 \times 400 + 4000 = 7600$ dollars.
2. So the mean is $7600/10 = 760$ dollars.
3. Ordering the values puts the median between the fifth and sixth, both of which are 400 dollars.
4. So the median is 400 dollars, which is what a typical worker actually earns.
5. The mean is pulled up by the owner's pay, so it describes the payroll, not the workers.

Final answer. mean 760 dollars, median 400 dollars; the median describes a typical worker

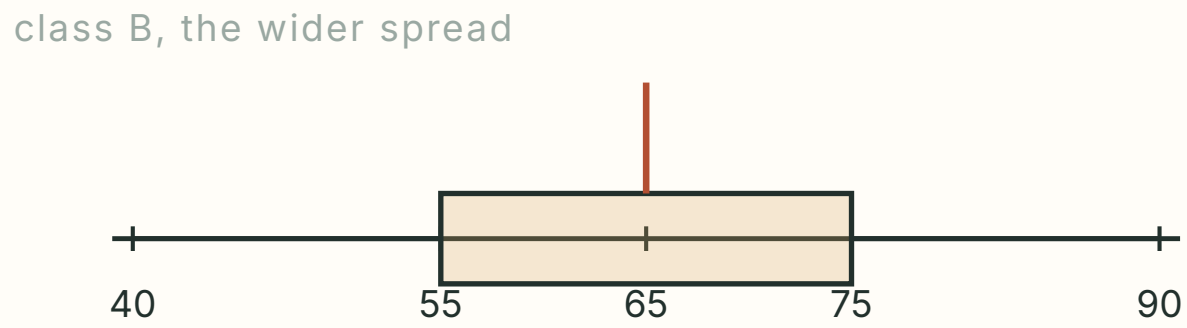
Interpretation. A single extreme value changed the mean by 90% and left the median unchanged, which is why the choice of average is a decision about what to describe.

CHAPTER 12 · 12.4 · WORKED EXAMPLE 2

Spread, outliers and robust measures

WORKED EXAMPLE 12.4.2

Problem. Two classes have mean test scores of 65. Class A has scores 60, 62, 65, 68, 70 and class B has 40, 55, 65, 75, 90. Compare their spreads.



Mathematical reasoning. Compute the range and the interquartile range for each.

1. Class A has range $70 - 60 = 10$ and interquartile range $68 - 62 = 6$.
2. Class B has range $90 - 40 = 50$ and interquartile range $75 - 55 = 20$.
3. Both classes have the same mean of 65.
4. Class B is far more spread, so its mean is less typical of any individual.
5. The same mean with a larger spread means the average describes fewer of the students.

Final answer. class A: range 10, interquartile range 6; class B: range 50, interquartile range 20

Interpretation. Two groups with the same average can be very different, and only the spread reveals the difference.

CHAPTER 12 · INVESTIGATION 12.5

Comparing distributions and interpreting critically

INQUIRY QUESTION

How is a statistic used to make a decision, and what does it cost the individual?

YOUR TASK

Find one real decision made from a statistic about a group: an insurance premium, a school place, a medical treatment or a loan. Model the group statistic and its spread, and write an account of what the statistic justifies and what it does not.

decision	group statistic	spread	justifies
car insurance	average claim by age	very wide	a group premium
school places	average test score by area	wide	a group prediction
medical treatment	success rate overall	narrow for some, wide for others	a group recommendation

CHAPTER 12 · INVESTIGATION 12.5

Comparing distributions and interpreting critically

What do you notice?

- Every decision was efficient for the group.
- Every spread was wide enough that many individuals differed from the average.
- The statistic justified a group decision but not a judgement about a person.
- A figure for the relevant subgroup would have been more useful to the individual.

Observation. A group statistic supports a decision about the group, and its spread is what limits what it can say about any individual member.

MATHEMATICAL CONCLUSION

A statistic about a group is used to plan for the group, and using it to judge an individual is an inference whose strength is limited by the spread of the group.

Your own words

CHAPTER 12 · 12.5

Comparing distributions and interpreting critically

A decision about an individual made from a group statistic is an inference, and it can be right for the group and wrong for the person.

A statistic about a group says nothing certain about any member of it, because the spread means individuals differ from the average.

Using group statistics to allocate resources is efficient; using them to judge individuals is unfair, and the difference is the difference between planning and discrimination.

A statistic can be presented to support a conclusion it does not support, and the reader's defence is to ask how the data were collected and what the spread was.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

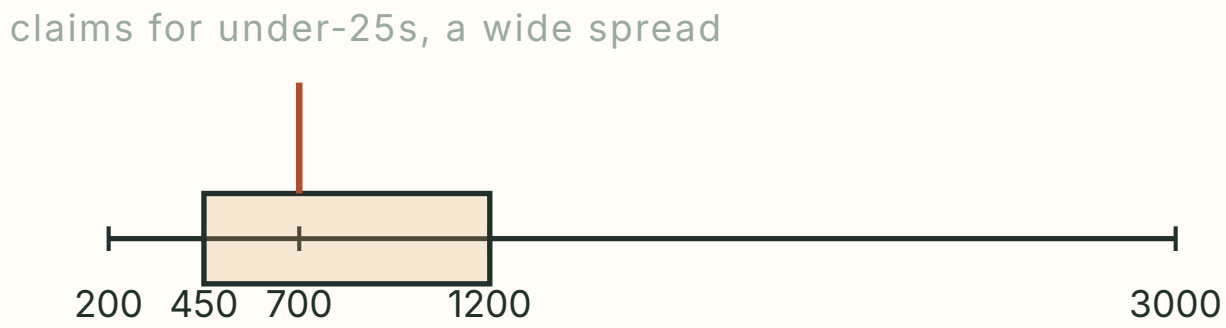
In your own words

CHAPTER 12 · 12.5 · WORKED EXAMPLE 1

Comparing distributions and interpreting critically

WORKED EXAMPLE 12.5.1

Problem. An insurer charges young drivers more because the average claim for under-25s is higher. A careful 20-year-old driver objects. Discuss the mathematics and the fairness.



Mathematical reasoning. Identify what the statistic says about the group and what it does not say about the individual.

1. The group statistic is the average claim for under-25s.
2. The spread is wide, so many young drivers claim far less than the average.
3. The statistic is true of the group and says nothing certain about this driver.
4. Charging by group is efficient for the insurer and unfair to the careful driver.
5. The mathematics supports the group claim, and the fairness question is not a mathematical one.

Final answer. the statistic is true of the group and silent about the individual

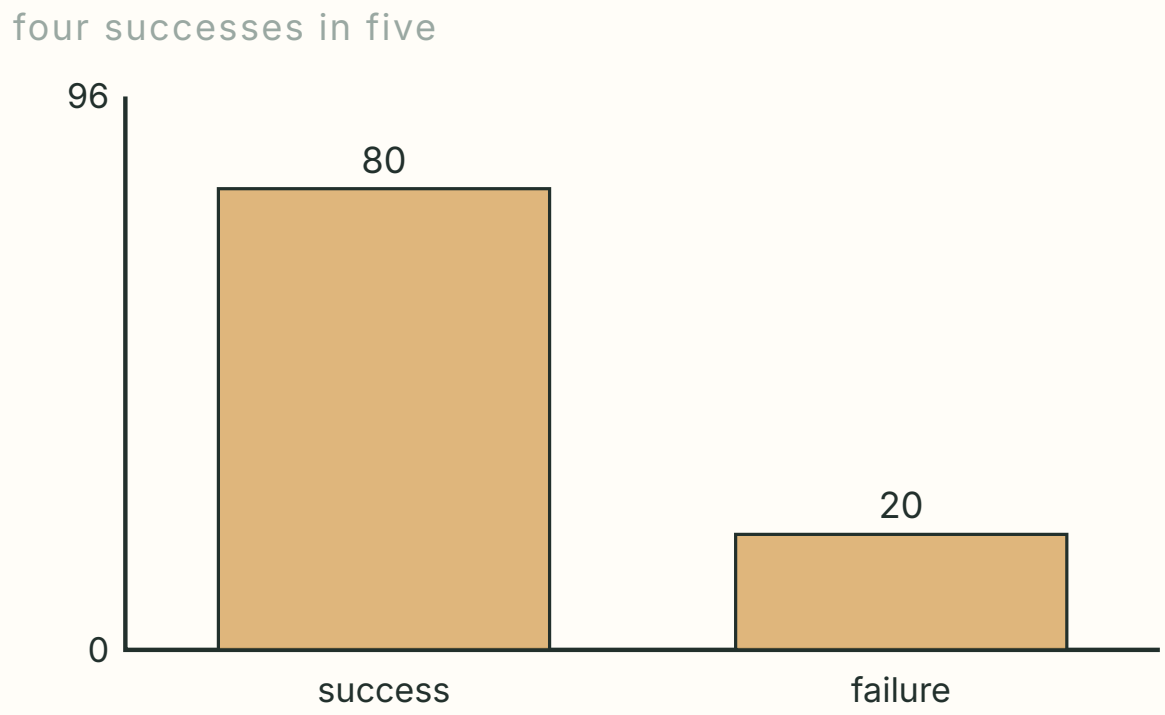
Interpretation. A group average is a planning tool, and using it to judge a person confuses efficiency with fairness.

CHAPTER 12 · 12.5 · WORKED EXAMPLE 2

Comparing distributions and interpreting critically

WORKED EXAMPLE 12.5.2

Problem. A hospital reports that a treatment succeeds in 80% of cases. A patient asks what this means for them. Answer using the spread.



Mathematical reasoning. State what the percentage says and what it does not.

1. The figure means that in the group treated, 80% had a successful outcome.
2. It does not mean this patient has an 80% chance, because outcomes depend on age, stage and other conditions.
3. The spread of outcomes across subgroups is not given by the single figure.
4. A more useful statement would give the rate for patients like this one.
5. The honest answer is that the figure describes the group, not the person.

Final answer. the figure describes the group; the patient's own chance depends on their subgroup

Interpretation. A percentage without its subgroups is a statement about a group, and a patient needs the figure for their own subgroup.

CHAPTER 12 · INVESTIGATION 12.6

Investigation: is the school day too long?

INQUIRY QUESTION

How can the same data support different conclusions, and what should a reader check?

YOUR TASK

Take one data set and present it three ways to support three different conclusions, using only true numbers. Then write the four questions a reader should ask, and answer them for your own presentations.

presentation	choice made	impression created
mean only	mean 62.1	the class does well
median and range	median 60, range 42	the class is very mixed
box plot	quartiles and outlier	one student is far ahead

CHAPTER 12 · INVESTIGATION 12.6

Investigation: is the school day too long?

What do you notice?

- All three presentations used true numbers.
- Each created a different impression of the same class.
- The choice of average did most of the work.
- The four questions exposed the manipulation in every case.

Observation. The same data support different conclusions according to the choices made in summarising and displaying them, and those choices are rarely stated.

MATHEMATICAL CONCLUSION

A reader of a statistic should ask how the data were collected, what the spread was, what the display's scale is, and what conclusion the display actually supports.

Your own words

CHAPTER 12 · 12.6

Investigation: is the school day too long?

The same data can be presented to support different conclusions by choosing different averages, different classes, different scales or different subsets.

A reader's defence is to ask four questions: how were the data collected, what is the spread, what is the scale of the display, and what conclusion does the display actually support?

A display with a truncated vertical axis exaggerates a small difference, and a display with unequal class widths distorts a shape unless frequency density is used.

Honest reporting states the sampling method, the spread and the limitations, and a report that omits them is not neutral even if every number in it is true.

> FORMAL DEVELOPMENT
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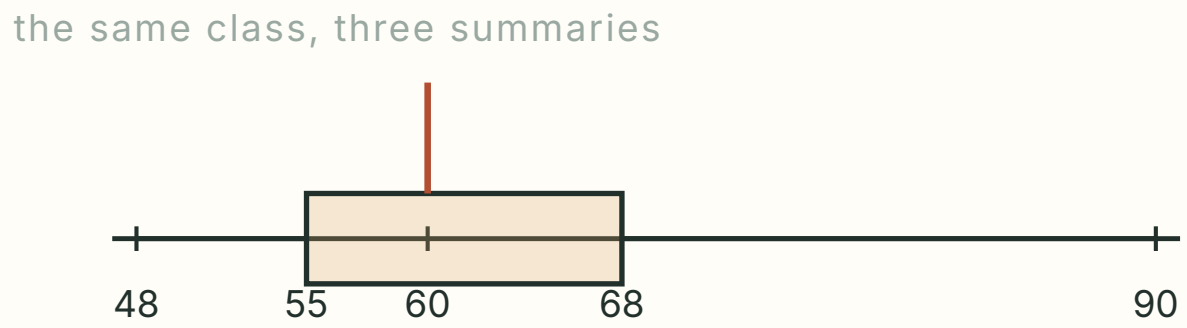
In your own words

CHAPTER 12 · 12.6 · WORKED EXAMPLE 1

Investigation: is the school day too long?

WORKED EXAMPLE 12.6.1

Problem. A class's scores are 48, 52, 55, 60, 62, 68, 90. Show how three different summaries describe the class.



Mathematical reasoning. Compute three different averages and compare what each emphasises.

1. The mean is $(48 + 52 + 55 + 60 + 62 + 68 + 90)/7 = 435/7 \approx 62.1$.
2. The median is the fourth value, 60.
3. The mode does not exist, since no value repeats.
4. The range is $90 - 48 = 42$, which is large compared with the mean.
5. So 'the class averages 62' and 'half the class scored 60 or below' are both true and sound different.

Final answer. mean about 62.1, median 60, no mode, range 42

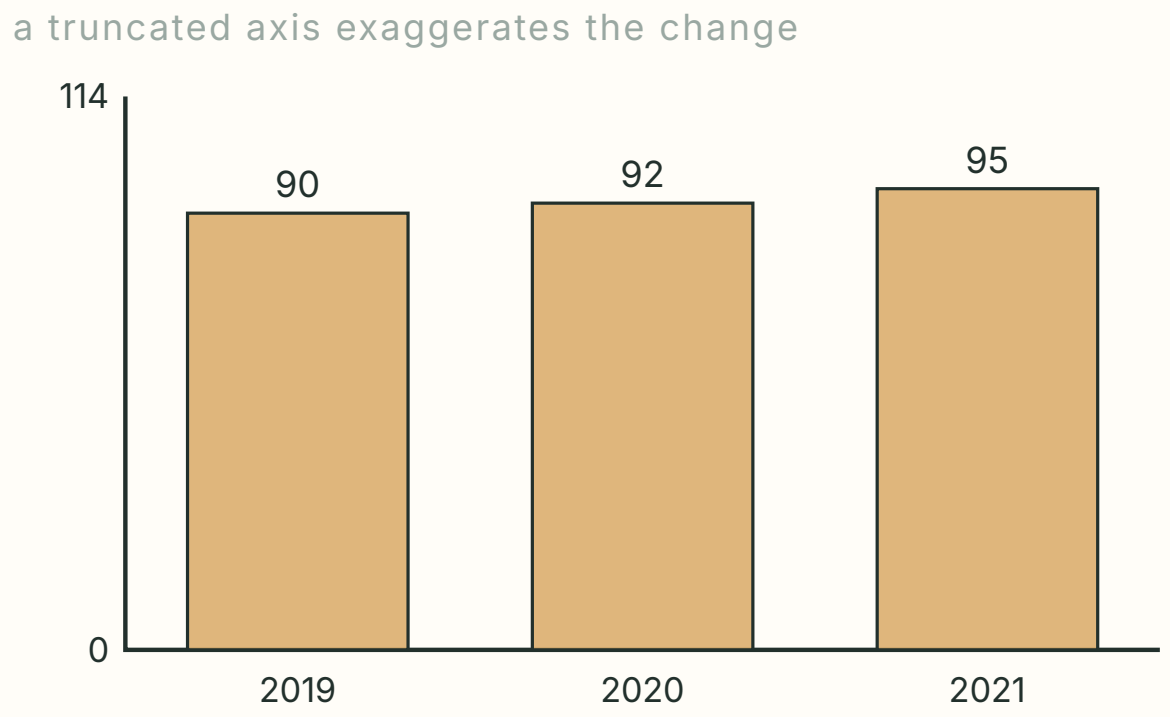
Interpretation. Two true statements about the same data can give different impressions, and the spread is what reconciles them.

CHAPTER 12 · 12.6 · WORKED EXAMPLE 2

Investigation: is the school day too long?

WORKED EXAMPLE 12.6.2

Problem. A bar chart of a school's results starts its vertical axis at 90% rather than 0%. Explain the effect on a reader.



Mathematical reasoning. Compare the visual impression with the actual change.

1. The axis runs from 90% to 95%, so the bars have very different heights.
2. The change from 90% to 95% is 5 percentage points, or about 6% relative.
3. Visually the last bar looks twice as tall as the first.
4. So the display exaggerates a real but modest improvement.
5. An axis starting at zero would show the bars as nearly equal.

Final answer. the display exaggerates a 5-point change into a visual doubling

Interpretation. A truncated axis is not a lie, because every number is correct, but it misleads by changing the visual proportions.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

12.A1 FLUENCY

Find the mean of 4, 7, 9, 12, 18.

12.A2 FLUENCY

Find the median of 3, 8, 12, 15, 19, 24.

12.A3 FLUENCY

Find the mode of 5, 6, 6, 7, 8, 8, 8, 9.

12.A4 FLUENCY

Find the range of 12, 15, 19, 24, 30.

12.A5 FLUENCY

Find the median of 2, 5, 8, 11, 14, 17, 20.

12.A6 FLUENCY

A data set has mean 20 for 6 values. Find the total.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

12.A7 FLUENCY

Find the upper quartile of 1, 3, 5, 7, 9, 11, 13, 15.

12.A8 FLUENCY

Find the interquartile range of 2, 4, 6, 8, 10, 12, 14, 16.

12.A9 FLUENCY

A sample of 50 is taken from 500. Find the sampling fraction.

12.A10 FLUENCY

A stratified sample of 40 is taken from groups of 60, 100 and 140. Find the number from the largest group.

12.A11 FLUENCY

A histogram class has frequency 18 and width 6. Find the frequency density.

12.A12 FLUENCY

A cumulative frequency reaches 45 at a value of 20. Estimate the median if the total is 90.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

12.B1 REASONING

The mean of five numbers is 12. Four of them are 9, 11, 14, 16. Find the fifth and justify.

12.B2 REASONING

Investigate the effect on the mean and median of adding one very large value to a symmetric data set.

12.B3 REASONING

Conjecture a rule for the median of an even number of ordered values and test it.

12.B4 REASONING

Investigate the pattern in the interquartile range of the first n positive integers.

12.B5 REASONING

Build a table of the mean, median and range of the squares of the first n integers and describe the pattern.

12.B6 REASONING

Conjecture a rule for the number of students in a stratified sample from a group of size g in a population of N, for a sample of size n.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

12.B7 REASONING

Investigate the effect of class width on the shape of a histogram of the same data.

12.B8 CHALLENGE

A data set has mean 50 and standard deviation 5. Investigate how many values lie within 10 of the mean for a bell-shaped set.

12.B9 REASONING

Find the pattern in the cumulative frequencies of a data set and explain why the diagram rises.

12.B10 REASONING

Investigate the effect of removing the largest value on the mean, median and range of a data set.

12.B11 CHALLENGE

Conjecture a rule for the sample size needed to halve the margin of error of a survey.

CRITERION C Communicating

Criterion C practice · Communicating

12.C1 REASONING
Explain in your own words the difference between the mean and the median.

12.C2 FLUENCY
Write a complete solution to: find the mean, median and range of 12, 15, 18, 22, 90.

12.C3 REASONING
Choose between the mean and the median to summarise house prices and justify.

12.C4 FLUENCY
Set out a full solution to: a stratified sample of 60 is taken from groups of 120, 180 and 300.

12.C5 REASONING
Explain why the range is a poor measure of spread.

12.C6 REASONING
Describe, with an example, the difference between a histogram and a bar chart.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

12.C7 REASONING
Explain why a biased sample cannot support a conclusion about the population.

12.C8 REASONING
Write the reasoning that shows a larger sample reduces random error but not bias.

12.C9 REASONING
Set out the argument that a stratified sample represents small groups.

12.C10 REASONING
Explain what a cumulative frequency diagram shows and how the median is read from it.

12.C11 REASONING
Describe the relationship between a box plot and the five-number summary.

12.C12 REASONING
Explain, with an example, when a statistic about a group is used unfairly.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

12.D1 APPLICATION

A class of 25 has mean score 68. One student transfers in with a score of 92. Find the new mean and comment on the change.

12.D2 APPLICATION

A survey samples 1 in 20 of a town of 24000. Find the sample size, and find the number from a district of 3600.

12.D3 APPLICATION

A histogram has classes of width 10 with frequency densities 1.2, 3.5, 4.8 and 2.1. Find the total frequency and the frequency of the largest class.

12.D4 APPLICATION

A company's staff salaries are 20000, 22000, 24000, 26000, 28000 and 120000. Find the mean and median and recommend which to publish.

12.D5 REASONING

A hospital reports a 90% success rate overall, but 70% for one subgroup. Investigate what a patient in that subgroup should be told.

12.D6 APPLICATION

A school takes a stratified sample of 80 from year groups of 150, 170, 160 and 220. Find the number from each and comment on the fairness.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

12.D7 REASONING

A poll of 400 people finds 52% support. Find the margin of error at the 95% level and state what it means.

12.D8 REASONING

Two classes have the same mean but interquartile ranges of 6 and 20. Investigate which class a new student should join and justify.

12.D9 REASONING

A shop's daily takings for a week are 420, 380, 510, 390, 460, 900, 350. Investigate the mean and median and say which describes a typical day.

12.D10 REASONING

A charity surveys its own donors about need in a district. Investigate whether the results describe the district and justify.

12.D11 REASONING

A test has mean 60 and the top score is 98. Investigate whether a student scoring 75 is above average and what else is needed.

12.D12 REASONING

Explain, with a survey of your own, when a reported average is misleading.

CHAPTER 12 · APPLICATION AND ASSESSMENT

Applying chapter 12

Each task involves a data set. State the sampling method, the measure you are using, and what the result can support.

A1 · The school report

A school reports that its students' reading ages have a mean of 10.4 years and a range of 6 years. A parent asks whether her child, with a reading age of 8, is behind. Using the mean, the median and the spread, explain what the school's figures do and do not support, and state what additional information would be needed to answer the parent's question.

Criteria C and D · use the measures of average and spread, and justify what the data can support.

A2 · The survey

A town of 36000 people is divided into four districts of 9000, 12000, 8000 and 7000. A stratified sample of 180 is to be taken to estimate the proportion who use the library. Find the number from each district, explain why a simple random sample might miss a district entirely, and state the margin of error of the estimate.

Criteria A and B · use the sampling fraction, justify the method, and estimate the error.

CHAPTER 12 · REVIEW

What you should now be able to do

Key ideas

- The mean uses every value; the median and mode do not.
- The mean is pulled by extremes and the median is not.
- The range is sensitive to one extreme value; the interquartile range is not.
- A histogram uses frequency density so that area represents frequency.
- A cumulative frequency diagram gives the quartiles by reading at fractions of the total.
- A box plot displays the five-number summary.
- A stratified sample takes each group in proportion to its share.
- A biased sample supports no conclusion about the population.

Vocabulary

population · sample · mean · median · mode · range · interquartile range · class width · frequency density · histogram · cumulative frequency · stratified sample

CHAPTER 12 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

12.R1 Find the mean and median of 6, 9, 11, 14, 21.

12.R2 Find the range and interquartile range of 3, 5, 8, 12, 15, 18, 20.

12.R3 A class has frequency 24 and width 8. Find the frequency density.

12.R4 A stratified sample of 50 is taken from groups of 200 and 300. Find the numbers.

12.R5 A cumulative frequency reaches 30 at a value of 15. Estimate the median if the total is 60.

12.R6 Explain why a survey of a gym's members cannot describe a town.

CHAPTER 12 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

A data set is a collection of quantities that describe a group, and its distribution is a representation of how those quantities are spread; the average and the spread are two different statements about the same group, and a decision about an individual made from a group statistic is an inference whose strength depends on how the data were collected.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of relationships and the related concepts of statistics, representation and quantity appear in your work?
- How is the global context of identities and relationships, and its exploration of how data about groups shape decisions about individuals, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a statistic for a decision about one person, and what would you still like to ask?

CHAPTER 12 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Data about a group is used to make decisions about individuals. Collect some, analyse it, and say what it does and does not justify.

Service as Action

COMMUNITY NEED	Decisions about individuals are made from group statistics without the spread.
MATHEMATICAL CONNECTION	Averages, spread and sampling.
PROPOSED ACTION	Collect one set of data about a community need and publish it with its spread and sampling method.
EXPECTED IMPACT	A reader can tell what the figures do and do not support.
STUDENT REFLECTION	Record the data, the method and the limits.

Community Project

PROJECT IDEA	A data audit of one figure the school publishes.
NEED	Averages are reported without their spread or their sampling method.
RESEARCH QUESTION	What does the school's own figure actually support, and what is missing?
MATHEMATICS	Averages, spread, displays and sampling.
ACTION OR OUTCOME	A one-page report with the figure, its spread and its limits.
BENEFICIARIES	Everyone who relies on the figure.
EVIDENCE	The report with the data and the method.

Personal Project connection · MYP 5

PERSONAL INTEREST	A sport, a game or a hobby with scores.
LEARNING GOAL	To collect data about something you enjoy and report it honestly.
PRODUCT OR OUTCOME	A data report with averages, spread and a display.
MATHEMATICS	Averages, spread and displays.
PROCESS	Collect at least 30 values, compute the measures, and display the distribution.
EVIDENCE	The data, the measures and the display.

CHAPTER 13 · PART IV · REASONING WITH DATA

13

Probability: Combined Events

Mutually exclusive events add. Independent events multiply. The two laws build every combined probability, and the sample space is what checks them.

ENGAGE

Two players throw two fair dice. Player A wins if the first die is even and the second odd. Player B wins if both dice show the same number and that number is 1 or 6. What is the probability that neither player wins?

Framing the inquiry

Key concept: Logic
Related concepts: Probability · Equivalence · Systems
Global context: Fairness and development
Global context exploration: How chance, risk and rules shape opportunities

Statement of inquiry: The probability of a combined event is built from the probabilities of its parts by the addition and multiplication laws, and the equivalence between a sample space, a Venn diagram and a tree diagram means the same result can be reached by three routes; a game is fair only when the probabilities of its outcomes match the rewards attached to them.

Questions to guide us

- **Factual:** What are the addition and multiplication laws, and how is a tree diagram used?
- **Conceptual:** Why does the probability of 'A and B' depend on whether the events are independent, and how does a Venn diagram show the same information as a tree?
- **Debatable:** If a game's probabilities are known to both players, can it still be unfair, and who is responsible for making the rules fair?

ATL · CRITICAL THINKING. Students test a probability claim by two independent methods, and identify the assumption of independence before applying the multiplication law.

CHAPTER 13 · INVESTIGATION 13.1

Sample spaces and probability rules

INQUIRY QUESTION

How is the probability of an event defined, and how is a sample space listed?

YOUR TASK

List the sample spaces of four experiments: one die, two dice, a coin and a die together, and a spinner with unequal sectors. For each, find the probability of two events of your own and check that the probabilities sum to 1.

experiment	outcomes	event	probability
one die	6	even	1/2
two dice	36	sum 7	1/6
coin and die	12	heads and even	1/4
spinner 1,2,3,4,6	5	prime	3/5

CHAPTER 13 · INVESTIGATION 13.1

Sample spaces and probability rules

What do you notice?

- Every probability was a count out of a listed sample space.
- The coin-and-die space had 12 outcomes, not 8, because the two experiments are independent.
- The unequal spinner was the only space in which the outcomes were not equally likely.
- The probabilities of each event and its complement always summed to 1.

Observation. A probability is a proportion of a listed sample space, and the requirement of equally likely outcomes is what makes counting valid.

MATHEMATICAL CONCLUSION

The probability of an event is the number of favourable equally likely outcomes divided by the total, and every event has a complement whose probability completes the sum to 1.

Your own words

CHAPTER 13 · 13.1

Sample spaces and probability rules

The sample space is the set of all equally likely outcomes, and the probability of an event is the number of favourable outcomes divided by the total number of outcomes.

Probabilities lie between 0 and 1, and the probabilities of all outcomes sum to 1. The complement of an event A has probability 1 − P(A).

An event and its complement exhaust the sample space, so 'not A' is often easier to count than A, and the two must always give answers that sum to 1.

Two events are mutually exclusive when they cannot both happen, in which case they have no outcomes in common and their probabilities add.

The addition law says that for any two events P(A or B) = P(A) + P(B) − P(A and B), and the subtraction removes the outcomes counted twice. When the events are mutually exclusive the overlap is empty and the probabilities simply add.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 13 · 13.1 · WORKED EXAMPLE 1

Sample spaces and probability rules

WORKED EXAMPLE 13.1.1

Problem. A fair die is thrown. Find the probability that the score is even, prime, greater than 4, and not prime.

event	favourable outcomes	probability
even	2, 4, 6	3/6 = 1/2
prime	2, 3, 5	3/6 = 1/2
greater than 4	5, 6	2/6 = 1/3
not prime	1, 4, 6	3/6 = 1/2

each event is a subset of the same sample space

Mathematical reasoning. Count the favourable outcomes in the sample space of six equally likely results.

1. The sample space is {1, 2, 3, 4, 5, 6}.
2. Even scores are 2, 4 and 6, so the probability is 3/6 = 1/2.
3. Prime scores are 2, 3 and 5, so the probability is 3/6 = 1/2.
4. Scores greater than 4 are 5 and 6, so the probability is 2/6 = 1/3.
5. Not prime means 1, 4 and 6, so the probability is 3/6 = 1/2, which is 1 − 1/2 as the complement rule requires.

Final answer. even 1/2, prime 1/2, greater than 4 is 1/3, not prime 1/2

Interpretation. Every probability is a count out of the sample space, and the complement rule is a check on the counting.

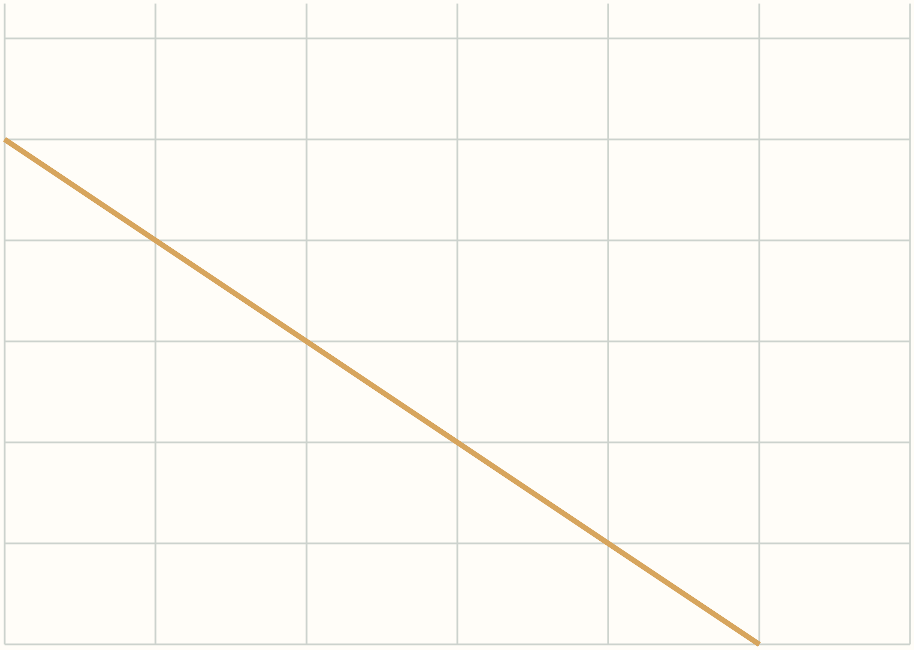
CHAPTER 13 · 13.1 · WORKED EXAMPLE 2

Sample spaces and probability rules

WORKED EXAMPLE 13.1.2

Problem. Two fair dice are thrown. Find the probability that the sum is 7.

the six outcomes with sum 7



Mathematical reasoning. List the sample space of 36 equally likely ordered pairs and count the favourable ones.

1. The sample space has $6 \times 6 = 36$ equally likely outcomes.
2. The pairs with sum 7 are (1,6), (2,5), (3,4), (4,3), (5,2) and (6,1).
3. That is 6 outcomes, so the probability is $6/36 = 1/6$.
4. Seven is the most likely sum, with six ways out of thirty-six.
5. The order matters, so (2,5) and (5,2) are different outcomes.

Final answer. 1/6

Interpretation. A grid of 36 outcomes makes the counting visible, and the symmetry about the diagonal is a check on the list.

CHAPTER 13 · INVESTIGATION 13.2

Two-way tables and Venn diagrams

INQUIRY QUESTION

How do a two-way table and a Venn diagram each organise two events, and what can be read from each?

YOUR TASK

Survey thirty people on two yes-or-no questions of your own, for example whether they walk to school and whether they play a sport. Record the results in a two-way table, draw a Venn diagram of the same data, and then compute P(A), P(B), P(A and B) and P(A or B). Check that the addition law holds, and find one conditional probability from each representation.

	plays a sport	does not	total
walks	8	7	15
does not walk	5	10	15
total	13	17	30

CHAPTER 13 · INVESTIGATION 13.2

Two-way tables and Venn diagrams

What do you notice?

- The two-way table gave the counts and the Venn diagram gave the same numbers as regions.
- P(A or B) was larger than either P(A) or P(B) but smaller than their sum.
- The overlap had been counted twice by adding, which is why the addition law subtracts it.
- The conditional probabilities were different from the unconditional ones.

Observation. A two-way table and a Venn diagram are two representations of the same two events, and each makes some probability readable at a glance.

MATHEMATICAL CONCLUSION

Counts, unions, intersections and conditional probabilities are all readable from either representation, and the addition law is the statement that the overlap is counted twice.

Your own words

CHAPTER 13 · 13.2

Two-way tables and Venn diagrams

A two-way table cross-classifies a sample by two variables, so each cell counts the members with one value of each, and the totals around the edge give the marginal counts.

A conditional probability is read from a two-way table by dividing a cell by the total of its row or column, which is the same as restricting the sample space to the given event.

A Venn diagram draws the sample space as a rectangle and the events as overlapping regions, so the overlap is the intersection, the union is everything covered, and the outside is the complement.

The addition law P(A or B) = P(A) + P(B) − P(A and B) subtracts the overlap because it has been counted twice, and a Venn diagram shows why the subtraction is needed.

Two representations of the same data answer different questions quickly: a table gives counts exactly and a Venn diagram shows the logical relations at a glance.

> FORMAL DEVELOPMENT
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> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

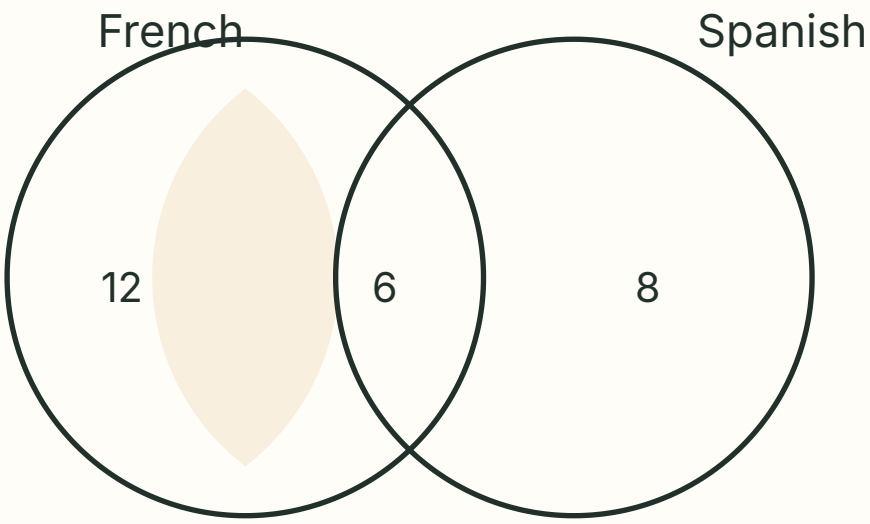
In your own words

CHAPTER 13 · 13.2 · WORKED EXAMPLE 1

Two-way tables and Venn diagrams

WORKED EXAMPLE 13.2.1

Problem. In a class of 30, 18 study French, 14 study Spanish and 6 study both. How many study neither, and what is the probability that a student chosen at random studies exactly one language?



Mathematical reasoning. Fill the Venn diagram from the overlap outwards.

1. The overlap is 6, so French only is $18 - 6 = 12$.
2. Spanish only is $14 - 6 = 8$.
3. So $12 + 6 + 8 = 26$ study at least one language.
4. Hence $30 - 26 = 4$ study neither.
5. The probability of exactly one language is $(12 + 8)/30 = 20/30 = 2/3$.

Final answer. 4 study neither; the probability is 2/3

Interpretation. The overlap is filled first, and every other region follows from it, which is why a Venn diagram is drawn from the inside out.

CHAPTER 13 · 13.2 · WORKED EXAMPLE 2

Two-way tables and Venn diagrams

WORKED EXAMPLE 13.2.2

Problem. A two-way table records 100 people by age group and whether they cycle. Given that a person is under 30, what is the probability that they cycle?

	cycles	does not cycle	total
under 30	30	20	50
30 or over	15	35	50
total	45	55	100

Mathematical reasoning. Restrict to the row for under 30 and divide by its total.

1. The condition 'under 30' selects the first row, of 50 people.
2. Of those, 30 cycle.
3. So the conditional probability is $30/50 = 0.6$.
4. The unconditional probability is $45/100 = 0.45$.
5. The two differ, which is the point of conditioning: the same question has a different answer once the sample space is restricted.

Final answer. 0.6

Interpretation. A conditional probability divides a cell by the total of its row or column, and the choice of row or column is the condition.

CHAPTER 13 · INVESTIGATION 13.3

Tree diagrams

INQUIRY QUESTION

How does a tree diagram represent a sequence of events, and how are its probabilities combined?

YOUR TASK

Build a tree diagram for a two-stage situation of your own with at least three branches at the first stage, for example a spinner followed by a die, or a bag of three colours followed by a second draw without replacement. Label every branch with its probability, check that the probabilities leaving each point sum to 1, list every complete path with its probability, and check that the paths sum to 1. Then use the tree to find the probability of an event that can happen in two different ways.

path	probability	event
red, heads	1/4	red and heads
red, tails	1/4	red and tails
blue, heads	1/6	blue and heads
blue, tails	1/6	blue and tails
green, heads	1/12	green and heads
green, tails	1/12	total = 1

CHAPTER 13 · INVESTIGATION 13.3

Tree diagrams

What do you notice?

- Every set of branches leaving a point summed to 1.
- Every path probability was a product along the path.
- The event 'red or heads' needed adding several paths and subtracting the overlap.
- The paths summed to exactly 1, which was the check on the whole diagram.

Observation. A tree diagram turns a sequence of stages into a list of paths, and the product rule gives a path while the sum rule gives an event.

MATHEMATICAL CONCLUSION

Multiply along a path and add across paths, and the two rules together turn any tree diagram into a probability.

Your own words

CHAPTER 13 · 13.3

Tree diagrams

A tree diagram shows the outcomes of successive stages, with a branch for each outcome and its probability written on the branch, and the probabilities leaving any point always sum to 1.

The probability of a complete path is the product of the probabilities along it, because the stages are combined by the multiplication law.

The probability of an event that can happen in several ways is the sum of the probabilities of the paths that lead to it, and the two rules of multiply and add are the whole method.

A tree with the same probabilities on both levels is the picture of independent events, and a tree whose second level differs from the first is the picture of dependent ones.

A tree diagram is a representation of a system of stages, and it is preferred when the stages are sequential rather than simultaneous.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

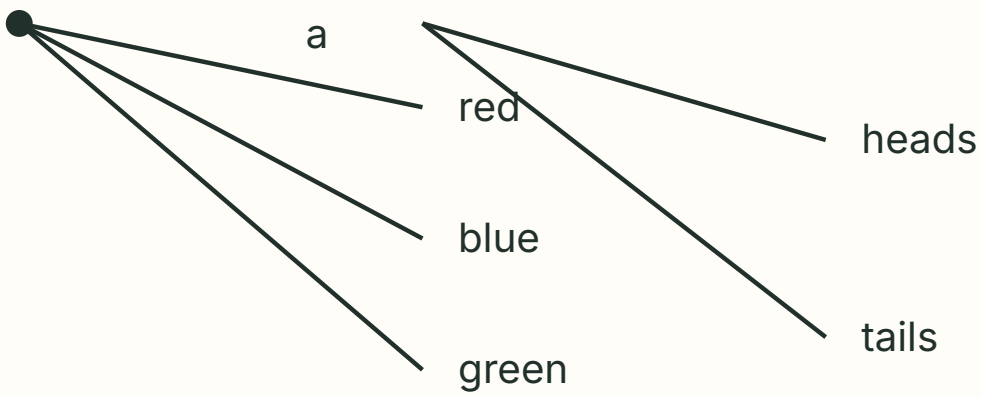
In your own words

CHAPTER 13 · 13.3 · WORKED EXAMPLE 1

Tree diagrams

WORKED EXAMPLE 13.3.1

Problem. A spinner has sectors red, blue and green with probabilities 1/2, 1/3 and 1/6. It is spun and then a fair coin is thrown. Find the probability of red and heads.



Mathematical reasoning. Multiply along the path red then heads.

1. The first stage has three branches: 1/2, 1/3 and 1/6, which sum to 1.
2. The second stage has two branches of 1/2 each.
3. The path red then heads has probability $(1/2) \times (1/2) = 1/4$.
4. The number of paths is $3 \times 2 = 6$, and their probabilities sum to 1.
5. The two stages are independent, so the second level repeats the first level's probabilities.

Final answer. 1/4

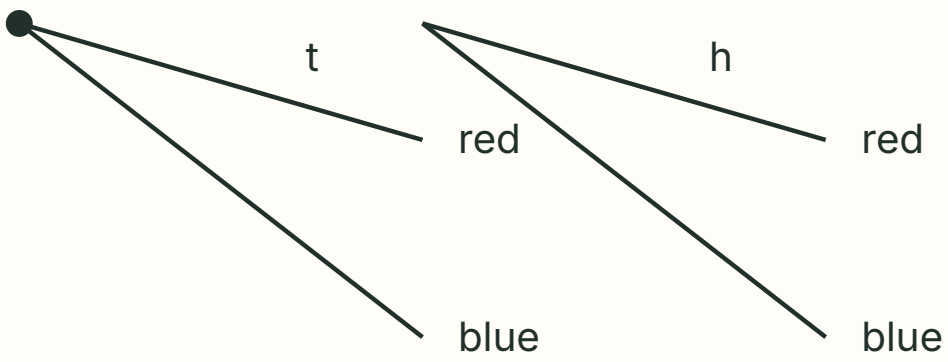
Interpretation. The product rule applies along a path, and the sum rule applies across paths, and every complete tree must sum to 1.

CHAPTER 13 · 13.3 · WORKED EXAMPLE 2

Tree diagrams

WORKED EXAMPLE 13.3.2

Problem. A bag holds 2 red and 3 blue counters. One is drawn and not replaced, and a second is then drawn. Find the probability that the two counters are different colours.



Mathematical reasoning. Add the two paths that give different colours.

1. P(red then blue) = (2/5) × (3/4) = 6/20.
2. P(blue then red) = (3/5) × (2/4) = 6/20.
3. The two paths are mutually exclusive, so add them.
4. The total is 6/20 + 6/20 = 12/20 = 3/5.
5. The second level has probabilities 3/4 and 2/4 after a red, which is what makes the events dependent.

Final answer. 3/5

Interpretation. An event that can happen in two ways is a sum of paths, and the order of the draws gives the two ways.

CHAPTER 13 · INVESTIGATION 13.4

Independent and dependent events

INQUIRY QUESTION

When does knowing that one event has happened change the probability of another, and how is that decided?

YOUR TASK

Take five pairs of events from a situation of your own: two throws of a die, two cards drawn with and without replacement, two mornings' weather, two students chosen from a class, and one pair of your own. For each, decide whether the events are independent, justify the decision from the situation, and compute P(A and B) using the correct form of the multiplication law.

pair of events	independent?	law used	P(A and B)
two throws of a die	yes	P(A) × P(B)	1/36
two cards, replaced	yes	P(A) × P(B)	1/16
two cards, not replaced	no	P(A) × P(B given A)	1/17
two students from a class	no	P(A) × P(B given A)	—
rain today and rain tomorrow	no	conditional	—

CHAPTER 13 · INVESTIGATION 13.4

Independent and dependent events

What do you notice?

- Replacement made the difference between independent and dependent.
- The conditional form was needed whenever the first event changed the sample space.
- The justification came from the situation, not from the numbers.
- The independent calculation always gave the larger answer for two reds.

Observation. Independence is decided by the situation, and the simple product is valid only when the first event leaves the sample space unchanged.

MATHEMATICAL CONCLUSION

Knowing one event has happened changes the probability of another exactly when the sample space has changed, and the multiplication law must then be used in its conditional form.

Your own words

CHAPTER 13 · 13.4

Independent and dependent events

Two events are independent when the probability of one is unchanged by knowing that the other has happened, so P(A given B) = P(A). Otherwise they are dependent.

For independent events the multiplication law is P(A and B) = P(A) × P(B), and this simple form is valid only when the events are independent.

For dependent events the conditional form P(A and B) = P(A) × P(B given A) is used, and the second factor is different from P(B) because the first draw has changed the sample space.

Drawing without replacement makes successive events dependent, and drawing with replacement keeps them independent, which is why the two procedures give different answers to the same question.

Independence is a property of the situation and not of the arithmetic, so it must be justified from the context. Assuming it when it is false is the commonest error in combined probability.

> FORMAL DEVELOPMENT
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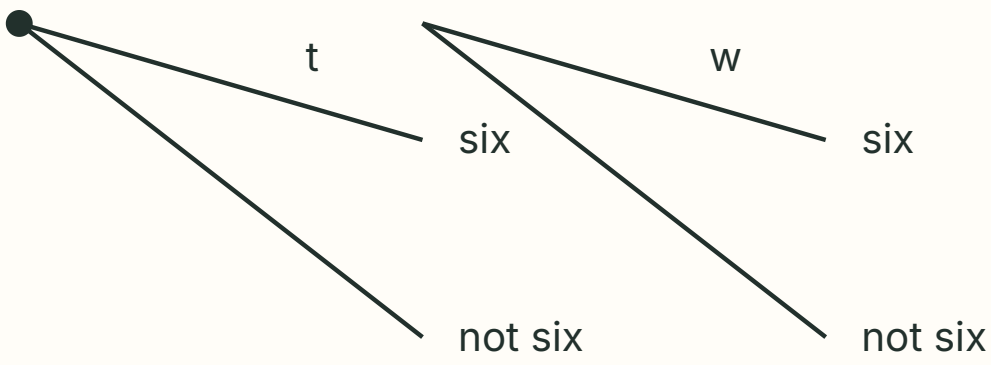
In your own words

CHAPTER 13 · 13.4 · WORKED EXAMPLE 1

Independent and dependent events

WORKED EXAMPLE 13.4.1

Problem. A fair die is thrown twice. Find the probability that both throws are sixes, and say whether the events are independent.



Mathematical reasoning. The throws do not affect each other, so multiply.

1. P(six on the first throw) = 1/6.
2. The second throw is unaffected by the first, so its probability is also 1/6.
3. So P(both sixes) = (1/6) × (1/6) = 1/36.
4. The events are independent because the die has no memory.
5. A tree diagram with the same branch probabilities on both levels is the picture of independence.

Final answer. 1/36; the events are independent

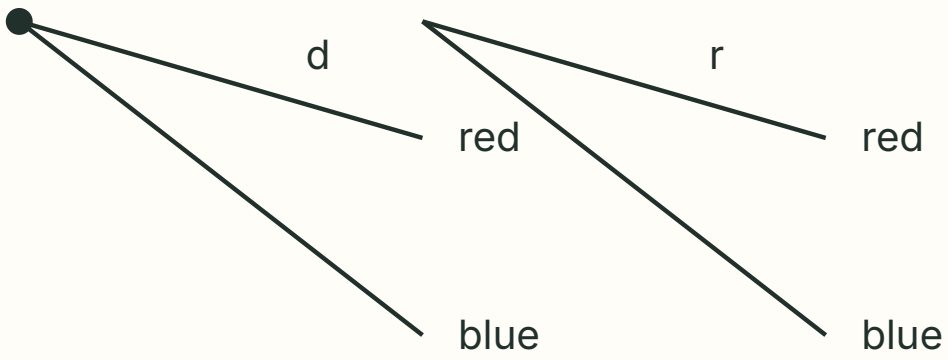
Interpretation. Independence is what allows the simple product, and the reason is physical: the second throw cannot know the first.

CHAPTER 13 · 13.4 · WORKED EXAMPLE 2

Independent and dependent events

WORKED EXAMPLE 13.4.2

Problem. A bag has 5 red and 3 blue counters. Two are drawn without replacement. Find the probability that both are red, and compare with drawing with replacement.



Mathematical reasoning. The first draw changes the contents, so the second probability is conditional.

1. P(first red) = 5/8.
2. Given the first was red, 4 red and 3 blue remain, so P(second red given first red) = 4/7.
3. So P(both red) = (5/8) × (4/7) = 20/56 = 5/14.
4. With replacement the answer would be (5/8) × (5/8) = 25/64.
5. The two differ because without replacement the events are dependent.

Final answer. 5/14 without replacement; 25/64 with replacement

Interpretation. The same question has two answers, and the difference is entirely due to whether the sample space is restored.

CHAPTER 13 · INVESTIGATION 13.5

Expected frequency, simulation and fairness

INQUIRY QUESTION

When is a game fair, and how is the expected value computed?

YOUR TASK

Analyse three games: one with dice, one with cards and one from your own invention. For each, list the outcomes and their probabilities, compute the expected value, and state the entry fee that would make the game fair.

game	expected prize	fair fee	verdict
die, win 6 for a six, stake 1	1	1	fair
two dice, win 12 for a double six, stake 1	12/36	1	biased against the player
coin, win 3 for heads, stake 2	1.5	1.5	biased against the player

CHAPTER 13 · INVESTIGATION 13.5

Expected frequency, simulation and fairness

What do you notice?

- Only the first game was fair at the stated stake.
- The double-six game was heavily biased against the player.
- The fair fee was always the expected prize.
- A fair game can still be a bad bet for a player who cannot afford the stake.

> **Observation.** The expected value of a game is the average gain per play, and a game is fair exactly when the entry fee equals the expected prize.

MATHEMATICAL CONCLUSION

A game is fair when its expected value is zero for every player, which happens when the entry fee equals the expected prize, and the expected value is a statement about the long run only.

Your own words

CHAPTER 13 · 13.5

Expected frequency, simulation and fairness

The expected value of a game is the sum of each outcome's value multiplied by its probability, and it is the average amount won per play over many plays.

A game is fair when the expected value is zero for every player, which means the rewards are proportional to the inverse of the probabilities.

A game with a positive expected value for one player and a negative one for the other is biased, and the bias can be measured exactly even though no single play is predictable.

Expected value is a long-run statement. It says nothing about any individual play, which is why a fair game can still be a bad bet for someone who cannot afford to lose.

The expected frequency of an event in n trials is n × P(event), and it is the number of times the event is expected, not the number it must occur.

A simulation uses random numbers to model a situation whose probabilities are known, and repeating it many times gives an experimental probability that approaches the theoretical one.

A game is one of many systems of rules that distribute chance, and fairness is a property of the rules rather than of any single play, which is why a rule can be changed to make the expected value zero.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 13 · 13.5 · WORKED EXAMPLE 1

Expected frequency, simulation and fairness

WORKED EXAMPLE 13.5.1

Problem. A game costs 2 dollars to play. A die is thrown and the player wins 5 dollars if the score is 6. Find the expected value and say whether the game is fair.

outcome	probability	gain
score 6	1/6	+3
not 6	5/6	−2

the gain is the prize minus the cost of playing

Mathematical reasoning. Multiply each gain by its probability and sum.

1. If the score is 6 the player gains $5 - 2 = 3$ dollars.
2. Otherwise the player loses the 2 dollars stake.
3. The expected value is $(1/6)(3) + (5/6)(-2) = 0.5 - 1.667$.
4. That is -1.167 dollars per play.
5. The game is not fair: the player loses about 1.17 dollars each time on average.

Final answer. expected value about −1.17 dollars per play, so the game is not fair

Interpretation. A negative expected value means the player loses in the long run, however many times they win in the short run.

CHAPTER 13 · 13.5 · WORKED EXAMPLE 2

Expected frequency, simulation and fairness

WORKED EXAMPLE 13.5.2

Problem. A fair coin is thrown and a player wins 4 dollars for heads and 2 dollars for tails. What entry fee makes the game fair?

outcome	probability	prize
heads	1/2	4
tails	1/2	2

the expected prize is 3 dollars

Mathematical reasoning. Set the expected prize equal to the entry fee.

1. The expected prize is (1/2)(4) + (1/2)(2) = 3 dollars.
2. For the game to be fair the entry fee must equal the expected prize.
3. So the fair entry fee is 3 dollars.
4. At any higher fee the player loses on average, and at any lower fee they gain.
5. The fairness is a statement about the long run, not about one throw.

Final answer. 3 dollars

Interpretation. A fair fee is the expected prize, and this is the principle behind every insurance premium and every lottery ticket.

CHAPTER 13 · INVESTIGATION 13.6

Investigation: is the game fair?

INQUIRY QUESTION

How is a game designed so that no player has an advantage, and how is that checked?

YOUR TASK

Design a two-player game using dice, cards or a spinner. State the rules precisely, compute each player's probability of winning and the probability that neither wins, and adjust the rules until the game is fair. Then compute the expected value of one play.

version	P(A)	P(B)	P(neither)	fair?
first attempt	9/36	2/36	25/36	no
second attempt	9/36	9/36	18/36	yes
with a stake	9/36	9/36	18/36	yes, if prizes match

CHAPTER 13 · INVESTIGATION 13.6

Investigation: is the game fair?

What do you notice?

- The first version was badly biased towards player A.
- The second gave each player the same number of outcomes.
- Neither player won in 18 of the 36 outcomes in the fair version.
- The expected value was zero only when the prizes were equal too.

Observation. A fair two-player game needs two disjoint winning events of equal probability, and the remainder is the probability that neither player wins.

MATHEMATICAL CONCLUSION

A game is designed fair by giving each player a disjoint set of outcomes of equal size, and it is checked by counting the sample space and by computing with the addition and multiplication laws.

Your own words

CHAPTER 13 · 13.6

Investigation: is the game fair?

A two-player game is fair when each player's probability of winning is the same, which requires the two winning events to have equal probability and to be disjoint.

The two winning events must also be exhaustive of the remaining outcomes, so that the probability of neither winning is what is left after the two.

A game is checked by listing the sample space and counting, and by computing with the laws, and the two methods must agree.

Designing a game is the reverse of analysing one: choose the probabilities first, then find rules that produce them, which is harder because the rules must be simple.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

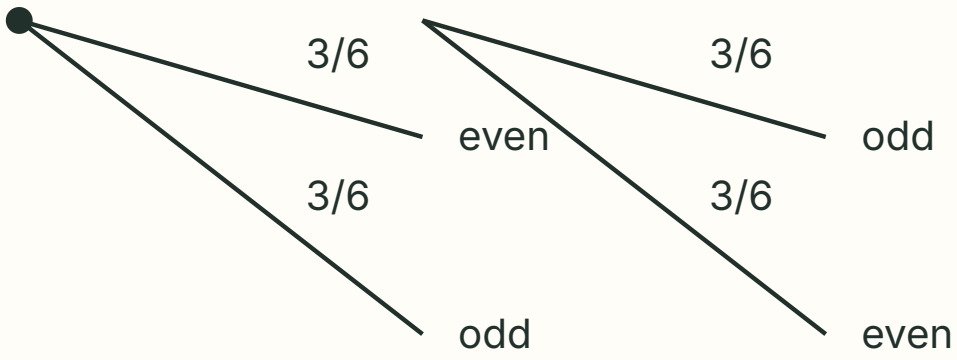
CHAPTER 13 · 13.6 · WORKED EXAMPLE 1

Investigation: is the game fair?

WORKED EXAMPLE 13.6.1

Problem. Two fair dice are thrown. Player A wins if the first die is even and the second is odd. Player B wins if both dice show the same number and that number is 1 or 6. Find the probability that neither player wins.

the two winning conditions are disjoint



Mathematical reasoning. Count the outcomes for each player and subtract from the total.

1. The sample space has 36 equally likely outcomes.
2. Player A wins when the first die is even and the second odd: $3 \times 3 = 9$ outcomes.
3. Player B wins on (1,1) and (6,6), which is 2 outcomes.
4. The two events are disjoint, because both dice cannot be the same and one odd.
5. So neither player wins in $36 - 9 - 2 = 25$ outcomes, a probability of $25/36$.

Final answer. $25/36$

Interpretation. The two winning conditions are disjoint, so their probabilities add, and the remainder is the probability that neither player wins.

CHAPTER 13 · 13.6 · WORKED EXAMPLE 2

Investigation: is the game fair?

WORKED EXAMPLE 13.6.2

Problem. Design a rule change that makes the game fair, and check it.

player	winning outcomes	probability
A (original)	first even, second odd	9/36
B (original)	both dice 1 or both 6	2/36
A (new rule)	first even, second odd	9/36
B (new rule)	first odd, second even	9/36

the new rule gives B the same probability as A

Mathematical reasoning. Give player B a rule with the same probability as player A's.

1. Player A wins on 9 outcomes, so a fair game gives B 9 too.
2. A simple rule is: B wins if the first die is odd and the second is even.
3. That is also $3 \times 3 = 9$ outcomes.
4. The two events are disjoint, because the first die cannot be both even and odd.
5. So each player wins with probability $9/36 = 1/4$, and neither wins with probability $18/36 = 1/2$.

Final answer. one fair rule gives each player 9 of the 36 outcomes, with 18 left over

Interpretation. Designing a fair rule means finding two disjoint events with equal probability, and the check is that the two counts are equal.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

13.A1 FLUENCY

Find the probability that a fair die shows an even number.

13.A2 FLUENCY

Find the probability that a fair die shows a prime number.

13.A3 FLUENCY

Find the probability that a fair die shows a number greater than 4.

13.A4 FLUENCY

Find the probability of not throwing a 6 with a fair die.

13.A5 FLUENCY

Find the probability that two fair dice sum to 7.

13.A6 FLUENCY

Find the probability of two heads when two fair coins are thrown.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

13.A7 FLUENCY

Find the probability of at least one head when two fair coins are thrown.

13.A8 FLUENCY

A bag has 4 red and 6 blue counters. Find the probability of drawing a red one.

13.A9 FLUENCY

Find the probability that a card drawn from a pack is a heart.

13.A10 FLUENCY

Find the probability that a card drawn from a pack is a king or a queen.

13.A11 FLUENCY

Two events are mutually exclusive with probabilities 0.3 and 0.4. Find the probability of either.

13.A12 FLUENCY

A fair coin is thrown three times. Find the probability of three heads.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

13.B1 REASONING

The probabilities of two independent events are 0.4 and 0.5. Find the probability of both and of either.

13.B2 REASONING

Investigate the pattern in the probability of at least one six in n throws of a die.

13.B3 CHALLENGE

Conjecture a rule for the probability that two people share a birthday in a group of n people.

13.B4 REASONING

Investigate the pattern in the probability of drawing two red cards without replacement from a pack.

13.B5 REASONING

Build a table of the probabilities of the sums of two dice and describe the pattern.

13.B6 REASONING

Conjecture a rule for the expected number of sixes in n throws of a die.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

13.B7 REASONING

Investigate the pattern in the probability that a family with two children has two girls, given that at least one is a girl.

13.B8 CHALLENGE

A test is 95% accurate and 2% of people are ill. Investigate the probability that a positive result means illness.

13.B9 REASONING

Find the pattern in the probability of winning a prize in a raffle as the number of tickets bought grows.

13.B10 REASONING

Investigate the expected value of a game that pays the square of the score on a fair die, for a stake of 10 dollars.

13.B11 CHALLENGE

Conjecture a rule for the number of throws needed for the probability of at least one six to exceed 0.9.

CRITERION C Communicating

Criterion C practice · Communicating

13.C1 REASONING
Explain in your own words what it means for two events to be independent.

13.C2 FLUENCY
Write a complete solution to: find the probability that two cards drawn without replacement are both aces.

13.C3 REASONING
Choose between a Venn diagram and a tree diagram for two events and justify.

13.C4 FLUENCY
Set out a full solution to: a bag has 3 red and 5 green counters. Two are drawn without replacement. Find the probability of one of each colour.

13.C5 REASONING
Explain why mutually exclusive events cannot be independent.

13.C6 REASONING
Describe, with an example, the difference between P(A and B) and P(A given B).

CRITERION C Communicating

Criterion C practice · Communicating (continued)

13.C7 REASONING

Explain why the probability of neither of two disjoint events is one minus the sum of their probabilities.

13.C8 REASONING

Write the reasoning that shows P(A or B) = P(A) + P(B) − P(A and B).

13.C9 REASONING

Set out the argument that the expected value of a game is the average gain per play.

13.C10 REASONING

Explain what makes a game fair in terms of expected value.

13.C11 REASONING

Describe the relationship between a rule and the probabilities it produces.

13.C12 REASONING

Explain, with an example, when a rule that looks neutral is biased in effect.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

13.D1 APPLICATION

A game costs 3 dollars. A die is thrown and the player wins 10 dollars for a six. Find the expected value and say whether the game is fair.

13.D2 APPLICATION

Two players throw a fair coin. The winner of a head takes 5 dollars from the other. Find each player's expected value.

13.D3 APPLICATION

A spinner has sectors of 30°, 90°, 120° and 120°. Find the probability of each sector and the expected score if the score is the sector number.

13.D4 APPLICATION

A bag has 5 red and 3 blue counters. Two are drawn without replacement. Find the probability of two of the same colour.

13.D5 REASONING

A medical test is 99% accurate for those who are ill and gives a false positive 5% of the time. One person in 100 is ill. Investigate the probability that a positive result means illness.

13.D6 CHALLENGE

A lottery sells 500 tickets and 3 prizes are drawn without replacement. Find the probability that one holder of 2 tickets wins exactly one prize.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

13.D7 REASONING

A school allocates 20 places among 400 applicants by lottery. Investigate whether a family with 3 applicants is favoured and justify.

13.D8 REASONING

A game uses two dice. Player A wins if the sum is even; player B wins if the sum is odd. Investigate whether the game is fair and what happens to the remaining probability.

13.D9 CHALLENGE

A biased coin has probability 0.6 of heads. Investigate the expected number of heads in 10 throws and the probability of more heads than tails.

13.D10 REASONING

A raffle sells 200 tickets and one holder buys 20. Investigate whether the raffle is fair if the tickets were given away equally.

13.D11 REASONING

Two players each roll a die; the higher score wins and ties are re-rolled. Investigate whether the game is fair.

13.D12 REASONING

Explain, with a game of your own, when a fair game is still a bad bet.

CHAPTER 13 · APPLICATION AND ASSESSMENT

Applying chapter 13

Each task involves combined events. Draw the sample space or the tree, state whether the events are independent, and justify the law you use.

A1 · The two dice

Two fair dice are thrown. Player A wins if the first die is even and the second is odd. Player B wins if both dice show the same number and that number is 1 or 6. Find the probability that A wins, that B wins, and that neither player wins. Then design one change to the rules that makes the game fair, and justify it by counting.

Criteria B and C · count the sample space, apply the addition law, and justify the new rule.

A2 · The game and the stake

A game costs 4 dollars to play. A spinner with four equal sectors pays 10 dollars on one sector, 6 dollars on two sectors and nothing on the fourth. Find the expected prize, the expected value of a play, and the entry fee that would make the game fair. Then explain why a player might still refuse to play the fair version.

Criteria A and D · compute the expected value, solve for the fair fee, and justify the answer with the meaning of expectation.

CHAPTER 13 · REVIEW

What you should now be able to do

Key ideas

- A probability is a proportion of a listed sample space.
- An event and its complement have probabilities that sum to 1.
- P(A or B) = P(A) + P(B) − P(A and B).
- P(A and B) = P(A) × P(B) only for independent events.
- A tree is multiplied along a path and summed across paths.
- P(A and B) = P(A) × P(B given A) works for dependent events.
- The expected value is the sum of value times probability.
- A game is fair when the entry fee equals the expected prize.

Vocabulary

sample space · mutually exclusive · independent · dependent · conditional probability · tree diagram · Venn diagram · expected value · fair game · complement · intersection · union

CHAPTER 13 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

13.R1 Find the probability that a fair die shows a number that is not prime.

13.R2 Find the probability that two fair coins both show tails.

13.R3 Find the probability that two cards drawn without replacement are both hearts.

13.R4 A bag has 6 red and 4 blue counters. Find the probability of one of each in two draws without replacement.

13.R5 A game pays 8 dollars with probability 1/4. Find the expected prize.

13.R6 Explain why two mutually exclusive events cannot be independent.

CHAPTER 13 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

The probability of a combined event is built from the probabilities of its parts by the addition and multiplication laws, and the equivalence between a sample space, a Venn diagram and a tree diagram means the same result can be reached by three routes; a game is fair only when the probabilities of its outcomes match the rewards attached to them.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of logic and the related concepts of probability, equivalence and systems appear in your work?
- How is the global context of fairness and development, and its exploration of how chance, risk and rules shape opportunities, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of an expected value for a decision about playing a game, and what would you still like to ask?

CHAPTER 13 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

Chance and rules shape who gets an opportunity. Model one real randomised process and say whether it is fair.

Service as Action

COMMUNITY NEED	Random allocations are decided by rules that are rarely examined.
MATHEMATICAL CONNECTION	Sample spaces, combined events and expected value.
PROPOSED ACTION	Analyse one real random allocation and publish whether it is fair in probability.
EXPECTED IMPACT	A group can see whether its rules give everyone the same chance.
STUDENT REFLECTION	Record the rules, the probabilities and the verdict.

Community Project

PROJECT IDEA	A fairness audit of one school or club allocation.
NEED	Places, roles and prizes are allocated by rules nobody has checked.
RESEARCH QUESTION	Does the rule give every participant the same probability, and is it fair in effect?
MATHEMATICS	Sample spaces, combined events and expected value.
ACTION OR OUTCOME	A one-page audit with the probability for each participant.
BENEFICIARIES	Everyone who takes part.
EVIDENCE	The audit with the rules and the probabilities.

Personal Project connection · MYP 5

PERSONAL INTEREST	A game, a sport or a competition you enjoy.
LEARNING GOAL	To analyse the rules of a game and say whether it is fair.
PRODUCT OR OUTCOME	An analysis with probabilities and a verdict.
MATHEMATICS	Sample spaces, combined events and expected value.
PROCESS	List the sample space, compute the probabilities, and compare the players.
EVIDENCE	The analysis and the verdict.

CHAPTER 14 · PART V · INTEGRATED MODELLING

14

Capstone: Designing a Sustainable Space

This chapter uses every idea in the book. A space is designed by measuring it, modelling it, costing it and checking it against the needs it must serve.

ENGAGE

Your school has a courtyard 7.2 m by 4.8 m that floods in the rainy season and is unused in the dry one. Design a space that captures the rain, provides shade, and costs less than the budget.

Framing the inquiry

Key concept: Form
Related concepts: Measurement · Space · Change
Global context: Globalisation and sustainability
Global context exploration: Designing places that meet present needs without exhausting future resources

Statement of inquiry: A design is a model of a space that must satisfy several constraints at once — area, capacity, cost, light and access — and each constraint is a measurement that the design must meet; choosing the form of the space is therefore a mathematical decision whose consequences change over time.

Questions to guide us

- **Factual:** Which measurements does a design of a space require, and which formulae from this course give them?
- **Conceptual:** How do the constraints on a space interact, and why does a change to one measurement change the others?
- **Debatable:** Is a design that meets every stated constraint a good design, or does goodness depend on needs the constraints do not express?

ATL · SELF-MANAGEMENT. Students organise a multi-part design task, set their own deadlines for the modelling, the calculation and the report, and manage the time needed to complete each part.

CHAPTER 14 · INVESTIGATION C.1

The brief and the modelling cycle

INQUIRY QUESTION

How is a design brief turned into a set of mathematical constraints?

YOUR TASK

Choose a real space you know: a classroom, a bedroom, a garden or a corner of a playground. Measure it, write a brief with at least four measurable constraints, and draw it to a stated scale.

requirement	constraint	measurement needed
floor area	$A \geq 12\ m^2$	length and width
seating	at least 8 places	$0.5\ m^2$ per place
light	window area $\geq 1.5\ m^2$	window dimensions
circulation	path width ≥ 0.9 m	clear width

CHAPTER 14 · INVESTIGATION C.1

The brief and the modelling cycle

What do you notice?

- Every requirement became a quantity with a bound.
- The seating requirement needed an assumption about space per person.
- The light requirement was an area, not a number of windows.
- The circulation requirement was a width, which is a length.

Observation. A design brief becomes a set of constraints when each requirement is written as a quantity with a bound, and each quantity needs a stated measurement.

MATHEMATICAL CONCLUSION

A space is designed by measuring it first and then writing each requirement as a measurable constraint, because a requirement that cannot be measured cannot be checked.

Your own words

CHAPTER 14 · C.1

The brief and the modelling cycle

A design brief states what the space must do, and each requirement becomes a constraint that the design must satisfy. A constraint with no number cannot be checked.

The site is measured first: its area, its perimeter, its levels, the direction of the sun and the direction of the drainage. Each measurement limits the design.

A constraint can be an equality, such as a fixed area, or an inequality, such as a maximum cost. The feasible designs are those that satisfy all of them at once.

Turning a wish into a constraint is a modelling decision, and a designer who leaves a requirement unstated will discover it later as a failure.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

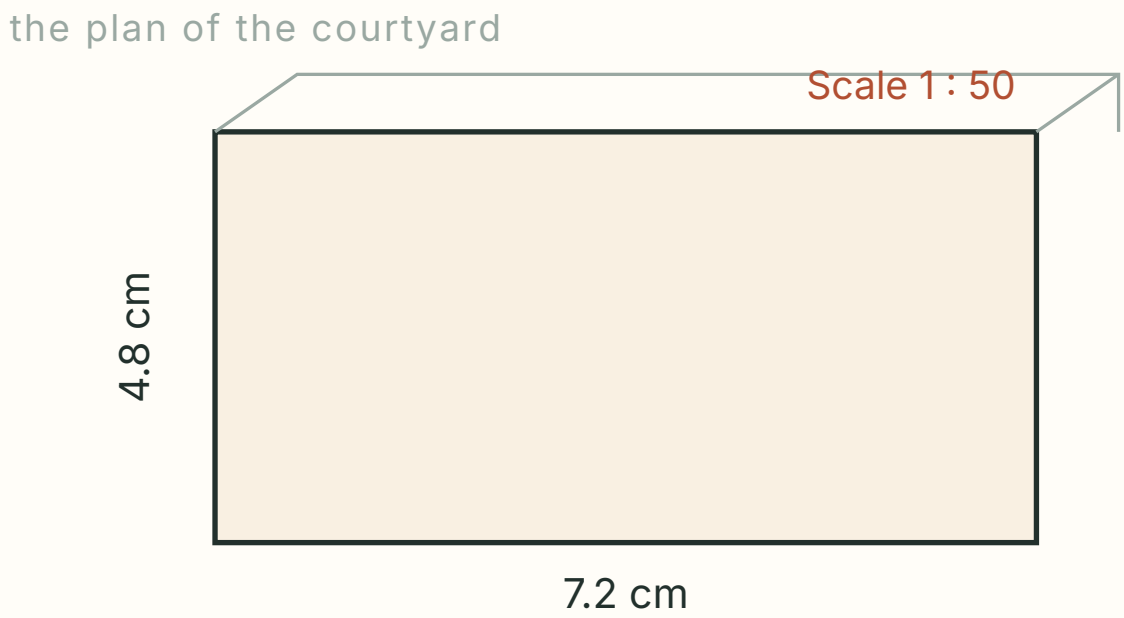
In your own words

CHAPTER 14 · C.1 · WORKED EXAMPLE 1

The brief and the modelling cycle

WORKED EXAMPLE C.1.1

Problem. A courtyard measures 7.2 m by 4.8 m on a plan drawn at a scale of 1 : 50. Find the real area of the courtyard and the area available for planting if paths take 20% of it.



Mathematical reasoning. Convert the plan dimensions using the scale, then compute the area.

1. On the plan the courtyard is 7.2 cm by 4.8 cm.
2. At a scale of 1 : 50 the real dimensions are $7.2 \times 50 = 360$ cm and $4.8 \times 50 = 240$ cm.
3. That is 3.6 m by 2.4 m, so the area is $3.6 \times 2.4 = 8.64$ m^2.
4. If paths take 20%, the planting area is 80% of $8.64 = 6.912$ m^2.
5. The plan gives the area directly, and the scale is what connects the drawing to the ground.

Final answer. 8.64 m² in total, of which 6.912 m² is available for planting

Interpretation. A scale drawing is a representation whose measurements are the real measurements divided by the scale factor, and the area needs the square of that factor.

CHAPTER 14 · C.1 · WORKED EXAMPLE 2

The brief and the modelling cycle

WORKED EXAMPLE C.1.2

Problem. A brief requires a seating area of at least 6 m^2, a water tank of at least 200 litres, and a total cost of at most 500 dollars. State these as constraints.

requirement	constraint	kind
seating area	$A \geq 6\ m^2$	inequality
water tank	$V \geq 200$ litres	inequality
total cost	$C \leq 500$ dollars	inequality
courtyard area	$A \leq 8.64\ m^2$	inequality

each requirement is a measurable bound

Mathematical reasoning. Write each requirement as a quantity and a bound.

1. The seating area is a quantity measured in square metres with a lower bound of 6.
2. The tank capacity is a quantity measured in litres with a lower bound of 200.
3. The cost is a quantity measured in dollars with an upper bound of 500.
4. The courtyard itself gives an upper bound on every area.
5. A design is feasible only when all four bounds hold at once.

Final answer. $A_{seating} \geq 6$, $V_{tank} \geq 200$, $C \leq 500$, $A_{total} \leq 8.64$

Interpretation. A constraint is a quantity with a bound, and a design is feasible when every bound is satisfied together.

CHAPTER 14 · INVESTIGATION C.2

Data, constraints and assumptions

INQUIRY QUESTION

What data does a design need, and how is each quantity turned into a constraint?

YOUR TASK

Design a rainwater system for one real roof: measure the roof area, find the local annual rainfall, choose an efficiency, and find the annual volume collected. Then choose a tank and find how many days of demand it covers.

quantity	value	source
roof area	30 m^2	measured
annual rainfall	900 mm	local records
efficiency	0.8	assumed
volume collected	21.6 m^3	computed
tank capacity	2.035 m^3	chosen
days of demand	51	computed

CHAPTER 14 · INVESTIGATION C.2

Data, constraints and assumptions

What do you notice?

- The annual volume came from three separate measurements.
- The efficiency was an assumption, not a measurement.
- The tank covered about 51 days, which is more than a typical dry season.
- A smaller tank would have been cheaper but riskier.

Observation. A water system is modelled by a supply volume, a capacity and a demand rate, and the duration is the capacity divided by the demand.

MATHEMATICAL CONCLUSION

A rainwater system is designed by matching the volume collected with the demand over the longest period in which supply falls short, and the tank size is the bridge between the two.

Your own words

CHAPTER 14 · C.2

Data, constraints and assumptions

A layout is a scale drawing in which every length is the real length divided by the scale factor, and every area is the real area divided by the square of the factor.

Drawing to scale makes the layout checkable: an area computed from the drawing must equal the area computed from the real dimensions, and the areas of the parts must sum to the area of the whole.

Circulation, planting, seating and water are all areas, and a plan that does not account for all of them is incomplete, so the first check on any plan is that its parts sum to its whole.

The capacity of a tank is its volume, and the volume of a cylinder is $\pi r^2 h$. A demand is a rate, and the time to fill or empty is the capacity divided by the rate.

A rainwater tank captures a volume equal to the roof area times the rainfall, and the fraction captured depends on the efficiency of the collection surface.

Modelling a water system means matching a supply rate with a demand rate, and the tank is what bridges the gap when the rates do not coincide.

Every quantity in a design rests on an assumption, and an assumption that is not stated cannot be questioned, which is why stating them is part of the modelling.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

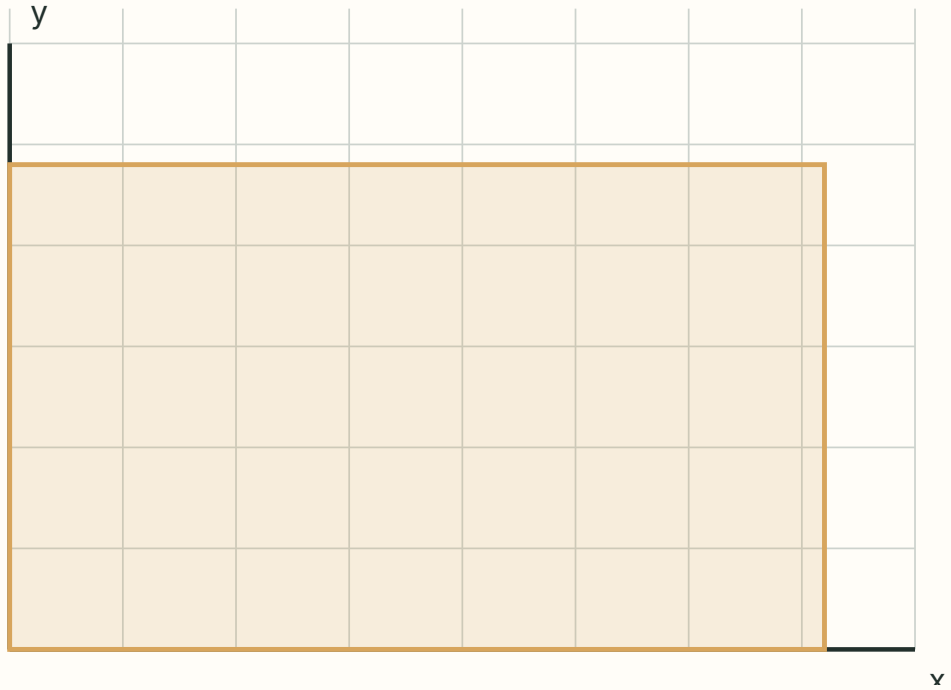
CHAPTER 14 · C.2 · WORKED EXAMPLE 1

Data, constraints and assumptions

WORKED EXAMPLE C.2.1

Problem. A plan of a courtyard is drawn at 1 : 50 and measures 7.2 cm by 4.8 cm. A seating area of 4.8 m² and paths of 1.92 m² are marked. Find the remaining area and check that the parts sum to the whole.

the plan divided into its parts



Mathematical reasoning. Compute the total from the plan and subtract the marked parts.

1. The real area is $3.6 \times 2.4 = 8.64$ m².
2. The seating area is 4.8 m² and the paths 1.92 m².
3. The remaining area is $8.64 - 4.8 - 1.92 = 1.92$ m².
4. The parts sum to $4.8 + 1.92 + 1.92 = 8.64$ m², which matches the whole.
5. So the plan accounts for the whole courtyard, which is the first check.

Final answer. 1.92 m² remaining, and the parts sum to 8.64 m²

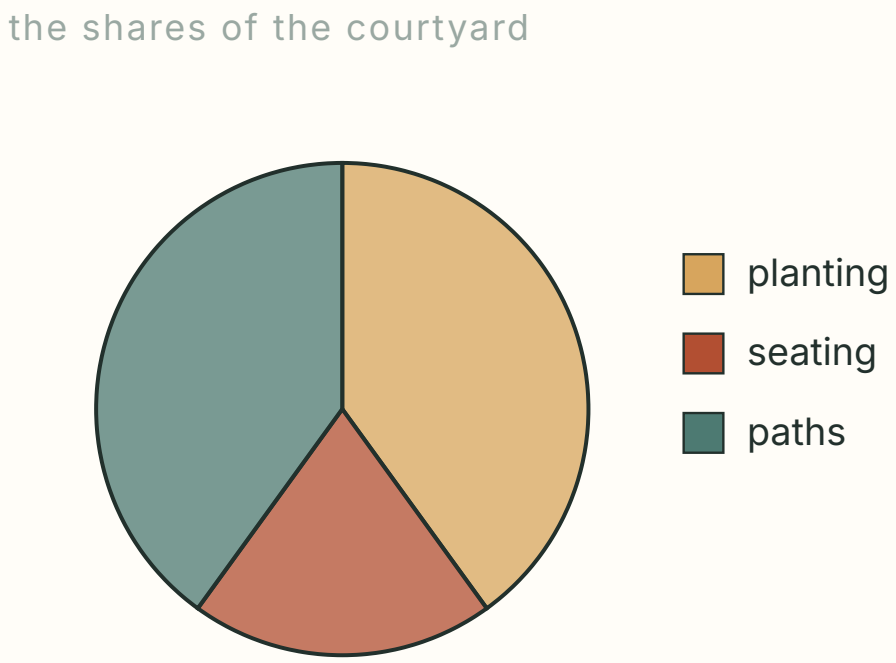
Interpretation. A plan whose parts do not sum to the whole has either an unmarked region or an arithmetic error, and either way it is not yet a design.

CHAPTER 14 · C.2 · WORKED EXAMPLE 2

Data, constraints and assumptions

WORKED EXAMPLE C.2.2

Problem. A designer wants the planting area to be at least 40% of the courtyard. Is the layout above acceptable?



Mathematical reasoning. Compare the planting share with the required share.

1. The planting area is 1.92 m^2 out of 8.64 m^2.
2. The share is 1.92/8.64 ≈ 22%.
3. The requirement is at least 40%, which would be 3.456 m^2.
4. So the layout is not acceptable as drawn.
5. The designer must reduce the seating or the paths, or increase the planting.

Final answer. no: planting is about 22%, below the required 40%

Interpretation. A percentage constraint is checked by comparing shares, and the shares must sum to 100% for the plan to be complete.

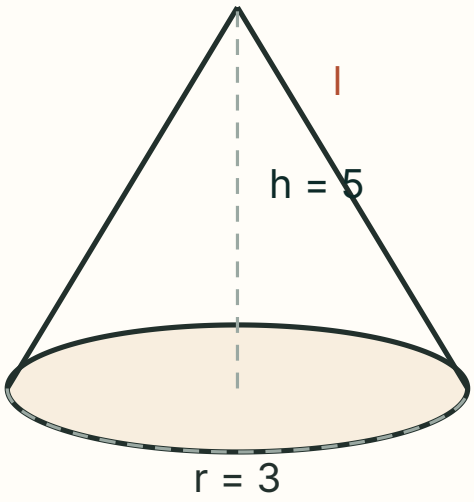
CHAPTER 14 · C.2 · WORKED EXAMPLE 3

Data, constraints and assumptions

WORKED EXAMPLE C.2.3

Problem. A roof of 30 m^2 collects rain. The annual rainfall is 900 mm and the collection efficiency is 0.8. Find the annual volume collected.

a tank fed by a 30 m^2 roof



Mathematical reasoning. Multiply the roof area by the rainfall depth and by the efficiency.

1. The rainfall depth is 900 mm = 0.9 m.
2. The volume falling on the roof is $30 \times 0.9 = 27$ m^3.
3. With an efficiency of 0.8 the volume collected is $27 \times 0.8 = 21.6$ m^3.
4. That is 21600 litres a year.
5. A tank of 2000 litres therefore stores about a month of the annual total.

Final answer. 21.6 m^3, or 21600 litres, a year

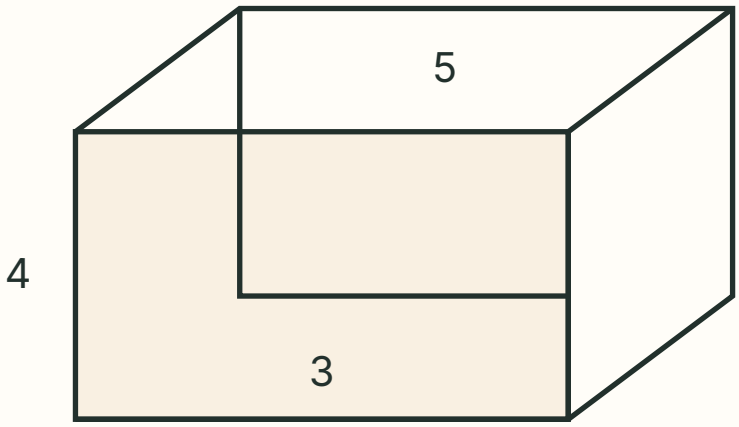
Interpretation. The collected volume is the roof area times the depth times the efficiency, and each of the three is a separate measurement.

CHAPTER 14 · C.2 · WORKED EXAMPLE 4

Data, constraints and assumptions

WORKED EXAMPLE C.2.4

Problem. A tank of radius 0.6 m and height 1.8 m is to hold the collected water. Find its capacity and say for how many days it would supply a garden using 40 litres a day.



a cylindrical tank of radius 0.6 m

Mathematical reasoning. Compute the volume and divide by the daily demand.

1. The capacity is $\pi \times 0.6^2 \times 1.8 = \pi \times 0.648 \approx 2.035$ m^3.
2. That is about 2035 litres.
3. At 40 litres a day the tank lasts 2035/40 ≈ 51 days.
4. So one filling covers about seven weeks of dry weather.
5. The tank is large enough for the dry season if it is filled before it begins.

Final answer. about 2035 litres, lasting about 51 days

Interpretation. The capacity and the demand are both measurements, and the duration is their ratio, which is the simplest model of a supply.

CHAPTER 14 · INVESTIGATION C.3

Analysis, evaluation and communication

INQUIRY QUESTION

How is a design analysed over time, evaluated against its brief, and communicated to a reader?

YOUR TASK

Review one real space against a brief of your own. Measure the quantities in the brief, survey at least 20 users, and write a review that states which constraints are met, which are not, and what the survey can and cannot support.

constraint	required	measured	verdict
seating area	≥ 6 m^2	5.5 m^2	not met
planting share	≥ 40%	22%	not met
path width	≥ 0.9 m	1.1 m	met
users wanting shade	survey	60% of 30	direction only

CHAPTER 14 · INVESTIGATION C.3

Analysis, evaluation and communication

What do you notice?

- Two of the three measurable constraints were not met.
- The survey gave a direction with a wide margin of error.
- The review identified precisely what to change.
- The unmet constraints were the ones nobody had measured before.

> **Observation.** A review measures each constraint and surveys the users, and the two kinds of evidence answer different questions about the design.

MATHEMATICAL CONCLUSION

A design is reviewed by measuring it against its constraints and by surveying its users, and the review states what each kind of evidence can support.

Your own words

CHAPTER 14 · C.3

Analysis, evaluation and communication

The cost of a design has a capital part, paid once, and a running part, paid each period. The total over n periods is the capital cost plus n times the running cost.

A cost that rises by a fixed percentage each year is exponential, and the total over n years is a geometric series, which is how inflation is modelled.

Comparing two designs means comparing their totals over the same period, and the cheaper capital cost is not always the cheaper total.

A design that saves money each year has a payback period, and a design whose savings grow is better in the long run than one whose savings are fixed.

A design is evaluated by measuring the space it creates and comparing the measurements with the constraints in the brief, so an unmeasured constraint cannot be evaluated.

The needs of the users are data, and collecting them means asking the people who will use the space and recording their answers as quantities rather than opinions.

A review compares what was intended with what was achieved, and the difference is the evidence for the next design, which is why a design that is never reviewed cannot improve.

Communication is part of the analysis: a result that cannot be explained to a reader has not been finished, and the explanation must name the model, the assumptions and the limits.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 14 · C.3 · WORKED EXAMPLE 1

Analysis, evaluation and communication

WORKED EXAMPLE C.3.1

Problem. A shade structure costs 400 dollars to build and 20 dollars a year to maintain, with maintenance rising 5% a year. Find the total cost over 5 years.

year	maintenance	running total
1	20.00	20.00
2	21.00	41.00
3	22.05	63.05
4	23.15	86.20
5	24.31	110.51

the maintenance grows geometrically

Mathematical reasoning. Model the maintenance as a geometric series and add the capital cost.

1. The maintenance is 20, 21, 22.05, ..., a geometric series with ratio 1.05.
2. The sum over 5 years is $20 \times (1.05^5 - 1)/0.05 \approx 110.51$ dollars.
3. Adding the capital cost gives $400 + 110.51 = 510.51$ dollars.
4. The running cost is about 22% of the capital cost over five years.
5. Over twenty years the running cost would exceed the capital cost.

Final answer. about 510.51 dollars over 5 years

Interpretation. A percentage rise in a running cost compounds, so the long-term total is much larger than five times the first year's cost.

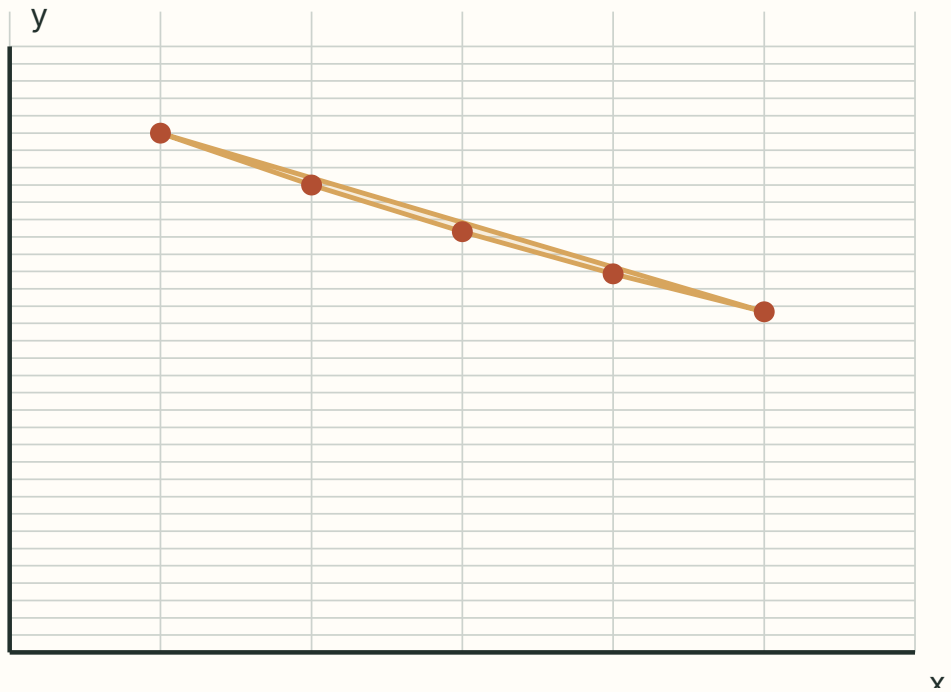
CHAPTER 14 · C.3 · WORKED EXAMPLE 2

Analysis, evaluation and communication

WORKED EXAMPLE C.3.2

Problem. A second design costs 520 dollars with maintenance falling by 10% a year from 30 dollars. Compare the two over 5 years.

the maintenance falls over time



Mathematical reasoning. Model the second maintenance as a decreasing geometric series and compare the totals.

1. The maintenance is 30, 27, 24.3, ..., a geometric series with ratio 0.9.
2. The sum over 5 years is $30 \times (1 - 0.9^5)/0.1 \approx 123.51$ dollars.
3. The total is $520 + 123.51 = 643.51$ dollars.
4. The first design costs about 510.51 dollars, so it is cheaper over five years.
5. But the second design's costs fall, so it would win over a longer period.

Final answer. the first design costs about 510.51 dollars and the second about 643.51 dollars over 5 years

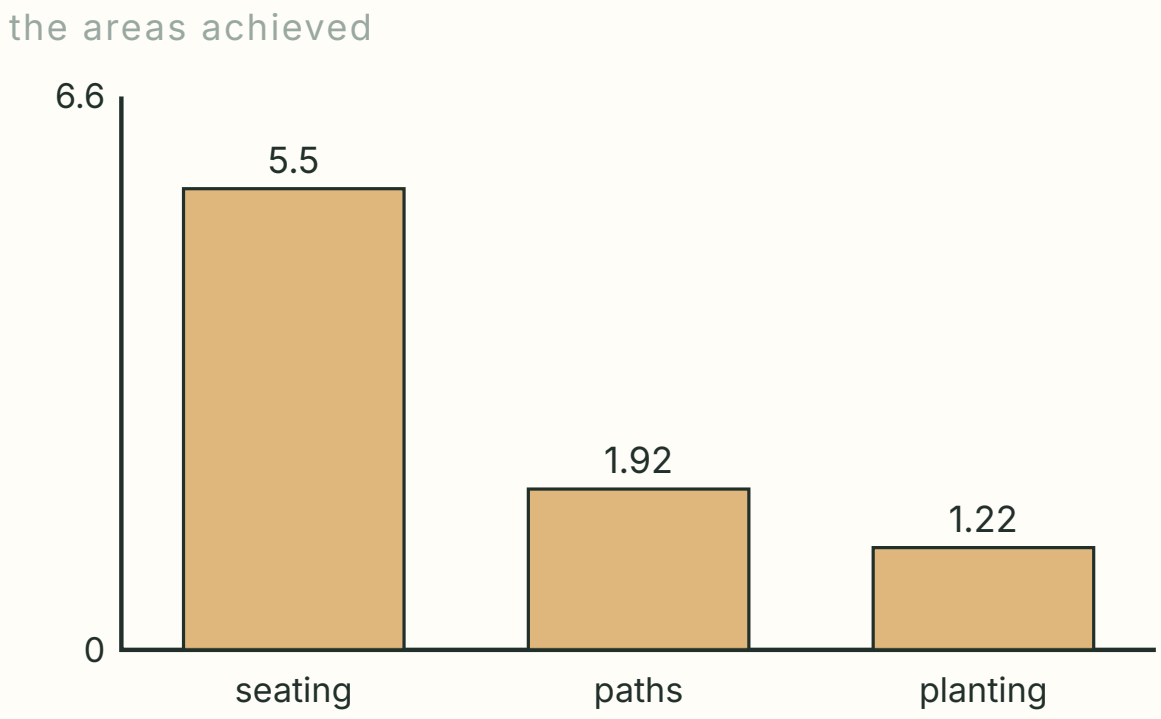
Interpretation. The comparison reverses over a longer period, which is why the time horizon must be stated before a design is chosen.

CHAPTER 14 · C.3 · WORKED EXAMPLE 3

Analysis, evaluation and communication

WORKED EXAMPLE C.3.3

Problem. A design provides 5.5 m^2 of seating in a courtyard of 8.64 m^2, against a requirement of at least 6 m^2 and at least 50%. Evaluate the design.



Mathematical reasoning. Test both constraints and report which are met.

1. The seating area is 5.5 m^2, which is below the required 6 m^2.
2. The share is 5.5/8.64 ≈ 64%, which is above the required 50%.
3. So one constraint is met and the other is not.
4. The design therefore fails the brief, even though it passes one test comfortably.
5. The review identifies exactly which constraint failed.

Final answer. the design fails: seating is 5.5 m^2 against a required 6 m^2, though the share of 64% passes

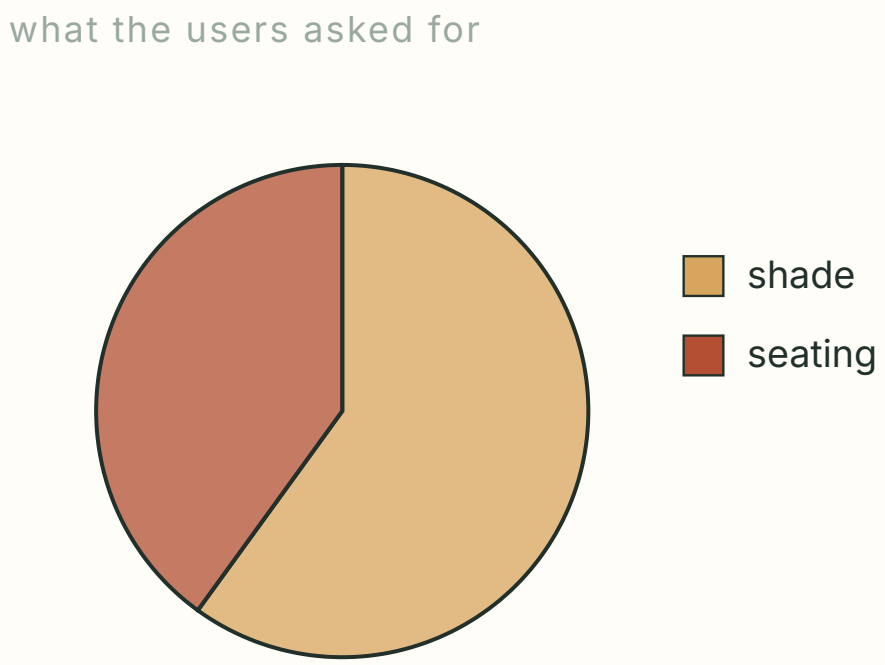
Interpretation. A review tests each constraint separately, and a design that fails one is not acceptable however well it meets the others.

CHAPTER 14 · C.3 · WORKED EXAMPLE 4

Analysis, evaluation and communication

WORKED EXAMPLE C.3.4

Problem. A survey of 30 users finds 18 want more shade and 12 want more seating. Find the percentages and state what a sample of 30 can support.



Mathematical reasoning. Convert the counts to percentages and state the limits of the sample.

1. 18 of 30 want more shade, which is 60%.
2. 12 of 30 want more seating, which is 40%.
3. The sample is 30 people, so the margin of error is about $1/\sqrt{30} \approx 18\%$.
4. So the true figures could be anywhere from about 42% to 78% for shade.
5. The survey shows a direction but not a precise proportion.

Final answer. 60% want shade and 40% want seating, with a margin of error of about 18%

Interpretation. A small sample shows a direction, and the margin of error is what limits how firmly the direction can be stated.

CHAPTER 14 · INVESTIGATION C.4

Assessment task and rubric

INQUIRY QUESTION

What must an assessment task ask for, and how is the work judged?

YOUR TASK

Write a one-page design report for a space you have modelled in this chapter. Include the brief, at least four constraints, three calculations in reportable form, one survey result, and a statement of the assumptions and the limits of the design.

section	content	checkable?
brief	four measurable constraints	yes
model	area, capacity, cost	yes
calculations	formula, substitution, units	yes
survey	30 users, 60% want shade	direction only
assumptions	demand, efficiency, rainfall	yes

CHAPTER 14 · INVESTIGATION C.4

Assessment task and rubric

What do you notice?

- Every calculation showed its formula, substitution and units.
- The survey result was reported with its margin of error.
- The assumptions were listed so the reader could judge them.
- The limits of the design were stated as plainly as its achievements.

Observation. A design report is checkable when every calculation shows its formula, substitution and units, and when the assumptions are stated.

MATHEMATICAL CONCLUSION

A design is presented as a chain of checkable calculations supported by stated assumptions, and the report is complete only when it also states what the design does not do.

Your own words

CHAPTER 14 · C.4

Assessment task and rubric

An assessment task asks for a model, a calculation and a judgement, and it is marked against criteria that name what each level of achievement looks like.

A rubric turns a judgement into a description: each criterion is stated, each level is described, and the evidence for the level is named, so that two markers reading the same work reach the same level.

The four criteria used in this course are Knowing and understanding, Investigating patterns, Communicating, and Applying mathematics in real-life contexts, and a task must give a student the chance to show all four.

A task that can be answered by copying a worked example tests nothing, so the numbers, the space and the question must be the student's own.

Every submission must state its assumptions and its limitations, because a model that hides them is not a model that can be evaluated.

> FORMAL DEVELOPMENT
>
> Work through the investigation above before reading this page: the statements here are the formal version of what you discovered.

In your own words

CHAPTER 14 · C.4 · WORKED EXAMPLE 1

Assessment task and rubric

WORKED EXAMPLE C.4.1

Problem. Write the calculation for the area of a courtyard 3.6 m by 2.4 m in the form a report should use.

element	content
quantity	area of the courtyard
formula	A = length × width
substitution	A = 3.6 × 2.4
result	8.64 m^2
check	3.6 × 2.4 ≈ 4 × 2.5 = 10, close enough

a calculation in reportable form

Mathematical reasoning. Show the formula, the substitution, the result with units and an estimate as a check.

1. State the quantity: the area of the courtyard.
2. State the formula: A = length × width.
3. Substitute: A = 3.6 × 2.4.
4. Give the result with units: 8.64 m^2.
5. Check by estimating: 3.6 × 2.4 is close to 4 × 2.5 = 10, and 8.64 is of that order.

Final answer. A = 3.6 × 2.4 = 8.64 m^2

Interpretation. A calculation is reportable when a reader can reproduce it, which requires the formula, the substitution and the units.

CHAPTER 14 · C.4 · WORKED EXAMPLE 2

Assessment task and rubric

WORKED EXAMPLE C.4.2

Problem. A report claims a tank will last 'about two months'. Rewrite the claim so that it can be checked.

element	content
claim	the tank lasts about 51 days
capacity	2.035 m^3 = 2035 litres
demand	40 litres a day
calculation	2035 ÷ 40 = 50.9 days
assumption	the demand stays at 40 litres a day

a claim that a reader can check

Mathematical reasoning. Replace the vague claim with the numbers that produce it.

1. State the capacity: 2.035 m^3, which is 2035 litres.
2. State the demand: 40 litres a day.
3. Show the calculation: 2035 ÷ 40 = 50.9 days.
4. State the assumption: the demand stays at 40 litres a day.
5. The claim 'about 51 days' can now be checked by anyone.

Final answer. the tank lasts 2035 ÷ 40 ≈ 51 days, assuming a demand of 40 litres a day

Interpretation. A vague claim cannot be checked, and a claim that cannot be checked cannot be trusted or improved.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding

14.A1 FLUENCY

A plan is drawn at 1 : 50. Find the real length of a wall measuring 7.2 cm on the plan.

14.A2 FLUENCY

A courtyard is 3.6 m by 2.4 m. Find its area.

14.A3 FLUENCY

A tank is a cylinder of radius 0.5 m and height 1.5 m. Find its capacity in litres.

14.A4 FLUENCY

A roof of 20 m^2 receives 100 mm of rain. Find the volume collected at an efficiency of 0.9.

14.A5 FLUENCY

A garden uses 30 litres a day. Find how long a 900-litre tank lasts.

14.A6 FLUENCY

A structure costs 300 dollars to build and 25 dollars a year to maintain. Find the 4-year total.

CRITERION A Knowing and understanding

Criterion A practice · Knowing and understanding (continued)

14.A7 FLUENCY

A maintenance cost of 20 dollars rises 10% a year. Find the cost in year 3.

14.A8 FLUENCY

A design provides 5 m^2 of seating in 10 m^2. Find the percentage.

14.A9 FLUENCY

A survey of 40 users finds 24 want more shade. Find the percentage.

14.A10 FLUENCY

A path must be at least 0.9 m wide. A measured path is 0.85 m. Is it acceptable?

14.A11 FLUENCY

A shade sail covers 6 m^2 of a 12 m^2 courtyard. Find the percentage covered.

14.A12 FLUENCY

A tank holds 500 litres and is filled at 20 litres a minute. Find the filling time.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns

14.B1 REASONING

The areas of the parts of a plan are 4.8, 1.92 and 1.92 m^2. Find the total and the share of the largest part.

14.B2 REASONING

Investigate the pattern in the capacity of cylindrical tanks of fixed height as the radius grows.

14.B3 REASONING

Conjecture a rule for the roof area needed to fill a 1000-litre tank from 100 mm of rain at an efficiency of 0.8.

14.B4 REASONING

Investigate the pattern in the 5-year total of a 400-dollar structure with maintenance rising 5% a year, for maintenance starting at 10, 20 and 30 dollars.

14.B5 REASONING

Build a table of the duration of tanks of 500, 1000 and 2000 litres at a demand of 40 litres a day.

14.B6 REASONING

Conjecture a rule for the payback period of a 600-dollar investment that saves 50 dollars a year.

CRITERION B Investigating patterns

Criterion B practice · Investigating patterns (continued)

14.B7 REASONING

Investigate the pattern in the margin of error of surveys of 10, 40 and 160 people.

14.B8 REASONING

A roof's collected volume is modelled with an efficiency of 0.8. Investigate the effect of an efficiency of 0.6 on the tank size needed.

14.B9 CHALLENGE

Find the pattern in the surface area of open-top cylindrical tanks of fixed volume as the radius grows.

14.B10 REASONING

Investigate the pattern in the number of days of water stored per square metre of roof.

14.B11 REASONING

Conjecture a rule for the total cost over n years of a capital cost C with maintenance rising at r a year from M.

CRITERION C Communicating

Criterion C practice · Communicating

14.C1 REASONING
Explain in your own words what a constraint is in a design brief.

14.C2 FLUENCY
Write a complete solution to: find the real area of a plan measuring 6 cm by 4 cm at a scale of 1 : 100.

14.C3 REASONING
Choose between a scale drawing and a formula to find the area of an irregular courtyard and justify.

14.C4 FLUENCY
Set out a full solution to: find the capacity in litres of a cylinder of radius 0.4 m and height 1.2 m.

14.C5 REASONING
Explain why a percentage constraint on an area is checked by comparing shares.

14.C6 REASONING
Describe, with an example, the difference between a capital cost and a running cost.

CRITERION C Communicating

Criterion C practice · Communicating (continued)

14.C7 REASONING

Explain why the duration of a tank is the capacity divided by the demand.

14.C8 REASONING

Write the reasoning that shows the volume collected from a roof is the area times the depth times the efficiency.

14.C9 REASONING

Set out the argument that a design that meets every constraint can still be a poor design.

14.C10 REASONING

Explain what a payback period is and how it is found.

14.C11 REASONING

Describe the relationship between the margin of error of a survey and its sample size.

14.C12 REASONING

Explain, with a design of your own, when a report is not checkable.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts

14.D1 APPLICATION

A roof of 45 m^2 collects rain with an efficiency of 0.75. Find the volume collected from 800 mm of rain and the tank needed for 60 days at 50 litres a day.

14.D2 APPLICATION

A design costs 600 dollars with maintenance of 40 dollars a year rising 8% a year. Find the total over 10 years.

14.D3 APPLICATION

Two designs cost 800 and 950 dollars. The first has running costs rising 10% a year from 50 dollars; the second has running costs falling 5% a year from 60 dollars. Find which is cheaper over 10 years.

14.D4 REASONING

A courtyard of 9 m^2 must have at least 45% planting, at most 30% paths and at least 2 m^2 of seating. Investigate whether these constraints can be met together.

14.D5 APPLICATION

A tank of 1500 litres is filled from a 25 m^2 roof at 0.8 efficiency. Find the rainfall needed to fill it and the number of fills in a 900 mm year.

14.D6 APPLICATION

A survey of 50 users finds 35 want more shade. Find the percentage and the margin of error, and state what the survey supports.

CRITERION D Applying mathematics in real-life contexts

Criterion D practice · Applying mathematics in real-life contexts (continued)

14.D7 REASONING

A shade structure costs 350 dollars and saves 60 dollars a year in cooling. Investigate the payback period and the effect of a 5% annual rise in the saving.

14.D8 APPLICATION

A plan at 1 : 25 shows a room 9.6 cm by 6.4 cm. Find the real dimensions and the area, and the number of 0.5 m^2 desks that fit.

14.D9 REASONING

A garden bed is 4 m by 1.5 m and needs 60 mm of water a week. Investigate the weekly volume and the roof area needed at 0.8 efficiency and 20 mm of weekly rain.

14.D10 CHALLENGE

A design's total cost over n years is $C = 500 + 30 \times 1.06^{n-1}$. Investigate the year in which the running cost first exceeds the capital cost.

14.D11 REASONING

A school courtyard of 10 m^2 must provide seating, planting, a path and a tank of at least 500 litres. Investigate whether a tank of radius 0.5 m and height 0.7 m fits within the 10% of the area allowed for it.

14.D12 REASONING

Explain, with a design of your own, when a design that meets every stated constraint is still unacceptable.

CHAPTER 14 · APPLICATION AND ASSESSMENT

Applying chapter 14

Each task asks you to model part of a design. State the constraint, the formula you use, and the units of your answer.

A1 · The courtyard design

A school courtyard measures 7.2 m by 4.8 m on a plan drawn at 1 : 50. The brief requires at least 45% planting, at most 30% paths, at least 6 m^2 of seating, and a rainwater tank of at least 500 litres. Design a layout that meets every constraint, give the area of each part, choose a tank and justify its size, and state the running cost over 5 years with a stated assumption about its change.

Criteria A and D · model the layout, check every constraint, and justify the choice of tank and the cost model.

A2 · The review

A design provides 5.5 m^2 of seating, 2.0 m^2 of paths and 1.14 m^2 of planting in a courtyard of 8.64 m^2. A survey of 30 users finds 18 who want more shade and 12 who want more seating. Test the design against the brief in task A1, report the survey with its margin of error, and write a recommendation that states what the evidence supports and what it does not.

Criteria B and C · test each constraint, interpret the survey within its limits, and justify the recommendation.

CHAPTER 14 · REVIEW

What you should now be able to do

Key ideas

- A constraint is a quantity with a bound, and a design must satisfy all of them.
- A scale drawing divides lengths by the scale factor and areas by its square.
- The areas of the parts of a plan must sum to the area of the whole.
- The capacity of a tank is its volume, and the duration is capacity divided by demand.
- The volume collected from a roof is area times depth times efficiency.
- A running cost that rises by a percentage is a geometric series.
- The comparison of two designs depends on the time horizon.
- A survey's margin of error is about 1/√n.

Vocabulary

brief · constraint · feasible · scale drawing · capacity · demand · capital cost · running cost · payback period · margin of error · assumption · review

CHAPTER 14 · REVIEW

Mixed practice · consolidate and extend

Do not use a calculator unless a question asks for one. Show a method where one is requested.

14.R1 A plan at 1 : 50 measures 7.2 cm by 4.8 cm. Find the real area.

14.R2 Find the capacity in litres of a cylinder of radius 0.5 m and height 1 m.

14.R3 A roof of 30 m^2 collects 800 mm of rain at 0.8 efficiency. Find the volume.

14.R4 A tank of 2000 litres supplies 50 litres a day. Find the duration.

14.R5 A cost of 400 dollars has maintenance of 20 dollars rising 5% a year. Find the 5-year total.

14.R6 Explain why a design that meets every stated constraint can still be unacceptable.

CHAPTER 14 · REVIEW

Returning to the statement of inquiry

STATEMENT OF INQUIRY

A design is a model of a space that must satisfy several constraints at once — area, capacity, cost, light and access — and each constraint is a measurement that the design must meet; choosing the form of the space is therefore a mathematical decision whose consequences change over time.

Reflect on the chapter

Answer in your own words, using evidence from this chapter.

- Write the statement of inquiry for this chapter in your own words.
- Which mathematical ideas from this chapter did you use to understand it? Name at least two.
- How did the key concept of form and the related concepts of measurement, space and change appear in your work?
- How is the global context of globalisation and sustainability, and its exploration of designing places that meet present needs without exhausting future resources, visible in the investigations and examples you completed?
- Use evidence from an investigation or a worked example to justify your answer.
- Has your understanding of the statement of inquiry changed since the start of the chapter? Explain how.
- What are the limitations of a mathematical model for a decision about a shared space, and what would you still like to ask?

CHAPTER 14 · PROJECTS, SERVICE AND ACTION

Acting on what you have learned

A design is a claim about the future. Model one and say what would have to stay true for your design to work.

Service as Action

COMMUNITY NEED	Shared spaces are designed and changed without measuring the needs they must serve.
MATHEMATICAL CONNECTION	Measurement, area, capacity, cost and data.
PROPOSED ACTION	Model one real space, check it against a written brief, and publish the review.
EXPECTED IMPACT	A group can see which constraints its space meets and which it fails.
STUDENT REFLECTION	Record the brief, the measurements and the verdict.

Community Project

PROJECT IDEA	A sustainable redesign of one community space.
NEED	A courtyard, a market corner or a waiting area is unused or uncomfortable.
RESEARCH QUESTION	What would a space that meets the users' measured needs look like, and what would it cost?
MATHEMATICS	Scale, area, capacity, cost and survey data.
ACTION OR OUTCOME	A design report with a scale plan, the calculations and a review.
BENEFICIARIES	Everyone who uses the space.
EVIDENCE	The report with the plan, the calculations and the survey.

Personal Project connection · MYP 5

PERSONAL INTEREST	A room, a desk or a corner that is yours.
LEARNING GOAL	To design a space that meets constraints you have measured yourself.
PRODUCT OR OUTCOME	A design report with a plan and a costed recommendation.
MATHEMATICS	Scale, area, capacity, cost and data.
PROCESS	Measure, model, cost and review, stating every assumption.
EVIDENCE	The report with the plan, the calculations and the review.

CHAPTER 1 · CRITERION A · KNOWING AND UNDERSTANDING

1.A1 **$\frac{5}{11}$**
Let $x = 0.4555\ldots$, then $10x - x = 4.1$.

1.A2 **$\sqrt{16} = 4$ rational; $\sqrt{20}$ irrational; $\sqrt[3]{8} = 2$ rational**
Simplify each root before classifying.

1.A3 **If it were rational r, then $(\sqrt{3} + \sqrt{2})^2 = 5 + 2\sqrt{6}$ would be rational, so $\sqrt{6}$ would be rational, which is false.**
Square the sum and use the closure of the rationals.

1.A4 **9**
Take the cube root first, then square.

1.A5 **$x = 3$**
Write both sides as powers of 2.

1.A6 **$3x^2y^2$**
Apply the power law then the quotient law.

1.A7 **4.2×10^{-7}**
Move the point seven places right.

1.A8 **$4\sqrt{3}$**
Extract the square factor 16.

1.A9 **$5\sqrt{7}/7$**
Multiply top and bottom by $\sqrt{7}$.

1.A10 **[3.395, 3.405) kg**
Half of the last unit either side.

1.A11 **Its decimal recurs, so no finite decimal equals it; only a fraction can be exact.**
A finite decimal is a fraction whose denominator is a product of 2s and 5s.

1.A12 **[15.4225, 17.1075) m^2**
Use the extreme dimensions for the extreme areas.

CHAPTER 1 · CRITERION B · INVESTIGATING PATTERNS

1.B1	**2^4, 2^5, 2^6; each term is the previous term multiplied by 2.** A geometric sequence has a constant multiplier.
1.B2	**Never: $n^2 < n^2 + n < (n + 1)^2$, so the root lies strictly between two consecutive integers.** Sandwich the value between consecutive squares.
1.B3	**The gaps decrease as n grows, because the square root function flattens.** Compute the differences and look at their trend.
1.B4	**The last digit cycles 2, 4, 8, 6, because $2^4 = 16$ ends in 6 and multiplying by 2 repeats the cycle.** Only the last digit of each power matters.
1.B5	**The maximum is at n = 3; the values fall after that.** Compute the values and compare.
1.B6	**Each perimeter is the previous one plus 1, because each new triangle adds a leg of length 1.** Track what is added at each step.
1.B7	**Yes: $n^3 - n = (n - 1)n(n + 1)$, the product of three consecutive integers, so it is divisible by both 2 and 3.** Factorise first.
1.B8	**The number of digits is the smallest integer greater than n log 2, so it grows roughly every three or four steps.** Compare with powers of 10.
1.B9	**The sums approach 1, because the total is $1 - 1/2^n$.** Write each sum with a common denominator.
1.B10	**The interval widens as n grows, because the square root function flattens.** Use the bounds of n and take roots.
1.B11	**The rule is $2n^2 - 2n + 1$, giving 61 for figure 6.** Take differences to identify a quadratic.

CHAPTER 1 · CRITERION C · COMMUNICATING

1.C1 **An index law is a rule for combining powers; it holds for negative exponents because $a^n \times a^{-n}$ must equal $a^0 = 1$, so a^{-n} is forced to be $1/a^n$.**
State the requirement, then the consequence.

1.C2 **$\sqrt{200} = 10\sqrt{2}$ and $\sqrt{72} = 6\sqrt{2}$, so the difference is $4\sqrt{2}$.**
Extract square factors, then collect like terms.

1.C3 **$2\sqrt{3}$ if the calculation continues or must be exact; 3.464 only to report a final value to a stated precision.**
Say what each form costs.

1.C4 **6×10^{-3}**
Multiply the coefficients and add the exponents.

1.C5 **Rounding gives 4.0 with error interval [3.965, 3.975); truncating gives 3.9 with interval [3.90, 3.91).**
State each interval.

1.C6 **[4.95, 5.05) cm, because the last unit is 0.1 cm and half of it is 0.05 cm either side.**
A number line makes the interval visible.

1.C7 **Because subtracting a smaller number gives a larger result, so the maximum of a − b uses the minimum of b.**
Test with numbers.

1.C8 **$\sqrt{2} = 2\text{^}(1/2)$, so $\sqrt{2} \times \sqrt{2} = 2\text{^}(1/2) \times 2\text{^}(1/2) = 2^1 = 2$.**
Convert to powers and apply the product law.

1.C9 **Let the numbers be a/b and c/d; their sum is (ad + bc)/bd, a fraction of integers with non-zero denominator, hence rational.**
Use the definition directly.

1.C10 **A number written as $a \times 10^n$ with $1 \le a < 10$; for example 3.2×10^8 and 4.5×10^{-9}.**
State the definition and two cases.

1.C11 **The number of significant figures states the precision claimed; more figures mean a narrower error interval.**
Link digits to interval width.

1.C12 **A universal claim says every case works; one case that fails makes it false, as $\sqrt{2} \times \sqrt{8} = 4$ disproves 'the product of two irrationals is irrational'.**
Contrast with needing infinitely many examples to prove one.

CHAPTER 1 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

1.D1 **One pipe: [2.495, 2.505) m; three pipes: [7.485, 7.515) m.**
Half of the last unit either side, then multiply.

1.D2 **4.6×10^{-2} kg, interval [0.0455, 0.0465) kg.**
Convert first, then bound.

1.D3 **[426, 534) litres, so 480 litres is possible but not certain.**
Extreme dimensions give extreme volumes; 1000 litres = 1 m^3.

1.D4 **About 3×10^8 m/s, to one significant figure.**
Divide distance by time in standard form.

1.D5 **1.3125 kg, interval about [1.265, 1.360) kg.**
Scale the interval with the multiplier.

1.D6 **About 21.3 litres, interval about [20.7, 21.9] litres.**
Multiply, then bound the inputs.

1.D7 **No: 2.98 m to the nearest centimetre could be as short as 2.975 m, which is below 3.00 m.**
Compare the lower bound with the requirement.

1.D8 **About 660 cm^2 to 690 cm^2, a spread of about 2%.**
Scale each dimension, then the interval.

1.D9 **It is not exact; the customer should ask for the exact balance and the rounding policy, because accumulated rounding is the bank's gain or the customer's loss.**
Compare rounding each step with rounding once.

1.D10 **Yes: the interval [2.45, 2.55) ml lies inside [2.4, 2.6] ml.**
Compare intervals, not single values.

1.D11 **[1.019, 1.027] hectares, a spread of about 0.8%, so the yield estimate carries the same relative error.**
Bound each side, then the area; 1 ha = 10 000 m^2.

1.D12 **Any answer is acceptable if it shows that the digits beyond the instrument's precision are not measured and imply an accuracy that does not exist.**
Link digits to error interval.

CHAPTER 2 · CRITERION A · KNOWING AND UNDERSTANDING

2.A1 **$x^2 + 2x - 15$**
Multiply every term of one bracket by every term of the other.

2.A2 **$(x - 3)(x - 4)$**
Two numbers with sum -7 and product 12.

2.A3 **$(x + 4)/(x - 4)$**
Factorise both parts and cancel.

2.A4 **$x = (y + 7)/3$**
Add 7, then divide by 3.

2.A5 **4**
Square the coefficient and multiply the indices.

2.A6 **16x**
Expand both squares and collect; the x^2 terms cancel.

2.A7 **$(2x + 7)(2x - 7)$**
A difference of two squares.

2.A8 **$(5x + 3)/(x(x + 1))$**
Use the common denominator $x(x + 1)$.

2.A9 **x^6**
Multiply the indices: $(-2)(-3) = 6$.

2.A10 **$a = (v - u)/t$**
Subtract u, then divide by t.

2.A11 **$4x^2 - 12xy + 9y^2$**
Use $a^2 - 2ab + b^2$ with $a = 2x$, $b = 3y$.

2.A12 **$2y/(3x^2)$**
Divide the coefficients and subtract the indices.

CHAPTER 2 · CRITERION B · INVESTIGATING PATTERNS

2.B1 **4n + 2, giving 22 cm for the fifth.**
The common difference is 4.

2.B2 **The areas are 2, 6, 12, 20, following $n^2 + n$.**
Take differences to identify a quadratic.

2.B3 **n(n − 3)/2, giving 2, 5, 9.**
Each vertex joins to n − 3 others; divide by 2.

2.B4 **The differences are the odd numbers, because $(n + 1)^2 - n^2 = 2n + 1$.**
Expand the difference of squares.

2.B5 **Those with $b^2 - 4c$ a perfect square: c = −5, 3, 4.**
Compute the discriminant.

2.B6 **n^2, because the odd numbers are the differences of consecutive squares.**
Relate to the pattern in 2.2.4.

2.B7 **It is prime for x = 1 to 39 but not for x = 40, where the value is 41^2.**
Test, then look for a counterexample.

2.B8 **x(x + 1)(x + 2)(x + 3), which is always divisible by 24.**
Four consecutive integers include a multiple of 4 and two evens.

2.B9 **n/(n + 1), because each term is 1/k − 1/(k + 1).**
Split the fraction into partial fractions.

2.B10 **The sum of the two smaller squares equals the largest, which is Pythagoras' theorem.**
Draw and measure for several triangles.

2.B11 **The last digit of n^5 equals the last digit of n.**
Only the last digit matters, so test ten cases.

CHAPTER 2 · CRITERION C · COMMUNICATING

2.C1 **They give the same value for every permitted input, which is stronger than agreeing at one value.**
State the definition and its strength.

2.C2 **$3(x + 2)(x - 2)$**
Take out the common factor, then the difference of squares.

2.C3 **$(x + 2)(x + 3)$, because a product is zero only when a factor is zero.**
Say what each form gives.

2.C4 **$(x + 5)/(x - 5)$, with $x \neq 5$.**
Factorise both parts and cancel.

2.C5 **Squaring a sum produces a cross term $2xy$, which is missing from the sum of the squares.**
Expand and compare.

2.C6 **A factor may be cancelled because it divides both parts; a term may not, because $(x + 3)/(x)$ does not simplify.**
Test with a number.

2.C7 **The original expression is undefined there even when the simplified form is not, so the two expressions differ at those points.**
Link to the definition of a function.

2.C8 **$x^n \times x^{-n} = x^0 = 1$, so x^{-n} is the reciprocal of x^n.**
Use the product law and the zero index.

2.C9 **The roots are $(-b \pm \sqrt{(b^2 - 4c)})/2$, and they are rational exactly when $b^2 - 4c$ is a perfect square.**
Use the quadratic formula.

2.C10 **The quantity isolated on one side; in $v = u + at$ the subject is v.**
State the definition and an example.

2.C11 **Factorising converts a product equal to zero into linear factors, each of which gives a solution.**
Use the zero-product property.

2.C12 **$x = -3$ becomes $x^2 = 9$, which also has $x = 3$ as a solution.**
Square and check both roots.

CHAPTER 2 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

2.D1 **$A = (2x + 6)(x - 1)$, zero at $x = -3$ or $x = 1$.**
A product is zero when a factor is zero.

2.D2 **Break-even at $x = 20$ and $x = 200$; the greatest profit is at $x = 110$.**
The maximum of a quadratic with roots 20 and 200 lies halfway between them.

2.D3 **$S = 2(x + 3)(x + 4)$, smallest at $x = -3.5$ with $S = 0.5$ cm^2.**
Complete the square or use the midpoint of the roots.

2.D4 **$n = (C - 20)/0.05$, so 300 minutes for 35 dollars.**
Rearrange, then substitute.

2.D5 **Zero only when $k = 0$, since $k^2 \geq 0$ and the frame factors are positive.**
A product is zero when a factor is zero.

2.D6 **The greatest volume is at $x = 1$, by symmetry about the midpoint of the roots 0 and 2.**
A cubic with roots 0, −3, 2 is symmetric about $x = 1$.

2.D7 **Yes, at $x = -4$ or $x = 5$, but only $x = 5$ is physically meaningful.**
A product is zero when a factor is zero.

2.D8 **$x = 20$ dollars, because the maximum lies halfway between the roots 0 and 40.**
Use the symmetry of a quadratic with two known roots.

2.D9 **The maximum is near $x = 4.2$, found by testing values rather than by symmetry, because the cubic has no axis of symmetry.**
Evaluate at several points.

2.D10 **$r = 1 - (1/2)\text{^}(1/3) \approx 0.206$, about 21% per year.**
Set $V = 6000$, divide, and take the cube root.

2.D11 **$r = \sqrt{(3V/(\pi h))}$; this form is best when V and h are known.**
Multiply by 3, divide by πh, then take the root.

2.D12 **Any answer is acceptable if it identifies a case where the factorised form hides the size of a coefficient or a gradient, as in $1000(x + 1)$.**
Give a specific counterexample.

CHAPTER 3 · CRITERION A · KNOWING AND UNDERSTANDING

3.A1	**$x = 7$** Add 7, then divide by 4.
3.A2	**$x = -4$** Collect, then divide by −3 and reverse the sign.
3.A3	**$x = 11/3$** Expand both sides, then collect.
3.A4	**$x = 12$** Multiply by 4, then subtract 8.
3.A5	**$x = 6, y = 3$** Add the equations to eliminate y.
3.A6	**$x = 4, y = 3$** Add the equations to eliminate y.
3.A7	**$x < 5$** Subtract 4, then divide by 3.
3.A8	**$x \leq -5$** Divide by −2 and reverse the sign.
3.A9	**$x = 3, y = 6$** Substitute the first into the second.
3.A10	**$x \geq 3$** Collect, then divide by −2 and reverse.
3.A11	**$x = 5.5$** Multiply through by 15, then collect.
3.A12	**$x = 9$** Expand, then collect.

CHAPTER 3 · CRITERION B · INVESTIGATING PATTERNS

3.B1 **n = 8; the other root is negative, so it is not a polygon.**
Solve the quadratic and test both roots.

3.B2 **The sums are 1, 4, 9, 16, so the sum of the first n odd numbers is n^2.**
Relate to the difference of consecutive squares.

3.B3 **$n(n-1)/2$, giving 3, 6, 10.**
Each pair shakes once.

3.B4 **$4n + 6$, an arithmetic sequence with common difference 4.**
Add the four sides.

3.B5 **The solutions fall by 1 in x and by 0.5 in y, all lying on one line.**
Solve for y and vary x.

3.B6 **$3n + 1$, because each new square shares one side.**
Count for n = 1, 2, 3.

3.B7 **$n(n-1)/2$, because each pair of lines meets once.**
Count for n = 2, 3, 4.

3.B8 **$(n+1)^2$, giving 64 for figure 7.**
Take differences to identify a quadratic.

3.B9 **The greatest area is 36, at the square 6 by 6.**
Write the area as a quadratic and find its vertex.

3.B10 **$n - 3$, because the vertex joins to all but itself and its two neighbours.**
Count for several polygons.

3.B11 **1, 3, 7, 15, 31, 63; each is one less than a power of two, and several are prime.**
Compute the values and test for factors.

CHAPTER 3 · CRITERION C · COMMUNICATING

3.C1 **To find every value of the unknown that makes the statement true, and to show that no other value does.**
State the definition and its strength.

3.C2 **x = 11.5**
Expand, collect, then divide.

3.C3 **Algebraically for exact values; graphically for a quick check or for a system of inequalities.**
Say what each method gives.

3.C4 **x = 14/5, y = 9/5**
Substitute x = y + 1 into the first equation.

3.C5 **Because the order of numbers is reversed by multiplying by a negative, so the larger side becomes the smaller.**
Test with numbers.

3.C6 **No solution: the lines are parallel, as x + y = 1 and x + y = 3. Infinitely many: the lines coincide, as x + y = 1 and 2x + 2y = 2.**
Compare the gradients and intercepts.

3.C7 **Because each inequality allows infinitely many values, and the system keeps those that satisfy all of them, which is a region.**
Draw a simple case.

3.C8 **If ax + by = p and ax + by = q with $p \neq q$, then subtracting gives 0 = p − q, which is false.**
Subtract the equations.

3.C9 **A linear function has no interior maximum, so its extreme values on a polygon occur where the boundary changes direction.**
Test with a triangle.

3.C10 **The value of the quantity when the variable is zero — the fixed charge, the starting amount, or the initial reading.**
Give a real example.

3.C11 **They are the same answer: the algebraic solution is the coordinates of the point where the two lines cross.**
Solve both ways for one pair.

3.C12 **Because an equation may allow negative or fractional values that the situation does not, as with a negative number of books.**
Model one case and check.

CHAPTER 3 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

3.D1 **The plans cost the same at 500 minutes; the first is cheaper for 400 minutes.**
Set the two linear models equal.

3.D2 **20 small and 20 large parcels.**
Two conditions, two equations.

3.D3 **100 kg of each.**
Conserve the total mass and the total cost.

3.D4 **The taxi is cheaper up to about 3.08 km, so for 6 km the bus is cheaper.**
Set the two models equal.

3.D5 **The feasible range is 30 to 60 type A jobs, depending on the mix.**
Write the total as an equation and the conditions as inequalities.

3.D6 **60 student and 60 adult meals.**
Two conditions, two equations.

3.D7 **Widths from 4 to 16 m give area at least 96 m^2; the greatest area is 100 m^2 at 10 m by 10 m.**
Write the area as a quadratic in the width.

3.D8 **The cost for 30 is 560 dollars, so it is affordable; the largest group is 41.**
Model the cost and solve the inequality.

3.D9 **Yes, 250 ml of each, because the target is exactly halfway between the two concentrations.**
Conserve volume and acid.

3.D10 **The feasible region runs from 40 to 60 hectares of maize.**
Write the land and labour constraints.

3.D11 **They are not: solving gives 120 loaves and 80 rolls, which does give 400 dollars, so they are consistent.**
Solve and check.

3.D12 **Any answer is acceptable if it identifies diminishing returns, a fixed capacity, or a threshold that a straight line cannot show.**
Give a specific counterexample.

CHAPTER 4 · CRITERION A · KNOWING AND UNDERSTANDING

4.A1 **(5, 2)**
Average each coordinate.

4.A2 **2**
Divide the change in y by the change in x.

4.A3 **−3**
Read it from the gradient-intercept form.

4.A4 **3**
Substitute x = 4.

4.A5 **4**
Set x = 0 and solve for y.

4.A6 **y = 3x − 2**
Use y = mx + c with c = −2.

4.A7 **−1/2**
The product of perpendicular gradients is −1.

4.A8 **x = 4**
Solve the equation 4x − 7 = 9.

4.A9 **−2**
Divide the change in y by the change in x.

4.A10 **75 km/h**
Divide distance by time.

4.A11 **3**
Set y = 0 and solve for x.

4.A12 **y = 4**
A horizontal line has gradient 0.

CHAPTER 4 · CRITERION B · INVESTIGATING PATTERNS

4.B1 **5, 8, 11, 14; and 32 for the tenth.**
Substitute n = 1, 2, 3, 4 and 10.

4.B2 **The gradients are 1, 2, 3, 4, so they rise by 1 each time.**
Compute each gradient and compare.

4.B3 **(n/2, n/2), which lies on the line y = x.**
Average each coordinate.

4.B4 **2n − 1, because the diagonal leaves each row and column once.**
Draw for n = 2, 3, 4.

4.B5 **The first differences are the odd numbers 1, 3, 5, 7, rising by 2.**
Subtract consecutive terms.

4.B6 **n√2, by Pythagoras' theorem on an isosceles right triangle.**
Apply the distance formula.

4.B7 **The gradients are 2, 3, 4, rising by 1, because the rule is quadratic.**
Compute each gradient and take differences.

4.B8 **The gradients are 1, 1/2, 1/3, …, halving and then continuing as 1/n.**
Compute each gradient.

4.B9 **The intercepts are 2, 1, 0, −1, −2, falling by 1 each time.**
Read the intercept from each equation.

4.B10 **The intersection moves along the first line as the second line rotates.**
Solve for several values of the coefficients.

4.B11 **n(n + 1)/2 + 1, because each new line cuts all the others.**
Count for n = 1, 2, 3, 4.

CHAPTER 4 · CRITERION C · COMMUNICATING

4.C1 **A rule that gives exactly one output for each permitted input.**
State the definition and give an example.

4.C2 **$y = -3x + 7$**
Substitute into $y = mx + c$ and solve for c.

4.C3 **The equation for an exact value; the graph for a comparison or a trend over a range.**
Say what each gives.

4.C4 **Gradient 2, intercepts $(0, -4)$ and $(2, 0)$.**
Rearrange into $y = mx + c$.

4.C5 **Because the ratio of the change in y to the change in x is constant for any two points on the line.**
Compute it from two pairs of points.

4.C6 **Interpolation predicts inside the range of the data, as predicting the cost at 3 hours from data covering 1 to 5 hours; extrapolation predicts outside it.**
Define both and give an example of each.

4.C7 **Because one value of x would be paired with many values of y, giving more than one output for one input.**
Apply the definition.

4.C8 **A rotation through 90° maps a segment of gradient m to one of gradient $-1/m$, so their product is -1.**
Draw two perpendicular segments.

4.C9 **If $y = kx$ then $y = 0$ when $x = 0$; and if the line passes through the origin its equation is $y = mx$, which is direct proportion.**
Use $y = mx + c$.

4.C10 **The distance from the start at time zero — how far away the journey began.**
Give a real example.

4.C11 **The gradient is the speed, because it is the change in distance per unit change in time.**
Read one from the other.

4.C12 **Any answer is acceptable if it identifies a rate that changes, such as a cooling cup, a growing tree, or a filling tank.**
Give a specific counterexample.

CHAPTER 4 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

4.D1 **C = 2.2d + 3; 36 dollars for 15 km; about 12.27 km for 30 dollars.**
Read the intercept and gradient, then solve.

4.D2 **P = 12h + 20; 440 dollars for 35 hours; 31.67 hours for 400 dollars.**
Read the intercept and gradient, then solve.

4.D3 **V = 12t; full at about 41.7 minutes; 240 litres after 20 minutes.**
Use a proportional model.

4.D4 **h = 25 − 1.5t; burns out at about 16.7 hours; 16 cm after 6 hours.**
The gradient is negative.

4.D5 **After 2 hours, at 180 km from the station.**
Set the two distance models equal.

4.D6 **N = 480 − 15t; 12 years to reach 300; the model cannot go below zero.**
Read the gradient, then solve.

4.D7 **F = 66.7p; about 733 g for 11; 22 people from 1.5 kg.**
Use a proportional model.

4.D8 **18 km/h; d = 18t; 45 km in 2.5 hours.**
Convert units first.

4.D9 **About 8% per hour, so zero at about 8.5 hours; real batteries fall faster near the end, so the model overestimates.**
Find the rate and solve.

4.D10 **4 cm per week; 95 cm after 20 weeks, which is unlikely for a real plant.**
Find the rate, then extrapolate.

4.D11 **After 1 hour, 14 km from the first cyclist's start.**
Set the sum of the distances equal to 30.

4.D12 **Any answer is acceptable if it shows that the digits beyond the model's reliability are not justified, as in predicting 95.3 cm for a plant.**
Link precision to reliability.

CHAPTER 5 · CRITERION A · KNOWING AND UNDERSTANDING

5.A1 **$x^2 + 3x - 10$**
Multiply every term of one bracket by every term of the other.

5.A2 **$4x^2 + 12x + 9$**
Use $a^2 + 2ab + b^2$ with $a = 2x$, $b = 3$.

5.A3 **$x = 4$ or $x = 5$**
Two numbers with sum 9 and product 20.

5.A4 **$x = 0$ or $x = 7$**
Take out the common factor x.

5.A5 **$x = \pm 5$**
A difference of two squares.

5.A6 **$(x + 4)^2 - 13$**
Add and subtract 16.

5.A7 **-4**
$b^2 - 4ac = 16 - 20$.

5.A8 **$x = -1$ or $x = -5$**
Two numbers with sum 6 and product 5.

5.A9 **$x = \pm 2$**
Divide by 3, then take roots.

5.A10 **$(3, -8)$**
Complete the square.

5.A11 **$x = 2$ or $x = -4$**
$b^2 - 4ac = 4 + 32 = 36$.

5.A12 **$x = 5$ or $x = -1$**
Take square roots of both sides.

CHAPTER 5 · CRITERION B · INVESTIGATING PATTERNS

5.B1 **n^2, giving 49 for the seventh.**
The areas are the square numbers.

5.B2 **$n(n+1)(2n+1)/6$, which can be proved by induction or by a telescoping sum.**
Compute the sums and look for a cubic rule.

5.B3 **$(n+1)^2 - n^2 = 2n + 1$, the odd numbers.**
Expand the difference.

5.B4 **$n^2(n+1)^2/4$, because a rectangle is fixed by two vertical and two horizontal lines.**
Count for $n = 2$ and $n = 3$.

5.B5 **The roots are always negative and get closer together as b grows.**
Use the formula.

5.B6 **The maximum is at equal numbers, because $x(S - x)$ is a parabola with vertex at $S/2$.**
Write the product as a quadratic.

5.B7 **$n^2 - 4n - 4$, which is non-negative for $n \geq 2 + 2\sqrt{2}$.**
Compute the discriminant and solve the inequality.

5.B8 **The areas approach 1 from above, so the limit is 1.**
Compute for $n = 1$ to 10.

5.B9 **The heights rise to a maximum at $t = 2$ then fall, as a parabola requires.**
Compute the values and compare with the vertex.

5.B10 **One root grows and the other falls, and the discriminant is $4 + 4k$, so real roots exist for all $k \geq -1$.**
Use the formula.

5.B11 **$n(n+1)/2 + 1$, because each new line is cut into $n + 1$ pieces.**
Count for $n = 1, 2, 3, 4$.

CHAPTER 5 · CRITERION C · COMMUNICATING

5.C1 **The quantity $b^2 - 4ac$, whose sign decides whether a quadratic has two, one or no real roots.**
State the definition and the three cases.

5.C2 **$x = 1$ or $x = -5$**
Rewrite as $(x + 2)^2 = 9$.

5.C3 **The formula, because the leading coefficient is not 1 and the factorisation is not obvious.**
Say what each method costs.

5.C4 **$x \approx 0.914$ or $x \approx -1.46$**
Use the formula and round once.

5.C5 **Because the square root of a negative number is not a real number, so the formula cannot produce a real root.**
Look at the formula.

5.C6 **A repeated root is one value counted twice, as in $(x - 2)^2 = 0$; two roots are different values, as in $x^2 - 4 = 0$.**
Compare the discriminants.

5.C7 **Because half the coefficient of x can always be squared and then subtracted, which leaves the value unchanged.**
Show the general step.

5.C8 **Completing the square gives $a(x + b/2a)^2 + \ldots$, whose square term is zero at $x = -b/(2a)$.**
Complete the square generally.

5.C9 **The product $x(S - x) = Sx - x^2$ is a downward parabola with vertex at $S/2$, so the maximum is at $x = S/2$.**
Use the symmetry of the roots.

5.C10 **The value of the variable at which the modelled quantity is zero — the break-even point, the landing time, or the empty moment.**
Give a real example.

5.C11 **They are the same: a root is a value of x for which $y = 0$, which is exactly an x-intercept.**
Set $y = 0$.

5.C12 **A negative time or a negative length is a root of the equation but not of the situation, as with $t = -1$ in a motion model.**
Check the domain.

CHAPTER 5 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

5.D1 **It hits the ground after about 4.85 s and reaches about 36.2 m.**
Solve $h = 0$ and find the vertex.

5.D2 **15 m by 15 m, giving 225 m^2.**
Write the area as a quadratic in one side.

5.D3 **Break-even at about 8.45 and 31.5 items; maximum profit 400 dollars at 20 items.**
Factorise or use the vertex.

5.D4 **Greatest height 8 m at $x = 20$; the base is 40 m wide.**
Find the vertex and the roots.

5.D5 **6 m by 8 m.**
Write $x(x + 2) = 48$ and solve.

5.D6 **It returns after 5 s and reaches 50 m.**
Solve $h = 0$ and find the vertex.

5.D7 **Yes: $x^2 - 30x + 200 = 0$ gives $x = 10$ or $x = 20$.**
Solve the quadratic and check both roots.

5.D8 **2 cm.**
Write $(20 + 2w)(15 + 2w) = 436$ and solve.

5.D9 **Yes: the maximum is 11.3 m, so it does not clear a 12 m wall.**
Compare the maximum with the wall height.

5.D10 **The maximum occurs when the two rectangles are equal, because the area is a downward parabola in the dividing position.**
Write the area as a quadratic.

5.D11 **$x = 10$ items, with profit 150 dollars.**
Write the profit as a quadratic and find the vertex.

5.D12 **Any answer is acceptable if it identifies air resistance, a fixed capacity, or a rate that is not constant.**
Give a specific counterexample.

CHAPTER 6 · CRITERION A · KNOWING AND UNDERSTANDING

6.A1 **(2, 3)**
Complete the square or use $x = -b/(2a)$.

6.A2 **3**
Set $x = 0$.

6.A3 **down**
The sign of a decides.

6.A4 **$x = -4$**
Use $x = -b/(2a)$.

6.A5 **$x = \pm 3$**
A difference of two squares.

6.A6 **a translation 5 units down**
The constant term shifts the graph vertically.

6.A7 **a translation 4 units left**
The bracket shifts the graph horizontally.

6.A8 **(2, 6)**
Read it from the completed square form.

6.A9 **3**
Set $x = 0$, since $2^0 = 1$.

6.A10 **2.56**
Multiply by the cube of the multiplier.

6.A11 **$x = 3$ or $x = -4$**
Factorise as $(x - 3)(x + 4)$.

6.A12 **4**
The vertex of a downward parabola is its maximum.

CHAPTER 6 · CRITERION B · INVESTIGATING PATTERNS

6.B1 **$2 \times 0.8^{n-1}$, giving about 0.655 m for the sixth.**
The common ratio is 0.8.

6.B2 **2n regions, because each line cuts all the others at one point.**
Draw for n = 1, 2, 3.

6.B3 **$2^n - 1$, because the sum telescopes or because each new term doubles the total and adds one.**
Compute the sums and prove by induction.

6.B4 **The areas are 1/3, 8/3 and 9, following $x^3/3$.**
Use the known formula for the area under a parabola.

6.B5 **The values double as x rises by 1 and halve as it falls by 1.**
Compute the values.

6.B6 **The width is $2/\sqrt{k}$, because $x^2 = 1/k$ at height 1.**
Set y = 1 and solve.

6.B7 **Two intersections for every positive n, at x = 0 and x = n.**
Set the two equal and solve.

6.B8 **The total is finite even though the number of bounces is infinite, because the heights form a convergent series.**
Sum the geometric series.

6.B9 **The gradients are 2, 4, 6, following 2x.**
Use chords that get shorter.

6.B10 **The maximum is n^2, because the vertex is at (n, n^2).**
Find the vertex for each n.

6.B11 **$2n^2 + 2n + 1$, because the parabola encloses one point for each integer pair inside.**
Count for n = 1, 2, 3.

CHAPTER 6 · CRITERION C · COMMUNICATING

6.C1 **The point at which the parabola turns, which is its maximum or minimum.**
State the definition and how it is found.

6.C2 **Vertex (1, −9), axis x = 1, intercepts (0, −8), (−2, 0), (4, 0).**
Complete the square and factorise.

6.C3 **The completed square form for an exact value; the graph for a comparison or a picture of the behaviour.**
Say what each gives.

6.C4 **Left 3, reflect in the x-axis, down 1.**
Read the equation from the inside out.

6.C5 **Because any quadratic can be obtained from $y = x^2$ by translations and stretches, which do not change the form of the curve.**
Use the transformation description.

6.C6 **A stretch multiplies distances from an axis, as in $y = 3x^2$; a translation moves every point by the same amount, as in $y = x^2 + 3$.**
Compare the equations.

6.C7 **Because 2^x is always positive for every real x, so y is never zero.**
Consider the sign of a power.

6.C8 **Completing the square gives $a(x + b/2a)^2 + \ldots$, whose square term vanishes at $x = -b/(2a)$.**
Complete the square generally.

6.C9 **A linear model adds a fixed amount each step while an exponential model multiplies, so the ratio of the two grows without bound.**
Compare a table of values.

6.C10 **The value of the variable at which the two models give the same output — the break-even point or the meeting time.**
Set the two equations equal.

6.C11 **A sketch shows the correct features at the correct places; a plotted graph shows accurate values at chosen points.**
Compare the two purposes.

6.C12 **Any answer is acceptable if it identifies unequal scales, a truncated axis, or a smooth curve drawn through scattered data.**
Give a specific example.

CHAPTER 6 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

6.D1 **Greatest height 7.2 m at $t = 1.2$ s; returns at $t = 2.4$ s.**
Find the vertex and the roots.

6.D2 **Span 40 m; greatest height 8 m.**
Find the roots and the vertex.

6.D3 **Profit for $4 < x < 26$ items; maximum profit 169 dollars at $x = 13$.**
Set the models equal and find the vertex.

6.D4 **$P = 400 \times 1.12^{n}$; about 705 after 5 years; doubles in about 6.1 years.**
Use an exponential model and solve.

6.D5 **$M = 80 \times 0.8^{n}$; about 32.8 g after 4 years; half-life about 3.1 years.**
Use an exponential model and solve.

6.D6 **Greatest height 30 m at $x = 10$; lands at $x = 20$.**
Find the vertex and the roots.

6.D7 **No: $900 \times 0.75^{6} \approx 191$ dollars, so it is worth more than 100 dollars after 6 years.**
Compute the model.

6.D8 **15 kg, giving 450 units.**
Find the vertex.

6.D9 **Yes: the maximum is 30 m exactly, reached at $t = 2$ s.**
Compare the vertex height with 30 m.

6.D10 **Compound is ahead from year 2 onward; after 10 years the difference is about 133 dollars.**
Compare the two models.

6.D11 **No: $h = 7.2 - 2.88 = 4.32$ m, so it does clear the wall.**
Evaluate the model at $x = 12$.

6.D12 **Any answer is acceptable if it identifies comparison, trend or the shape of behaviour beyond the data.**
Give a specific example.

CHAPTER 7 · CRITERION A · KNOWING AND UNDERSTANDING

7.A1 **39**
Use $a + (n - 1)d$ with $a = 3$, $d = 4$.

7.A2 **−1**
Use $a + (n - 1)d$ with $a = 20$, $d = -3$.

7.A3 **4n + 1**
The common difference is 4 and the first term is 5.

7.A4 **222**
Use $n/2 \times (\text{first} + \text{last})$.

7.A5 **96**
The common ratio is 2.

7.A6 3^{5-n}
The common ratio is 1/3.

7.A7 **25, 36**
These are the square numbers.

7.A8 **30**
The first differences are 4, 6, 8, so the next is 10.

7.A9 **63**
Use $a(r^n - 1)/(r - 1)$ with $a = 1$, $r = 2$.

7.A10 **36.45**
The common ratio is 0.9.

7.A11 **5n + 2**
The common difference is 5.

7.A12 **81, −243**
The common ratio is −3.

CHAPTER 7 · CRITERION B · INVESTIGATING PATTERNS

7.B1 **2, 6, 12, 20; and 72 for figure 8.**
Substitute n = 1 to 4 and 8.

7.B2 **The sums are 1, 3, 6, 10, following $n(n + 1)/2$.**
Pair the first and last terms.

7.B3 **$n(n - 3)/2$, giving 5, 9, 14.**
Count from each vertex and divide by 2.

7.B4 **$n(n + 1)/2$, the triangular numbers.**
Count the cubes layer by layer.

7.B5 **Second differences are 2, so the rule is $n^2 + 1$.**
Take differences and fit.

7.B6 **$5n + 1$, because each new hexagon shares one side.**
Count for n = 1, 2, 3.

7.B7 **The sums approach 1, because the total is $1 - 1/2^n$.**
Write with a common denominator.

7.B8 **18 m, from the geometric series.**
Sum the two journeys per bounce.

7.B9 **The cycle is 3, 9, 7, 1, so 3^{50} ends in 9.**
Divide by the period.

7.B10 **$n(n + 1)(2n + 1)/6$, the sum of the squares.**
Count squares of each size.

7.B11 **$n(n - 1)(n - 2)/6$, because each triple of lines makes one triangle.**
Count for n = 3, 4, 5.

CHAPTER 7 · CRITERION C · COMMUNICATING

7.C1 **A sequence in which the difference between consecutive terms is constant.**
State the definition and give an example.

7.C2 **$7n - 3$, and the 30th term is 207.**
Use the common difference.

7.C3 **An explicit rule, because a recursive rule needs all the previous terms.**
Say what each rule requires.

7.C4 **5115, using $a(r^n - 1)/(r - 1)$.**
Identify a and r first.

7.C5 **Because the first differences of a quadratic form an arithmetic sequence, so their differences are constant.**
Take differences of n^2, $n^2 + n$ and so on.

7.C6 **An arithmetic sequence adds a constant, as 2, 5, 8; a geometric sequence multiplies by one, as 2, 6, 18.**
Compare the differences and the ratios.

7.C7 **Because the terms shrink fast enough that the running total approaches a limit, as with $1 + 1/2 + 1/4 + \ldots = 2$.**
Use a geometric series with $|r| < 1$.

7.C8 **The odd numbers are the differences of consecutive squares, so they telescope to n^2.**
Relate to $(n + 1)^2 - n^2$.

7.C9 **Each vertex joins to $n - 3$ others, giving $n(n - 3)$ endpoints, and each diagonal has two endpoints.**
Count and divide by 2.

7.C10 **The multiplier applied at each step — the growth factor, the rebound fraction, or the decay rate.**
Give a real example.

7.C11 **Constant first differences give a linear rule, constant second differences a quadratic, and so on.**
Compare the tables.

7.C12 **Any answer is acceptable if it identifies a rule that fails later, as with $n^2 + n + 41$ being prime, or a cycle that looks like a trend.**
Give a specific counterexample.

CHAPTER 7 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

7.D1 **$P = 500 \times 1.08^n$; about 1079 after 10 years; doubles in about 9 years.**
Use a geometric model and solve.

7.D2 **$V = 15000 \times 0.88^n$; about 7913 after 5 years; below 5000 after about 9 years.**
Use a geometric model and solve.

7.D3 **200 + 200 + 180 = 580 dollars for 8 hours.**
Split the job at the 5-hour point.

7.D4 **75 dollars in month 12; 570 dollars in total.**
Use an arithmetic model.

7.D5 **38 m, from $10 + 2 \times 10 \times 0.6/(1 - 0.6)$.**
Sum the geometric series.

7.D6 **37 dollars in month 12; 312 dollars in total.**
Use an arithmetic model.

7.D7 **About 17.7 years, because $1.04^n = 2$ gives $n = \ln 2/\ln 1.04 \approx 17.7$.**
Solve the exponential equation.

7.D8 **$P = 200 \times 2^{(t/3)}$; about 25600 after 24 hours; the model cannot continue indefinitely.**
Use an exponential model.

7.D9 **44 steps, because $44 \times 45/2 = 990$ and $45 \times 46/2 = 1035$.**
Solve the quadratic and round down.

7.D10 **About 616 dollars, because the interest is small but compounds on a growing balance.**
Model month by month.

7.D11 **1.2 + 2 + 1 = 4.2 m after 10 years.**
Split the model at year 5.

7.D12 **Any answer is acceptable if it identifies a mechanism that changes, such as a population reaching a limit or a learner plateauing.**
Give a specific counterexample.

CHAPTER 8 · CRITERION A · KNOWING AND UNDERSTANDING

8.A1 **78.20**
Multiply by 1.15 then by 0.85.

8.A2 **0.99**
$1.1 \times 0.9 = 0.99$.

8.A3 **0.75**
$1 - 0.25$.

8.A4 **2676.45**
Apply the compound formula.

8.A5 **600 dollars**
$3000 \times 0.05 \times 4$.

8.A6 **2560 dollars**
5000×0.8^3.

8.A7 **1.04**
$1 + 0.04$.

8.A8 **66 dollars**
$50 \times 1.2 \times 1.1$.

8.A9 **60.90 dollars**
$1000 \times 1.03^2 - 1000$.

8.A10 **25%**
$(12000 - 9000)/12000$.

8.A11 **882 dollars**
800×1.05^2.

8.A12 **a rise of 8%**
$1.2 \times 0.9 = 1.08$.

CHAPTER 8 · CRITERION B · INVESTIGATING PATTERNS

8.B1	**1000×0.9^n, giving 590.49 in year 5.** Use a geometric model.
8.B2	**The balance falls by 100, 90, 80, ..., so the interest portion falls each month.** Model month by month.
8.B3	**There is no simple closed form; the balance is $P(1 + r)^n - M(((1 + r)^n - 1)/r)$.** Model recursively and solve.
8.B4	**The difference grows slowly at first and then faster, following the compound term.** Compute both for each year.
8.B5	**The values multiply by 1.03 each month, so the growth accelerates.** Compute the values.
8.B6	**About 72/r years, the rule of 72, accurate for rates between 4% and 12%.** Solve $(1 + r)^n = 2$.
8.B7	**The total rises faster than the rate, because the interest compounds.** Model for several rates.
8.B8	**$1.01^{12} - 1 \approx 12.68\%$, larger than 12%.** Compound the monthly rate.
8.B9	**20000×0.85^n, giving about 3922 dollars after 10 years.** Use a geometric model.
8.B10	**The interest falls each year while the capital repaid rises, keeping the payment constant.** Model year by year.
8.B11	**About $P \times r \times n/2$, because the average balance is about half the principal.** Model and approximate.

CHAPTER 8 · CRITERION C · COMMUNICATING

8.C1 **Simple interest is a fixed percentage of the original principal; compound interest is a percentage of the current amount, so the interest earns interest.**
State both definitions.

8.C2 **1687.30 dollars**
Use $A = P(1 + r)^n$.

8.C3 **Either is defensible; compound is correct for interest paid into the account, simple for a flat bonus.**
Say what each assumes.

8.C4 **2160.90 dollars**
Use $V_0(1 - r)^n$.

8.C5 **Because the multipliers are 1.2 and 0.8, whose product is 0.96, a fall of 4%.**
Multiply the multipliers.

8.C6 **A multiplier is the factor applied to the quantity, as 1.15; a percentage is the change relative to the quantity, as 15%.**
Compare the two.

8.C7 **Because each month's interest is charged on a balance that already includes the previous interest, so the rate compounds.**
Compound twelve times.

8.C8 **Because the interest is a percentage of a balance that falls with each payment, so the interest falls too.**
Model two months.

8.C9 **Because each year multiplies the previous value by a factor strictly between 0 and 1, so the value stays positive.**
Consider the limit.

8.C10 **The period of use at which the two options cost the same, so neither is better before it.**
Set the two models equal.

8.C11 **The total cost rises faster than the rate, because a higher rate both increases each interest charge and lengthens the repayment.**
Model at two rates.

8.C12 **Any answer is acceptable if it identifies a monthly figure that hides the total, or a rate that changes without notice.**
Give a specific example.

CHAPTER 8 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

8.D1 **Surplus 450 dollars; 20 months.**
Divide the goal by the surplus.

8.D2 **About 1122.53 dollars after 3 months.**
Model month by month.

8.D3 **1200 dollars in total; about 44% a year.**
Compare the total with the fee.

8.D4 **About 58773 dollars in year 5; about 263700 dollars in total.**
Use a geometric model and the series formula.

8.D5 **About 9230 dollars repaid, of which about 1230 is interest.**
Model the repayments.

8.D6 **About 331.50 dollars after 3 years; a loss of about 268.50 dollars.**
Use the depreciation model.

8.D7 **The second, at 100 dollars against 80 dollars — no, the first is cheaper at 80 dollars.**
Model both and compare.

8.D8 **About 243 dollars repaid; an effective annual rate of about 79%.**
Model the repayments and compound the rate.

8.D9 **Yes: $1.005^{36} - 1 \approx 19.7\%$, against 18% simple.**
Compound the monthly rate.

8.D10 **After about 9 years, because $500000 \times 0.95^{n} < 300000$ gives $n > 8.6$.**
Solve the exponential inequality.

8.D11 **About 31.2 tonnes; the model assumes no limit on land or water, which is unrealistic.**
Use a geometric model.

8.D12 **Any answer is acceptable if it shows that the monthly figure hides the total, as with 84 dollars a month for a 1000 dollar loan.**
Compare the monthly figure with the total.

CHAPTER 9 · CRITERION A · KNOWING AND UNDERSTANDING

9.A1 **(1, 7)**
Add the vector to the point.

9.A2 **(4, 1)**
Change the sign of y.

9.A3 **(−4, −1)**
Change the sign of x.

9.A4 **No: the third sides differ, so SSS fails.**
Compare all three sides.

9.A5 **Yes, by SAS.**
Two sides and the included angle is a test.

9.A6 **9 : 25**
Square the length ratio.

9.A7 **8 : 27**
Cube the length ratio.

9.A8 **1 km**
4 × 25000 cm = 100000 cm.

9.A9 **(6, 9)**
Multiply each coordinate by 3.

9.A10 **(0, 1)**
A quarter turn maps (x, y) to (−y, x).

9.A11 **30 m**
15 × 200 cm.

9.A12 **Yes, by RHS or SSS.**
The third side is fixed by Pythagoras.

CHAPTER 9 · CRITERION B · INVESTIGATING PATTERNS

9.B1 **4 : 49, because area scales as the square of the length ratio.**
Square the ratio.

9.B2 **The areas are 1, 4, 9, 16 times the original.**
Square each factor.

9.B3 **1 : 64, because volume scales as the cube.**
Cube the ratio.

9.B4 **No: AAA gives similarity only, and the size is not fixed.**
Equal angles force proportion, not equality.

9.B5 **Reflecting in both axes is a rotation of 180° about the origin.**
Apply the reflections in turn.

9.B6 **√2, because the area scales as the square of the factor.**
Solve $k^2 = 2$.

9.B7 **Two copies tile a rectangle if the triangle is halved along an altitude.**
Draw and count.

9.B8 **A single enlargement by factor km, because the multiplications compose.**
Track one point through both steps.

9.B9 **The sides are 1, ½, $\frac{1}{4}$, halving each time.**
Multiply by the factor.

9.B10 **9 : 1, because surface area scales as the square.**
Square the ratio.

9.B11 **n lines, one through each vertex and the opposite side midpoint.**
Draw for n = 3, 4, 5.

CHAPTER 9 · CRITERION C · COMMUNICATING

9.C1 **That one can be moved onto the other by a rigid transformation, so all measurements match.**
State the definition.

9.C2 **(2, 2), (8, 2), (2, 6)**
Multiply each coordinate by 2.

9.C3 **Transforming, because it is exact where measurement is approximate.**
Say what each gives.

9.C4 **3, because 108/12 = 9 and √9 = 3.**
Take the square root of the area ratio.

9.C5 **Because two different triangles can share two sides and a non-included angle, as the ambiguous case shows.**
Try to construct two.

9.C6 **Congruent shapes are identical in size and shape, as two copies of the same triangle; similar shapes have the same shape but may differ in size, as two photographs of different sizes.**
Compare the definitions.

9.C7 **Because an area is a product of two lengths, and each length is multiplied by the factor.**
Consider a rectangle.

9.C8 **Reflecting in the y-axis maps (x, y) to (−x, y) and then in the x-axis to (−x, −y), which is a rotation of 180° about the origin.**
Track one point.

9.C9 **The hypotenuse and one leg fix the third leg by Pythagoras, so the test reduces to SSS.**
Apply Pythagoras.

9.C10 **The permitted difference between a measurement and its specification, stated as a band either side of the nominal value.**
Give an example.

9.C11 **They are similar, with the scale factor connecting every corresponding length, its square connecting areas and its cube connecting volumes.**
State the three connections.

9.C12 **Any answer is acceptable if it identifies a property a model cannot show, such as weight, strength, cost or the feel of a material.**
Give a specific example.

CHAPTER 9 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

9.D1 **600 hectares, because $6 \times 4 \times 50000^2$ cm^2 $= 6 \times 10^{10}$ cm^2.**
Scale both lengths, then the area.

9.D2 **1687.5 ml, because the volume factor is $(18/12)^3 = 3.375$.**
Cube the length ratio.

9.D3 **4 m long; the area ratio is 400 : 1.**
Scale the length, then square.

9.D4 **Scale factor 2.5; the area ratio is 6.25 : 1.**
Scale the length, then square.

9.D5 **24.95 is acceptable; 25.12 is not.**
Compare with the interval.

9.D6 **12 m, assuming level ground and parallel rays.**
Use similar triangles.

9.D7 **1 : 16, because the length ratio is the cube root of 64, which is 4, and the area ratio is 4^2.**
Take the cube root, then square.

9.D8 **200 m^2; the model says nothing about the height of the ceiling or the materials.**
Square the scale factor.

9.D9 **About 11.7 m, subject to the distortion of the lens and the level of the camera.**
Multiply the reference height by the ratio.

9.D10 **Yes for the surface area, but only 3.375 times the capacity, because area scales as k^2 and volume as k^3.**
Square and cube the ratio.

9.D11 **140 km^2, but the outline on a map is a simplification of the real shore.**
Scale both lengths, then the area.

9.D12 **Any answer is acceptable if it shows that areas need the square and volumes the cube, as with a model wing area at 1 : 50.**
Give a specific counterexample.

CHAPTER 10 · CRITERION A · KNOWING AND UNDERSTANDING

10.A1	**10 cm** √(36 + 64) = √100.
10.A2	**15 cm** √(81 + 144) = √225.
10.A3	**12 cm** √(169 − 25) = √144.
10.A4	**0.5** The ratio of the opposite side to the hypotenuse in a 30-60-90 triangle.
10.A5	**0.5** The ratio of the adjacent side to the hypotenuse.
10.A6	**1** The opposite and adjacent sides are equal.
10.A7	**5** 10 × sin 30°.
10.A8	**6.71** 8 × tan 40°.
10.A9	**30°** The inverse sine of 0.5.
10.A10	**4 m** √(25 − 9).
10.A11	**317°** Add 180° to the forward bearing.
10.A12	**25 + 144 = 169, so the converse of Pythagoras applies.** Check $a^2 + b^2 = c^2$.

CHAPTER 10 · CRITERION B · INVESTIGATING PATTERNS

10.B1 **5 cm and 12 cm, because 25 + 144 = 169.**
Scale the 5-12-13 triangle.

10.B2 **sin rises 0, 0.5, 0.707, 0.866, 1 while cos falls 1, 0.866, 0.707, 0.5, 0.**
Compute the ratios in special triangles.

10.B3 **$\sin^2 \theta + \cos^2 \theta = 1$, by dividing Pythagoras by the hypotenuse squared.**
Use Pythagoras on the ratios.

10.B4 **The height falls from 5 m to 0 m, following $\sqrt{(25 - x^2)}$.**
Apply Pythagoras for each position.

10.B5 **The gradients are the tangents, rising from 0 to infinity.**
Compute tan θ for each angle.

10.B6 **There is no such angle, because the height is proportional to tan θ, not to θ.**
Compare tan θ for several angles.

10.B7 **The area is greatest for the isosceles triangle, at 25 cm^2.**
Write the area as a quadratic in one leg.

10.B8 **50 km on a bearing of 233°.**
Use Pythagoras and the tangent.

10.B9 **180°, 225°, 270°, 315°, 000°; each differs by 180°.**
Add 180° and reduce modulo 360°.

10.B10 **The error is large, because tan θ is steep near 90°.**
Compare tan 89° and tan 88°.

10.B11 **About $3.57\sqrt{h}$ km, from the tangent to a sphere of radius 6371 km.**
Use Pythagoras with the Earth's radius.

CHAPTER 10 · CRITERION C · COMMUNICATING

10.C1 **That in a right-angled triangle the square on the hypotenuse equals the sum of the squares on the other two sides.**
State the theorem.

10.C2 **25 cm**
$\sqrt{(49 + 576)} = \sqrt{625}$.

10.C3 **Pythagoras when two sides are known; a ratio when one side and one acute angle are known.**
Say what each needs.

10.C4 **About 21.8°, from arctan(2/5).**
Use the inverse tangent.

10.C5 **Because sine involves the hypotenuse, cosine involves the hypotenuse and tangent does not, so the ratio must match the sides available.**
List the three ratios.

10.C6 **Elevation is measured upwards from the horizontal to an object above; depression is measured downwards to an object below.**
Draw both.

10.C7 **Because the horizontal lines are parallel and the line of sight is a transversal, so the alternate angles are equal.**
Draw the two horizontals.

10.C8 **Dividing $a^2 + b^2 = c^2$ by c^2 gives $(a/c)^2 + (b/c)^2 = 1$, which is $\sin^2\theta + \cos^2\theta$.**
Divide Pythagoras by c^2.

10.C9 **By the converse of Pythagoras, the angle opposite c is a right angle, because the triangle with those sides must be the right-angled one.**
Use the converse.

10.C10 **An angle measured clockwise from north; three figures prevent 05° and 50° from being confused.**
State the convention.

10.C11 **They differ by 180°, because reversing a direction turns it through a half turn.**
Add or subtract 180°.

10.C12 **Any answer is acceptable if it identifies a safety limit that the error interval crosses, as with a slope measured at 30° against a limit of 28°.**
Give a specific example.

CHAPTER 10 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

10.D1 **About 12.3 m, from 18 × tan 31° + 1.5.**
Use the tangent and add the eye level.

10.D2 **About 284 m, from 45/tan 9°.**
Use the tangent.

10.D3 **About 4.8°; the ramp is about 10.8 m long.**
Use the inverse tangent and Pythagoras.

10.D4 **About 30 km on a bearing of about 079°.**
Resolve each leg into components.

10.D5 **About 56.4 m high; the wire is about 64 m long.**
Use the tangent and then Pythagoras.

10.D6 **500 km on a bearing of about 127°.**
Use Pythagoras and the tangent.

10.D7 **No: the pitch is arctan(0.4) ≈ 21.8°, so it does meet the rule.**
Compare the angle with the limit.

10.D8 **$25^2 + 39^2 = 625 + 1521 = 2146$, and $56^2 = 3136$, so no right angle is present.**
Test Pythagoras on the longest side.

10.D9 **Yes: √(25 + 3.24) = 5.31 m, so a 5.5 m ladder reaches with the foot at 1.8 m.**
Apply Pythagoras.

10.D10 **About 288 m north and 35 m west of the start, so the displacement is about 290 m on a bearing of about 353°.**
Resolve into components.

10.D11 **The height changes from 17.6 m to 19.4 m, an error of about 1.8 m or 10%.**
Compare tan 20° and tan 21°.

10.D12 **Any answer is acceptable if it shows that the error interval crosses the limit that the decision depends on.**
Give a specific example.

CHAPTER 11 · CRITERION A · KNOWING AND UNDERSTANDING

11.A1 **60 cm³**
Multiply the three dimensions.

11.A2 **216 cm²**
6×6^2.

11.A3 **198 cm³**
$\pi \times 9 \times 7$.

11.A4 **226 cm²**
$2\pi \times 4 \times 9$.

11.A5 **75.4 cm³**
$(1/3)\pi \times 9 \times 8$.

11.A6 **905 cm³**
$(4/3)\pi \times 216$.

11.A7 **314 cm²**
$4\pi \times 25$.

11.A8 **96 cm³**
Area times length.

11.A9 **2500 cm³**
1 litre = 1000 cm^3.

11.A10 **13 cm**
$\sqrt{(25 + 144)}$.

11.A11 **144 cm³**
$(1/3) \times 48 \times 9$.

11.A12 **6 cm**
Divide the volume by the base area.

CHAPTER 11 · CRITERION B · INVESTIGATING PATTERNS

11.B1 **8 : 27, because the length ratio is 2 : 3 and volume scales as its cube.**
Take the square root, then cube.

11.B2 **The ratios are 6, 3, 2, 1.5, 1.2, 1, halving and then continuing as 6/a.**
Compute both for each cube.

11.B3 **The height equals the diameter, because the optimum has h = 2r.**
Write the area in terms of r and minimise.

11.B4 **The volume rises to a maximum and then falls, with the maximum at h = 2r.**
Compute for several radii.

11.B5 **The volumes rise faster than the cube of the radius alone suggests, following $(4/3)\pi r^3$.**
Compute each volume.

11.B6 **3/r, because $4\pi r^2$ divided by $(4/3)\pi r^3$ is 3/r.**
Divide the two formulae.

11.B7 **The sphere needs least, then the optimal cylinder, then the cube, then the long box.**
Compute each surface area at 1000 cm^3.

11.B8 **The ratio halves, because it is proportional to 1/r for similar shapes.**
Compute the ratio at two sizes.

11.B9 **n^3, because the volume is the cube of the side.**
Count for n = 1, 2, 3.

11.B10 **The area is $6n - 2(n - 1) = 4n + 2$, because each new cube hides two faces.**
Count for n = 1, 2, 3.

11.B11 **A cone of volume $(1/3)\pi r^2 h$, with r and h the two legs.**
Compare with the cone formula.

CHAPTER 11 · CRITERION C · COMMUNICATING

11.C1 **Surface area measures the material needed to make the surface; volume measures the space inside.**
State both definitions.

11.C2 **Volume 502.7 cm^3; surface area 351.9 cm^2.**
Apply the cylinder formulae.

11.C3 **Volume, because capacity is a quantity of space and surface area is a quantity of material.**
Say what each measures.

11.C4 **204 cm^2, from $\pi \times 5 \times 13$.**
Use πrl.

11.C5 **Because surface area scales as the square of the length and volume as the cube, so the ratio scales as 1/length.**
Compare the powers.

11.C6 **A prism has a polygonal cross-section and a cylinder a circular one, but both have volume equal to cross-section area times length.**
Compare the two formulae.

11.C7 **Because a cone tapers to a point, and experiment or integration shows it fills exactly one third of the cylinder.**
Compare the two formulae.

11.C8 **The slant height is the hypotenuse of the right triangle formed by the radius, the height and the slant, so $l = \sqrt{(r^2 + h^2)}$.**
Apply Pythagoras.

11.C9 **Any answer is acceptable if it uses the fact that the sphere is the isoperimetric optimum, verified by comparing the ratio 3/r with larger ratios for other shapes.**
Compare the ratios.

11.C10 **Mass per unit volume, so mass = density × volume, and the units must match.**
State the definition.

11.C11 **The material mass is the surface area times the mass per unit area of the material, so a smaller area means a lighter container.**
Multiply area by mass per area.

11.C12 **Any answer is acceptable if it identifies a constraint such as standing, stacking, holding or printing that the material model ignores.**
Give a specific example.

CHAPTER 11 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

11.D1 **About 3016 litres; the curved surface is about 7.54 m^2.**
Use the cylinder formulae.

11.D2 **9000 cm^3; 2700 cm^2; about 297 m^2 for 100 boxes.**
Apply the cuboid formulae and add 10%.

11.D3 **Radius about 1.34 m; surface about 22.5 m^2.**
Solve $(4/3)\pi r^3 = 10$.

11.D4 **About 377 ml; the curved surface is about 204 cm^2.**
Use the cone formulae and 1 ml = 1 cm^3.

11.D5 **Cube about 934 cm^2; cylinder about 774 cm^2; sphere about 484 cm^2. The cylinder is recommended because it stands and stacks.**
Compute each surface area at 2000 cm^3.

11.D6 **Empty 240 g; the glass is about 24% of the full mass of about 990 g.**
Add the contents mass to the glass mass.

11.D7 **Volume about 1.13 m^3; mass about 2707 kg.**
Subtract the inner cylinder from the outer.

11.D8 **About 282.7 m^3; the external area is about 197.8 m^2.**
Add the cylinder and hemisphere formulae.

11.D9 **The ball occupies about 52% of the box, so about 48% of the shipped volume is air.**
Compare the two volumes.

11.D10 **Radius about 0.86 m and height about 1.72 m, so h = 2r.**
Write the area in terms of r and minimise.

11.D11 **The new box is about 19.2 cm by 12.5 cm by 5 cm, saving about 60 cm^2 of cardboard.**
Keep the volume fixed and recompute.

11.D12 **Any answer is acceptable if it identifies transport, handling, stacking or recycling as constraints the model omits.**
Give a specific example.

CHAPTER 12 · CRITERION A · KNOWING AND UNDERSTANDING

12.A1	**10** Divide the sum by the number of values.
12.A2	**13.5** Average the two middle values.
12.A3	**8** The most frequent value.
12.A4	**18** Largest minus smallest.
12.A5	**11** The middle value of seven ordered values.
12.A6	**120** Mean times the number of values.
12.A7	**11** The median of the upper half.
12.A8	**8** Upper quartile minus lower quartile.
12.A9	**0.1** 50 divided by 500.
12.A10	**20** 140/300 × 40.
12.A11	**3** Frequency divided by class width.
12.A12	**20** The median is at half the total frequency.

CHAPTER 12 · CRITERION B · INVESTIGATING PATTERNS

12.B1 **10, because the total is 60 and the others sum to 50.**
Use the mean formula.

12.B2 **The mean moves towards the large value and the median moves only slightly.**
Compute both before and after.

12.B3 **The average of the two middle values.**
Order and take the middle pair.

12.B4 **About n/2, because the quartiles are at n/4 and 3n/4.**
Compute for several n.

12.B5 **The mean grows as $n^2/3$ and the range as $n^2 - 1$.**
Compute for n = 2 to 6.

12.B6 **ng/N, the group's share of the population times the sample size.**
Use the sampling fraction.

12.B7 **Wider classes give a smoother but less detailed histogram; the area always represents the frequency.**
Redraw at two widths.

12.B8 **About 95%, by the empirical rule.**
Compare with a normal distribution.

12.B9 **The running total never falls, so the diagram rises monotonically.**
Compute the running total.

12.B10 **The mean falls, the median may not change, and the range falls.**
Compute all three before and after.

12.B11 **Four times the size, because the error is proportional to $1/\sqrt{n}$.**
Compare the errors at two sizes.

CHAPTER 12 · CRITERION C · COMMUNICATING

12.C1 **The mean uses every value and is pulled by extremes; the median is the middle value and is not.**
State both definitions.

12.C2 **Mean 31.4, median 18, range 78.**
Compute each measure.

12.C3 **The median, because a few very expensive houses pull the mean above what a typical house costs.**
Say what each resists.

12.C4 **12, 18 and 30 students.**
Use the sampling fraction.

12.C5 **Because it depends only on the two extreme values, so one outlier makes it large however typical the rest are.**
Give an example.

12.C6 **A histogram has frequency density on the vertical axis and represents frequency by area; a bar chart has frequency and a gap between bars.**
Compare the axes.

12.C7 **Because the sample does not represent the population, so its statistics describe the sample and not the population.**
Give an example.

12.C8 **Random error falls as $1/\sqrt{n}$, but bias comes from the selection method and is unchanged by taking more of the same kind of sample.**
Compare the two effects.

12.C9 **Because each group is sampled in proportion to its share, every group appears in the sample, which a simple random sample does not guarantee.**
Use the sampling fraction.

12.C10 **The running total against the upper class boundary; the median is read at half the total frequency.**
State the method.

12.C11 **The box plot is a picture of the minimum, lower quartile, median, upper quartile and maximum.**
List the five numbers.

12.C12 **Any answer is acceptable if it identifies a judgement about an individual made from a group average with a wide spread.**
Give a specific example.

CHAPTER 12 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

12.D1 **The new mean is 68.96, a rise of about 1.**
Use the total.

12.D2 **1200 in total, and 180 from the district.**
Use the sampling fraction.

12.D3 **Total 116; the largest class has 48.**
Multiply density by width.

12.D4 **Mean 40000, median 25000; the median is fairer to publish.**
Compute both and compare.

12.D5 **The subgroup rate of 70%, because the overall figure hides the difference.**
Compare the subgroup with the whole.

12.D6 **24, 27.2, 25.6 and 35.2, so about 24, 27, 26 and 35.**
Use the sampling fraction and round.

12.D7 **About ±5%, so the true support is likely between 47% and 57%.**
Use $1/\sqrt{n}$ for the approximate error.

12.D8 **Either is defensible; the smaller range means more predictable teaching, the larger means more variety.**
Compare the spreads.

12.D9 **Mean 487, median 420; the median, because Saturday is exceptional.**
Compute both.

12.D10 **No: donors are a self-selected group with an interest, so the sample is biased.**
Compare the sampling frame with the population.

12.D11 **Yes, 75 is above the mean, but the spread is needed to say how unusual it is.**
Ask for the standard deviation or the quartiles.

12.D12 **Any answer is acceptable if it identifies a skewed distribution, a biased sample, or a truncated display axis.**
Give a specific example.

CHAPTER 13 · CRITERION A · KNOWING AND UNDERSTANDING

13.A1	**1/2** Three favourable outcomes out of six.
13.A2	**1/2** The primes are 2, 3 and 5.
13.A3	**1/3** The outcomes are 5 and 6.
13.A4	**5/6** The complement of 1/6.
13.A5	**1/6** Six favourable outcomes out of 36.
13.A6	**1/4** Multiply 1/2 by 1/2.
13.A7	**3/4** 1 − P(no heads) = 1 − 1/4.
13.A8	**0.4** 4 out of 10.
13.A9	**1/4** 13 hearts out of 52.
13.A10	**2/13** 8 favourable cards out of 52.
13.A11	**0.7** Add the probabilities.
13.A12	**1/8** Multiply 1/2 three times.

CHAPTER 13 · CRITERION B · INVESTIGATING PATTERNS

13.B1 **Both 0.2; either 0.7.**
Multiply for both; add and subtract the intersection for either.

13.B2 **$1 - (5/6)^n$, which rises towards 1.**
Use the complement.

13.B3 **$1 - (365/365)(364/365)\ldots$, which passes 0.5 at n = 23.**
Use the complement and multiply.

13.B4 **$26/52 \times 25/51 \approx 0.245$, less than 0.25 because the first draw changes the second.**
Multiply the conditional probabilities.

13.B5 **The probabilities rise to 1/6 at 7 and then fall symmetrically.**
Count the outcomes for each sum.

13.B6 **n/6, because the expectation is linear.**
Multiply n by 1/6.

13.B7 **1/3, because the sample space reduces to three outcomes.**
Restrict the sample space.

13.B8 **About 28%, because the false positives outnumber the true positives.**
Use a tree and the conditional probability.

13.B9 **The probability is proportional to the tickets, so it rises linearly to 1.**
Divide the tickets by the total.

13.B10 **Expected prize $91/6 \approx 15.17$, so the game favours the player.**
Sum score$^2 \times 1/6$.

13.B11 **13 throws, because $(5/6)^n < 0.1$ gives $n > 12.6$.**
Solve the inequality.

CHAPTER 13 · CRITERION C · COMMUNICATING

13.C1 **That the outcome of one does not change the probability of the other, so the probabilities multiply.**
State the definition.

13.C2 **4/52 × 3/51 = 1/221.**
Multiply the conditional probabilities.

13.C3 **A Venn diagram for a single stage with overlap; a tree for successive stages or for conditional probabilities.**
Say what each shows.

13.C4 **30/56 = 15/28, from the two paths red-green and green-red.**
Add the two path probabilities.

13.C5 **Because if they are mutually exclusive, knowing one happened tells you the other did not, so the probability changes to zero.**
Apply the definitions.

13.C6 **P(A and B) is the probability of both in the whole sample space; P(A given B) restricts the sample space to B.**
Compare the denominators.

13.C7 **Because the three regions exhaust the sample space, so their probabilities sum to 1.**
Use the complement rule.

13.C8 **Adding P(A) and P(B) counts the intersection twice, so it must be subtracted once.**
Use a Venn diagram.

13.C9 **Because over many plays each outcome occurs in proportion to its probability, so the average gain tends to the sum of value times probability.**
Use the law of large numbers.

13.C10 **That the expected value is zero for every player, which happens when the entry fee equals the expected prize.**
State the condition.

13.C11 **The rule decides which outcomes belong to which player, so it decides the probabilities, and a fair rule gives equal probabilities.**
Compare two rules.

13.C12 **Any answer is acceptable if it identifies a rule giving equal chances per ticket while the tickets themselves were distributed unequally.**
Give a specific example.

CHAPTER 13 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

13.D1 **Expected value $10/6 - 3 \approx -1.33$ dollars, so it is not fair.**
Compute the expected prize and subtract the stake.

13.D2 **Both zero, because each wins with probability $1/2$ and gains or loses 5.**
Compute the expected gain for each.

13.D3 **$1/12$, $1/4$, $1/3$, $1/3$; the expected score is $1/12 + 2/4 + 3/3 + 4/3 \approx 2.92$.**
Divide each angle by $360°$.

13.D4 **$(5/8)(4/7) + (3/8)(2/7) = 26/56 = 13/28$.**
Add the two matching paths.

13.D5 **About 17%, because the false positives outnumber the true positives.**
Use a tree and divide.

13.D6 **About 0.0235, from $2/500 \times 498/499 \times 497/498 \times 3$ arrangements.**
Use the hypergeometric count.

13.D7 **Each applicant has probability 0.05, so the family's expected places are 0.15, which is proportional and therefore fair in probability.**
Compare the probabilities.

13.D8 **Each player wins with probability $1/2$, so the game is fair and neither wins never happens.**
Count the outcomes.

13.D9 **Expected 6 heads; the probability of more heads than tails is about 0.63.**
Use the binomial model.

13.D10 **The holder has probability 0.1, ten times any other, so it is fair only if everyone had the same chance to acquire tickets.**
Compare the probabilities.

13.D11 **Yes: by symmetry each player wins with probability $15/36$ before ties, and ties are symmetric too.**
Count the outcomes.

13.D12 **Any answer is acceptable if it identifies a stake that the player cannot afford to lose, however small the expected loss.**
Give a specific example.

CHAPTER 14 · CRITERION A · KNOWING AND UNDERSTANDING

14.A1 **3.6 m**
Multiply by 50 and convert to metres.

14.A2 **8.64 m^2**
Multiply the two dimensions.

14.A3 **1178 litres**
$\pi \times 0.25 \times 1.5$ m^3, times 1000.

14.A4 **1.8 m^3**
$20 \times 0.1 \times 0.9$.

14.A5 **30 days**
Divide the capacity by the demand.

14.A6 **400 dollars**
Add the capital cost to four years of running cost.

14.A7 **24.20 dollars**
20×1.1^2.

14.A8 **50%**
Divide and multiply by 100.

14.A9 **60%**
24 out of 40.

14.A10 **No**
Compare with the bound.

14.A11 **50%**
Divide the areas.

14.A12 **25 minutes**
Divide the capacity by the rate.

CHAPTER 14 · CRITERION B · INVESTIGATING PATTERNS

14.B1 **Total 8.64 m^2; the largest share is about 56%.**
Sum and divide.

14.B2 **The capacity grows as the square of the radius, because $V = \pi r^2 h$.**
Compute for several radii.

14.B3 **12.5 m^2, because area = volume/(depth × efficiency).**
Rearrange the collection formula.

14.B4 **The totals are 455.26, 510.51 and 565.77, rising linearly with the starting maintenance.**
Sum the geometric series.

14.B5 **12.5, 25 and 50 days, proportional to the capacity.**
Divide capacity by demand.

14.B6 **12 years, from 600/50.**
Divide the cost by the annual saving.

14.B7 **About 32%, 16% and 8%, halving as the sample quadruples.**
Use $1/\sqrt{n}$.

14.B8 **The tank must be 25% larger, because the volume falls in the same proportion.**
Compare the two volumes.

14.B9 **The area falls to a minimum at $h = r$ and then rises.**
Write the area in terms of r and minimise.

14.B10 **About 0.72 days per m^2 for 900 mm of rain at 0.8 efficiency and 40 litres a day.**
Compute the ratio.

14.B11 **$C + M((1 + r)^n - 1)/r$, a capital cost plus a geometric series.**
Sum the series.

CHAPTER 14 · CRITERION C · COMMUNICATING

14.C1 **A requirement written as a quantity with a bound, so that a design can be tested against it.**
State the definition.

14.C2 **24 m², from 6 m by 4 m.**
Scale both lengths, then multiply.

14.C3 **A formula for a regular shape; a scale drawing with a grid for an irregular one, counting squares.**
Say what each suits.

14.C4 **About 603 litres.**
Use $\pi r^2 h$ and convert.

14.C5 **Because a percentage is a share of the whole, and the shares must sum to 100% for the plan to be complete.**
Compare the shares.

14.C6 **A capital cost is paid once to build, as 400 dollars for a structure; a running cost is paid each period, as 20 dollars a year.**
Compare the two.

14.C7 **Because the tank empties at the demand rate, so the time is the quantity divided by the rate.**
Use the rate formula.

14.C8 **The volume falling on the roof is the area times the depth, and only the efficiency fraction of it is collected.**
Multiply the three factors.

14.C9 **Because the constraints are the requirements that were thought of, and a need that was not written down cannot be met by the model.**
Give an example.

14.C10 **The time for the savings of a design to equal its capital cost, found by dividing the cost by the annual saving.**
State the definition.

14.C11 **The margin of error is about 1/√n, so it halves when the sample size quadruples.**
Compare two sizes.

14.C12 **Any answer is acceptable if it identifies a number without units, a calculation without a formula, or an assumption left unstated.**
Give a specific example.

CHAPTER 14 · CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

14.D1 **27 m^3 collected; a 3000-litre tank covers 60 days.**
Multiply for the supply and divide for the demand.

14.D2 **About 1259 dollars, from $600 + 40 \times (1.08^{10} - 1)/0.08$.**
Sum the geometric series.

14.D3 **The second, at about 1663 dollars against about 1794 dollars.**
Sum both series and compare.

14.D4 **Yes: 4.05 m^2 planting, 2.7 m^2 paths and 2.25 m^2 seating sum to 9 m^2 and meet every bound.**
Test the shares.

14.D5 **75 mm of rain; about 12 fills a year.**
Rearrange and divide.

14.D6 **70% with a margin of about 14%, so the true figure is likely between 56% and 84%.**
Compute the share and $1/\sqrt{n}$.

14.D7 **About 5.8 years without the rise, and about 4.8 years with it.**
Solve for the year in which the savings exceed the cost.

14.D8 **2.4 m by 1.6 m, area 3.84 m^2, so 7 desks with space left over.**
Scale, then divide.

14.D9 **360 litres a week; a roof of 22.5 m^2 is needed.**
Multiply and rearrange.

14.D10 **After about 23 years, when the series passes 500.**
Solve the geometric sum inequality.

14.D11 **The tank's base is about 0.785 m^2, which is 7.9% of 10 m^2, so it fits.**
Compare the base area with the allowed share.

14.D12 **Any answer is acceptable if it identifies a need that was never written down as a constraint, such as safety, privacy or maintenance access.**
Give a specific example.

Formula bank (1 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

NUMBER, ACCURACY AND FORM

Index laws

$a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$, for $a \neq 0$.

Zero and negative indices

$a^0 = 1$ for $a \neq 0$; $a^{-n} = 1/a^n$.

Rational index

a^(1/n) is the nth root of a; a^(m/n) = $(\sqrt[n]{a})^m$.

Standard form

$a \times 10^n$ with $1 \leq a < 10$.

Surd rules

$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$; $\sqrt{a} \div \sqrt{b} = \sqrt{a/b}$; $(\sqrt{a})^2 = a$.

Rationalising

multiply by the conjugate so that the denominator becomes rational.

Error interval

a value given to the nearest unit lies within half a unit either side.

Bounds of a calculation

the lower bound uses the lower bounds of subtracted terms.

Percentage multiplier

an increase of r% multiplies by 1 + r/100; a decrease by 1 − r/100.

Formula bank (2 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

ALGEBRA AND EQUATIONS

Expanding

$a(b + c) = ab + ac$.

Difference of squares

$a^2 - b^2 = (a - b)(a + b)$.

Perfect square

$a^2 + 2ab + b^2 = (a + b)^2$; $a^2 - 2ab + b^2 = (a - b)^2$.

Quadratic equation

$ax^2 + bx + c = 0$ with $a \neq 0$.

Quadratic formula

$x = (-b \pm \sqrt{(b^2 - 4ac)}) \div 2a$.

Discriminant

$b^2 - 4ac > 0$ gives two real roots, $= 0$ one repeated root, < 0 none.

Completing the square

$ax^2 + bx + c = a(x + b/2a)^2 + c - b^2/4a$.

Algebraic fraction

cancel only common factors, and state the excluded values first.

Inequality rule

multiplying or dividing by a negative number reverses the sign.

Function

a rule giving exactly one output for each input of its domain.

Formula bank (3 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

FUNCTIONS, GRAPHS AND SEQUENCES

Straight line

$y = mx + c$, with gradient m and y-intercept c.

Gradient

$m = (y_2 - y_1) \div (x_2 - x_1)$.

Parallel and perpendicular

parallel lines share a gradient; perpendicular gradients multiply to -1.

Distance and midpoint

$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$; midpoint $= ((x_1 + x_2)/2, (y_1 + y_2)/2)$.

Parabola

$y = ax^2 + bx + c$; vertex form $y = a(x - h)^2 + k$ has vertex (h, k).

Axis of symmetry

$x = -b \div 2a$.

nth term, linear

a constant first difference gives term $= mn + c$.

nth term, quadratic

a constant second difference $2d$ gives term $= dn^2 + en + f$.

Recursive rule

each term is defined from the one before it; an initial value is required.

Compound change

final = initial × multiplier^n.

Direct variation

$y = kx^n$; inverse variation $y = k/x^n$.

Formula bank (4 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

SHAPE, SPACE AND MEASUREMENT

Congruence

SSS, SAS, ASA, RHS for triangles.

Similarity

corresponding angles equal and corresponding sides in one ratio k.

Scale laws

lengths scale by k, areas by k^2 and volumes by k^3.

Pythagoras

$a^2 + b^2 = c^2$ for the hypotenuse c of a right-angled triangle.

Trigonometric ratios

$\sin\theta$ = opposite/hypotenuse; $\cos\theta$ = adjacent/hypotenuse; $\tan\theta$ = opposite/adjacent.

Angles of elevation and depression

measured from the horizontal, and equal by alternate angles.

Sine rule

$a/\sin A = b/\sin B = c/\sin C$.

Cosine rule

$a^2 = b^2 + c^2 - 2bc\cos A$.

Area of a triangle

$\frac{1}{2}ab\sin C$.

Prism

volume = area of cross-section × length.

Cylinder

volume = $\pi r^2 h$; curved surface area = $2\pi rh$.

Cone

volume = $\frac{1}{3}\pi r^2 h$; curved surface area = πrl for slant height l.

Sphere

volume = $\frac{4}{3}\pi r^3$; surface area = $4\pi r^2$.

Frustum

a cone or pyramid with the top cut off parallel to the base.

Density

density = mass ÷ volume.

Formula bank (5 of 5)

Key facts and formulas, chapter by chapter. Use them to check work, not to replace reasoning.

DATA, PROBABILITY AND EVIDENCE

Frequency density

frequency ÷ class width; it is the height of a histogram bar.

Cumulative frequency

the running total of frequencies up to the upper class boundary.

Median and quartiles

the median halves the data; the quartiles quarter it.

Interquartile range

Q3 − Q1, a measure of spread that ignores the extremes.

Box plot

minimum, Q1, median, Q3, maximum on one axis.

Sample and population

a sample estimates a population parameter; bias makes it unreliable.

Stratified sampling

sample each group in proportion to its size in the population.

Probability of an event

P(event) = favourable outcomes ÷ total equally likely outcomes.

Addition rule

P(A or B) = P(A) + P(B) − P(A and B).

Multiplication rule

P(A and B) = P(A) × P(B) for independent events.

Conditional probability

P(A given B) = P(A and B) ÷ P(B).

Expected frequency

expected count = probability × number of trials.

Glossary (1 of 3)

('Algebraic fraction', 'A fraction whose numerator and denominator are algebraic expressions.', 2) ch. 2
A fraction whose numerator and denominator are algebraic expressions.

('Bounds', 'The greatest and least values a rounded quantity could have had.', 1) ch. 1
The greatest and least values a rounded quantity could have had.

('Box plot', 'A display of the minimum, lower quartile, median, upper quartile and maximum.', 12) ch. 12
A display of the minimum, lower quartile, median, upper quartile and maximum.

('Class width', 'The difference between the upper and lower boundaries of a class interval.', 12) ch. 12
The difference between the upper and lower boundaries of a class interval.

('Common factor', 'A factor shared by every term of an expression.', 2) ch. 2
A factor shared by every term of an expression.

('Completing the square', 'Rewriting a quadratic as a(x + p)² + q so that its vertex is visible.', 5) ch. 5
Rewriting a quadratic as $a(x + p)^2 + q$ so that its vertex is visible.

('Compound interest', 'Interest earned on interest, modelled by repeated multiplication.', 8) ch. 8
Interest earned on interest, modelled by repeated multiplication.

('Conditional probability', 'The probability of an event given that another event has occurred.', 13) ch. 13
The probability of an event given that another event has occurred.

('Congruent', 'Identical in shape and size; for triangles, SSS, SAS, ASA or RHS proves it.', 9) ch. 9
Identical in shape and size; for triangles, SSS, SAS, ASA or RHS proves it.

('Conjecture', 'A mathematical claim believed to be true but not yet justified.', 7) ch. 7
A mathematical claim believed to be true but not yet justified.

('Conjugate', 'The pair a + √b and a − √b, whose product is rational.', 1) ch. 1
The pair $a + \sqrt{b}$ and $a - \sqrt{b}$, whose product is rational.

('Correlation', 'The strength and direction of a linear association between two variables.', 12) ch. 12
The strength and direction of a linear association between two variables.

('Cumulative frequency', 'The running total of frequencies up to a class boundary.', 12) ch. 12
The running total of frequencies up to a class boundary.

('Dependent events', 'Events where the outcome of one changes the probability of the other.', 13) ch. 13
Events where the outcome of one changes the probability of the other.

('Depreciation', 'A loss of value, modelled by repeated multiplication by a factor below 1.', 8) ch. 8
A loss of value, modelled by repeated multiplication by a factor below 1.

('Discriminant', 'b² − 4ac, which decides how many real roots a quadratic has.', 5) ch. 5
$b^2 - 4ac$, which decides how many real roots a quadratic has.

('Domain', 'The set of permitted inputs of a function.', 4) ch. 4
The set of permitted inputs of a function.

('Enlargement', 'A transformation that multiplies every distance from a centre by one factor.', 9) ch. 9
A transformation that multiplies every distance from a centre by one factor.

('Error interval', 'The interval of values that would round to a stated value.', 1) ch. 1
The interval of values that would round to a stated value.

('Expected frequency', 'How often an event is expected to occur in a given number of trials.', 13) ch. 13
How often an event is expected to occur in a given number of trials.

('Factorise', 'Rewrite an expression as a product of simpler expressions.', 2) ch. 2
Rewrite an expression as a product of simpler expressions.

('Frequency density', 'Frequency divided by class width, used as the height of a histogram bar.', 12) ch. 12
Frequency divided by class width, used as the height of a histogram bar.

('Function', 'A rule that gives exactly one output for each permitted input.', 4) ch. 4
A rule that gives exactly one output for each permitted input.

('Gradient', 'The change in y per unit change in x; the steepness of a line.', 4) ch. 4
The change in y per unit change in x; the steepness of a line.

Glossary (2 of 3)

('Histogram', 'A display of grouped continuous data in which area represents frequency.', 12) ch. 12
A display of grouped continuous data in which area represents frequency.

('Independent events', 'Events where neither changes the probability of the other.', 13) ch. 13
Events where neither changes the probability of the other.

('Index', 'The power to which a base is raised; plural indices.', 1) ch. 1
The power to which a base is raised; plural indices.

('Interquartile range', 'The difference between the upper and lower quartiles.', 12) ch. 12
The difference between the upper and lower quartiles.

('Intersection', 'The set of outcomes in both of two sets; for events, A and B.', 13) ch. 13
The set of outcomes in both of two sets; for events, A and B.

('Inverse variation', 'A relationship y = k/x^n in which one variable falls as the other rises.', 8) ch. 8
A relationship y = k/x^n in which one variable falls as the other rises.

('Lower quartile', 'The value below which a quarter of the data lies.', 12) ch. 12
The value below which a quarter of the data lies.

('Midpoint', 'The point halfway between two points, found by averaging each coordinate.', 4) ch. 4
The point halfway between two points, found by averaging each coordinate.

('Model', 'A mathematical description of a situation, built on stated assumptions.', 8) ch. 8
A mathematical description of a situation, built on stated assumptions.

('Multiplier', 'The single factor by which a quantity is repeatedly multiplied.', 8) ch. 8
The single factor by which a quantity is repeatedly multiplied.

('Mutually exclusive', 'Events that cannot both happen, so P(A or B) = P(A) + P(B).', 13) ch. 13
Events that cannot both happen, so P(A or B) = P(A) + P(B).

('Outlier', 'A value that lies well away from the rest of the data.', 12) ch. 12
A value that lies well away from the rest of the data.

('Parabola', 'The graph of a quadratic function, symmetric about a vertical axis.', 6) ch. 6
The graph of a quadratic function, symmetric about a vertical axis.

('Perpendicular', 'At right angles; two lines are perpendicular when their gradients multiply to −1.', 4) ch. 4
At right angles; two lines are perpendicular when their gradients multiply to −1.

('Population', 'The whole group a claim is about.', 12) ch. 12
The whole group a claim is about.

('Quadratic', 'An expression of degree two, containing a term in x² and no higher power.', 5) ch. 5
An expression of degree two, containing a term in x² and no higher power.

('Range (of a function)', 'The set of outputs a function can produce.', 4) ch. 4
The set of outputs a function can produce.

('Recursive rule', 'A rule that gives each term of a sequence from the term before it.', 7) ch. 7
A rule that gives each term of a sequence from the term before it.

('Sample space', 'The set of all possible outcomes of an experiment.', 13) ch. 13
The set of all possible outcomes of an experiment.

('Similar', 'Same shape, possibly different size: corresponding angles equal, sides in ratio.', 9) ch. 9
Same shape, possibly different size: corresponding angles equal, sides in ratio.

('Standard form', 'A number written as a × 10^n with 1 ≤ a < 10.', 1) ch. 1
A number written as a × 10^n with 1 ≤ a < 10.

('Stratified sample', 'A sample that takes each group in proportion to its share of the population.', 12) ch. 12
A sample that takes each group in proportion to its share of the population.

('Surd', 'An exact irrational root left in root form, such as √2 or 3√5.', 1) ch. 1
An exact irrational root left in root form, such as √2 or 3√5.

('Transformation', 'A rule that moves or changes a shape: translation, reflection, rotation, enlargement.', 9) ch. 9
A rule that moves or changes a shape: translation, reflection, rotation, enlargement.

Glossary (3 of 3)

('Union', 'The set of outcomes in either of two sets; for events, A or B.', 13) ch. 13
The set of outcomes in either of two sets; for events, A or B.

('Upper quartile', 'The value below which three quarters of the data lies.', 12) ch. 12
The value below which three quarters of the data lies.

('Vertex', 'The turning point of a parabola, written (h, k) in vertex form.', 6) ch. 6
The turning point of a parabola, written (h, k) in vertex form.

('y-intercept', 'The value of y where a line crosses the y-axis.', 4) ch. 4
The value of y where a line crosses the y-axis.

REFERENCES

References and acknowledgements

Official IB sources

- IB Middle Years Programme curriculum — https://www.ibo.org/programmes/middle-years-programme/curriculum/ (page states last updated 19 February 2026). Used for: at least 50 hours per subject per year, global contexts, conceptual understanding, approaches to learning and service as action.
- IB MYP mathematics subject page — https://www.ibo.org/programmes/middle-years-programme/curriculum/mathematics/ Used for: the four branches of MYP mathematics, standard and extended mathematics, and the inquiry and application focus.
- IB MYP projects — https://www.ibo.org/programmes/middle-years-programme/curriculum/myp-projects/ (page states last updated 19 February 2026). Used for: the Community Project completed by students finishing MYP in Year 3 or Year 4, the Personal Project completed by Year 5 students, and the student-centred nature of both.
- IB MYP general regulations (2026 assessment procedures) — https://www.ibo.org/globalassets/new-structure/become-an-ib-school/pdfs/myp-2026-general-regulations.pdf Used for: the required use of MYP objectives and assessment criteria, minimum teaching hours, and Year 5 eAssessment eligibility.

Principal content reference

- Pre-rewrite MYP 4 chapter and subtopic audit (second-edition/audit/chapter-audits/myp4-audit.md with the chapter and subtopic registers). Used as the principal content reference for chapter scope, teaching order and the key concepts, related concepts, global contexts and inquiry questions of each chapter. The audit records that the earlier edition's course map and its chapter openers disagreed on the key concept and global context in eight chapters; the opener labels are used here, and each reconciliation is listed in the alignment review.

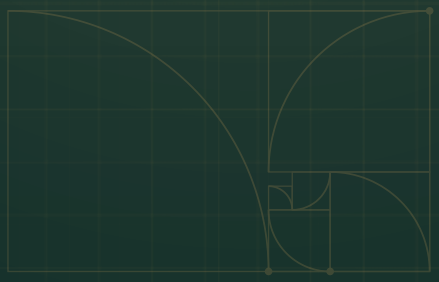
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- All classroom data sets are simulated and are labelled in the text. No claim in this book relies on unverified real-world data.

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ABOUT THIS BOOK

Thirteen chapters and a capstone that take number foundations, algebraic expressions and factorisation, linear equations and systems, functions and linear models, quadratics, sequences and generalisation, growth and financial mathematics, congruence and similarity, Pythagoras and trigonometry, surface area and volume, statistics and combined probability from first ideas to justified use. Each chapter opens with a task: you investigate, notice a pattern, state a mathematical conclusion you can defend, and only then meet the formal explanation — with worked examples that model a complete solution and practice separated into Criterion A, B, C and D sets so you can see which skill each task trains.



Year 4 reasoning is applied, not just correct: models are built from stated assumptions, checked against data and evaluated for their limitations, and Criterion B tasks ask you to justify a generalisation rather than to list cases. Every chapter closes with a reflection that returns to the statement of inquiry.



The book closes with a capstone: a sustained modelling task in which a design claim about space, cost, shade and water becomes defensible only when it is modelled, tested and evaluated. Every chapter also closes with a Service as Action opportunity, a Community Project connection and a Personal Project extension



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